

DRL Assignment 2 - Robust Reinforcement Learning under Stochastic Action Failure

Q-learning, DQN & DDQN on a modified LunarLander-v3

Group 165 - Contribution Declaration

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1. Environment Setup

We import the required libraries, define a global seeding utility (for full reproducibility across `random`, NumPy, PyTorch and the Gymnasium environment) and set the compute device.

```
In [18]: # -----  
# Imports  
# -----  
  
import os  
import random  
import time  
import math  
from collections import deque, namedtuple  
  
import numpy as np  
import torch  
import torch.nn as nn  
import torch.nn.functional as F  
import matplotlib.pyplot as plt  
  
import gymnasium as gym  
  
print("Library versions")  
print(" gymnasium :", gym.__version__)  
print(" torch      :", torch.__version__)  
print(" numpy       :", np.__version__)
```

```

# -----
# Global reproducibility helper
# -----
def set_global_seed(seed: int) -> None:
    """Seed every source of randomness so that experiments are reproducible.

    Seeds Python's ``random``, NumPy and PyTorch, and forces deterministic cuDNN
    Call this once before each agent is built so that all four configurations
    start from an identical state.
    """
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    if torch.cuda.is_available():
        torch.cuda.manual_seed_all(seed)
    torch.backends.cudnn.deterministic = True
    torch.backends.cudnn.benchmark = False

# -----
# Compute device
# -----
# The network is tiny and the batch is small, MLP (8 -> 128 -> 128 -> 4) so the
# CPU is as fast as a GPU.
DEVICE = torch.device("cpu")
print(f"Using device: {DEVICE}")

```

Library versions
 gymnasium : 1.3.0
 torch : 2.13.0+cpu
 numpy : 2.5.1
 Using device: cpu

2. Modified Environment: Implementation & Verification

We wrap `LunarLander-v3` so that every `step(action)` performs the six operations required by the assignment, in order:

1. **Receive & store** the agent's selected action `a`.
2. **Simulate intermittent engine failure:** if `a` in `{1, 2, 3}`, draw `r ~ U[0, 1)`; if `r < 0.15` replace the executed action with `0` (*Do Nothing*), otherwise keep `a`. `a = 0` is never modified. The executed action is `a_exec`.
3. **Execute** `a_exec` in the base environment.
4. **Modified reward:** $R = R_{base} - 0.3 \cdot 1_{a \neq 0} + B$. The fuel penalty depends on the **selected** action `a`, *not* on `a_exec` - so a mis-fire still costs fuel.
5. **Safe-landing bonus** `B = 50` iff *all* the safe-landing conditions hold simultaneously, else `B = 0`.
6. **Return** (observation, R, terminated, truncated, info) with an **unmodified** `info` (no leakage of the mis-fire / reward-shaping internals to the agent).

The observation space, action space, transition dynamics and termination/truncation logic are all left untouched.

```
In [19]: # =====
# Safe-landing predicate (pure function -> reusable & unit-testable)
# =====
# Observation layout of LunarLander-v3:
# [0]=x pos, [1]=y pos, [2]=x velocity, [3]=y velocity,
# [4]=angle, [5]=angular velocity, [6]=left-Leg contact, [7]=right-Leg contact
SAFE_VEL_TOL = 0.10 # |vx|, |vy| tolerance
SAFE_ANGLE_TOL = 0.10 # |angle| tolerance (radians)

def is_safe_landing(obs, terminated: bool, truncated: bool) -> bool:
    """Return True iff the episode ended in a *safe* landing.

    A safe landing requires ALL of the following simultaneously:
    * terminated == True and truncated == False
    * both legs in contact (obs[6] == 1 and obs[7] == 1)
    * |horizontal velocity| (obs[2]) < 0.10
    * |vertical velocity| (obs[3]) < 0.10
    * |orientation angle| (obs[4]) < 0.10 rad

    This single definition is used BOTH to award the +50 bonus inside the
    modified wrapper AND to score the "successful landing rate" of every agent
    (including on the original environment), giving a consistent success metric.
    """
    if not (terminated and not truncated):
        return False
    return bool(
        obs[6] == 1 and obs[7] == 1
        and abs(obs[2]) < SAFE_VEL_TOL
        and abs(obs[3]) < SAFE_VEL_TOL
        and abs(obs[4]) < SAFE_ANGLE_TOL
    )

# =====
# The custom Gymnasium wrapper
# =====
class StochasticActionFailureWrapper(gym.Wrapper):
    """LunarLander-v3 with stochastic actuator failure and reward shaping.

    Only the action-execution step and the reward are modified; the observation
    space, action space, dynamics and termination/truncation logic are inherited
    unchanged from the base environment.
    """

    MISFIRE_PROB = 0.15 # probability a thruster command mis-fires
    FUEL_PENALTY = 0.30 # penalty per attempted thruster action (selected, no
    LANDING_BONUS = 50.0 # bonus for a verified safe landing

    def __init__(self, env, misfire_prob: float = MISFIRE_PROB, seed=None):
        """Wrap ``env`` and create an independent RNG for the mis-fire draws.

        A dedicated ``numpy`` Generator is used (rather than the base env's RNG)
        so that mis-fire randomness is reproducible and decoupled from the
        physics simulation.
        """
```

```

super().__init__(env)
self.misfire_prob = misfire_prob
self._rng = np.random.default_rng(seed)
# Counters + last-step trace used only for our own verification (Task a)
# These are internal attributes; they are NEVER placed in the info dict.
self.reset_counters()
self.last_trace = {}

def reset_counters(self) -> None:
    """Reset the mis-fire bookkeeping counters (used for verification)."""
    self.n_thruster_attempts = 0 # how many times a thruster was selected
    self.n_misfires = 0 # how many of those were replaced by Do-N

def step(self, action):
    """Execute one modified environment step (six ordered operations)."""
    # --- Step 1: receive & store the agent's selected action -----
    a = int(action)

    # --- Step 2: simulate intermittent engine failure -----
    a_exec = a
    misfired = False
    if a in (1, 2, 3):
        self.n_thruster_attempts += 1
        r = float(self._rng.random()) # r ~U[0,1)
        if r < self.misfire_prob: # r < 0.15 -> mis-fire
            a_exec = 0
            misfired = True
            self.n_misfires += 1
    else:
        r = None

    # --- Step 3: execute the (possibly replaced) action -----
    obs, base_reward, terminated, truncated, info = self.env.step(a_exec)

    # --- Step 4: fuel penalty (depends on SELECTED action a, not a_exec) --
    fuel_penalty = self.FUEL_PENALTY if a != 0 else 0.0

    # --- Step 5: safe-landing bonus -----
    bonus = self.LANDING_BONUS if is_safe_landing(obs, terminated, truncated)

    # --- Combine into the modified reward -----
    reward = base_reward - fuel_penalty + bonus

    # Store a private trace for verification (NOT returned to the agent).
    self.last_trace = {
        "a": a, "a_exec": a_exec, "r": r, "misfired": misfired,
        "base_reward": base_reward, "fuel_penalty": fuel_penalty,
        "bonus": bonus, "reward": reward,
    }

    # --- Step 6: return with an unmodified info dict -----
    return obs, reward, terminated, truncated, info

# =====
# Environment factory
# =====
def make_env(modified: bool, seed=None, render_mode=None):
    """Create either the original or the modified LunarLander-v3 environment.

```

```

Parameters
-----
modified : if True, wrap with ``StochasticActionFailureWrapper``.
seed      : seed forwarded to the mis-fire RNG (modified env only).
"""

base = gym.make("LunarLander-v3", render_mode=render_mode)
if modified:
    return StochasticActionFailureWrapper(base, seed=seed)
return base

print("Environment classes defined.")

```

Environment classes defined.

2a. Unit tests for the safe-landing predicate

Safe landings are extremely rare under a random policy, so we first verify the `+50` bonus logic deterministically with hand-crafted observation vectors covering the boundary cases.

```

In [20]: def _obs(vx=0.0, vy=0.0, angle=0.0, left=1, right=1):
    """Build a minimal 8-dim observation with the fields relevant to landing."""
    o = np.zeros(8, dtype=np.float32)
    o[2], o[3], o[4], o[6], o[7] = vx, vy, angle, left, right
    return o

# (description, obs, terminated, truncated, expected_result)
_cases = [
    ("perfect landing", _obs(), True,
    ("too fast horizontally", _obs(vx=0.2), True,
    ("too fast vertically", _obs(vy=-0.5), True,
    ("tilted too much", _obs(angle=0.3), True,
    ("only one leg down", _obs(left=1, right=0), True,
    ("truncated (timeout)", _obs(), False,
    ("not terminated", _obs(), False,
    ("boundary vel (0.099 ok)", _obs(vx=0.099, vy=0.099, angle=0.099), True,
    ("boundary vel (0.10 fails)", _obs(vx=0.10), True,
]

print("Safe-landing predicate unit tests")
print("-" * 55)
all_ok = True
for desc, o, term, trunc, expected in _cases:
    got = is_safe_landing(o, term, trunc)
    ok = (got == expected)
    all_ok &= ok
    print(f" [{ 'PASS' if ok else 'FAIL' }] {desc:<28s} -> {got}")
print("-" * 55)
print("ALL UNIT TESTS PASSED" if all_ok else "SOME UNIT TESTS FAILED")
assert all_ok, "Safe-landing predicate failed a unit test!"

```

```

[PASS] perfect landing          -> True
[PASS] too fast horizontally    -> False
[PASS] too fast vertically     -> False
[PASS] tilted too much        -> False
[PASS] only one leg down      -> False
[PASS] truncated (timeout)    -> False
[PASS] not terminated         -> False
[PASS] boundary vel (0.099 ok) -> True
[PASS] boundary vel (0.10 fails) -> False

```

ALL UNIT TESTS PASSED

2b. Statistical verification with a random policy

We run many random-policy episodes on the modified environment and record, for **every step**, the private trace `(a, a_exec, base_reward, fuel_penalty, bonus, reward)`. From these traces we verify the three required properties:

1. **~15 %** of attempted thruster actions are replaced by *Do Nothing*.
2. The **0.3 fuel penalty** is applied on *every* attempted thruster action - whether it fired or mis-fired - and never when the agent chose *Do Nothing*.
3. The **reward decomposition** $R = R_{base} - 0.3 \cdot 1_{a \neq 0} + B$ holds exactly at every step (checked numerically).

```

In [21]: def verify_modified_env(n_episodes: int = 300, seed: int = 165):
    """Roll out a random policy and collect statistics that verify the wrapper.

    Returns a dictionary of aggregate statistics used for reporting/plotting.
    """
    env = make_env(modified=True, seed=seed)
    rng = np.random.default_rng(seed)          # independent policy RNG
    env.reset(seed=seed)
    env.action_space.seed(seed)

    # Per-action-type mis-fire counters.
    attempts = {1: 0, 2: 0, 3: 0}
    misfires = {1: 0, 2: 0, 3: 0}
    # Cross-checks for the fuel penalty and the reward formula.
    penalty_when_thruster_ok = True           # penalty present for every a in {1,2,3}
    penalty_when_donothing_ok = True          # NO penalty when a == 0
    reward_formula_ok = True
    misfire_examples = []                    # a few (a, r, a_exec) rows for the repo

    for ep in range(n_episodes):
        obs, _ = env.reset()
        done = False
        while not done:
            a = int(rng.integers(0, env.action_space.n)) # uniform random action
            obs, reward, terminated, truncated, info = env.step(a)
            done = terminated or truncated
            tr = env.last_trace

            # (1) mis-fire accounting
            if a in (1, 2, 3):
                attempts[a] += 1

```

```

        if tr["misfired"]:
            misfires[a] += 1
            if len(misfire_examples) < 5:
                misfire_examples.append((tr["a"], round(tr["r"], 3), tr["r"] - tr["a"]))

# (2) fuel-penalty checks
if a in (1, 2, 3) and tr["fuel_penalty"] != 0.30:
    penalty_when_thruster_ok = False
if a == 0 and tr["fuel_penalty"] != 0.0:
    penalty_when_donothing_ok = False

# (3) reward-formula check (guard against float noise)
expected = tr["base_reward"] - tr["fuel_penalty"] + tr["bonus"]
if abs(expected - reward) > 1e-6:
    reward_formula_ok = False

# (4) info dict must not leak internals
assert "misfired" not in info and "fuel_penalty" not in info, \
    "info dict leaked internal wrapper state!"

total_attempts = sum(attempts.values())
total_misfires = sum(misfires.values())
env.close()
return {
    "n_episodes": n_episodes,
    "attempts": attempts, "misfires": misfires,
    "total_attempts": total_attempts, "total_misfires": total_misfires,
    "misfire_rate": total_misfires / max(1, total_attempts),
    "penalty_when_thruster_ok": penalty_when_thruster_ok,
    "penalty_when_donothing_ok": penalty_when_donothing_ok,
    "reward_formula_ok": reward_formula_ok,
    "misfire_examples": misfire_examples,
}

stats = verify_modified_env(n_episodes=300, seed=165)

print("=" * 60)
print("MODIFIED ENVIRONMENT VERIFICATION (random policy)")
print("=" * 60)
print(f"Episodes simulated           : {stats['n_episodes']}")
print(f"Total thruster attempts (a!=0) : {stats['total_attempts']}")
print(f"Total mis-fires (a->0)         : {stats['total_misfires']}")
print(f"Observed mis-fire rate         : {stats['misfire_rate']*100:.2f} % (t")
print("-" * 60)
print("Per-action mis-fire rate:")
for a in (1, 2, 3):
    att, mis = stats["attempts"][a], stats["misfires"][a]
    print(f"  action {a}: {mis:5d}/{att:5d} = {mis/max(1,att)*100:5.2f} %")
print("-" * 60)
print(f"Fuel penalty applied for EVERY thruster action : {stats['penalty_when_th")
print(f"No fuel penalty when action == 0               : {stats['penalty_when_do")
print(f"Reward formula R = R_base - 0.3*1[a!=0] + B      : {stats['reward_formula")
print("-" * 60)
print("Example mis-fire traces (selected a, random r, executed a_exec):")
for a, r, a_exec in stats["misfire_examples"]:
    print(f"  a={a}, r={r:.3f} < 0.15 -> a_exec={a_exec}")
print("=" * 60)

```

```

=====
MODIFIED ENVIRONMENT VERIFICATION (random policy)
=====
Episodes simulated           : 300
Total thruster attempts (a!=0) : 20685
Total mis-fires (a->0)       : 3078
Observed mis-fire rate       : 14.88 % (target = 15.00 %)
-----
Per-action mis-fire rate:
    action 1: 1046/ 7015 = 14.91 %
    action 2: 1001/ 6837 = 14.64 %
    action 3: 1031/ 6833 = 15.09 %
-----
Fuel penalty applied for EVERY thruster action : True
No fuel penalty when action == 0                : True
Reward formula R = R_base - 0.3*1[a!=0] + B      : True
-----
Example mis-fire traces (selected a, random r, executed a_exec):
    a=3, r=0.118 < 0.15 -> a_exec=0
    a=3, r=0.113 < 0.15 -> a_exec=0
    a=2, r=0.128 < 0.15 -> a_exec=0
    a=2, r=0.045 < 0.15 -> a_exec=0
    a=2, r=0.086 < 0.15 -> a_exec=0
=====

```

2c. Visual evidence of the ~15 % mis-fire rate

```

In [22]: # Bar chart: observed mis-fire rate per thruster action vs. the 15% target.
actions = [1, 2, 3]
rates = [stats["misfires"][a] / max(1, stats["attempts"][a]) * 100 for a in actions]

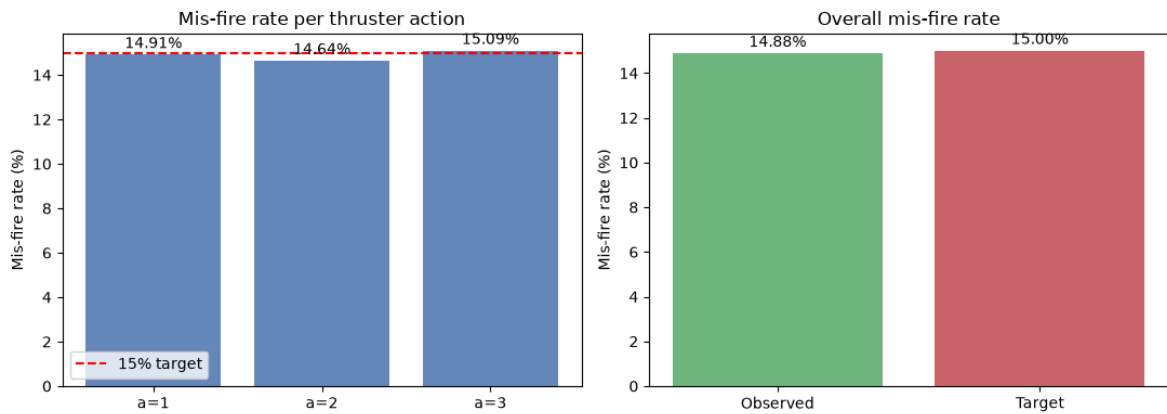
fig, ax = plt.subplots(1, 2, figsize=(11, 4))

ax[0].bar([f"a={a}" for a in actions], rates, color="#4C72B0", alpha=0.85)
ax[0].axhline(15.0, color="red", ls="--", label="15% target")
ax[0].set_ylabel("Mis-fire rate (%)")
ax[0].set_title("Mis-fire rate per thruster action")
ax[0].legend()
for i, v in enumerate(rates):
    ax[0].text(i, v + 0.3, f"{v:.2f}%", ha="center")

# Overall observed vs target.
ax[1].bar(["Observed", "Target"],
          [stats["misfire_rate"] * 100, 15.0],
          color=["#55A868", "#C44E52"], alpha=0.85)
ax[1].set_ylabel("Mis-fire rate (%)")
ax[1].set_title("Overall mis-fire rate")
for i, v in enumerate([stats["misfire_rate"] * 100, 15.0]):
    ax[1].text(i, v + 0.3, f"{v:.2f}%", ha="center")

plt.tight_layout()
plt.show()

```



Verification conclusion. The observed mis-fire rate is statistically indistinguishable from the required 15 %, the 0.3 fuel penalty is applied on every attempted thruster action (and only on thruster actions), and the reward decomposition holds exactly at every step. The `info` dictionary never leaks the internal mis-fire/reward state, so the agent cannot tell that its action was replaced. The wrapper is therefore correct.

3. DQN and DDQN Agents

Both agents share **identical** components:

- **Q-network:** MLP 8 -> 128 -> 128 -> 4 with ReLU activations.
- **Experience replay buffer:** uniform sampling, capacity 100 000 .
- **epsilon-greedy exploration:** epsilon decays exponentially from 1.0 to 0.02 .
- **Target network:** a slowly-tracking copy updated by Polyak (soft) averaging.

The single difference is the **target Q-value**:

$$y^{\text{DQN}} = r + \gamma (1 - d) \max_{a'} Q_{\theta^-}(s', a')$$

$$y^{\text{DDQN}} = r + \gamma (1 - d) Q_{\theta^-}(s', \arg \max_{a'} Q_{\theta}(s', a'))$$

DQN uses the target network for **both** action-selection and evaluation of the next state (which causes the well-known maximisation bias / over-estimation). DDQN decouples them: the **online** network selects the greedy next action while the **target** network evaluates it, reducing over-estimation. In code this is a single `if self.double:` branch.

```
In [23]: # =====
# Q-network
# =====
class QNetwork(nn.Module):
    """Feed-forward Q-network: state (8) -> hidden -> hidden -> Q-values (n_acti

    def __init__(self, state_dim: int, n_actions: int, hidden: int = 128):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(state_dim, hidden), nn.ReLU(),
            nn.Linear(hidden, hidden), nn.ReLU(),
            nn.Linear(hidden, n_actions),
```

```

    )

    def forward(self, x):
        """Map a batch of states to a batch of action-value vectors."""
        return self.net(x)

# =====
# Experience replay buffer
# =====
Transition = namedtuple("Transition", ["state", "action", "reward", "next_state"])

class ReplayBuffer:
    """Fixed-capacity circular buffer storing (s, a, r, s', done) transitions."""

    def __init__(self, capacity: int, seed: int = 0):
        self.buffer = deque(maxlen=capacity)
        self._rng = random.Random(seed) # dedicated RNG -> reproducible sampling

    def push(self, state, action, reward, next_state, done):
        """Append a single transition to the buffer (oldest is dropped when full)
        self.buffer.append(Transition(state, action, reward, next_state, done))

    def sample(self, batch_size: int):
        """Uniformly sample a mini-batch and return it as stacked NumPy arrays."""
        batch = self._rng.sample(self.buffer, batch_size)
        states = np.array([t.state for t in batch], dtype=np.float32)
        actions = np.array([t.action for t in batch], dtype=np.int64)
        rewards = np.array([t.reward for t in batch], dtype=np.float32)
        next_states = np.array([t.next_state for t in batch], dtype=np.float32)
        dones = np.array([t.done for t in batch], dtype=np.float32)
        return states, actions, rewards, next_states, dones

    def __len__(self):
        return len(self.buffer)

```

```

In [24]: # =====
# Unified value-based agent (DQN when double=False, DDQN when double=True)
# =====
class ValueBasedAgent:
    """A DQN / DDQN agent. The ONLY behavioural difference is the target Q-value

    Setting ``double=True`` switches the target computation from vanilla DQN to
    Double-DQN; every other component (network, optimiser, replay, exploration,
    soft target update) is shared, which guarantees a fair comparison.
    """

    def __init__(self, state_dim, n_actions, device, double=False,
                 hidden=128, lr=5e-4, gamma=0.99,
                 buffer_size=100_000, batch_size=64,
                 tau=5e-3, learn_start=1_000, update_every=4, seed=0):
        self.device = device
        self.n_actions = n_actions
        self.double = double # <-- DQN vs DDQN switch
        self.gamma = gamma
        self.batch_size = batch_size
        self.tau = tau # Polyak soft-update coefficient
        self.learn_start = learn_start # min transitions before learning starts
        self.update_every = update_every

```

```

self._step = 0

# Online and target networks (target starts as an exact copy).
self.q_online = QNetwork(state_dim, n_actions, hidden).to(device)
self.q_target = QNetwork(state_dim, n_actions, hidden).to(device)
self.q_target.load_state_dict(self.q_online.state_dict())
self.q_target.eval()

self.optimizer = torch.optim.Adam(self.q_online.parameters(), lr=lr)
self.buffer = ReplayBuffer(buffer_size, seed=seed)

# RNG for epsilon-greedy exploration (independent & reproducible).
self._rng = np.random.default_rng(seed)

# ----- acting
def select_action(self, state, epsilon: float) -> int:
    """epsilon-greedy action selection: random with prob epsilon, else greed
    if self._rng.random() < epsilon:
        return int(self._rng.integers(0, self.n_actions))
    return self.act_greedy(state)

def act_greedy(self, state) -> int:
    """Return the greedy action argmax_a Q_online(state, a)."""
    with torch.no_grad():
        s = torch.as_tensor(state, dtype=torch.float32, device=self.device).
        return int(self.q_online(s).argmax(dim=1).item())

# ----- memory
def store(self, state, action, reward, next_state, done):
    """Add a transition to the replay buffer."""
    self.buffer.push(state, action, reward, next_state, done)

# ----- Learnin
def learn_step(self):
    """Perform one gradient update every ``update_every`` environment steps.

    Returns the scalar loss when an update happens, else ``None``.
    """
    self._step += 1
    # Wait for enough warm-up transitions (and at least one full batch) befo
    if (len(self.buffer) < max(self.learn_start, self.batch_size)
        or self._step % self.update_every != 0):
        return None

    # Sample a mini-batch and move it to the compute device.
    states, actions, rewards, next_states, dones = self.buffer.sample(self.b
    states = torch.as_tensor(states, device=self.device)
    actions = torch.as_tensor(actions, device=self.device).unsqueeze(1)
    rewards = torch.as_tensor(rewards, device=self.device).unsqueeze(1)
    next_states = torch.as_tensor(next_states, device=self.device)
    dones = torch.as_tensor(dones, device=self.device).unsqueeze(1)

    # Current Q(s,a) for the taken actions.
    q_sa = self.q_online(states).gather(1, actions)

    # ---- Target Q-value : the ONLY difference between DQN and DDQN -----
    with torch.no_grad():
        if self.double:
            # DDQN: online net picks the action, target net evaluates it.
            next_actions = self.q_online(next_states).argmax(dim=1, keepdim=

```

```

        next_q = self.q_target(next_states).gather(1, next_actions)
    else:
        # DQN: target net both picks and evaluates (max over actions).
        next_q = self.q_target(next_states).max(dim=1, keepdim=True)[0]
        target = rewards + self.gamma * next_q * (1.0 - dones)
# -----

    loss = F.smooth_l1_loss(q_sa, target) # Huber Loss (robust to outliers)

    self.optimizer.zero_grad()
    loss.backward()
    nn.utils.clip_grad_norm_(self.q_online.parameters(), 10.0) # stabilise
    self.optimizer.step()

    self._soft_update()
    return float(loss.item())

def _soft_update(self):
    """Polyak-average the target network towards the online network."""
    with torch.no_grad():
        for tp, op in zip(self.q_target.parameters(), self.q_online.parameters()):
            tp.mul_(1.0 - self.tau).add_(self.tau * op)

# ----- metrics
@torch.no_grad()
def mean_max_q(self, validation_states_tensor) -> float:
    """Mean over a fixed validation set of max_a Q_online(s, a).

    This is the "average predicted Q-value" tracked during training. Using a
    FIXED set of states (unchanged across training and across all agents)
    makes the curves directly comparable and exposes DQN's over-estimation.
    """
    q = self.q_online(validation_states_tensor) # (N, n_actions)
    return float(q.max(dim=1)[0].mean().item())

```

Shared hyper-parameters & fixed validation set

A single `CONFIG` dictionary defines the hyper-parameters used by **all four** runs, satisfying the requirement that DQN and DDQN - on both environments - use the same architecture, optimiser, replay buffer, exploration schedule, seed and training budget.

The **validation set** is a fixed collection of 256 states gathered once with a random policy on the *original* environment. The same set is reused for every agent so that the "average predicted Q-value" curves are directly comparable.

```

In [25]: # ----- shared config
CONFIG = dict(
    state_dim=8,
    n_actions=4,
    hidden=128,
    lr=5e-4,
    gamma=0.99,
    buffer_size=100_000,
    batch_size=64,
    tau=5e-3, # soft target-update coefficient
    learn_start=1_000, # warm-up transitions before learning
    update_every=4, # one gradient step every 4 env steps

```

```

)
SEED = 165 # group number -> reproducible & memorable
NUM_EPISODES = 1000 # training budget per configuration
EPS_START, EPS_END, EPS_DECAY = 1.0, 0.02, 0.995 # epsilon-greedy schedule (pe
VALIDATION_SIZE = 256 # fixed validation set size

def collect_validation_states(n_states: int, seed: int) -> np.ndarray:
    """Collect a fixed set of states via a random policy on the ORIGINAL env.

    The set is deliberately drawn from the unmodified environment so it is a
    neutral reference shared by all four agents.
    """
    env = make_env(modified=False)
    rng = np.random.default_rng(seed)
    env.reset(seed=seed)
    env.action_space.seed(seed)
    states = []
    obs, _ = env.reset()
    while len(states) < n_states:
        states.append(obs.copy())
        a = int(rng.integers(0, env.action_space.n))
        obs, _, terminated, truncated, _ = env.step(a)
        if terminated or truncated:
            obs, _ = env.reset()
    env.close()
    return np.array(states[:n_states], dtype=np.float32)

VALIDATION_STATES = collect_validation_states(VALIDATION_SIZE, SEED)
VALIDATION_TENSOR = torch.as_tensor(VALIDATION_STATES, device=DEVICE)
print(f"Collected fixed validation set: {VALIDATION_STATES.shape}")

```

Collected fixed validation set: (256, 8)

```

In [26]: # =====
# Training loop (shared by all four configurations)
# =====
def train_agent(modified: bool, double: bool, tag: str,
                num_episodes: int = NUM_EPISODES, seed: int = SEED,
                log_every: int = 1):
    """Train one agent on one environment and return per-episode metric arrays.

    Parameters
    -----
    modified : train on the modified (True) or original (False) environment.
    double   : DDQN (True) or DQN (False).
    tag      : short label used in logs and plots.

    Returns a dict with per-episode lists:
        rewards, avg_q, landings (0/1), thrusters, steps.
    """
    # Reset ALL randomness so every configuration is identically initialised.
    set_global_seed(seed)

    env = make_env(modified=modified, seed=seed)
    env.reset(seed=seed)
    env.action_space.seed(seed)

    agent = ValueBasedAgent(device=DEVICE, double=double, seed=seed, **CONFIG)

```

```

ep_rewards, ep_avg_q, ep_landings, ep_thrusters, ep_steps = [], [], [], [],
epsilon = EPS_START
t_start = time.time()

for ep in range(1, num_episodes + 1):
    obs, _ = env.reset()
    done = False
    ep_reward, thrusters, steps = 0.0, 0, 0
    terminated = truncated = False

    while not done:
        # epsilon-greedy action; count thruster activations (selected, not e
        a = agent.select_action(obs, epsilon)
        if a in (1, 2, 3):
            thrusters += 1

        next_obs, reward, terminated, truncated, _ = env.step(a)
        done = terminated or truncated

        # Bootstrap only on genuine termination, not on time-limit truncatio
        agent.store(obs, a, reward, next_obs, float(terminated))
        agent.learn_step()

        obs = next_obs
        ep_reward += reward
        steps += 1

    # --- per-episode metrics -----
    landed = 1 if is_safe_landing(obs, terminated, truncated) else 0
    avg_q = agent.mean_max_q(VALIDATION_TENSOR)

    ep_rewards.append(ep_reward)
    ep_avg_q.append(avg_q)
    ep_landings.append(landed)
    ep_thrusters.append(thrusters)
    ep_steps.append(steps)

    # Decay exploration once per episode.
    epsilon = max(EPS_END, epsilon * EPS_DECAY)

    # --- per-episode logging (prints every iteration; the notebook UI keeps
    # the long output in a scrollable region) ---
    if ep % log_every == 0 or ep == num_episodes:
        avg_r100 = np.mean(ep_rewards[-100:])
        land100 = np.mean(ep_landings[-100:])
        print(f"[{tag}] Ep {ep:4d} | R {ep_reward:8.2f} | avgR100 {avg_r100:
              f"| eps {epsilon:.3f} | Qval {avg_q:7.2f} | land100 {land100:4
              f"| thr {thrusters:3d}")

env.close()
elapsed = time.time() - t_start
print(f"[{tag}] finished {num_episodes} episodes in {elapsed/60:.1f} min "
      f"(final avgR100 = {np.mean(ep_rewards[-100:]):.2f})\n")

return dict(tag=tag, modified=modified, double=double,
            rewards=ep_rewards, avg_q=ep_avg_q, landings=ep_landings,
            thrusters=ep_thrusters, steps=ep_steps, minutes=elapsed / 60,
            agent=agent)

```

4. Train all four configurations

We train the four required agents with **identical** settings (seed, network, optimiser, replay, epsilon-schedule and 1000-episode budget), differing only in *algorithm* (DQN/DDQN) and *environment* (original/modified):

1. **DQN - Original**
2. **DDQN - Original**
3. **DQN - Modified**
4. **DDQN - Modified**

Each configuration is trained in its **own cell** below, and **every episode** is printed (reward, 100-episode moving averages, epsilon, predicted Q-value, landing rate, thruster count) for full transparency; the notebook keeps the long output in a scrollable region. The full per-episode arrays are also retained in **RESULTS** for the plots and summary table.

```
In [27]: # Each configuration is trained in its OWN cell so the first/last 10 episodes
# of that run are easy to read. RESULTS is initialised here and extended below.
RESULTS = {}

# ---- Configuration 1 of 4: DQN on the ORIGINAL environment ----
RESULTS["DQN-Orig"] = train_agent(modified=False, double=False, tag="DQN-Orig")
```

[DQN-Orig]	Ep 1	R	-76.76	avgR100	-76.76	eps 0.995	Qval	0.05	1
and100 0.00	thr 43								
[DQN-Orig]	Ep 2	R	-111.39	avgR100	-94.07	eps 0.990	Qval	0.05	1
and100 0.00	thr 65								
[DQN-Orig]	Ep 3	R	-248.17	avgR100	-145.44	eps 0.985	Qval	0.05	1
and100 0.00	thr 64								
[DQN-Orig]	Ep 4	R	-122.59	avgR100	-139.73	eps 0.980	Qval	0.05	1
and100 0.00	thr 56								
[DQN-Orig]	Ep 5	R	-292.21	avgR100	-170.22	eps 0.975	Qval	0.05	1
and100 0.00	thr 73								
[DQN-Orig]	Ep 6	R	-118.75	avgR100	-161.64	eps 0.970	Qval	0.05	1
and100 0.00	thr 79								
[DQN-Orig]	Ep 7	R	-500.86	avgR100	-210.10	eps 0.966	Qval	0.05	1
and100 0.00	thr 74								
[DQN-Orig]	Ep 8	R	-112.02	avgR100	-197.84	eps 0.961	Qval	0.05	1
and100 0.00	thr 51								
[DQN-Orig]	Ep 9	R	-95.80	avgR100	-186.51	eps 0.956	Qval	0.05	1
and100 0.00	thr 59								
[DQN-Orig]	Ep 10	R	-224.33	avgR100	-190.29	eps 0.951	Qval	0.05	1
and100 0.00	thr 79								
[DQN-Orig]	Ep 11	R	-289.42	avgR100	-199.30	eps 0.946	Qval	0.05	1
and100 0.00	thr 81								
[DQN-Orig]	Ep 12	R	-102.24	avgR100	-191.21	eps 0.942	Qval	0.09	1
and100 0.00	thr 55								
[DQN-Orig]	Ep 13	R	-48.74	avgR100	-180.25	eps 0.937	Qval	0.15	1
and100 0.00	thr 63								
[DQN-Orig]	Ep 14	R	-498.84	avgR100	-203.01	eps 0.932	Qval	0.25	1
and100 0.00	thr 104								
[DQN-Orig]	Ep 15	R	-93.01	avgR100	-195.67	eps 0.928	Qval	0.15	1
and100 0.00	thr 56								
[DQN-Orig]	Ep 16	R	-185.01	avgR100	-195.01	eps 0.923	Qval	0.18	1
and100 0.00	thr 51								
[DQN-Orig]	Ep 17	R	-183.11	avgR100	-194.31	eps 0.918	Qval	0.15	1
and100 0.00	thr 106								
[DQN-Orig]	Ep 18	R	-301.05	avgR100	-200.24	eps 0.914	Qval	0.03	1
and100 0.00	thr 94								
[DQN-Orig]	Ep 19	R	-401.95	avgR100	-210.85	eps 0.909	Qval	-0.19	1
and100 0.00	thr 97								
[DQN-Orig]	Ep 20	R	-278.16	avgR100	-214.22	eps 0.905	Qval	-0.16	1
and100 0.00	thr 75								
[DQN-Orig]	Ep 21	R	-187.16	avgR100	-212.93	eps 0.900	Qval	-0.11	1
and100 0.00	thr 109								
[DQN-Orig]	Ep 22	R	-84.46	avgR100	-207.09	eps 0.896	Qval	0.24	1
and100 0.00	thr 50								
[DQN-Orig]	Ep 23	R	-242.43	avgR100	-208.63	eps 0.891	Qval	0.46	1
and100 0.00	thr 79								
[DQN-Orig]	Ep 24	R	29.91	avgR100	-198.69	eps 0.887	Qval	0.33	1
and100 0.00	thr 80								
[DQN-Orig]	Ep 25	R	-297.48	avgR100	-202.64	eps 0.882	Qval	0.50	1
and100 0.00	thr 75								
[DQN-Orig]	Ep 26	R	-120.85	avgR100	-199.50	eps 0.878	Qval	0.59	1
and100 0.00	thr 68								
[DQN-Orig]	Ep 27	R	-104.17	avgR100	-195.96	eps 0.873	Qval	0.44	1
and100									

[DQN-Orig]	Ep 31	R	-112.60	avgR100	-190.73	eps 0.856	Qval	0.77	1
and100 0.00	thr 46								
[DQN-Orig]	Ep 32	R	-96.21	avgR100	-187.78	eps 0.852	Qval	0.95	1
and100 0.00	thr 88								
[DQN-Orig]	Ep 33	R	-46.33	avgR100	-183.49	eps 0.848	Qval	0.95	1
and100 0.00	thr 56								
[DQN-Orig]	Ep 34	R	-192.85	avgR100	-183.77	eps 0.843	Qval	1.15	1
and100 0.00	thr 95								
[DQN-Orig]	Ep 35	R	-174.32	avgR100	-183.50	eps 0.839	Qval	1.10	1
and100 0.00	thr 86								
[DQN-Orig]	Ep 36	R	-111.92	avgR100	-181.51	eps 0.835	Qval	1.21	1
and100 0.00	thr 83								
[DQN-Orig]	Ep 37	R	-205.51	avgR100	-182.16	eps 0.831	Qval	1.30	1
and100 0.00	thr 74								
[DQN-Orig]	Ep 38	R	-64.97	avgR100	-179.07	eps 0.827	Qval	1.46	1
and100 0.00	thr 59								
[DQN-Orig]	Ep 39	R	-102.08	avgR100	-177.10	eps 0.822	Qval	1.44	1
and100 0.00	thr 76								
[DQN-Orig]	Ep 40	R	-149.77	avgR100	-176.42	eps 0.818	Qval	1.28	1
and100 0.00	thr 92								
[DQN-Orig]	Ep 41	R	-206.38	avgR100	-177.15	eps 0.814	Qval	1.69	1
and100 0.00	thr 89								
[DQN-Orig]	Ep 42	R	-102.53	avgR100	-175.37	eps 0.810	Qval	1.68	1
and100 0.00	thr 80								
[DQN-Orig]	Ep 43	R	-93.98	avgR100	-173.48	eps 0.806	Qval	1.66	1
and100 0.00	thr 53								
[DQN-Orig]	Ep 44	R	-58.83	avgR100	-170.87	eps 0.802	Qval	1.92	1
and100 0.00	thr 65								
[DQN-Orig]	Ep 45	R	-75.51	avgR100	-168.75	eps 0.798	Qval	1.64	1
and100 0.00	thr 75								
[DQN-Orig]	Ep 46	R	-112.71	avgR100	-167.54	eps 0.794	Qval	1.95	1
and100 0.00	thr 194								
[DQN-Orig]	Ep 47	R	-363.17	avgR100	-171.70	eps 0.790	Qval	2.10	1
and100 0.00	thr 107								
[DQN-Orig]	Ep 48	R	-112.66	avgR100	-170.47	eps 0.786	Qval	1.90	1
and100 0.00	thr 81								
[DQN-Orig]	Ep 49	R	-124.73	avgR100	-169.53	eps 0.782	Qval	1.95	1
and100 0.00	thr 98								
[DQN-Orig]	Ep 50	R	-133.19	avgR100	-168.81	eps 0.778	Qval	2.16	1
and100 0.00	thr 66								
[DQN-Orig]	Ep 51	R	-233.02	avgR100	-170.07	eps 0.774	Qval	2.15	1
and100 0.00	thr 105								
[DQN-Orig]	Ep 52	R	-81.03	avgR100	-168.35	eps 0.771	Qval	2.39	1
and100 0.00	thr 70								
[DQN-Orig]	Ep 53	R	-64.28	avgR100	-166.39	eps 0.767	Qval	2.23	1
and100 0.00	thr 96								
[DQN-Orig]	Ep 54	R	-77.10	avgR100	-164.74	eps 0.763	Qval	2.30	1
and100 0.00	thr 92								
[DQN-Orig]	Ep 55	R	-109.81	avgR100	-163.74	eps 0.759	Qval	2.79	1
and100 0.00	thr 94								
[DQN-Orig]	Ep 56	R	-51.20	avgR100	-161.73	eps 0.755	Qval	3.08	1
and100 0.00	thr 91								
[DQN-Orig]	Ep 57	R	-215.82	avgR100	-162.68	eps 0.751	Qval	2.67	1

[DQN-Orig]	Ep 61 R	-109.44	avgR100	-159.57	eps 0.737	Qval	3.04	1
and100 0.00	thr 99							
[DQN-Orig]	Ep 62 R	-76.78	avgR100	-158.23	eps 0.733	Qval	3.24	1
and100 0.00	thr 63							
[DQN-Orig]	Ep 63 R	-99.44	avgR100	-157.30	eps 0.729	Qval	3.32	1
and100 0.00	thr 86							
[DQN-Orig]	Ep 64 R	-182.40	avgR100	-157.69	eps 0.726	Qval	3.42	1
and100 0.00	thr 107							
[DQN-Orig]	Ep 65 R	-126.75	avgR100	-157.21	eps 0.722	Qval	3.48	1
and100 0.00	thr 73							
[DQN-Orig]	Ep 66 R	-241.92	avgR100	-158.50	eps 0.718	Qval	3.54	1
and100 0.00	thr 107							
[DQN-Orig]	Ep 67 R	-112.43	avgR100	-157.81	eps 0.715	Qval	3.67	1
and100 0.00	thr 101							
[DQN-Orig]	Ep 68 R	-67.17	avgR100	-156.48	eps 0.711	Qval	3.93	1
and100 0.00	thr 96							
[DQN-Orig]	Ep 69 R	-117.35	avgR100	-155.91	eps 0.708	Qval	3.65	1
and100 0.00	thr 77							
[DQN-Orig]	Ep 70 R	-44.94	avgR100	-154.33	eps 0.704	Qval	4.04	1
and100 0.00	thr 113							
[DQN-Orig]	Ep 71 R	-140.93	avgR100	-154.14	eps 0.701	Qval	4.10	1
and100 0.00	thr 101							
[DQN-Orig]	Ep 72 R	-61.34	avgR100	-152.85	eps 0.697	Qval	4.01	1
and100 0.00	thr 79							
[DQN-Orig]	Ep 73 R	-21.64	avgR100	-151.05	eps 0.694	Qval	4.36	1
and100 0.00	thr 116							
[DQN-Orig]	Ep 74 R	-90.49	avgR100	-150.23	eps 0.690	Qval	4.37	1
and100 0.00	thr 96							
[DQN-Orig]	Ep 75 R	-106.37	avgR100	-149.65	eps 0.687	Qval	4.61	1
and100 0.00	thr 97							
[DQN-Orig]	Ep 76 R	-86.73	avgR100	-148.82	eps 0.683	Qval	4.62	1
and100 0.00	thr 91							
[DQN-Orig]	Ep 77 R	-35.63	avgR100	-147.35	eps 0.680	Qval	4.56	1
and100 0.00	thr 82							
[DQN-Orig]	Ep 78 R	-90.59	avgR100	-146.62	eps 0.676	Qval	4.57	1
and100 0.00	thr 60							
[DQN-Orig]	Ep 79 R	-38.73	avgR100	-145.26	eps 0.673	Qval	4.88	1
and100 0.00	thr 69							
[DQN-Orig]	Ep 80 R	-62.12	avgR100	-144.22	eps 0.670	Qval	4.75	1
and100 0.00	thr 95							
[DQN-Orig]	Ep 81 R	-114.61	avgR100	-143.85	eps 0.666	Qval	5.00	1
and100 0.00	thr 105							
[DQN-Orig]	Ep 82 R	-68.70	avgR100	-142.93	eps 0.663	Qval	5.24	1
and100 0.00	thr 66							
[DQN-Orig]	Ep 83 R	-63.27	avgR100	-141.97	eps 0.660	Qval	5.19	1
and100 0.00	thr 154							
[DQN-Orig]	Ep 84 R	-144.46	avgR100	-142.00	eps 0.656	Qval	5.53	1
and100 0.00	thr 166							
[DQN-Orig]	Ep 85 R	-93.25	avgR100	-141.43	eps 0.653	Qval	5.46	1
and100 0.00	thr 70							
[DQN-Orig]	Ep 86 R	-69.36	avgR100	-140.59	eps 0.650	Qval	5.57	1
and100 0.00	thr 72							
[DQN-Orig]	Ep 87 R	-152.27	avgR100	-140.73	eps 0.647	Qval	5.84	1
and100 0.00	thr 103							
[DQN-Orig]	Ep 88 R	-67.92	avgR100	-139.90	eps 0.643	Qval	5.57	1
and100 0.00	thr 110							
[DQN-Orig]	Ep 89 R	-54.45	avgR100	-138.94	eps 0.640	Qval	5.89	1
and100 0.00	thr 56							
[DQN-Orig]	Ep 90 R	-123.01	avgR100	-138.76	eps 0.637	Qval	6.08	1
and100 0.00	thr 94							

[DQN-Orig]	Ep 91 R	-52.19	avgR100	-137.81	eps 0.634	Qval	6.26	1
and100 0.00	thr 103							
[DQN-Orig]	Ep 92 R	-64.84	avgR100	-137.02	eps 0.631	Qval	6.21	1
and100 0.00	thr 71							
[DQN-Orig]	Ep 93 R	-58.67	avgR100	-136.18	eps 0.627	Qval	6.55	1
and100 0.00	thr 93							
[DQN-Orig]	Ep 94 R	-12.35	avgR100	-134.86	eps 0.624	Qval	6.49	1
and100 0.00	thr 140							
[DQN-Orig]	Ep 95 R	-79.13	avgR100	-134.27	eps 0.621	Qval	6.71	1
and100 0.00	thr 105							
[DQN-Orig]	Ep 96 R	-13.36	avgR100	-133.01	eps 0.618	Qval	6.96	1
and100 0.00	thr 139							
[DQN-Orig]	Ep 97 R	-52.51	avgR100	-132.18	eps 0.615	Qval	7.48	1
and100 0.00	thr 110							
[DQN-Orig]	Ep 98 R	-43.36	avgR100	-131.28	eps 0.612	Qval	7.41	1
and100 0.00	thr 145							
[DQN-Orig]	Ep 99 R	-58.29	avgR100	-130.54	eps 0.609	Qval	7.84	1
and100 0.00	thr 196							
[DQN-Orig]	Ep 100 R	-64.56	avgR100	-129.88	eps 0.606	Qval	7.79	1
and100 0.00	thr 73							
[DQN-Orig]	Ep 101 R	-222.96	avgR100	-131.34	eps 0.603	Qval	8.48	1
and100 0.00	thr 133							
[DQN-Orig]	Ep 102 R	-41.65	avgR100	-130.64	eps 0.600	Qval	8.37	1
and100 0.00	thr 139							
[DQN-Orig]	Ep 103 R	-92.04	avgR100	-129.08	eps 0.597	Qval	8.38	1
and100 0.00	thr 160							
[DQN-Orig]	Ep 104 R	-110.35	avgR100	-128.96	eps 0.594	Qval	9.06	1
and100 0.00	thr 296							
[DQN-Orig]	Ep 105 R	-7.70	avgR100	-126.11	eps 0.591	Qval	9.06	1
and100 0.00	thr 74							
[DQN-Orig]	Ep 106 R	-67.65	avgR100	-125.60	eps 0.588	Qval	9.09	1
and100 0.00	thr 106							
[DQN-Orig]	Ep 107 R	-41.07	avgR100	-121.01	eps 0.585	Qval	9.30	1
and100 0.00	thr 75							
[DQN-Orig]	Ep 108 R	-44.08	avgR100	-120.33	eps 0.582	Qval	9.66	1
and100 0.00	thr 99							
[DQN-Orig]	Ep 109 R	-68.10	avgR100	-120.05	eps 0.579	Qval	10.00	1
and100 0.00	thr 121							
[DQN-Orig]	Ep 110 R	-73.15	avgR100	-118.54	eps 0.576	Qval	10.01	1
and100 0.00	thr 114							
[DQN-Orig]	Ep 111 R	-37.58	avgR100	-116.02	eps 0.573	Qval	10.27	1
and100 0.00	thr 62							
[DQN-Orig]	Ep 112 R	-109.32	avgR100	-116.09	eps 0.570	Qval	9.96	1
and100 0.00	thr 104							
[DQN-Orig]	Ep 113 R	-33.58	avgR100	-115.94	eps 0.568	Qval	10.13	1
and100 0.00	thr 135							
[DQN-Orig]	Ep 114 R	-35.60	avgR100	-111.31	eps 0.565	Qval	10.42	1
and100 0.00	thr 101							
[DQN-Orig]	Ep 115 R	-83.69	avgR100	-111.21	eps 0.562	Qval	10.56	1
and100 0.00	thr 127							
[DQN-Orig]	Ep 116 R	-68.56	avgR100	-110.05	eps 0.559	Qval	10.65	1
and100 0.00	thr 55							
[DQN-Orig]	Ep 117 R	-122.48	avgR100	-109.44	eps 0.556	Qval	10.50	1
and100 0.00	thr 100							
[DQN-Orig]	Ep 118 R	18.89	avgR100	-106.24	eps 0.554	Qval	11.31	1
and100 0.00	thr 193							
[DQN-Orig]	Ep 119 R	-31.25	avgR100	-102.54	eps 0.551	Qval	11.72	1
and100 0.00	thr 125							
[DQN-Orig]	Ep 120 R	-54.63	avgR100	-100.30	eps 0.548	Qval	11.75	1
and100 0.00	thr 161							

[DQN-Orig]	Ep 121 R	-49.99	avgR100	-98.93	eps 0.545	Qval 11.94	1
and100 0.00	thr 172						
[DQN-Orig]	Ep 122 R	22.22	avgR100	-97.86	eps 0.543	Qval 12.44	1
and100 0.00	thr 202						
[DQN-Orig]	Ep 123 R	-18.29	avgR100	-95.62	eps 0.540	Qval 12.46	1
and100 0.00	thr 111						
[DQN-Orig]	Ep 124 R	19.05	avgR100	-95.73	eps 0.537	Qval 12.99	1
and100 0.00	thr 147						
[DQN-Orig]	Ep 125 R	-26.63	avgR100	-93.02	eps 0.534	Qval 13.34	1
and100 0.00	thr 164						
[DQN-Orig]	Ep 126 R	-66.35	avgR100	-92.48	eps 0.532	Qval 13.22	1
and100 0.00	thr 108						
[DQN-Orig]	Ep 127 R	-30.76	avgR100	-91.74	eps 0.529	Qval 13.66	1
and100 0.00	thr 127						
[DQN-Orig]	Ep 128 R	-193.05	avgR100	-90.94	eps 0.526	Qval 13.67	1
and100 0.00	thr 118						
[DQN-Orig]	Ep 129 R	-25.45	avgR100	-89.87	eps 0.524	Qval 14.01	1
and100 0.00	thr 70						
[DQN-Orig]	Ep 130 R	-15.44	avgR100	-88.99	eps 0.521	Qval 14.27	1
and100 0.00	thr 115						
[DQN-Orig]	Ep 131 R	20.91	avgR100	-87.65	eps 0.519	Qval 14.76	1
and100 0.00	thr 154						
[DQN-Orig]	Ep 132 R	-38.82	avgR100	-87.08	eps 0.516	Qval 14.54	1
and100 0.00	thr 152						
[DQN-Orig]	Ep 133 R	13.63	avgR100	-86.48	eps 0.513	Qval 14.85	1
and100 0.00	thr 131						
[DQN-Orig]	Ep 134 R	-35.20	avgR100	-84.90	eps 0.511	Qval 14.99	1
and100 0.00	thr 140						
[DQN-Orig]	Ep 135 R	29.26	avgR100	-82.87	eps 0.508	Qval 15.18	1
and100 0.00	thr 108						
[DQN-Orig]	Ep 136 R	-6.74	avgR100	-81.82	eps 0.506	Qval 15.37	1
and100 0.00	thr 102						
[DQN-Orig]	Ep 137 R	13.72	avgR100	-79.63	eps 0.503	Qval 17.10	1
and100 0.00	thr 854						
[DQN-Orig]	Ep 138 R	-43.12	avgR100	-79.41	eps 0.501	Qval 17.21	1
and100 0.00	thr 95						
[DQN-Orig]	Ep 139 R	62.09	avgR100	-77.76	eps 0.498	Qval 18.55	1
and100 0.00	thr 869						
[DQN-Orig]	Ep 140 R	32.37	avgR100	-75.94	eps 0.496	Qval 19.84	1
and100 0.00	thr 866						
[DQN-Orig]	Ep 141 R	43.83	avgR100	-73.44	eps 0.493	Qval 21.65	1
and100 0.00	thr 869						
[DQN-Orig]	Ep 142 R	-17.04	avgR100	-72.59	eps 0.491	Qval 21.99	1
and100 0.00	thr 129						
[DQN-Orig]	Ep 143 R	33.63	avgR100	-71.31	eps 0.488	Qval 23.17	1
and100 0.00	thr 814						
[DQN-Orig]	Ep 144 R	12.46	avgR100	-70.60	eps 0.486	Qval 23.38	1
and100 0.00	thr 130						
[DQN-Orig]	Ep 145 R	-331.03	avgR100	-73.15	eps 0.483	Qval 23.67	1
and100 0.00	thr 152						
[DQN-Orig]	Ep 146 R	-77.59	avgR100	-72.80	eps 0.481	Qval 25.03	1
and100 0.00	thr 875						
[DQN-Orig]	Ep 147 R	-31.16	avgR100	-69.48	eps 0.479	Qval 25.27	1
and100 0.00	thr 280						
[DQN-Orig]	Ep 148 R	-0.23	avgR100	-68.36	eps 0.476	Qval 25.80	1
and100 0.00	thr 84						
[DQN-Orig]	Ep 149 R	8.13	avgR100	-67.03	eps 0.474	Qval 26.74	1
and100 0.00	thr 784						
[DQN-Orig]	Ep 150 R	-69.73	avgR100	-66.39	eps 0.471	Qval 27.87	1
and100 0.00	thr 834						

[DQN-Orig]	Ep 151 R	45.05 avgR100	-63.61 eps 0.469 Qval	28.01 1
and100 0.00	thr 109			
[DQN-Orig]	Ep 152 R	28.11 avgR100	-62.52 eps 0.467 Qval	27.59 1
and100 0.00	thr 109			
[DQN-Orig]	Ep 153 R	-63.78 avgR100	-62.52 eps 0.464 Qval	27.90 1
and100 0.00	thr 190			
[DQN-Orig]	Ep 154 R	-36.82 avgR100	-62.11 eps 0.462 Qval	28.95 1
and100 0.00	thr 872			
[DQN-Orig]	Ep 155 R	-21.72 avgR100	-61.23 eps 0.460 Qval	29.84 1
and100 0.00	thr 849			
[DQN-Orig]	Ep 156 R	-127.48 avgR100	-62.00 eps 0.458 Qval	30.04 1
and100 0.00	thr 142			
[DQN-Orig]	Ep 157 R	-53.95 avgR100	-60.38 eps 0.455 Qval	30.72 1
and100 0.00	thr 860			
[DQN-Orig]	Ep 158 R	-34.92 avgR100	-59.88 eps 0.453 Qval	30.24 1
and100 0.00	thr 80			
[DQN-Orig]	Ep 159 R	-22.11 avgR100	-59.27 eps 0.451 Qval	30.48 1
and100 0.00	thr 252			
[DQN-Orig]	Ep 160 R	-37.66 avgR100	-57.81 eps 0.448 Qval	31.67 1
and100 0.00	thr 861			
[DQN-Orig]	Ep 161 R	-41.34 avgR100	-57.13 eps 0.446 Qval	31.30 1
and100 0.00	thr 95			
[DQN-Orig]	Ep 162 R	-34.46 avgR100	-56.70 eps 0.444 Qval	30.99 1
and100 0.00	thr 142			
[DQN-Orig]	Ep 163 R	7.92 avgR100	-55.63 eps 0.442 Qval	31.59 1
and100 0.00	thr 97			
[DQN-Orig]	Ep 164 R	-41.87 avgR100	-54.23 eps 0.440 Qval	31.89 1
and100 0.00	thr 874			
[DQN-Orig]	Ep 165 R	-30.35 avgR100	-53.26 eps 0.437 Qval	31.71 1
and100 0.00	thr 117			
[DQN-Orig]	Ep 166 R	-40.69 avgR100	-51.25 eps 0.435 Qval	31.56 1
and100 0.00	thr 57			
[DQN-Orig]	Ep 167 R	-298.62 avgR100	-53.11 eps 0.433 Qval	31.55 1
and100 0.00	thr 703			
[DQN-Orig]	Ep 168 R	-218.49 avgR100	-54.62 eps 0.431 Qval	31.57 1
and100 0.00	thr 161			
[DQN-Orig]	Ep 169 R	-29.94 avgR100	-53.75 eps 0.429 Qval	31.95 1
and100 0.00	thr 139			
[DQN-Orig]	Ep 170 R	-159.87 avgR100	-54.90 eps 0.427 Qval	31.92 1
and100 0.00	thr 786			
[DQN-Orig]	Ep 171 R	-1.86 avgR100	-53.51 eps 0.424 Qval	32.04 1
and100 0.00	thr 113			
[DQN-Orig]	Ep 172 R	-6.62 avgR100	-52.96 eps 0.422 Qval	31.35 1
and100 0.00	thr 856			
[DQN-Orig]	Ep 173 R	-2.90 avgR100	-52.77 eps 0.420 Qval	31.81 1
and100 0.00	thr 130			
[DQN-Orig]	Ep 174 R	-8.28 avgR100	-51.95 eps 0.418 Qval	31.69 1
and100 0.00	thr 867			
[DQN-Orig]	Ep 175 R	-25.16 avgR100	-51.14 eps 0.416 Qval	31.77 1
and100 0.00	thr 69			
[DQN-Orig]	Ep 176 R	-140.71 avgR100	-51.68 eps 0.414 Qval	31.48 1
and100 0.00	thr 134			
[DQN-Orig]	Ep 177 R	-8.81 avgR100	-51.41 eps 0.412 Qval	31.56 1
and100 0.00	thr 111			
[DQN-Orig]	Ep 178 R	-44.39 avgR100	-50.95 eps 0.410 Qval	31.78 1
and100 0.00	thr 856			
[DQN-Orig]	Ep 179 R	-33.12 avgR100	-50.89 eps 0.408 Qval	31.36 1
and100 0.00	thr 87			
[DQN-Orig]	Ep 180 R	-21.61 avgR100	-50.49 eps 0.406 Qval	31.03 1
and100 0.00	thr 862			

[DQN-Orig]	Ep 181 R	-3.94	avgR100	-49.38	eps 0.404	Qval	31.11	1
and100 0.00	thr 90							
[DQN-Orig]	Ep 182 R	-46.75	avgR100	-49.16	eps 0.402	Qval	30.75	1
and100 0.00	thr 100							
[DQN-Orig]	Ep 183 R	-26.72	avgR100	-48.80	eps 0.400	Qval	31.57	1
and100 0.00	thr 73							
[DQN-Orig]	Ep 184 R	-57.39	avgR100	-47.93	eps 0.398	Qval	31.13	1
and100 0.00	thr 241							
[DQN-Orig]	Ep 185 R	-15.70	avgR100	-47.15	eps 0.396	Qval	31.32	1
and100 0.00	thr 76							
[DQN-Orig]	Ep 186 R	-15.64	avgR100	-46.61	eps 0.394	Qval	31.13	1
and100 0.00	thr 120							
[DQN-Orig]	Ep 187 R	-59.28	avgR100	-45.68	eps 0.392	Qval	30.89	1
and100 0.00	thr 833							
[DQN-Orig]	Ep 188 R	16.48	avgR100	-44.84	eps 0.390	Qval	29.81	1
and100 0.00	thr 849							
[DQN-Orig]	Ep 189 R	-44.04	avgR100	-44.74	eps 0.388	Qval	29.68	1
and100 0.00	thr 865							
[DQN-Orig]	Ep 190 R	-118.41	avgR100	-44.69	eps 0.386	Qval	29.49	1
and100 0.00	thr 150							
[DQN-Orig]	Ep 191 R	-20.28	avgR100	-44.37	eps 0.384	Qval	28.99	1
and100 0.00	thr 875							
[DQN-Orig]	Ep 192 R	4.91	avgR100	-43.67	eps 0.382	Qval	28.26	1
and100 0.00	thr 116							
[DQN-Orig]	Ep 193 R	-34.23	avgR100	-43.43	eps 0.380	Qval	27.87	1
and100 0.00	thr 824							
[DQN-Orig]	Ep 194 R	-3.42	avgR100	-43.34	eps 0.378	Qval	27.76	1
and100 0.00	thr 841							
[DQN-Orig]	Ep 195 R	-1.21	avgR100	-42.56	eps 0.376	Qval	27.21	1
and100 0.00	thr 867							
[DQN-Orig]	Ep 196 R	-13.91	avgR100	-42.57	eps 0.374	Qval	26.26	1
and100 0.00	thr 826							
[DQN-Orig]	Ep 197 R	-69.26	avgR100	-42.73	eps 0.373	Qval	26.22	1
and100 0.00	thr 862							
[DQN-Orig]	Ep 198 R	-33.93	avgR100	-42.64	eps 0.371	Qval	25.28	1
and100 0.00	thr 865							
[DQN-Orig]	Ep 199 R	-110.69	avgR100	-43.16	eps 0.369	Qval	24.01	1
and100 0.00	thr 885							
[DQN-Orig]	Ep 200 R	-28.07	avgR100	-42.80	eps 0.367	Qval	23.91	1
and100 0.00	thr 870							
[DQN-Orig]	Ep 201 R	-7.72	avgR100	-40.65	eps 0.365	Qval	22.66	1
and100 0.00	thr 873							
[DQN-Orig]	Ep 202 R	-58.00	avgR100	-40.81	eps 0.363	Qval	21.88	1
and100 0.00	thr 861							
[DQN-Orig]	Ep 203 R	-140.43	avgR100	-41.29	eps 0.361	Qval	22.28	1
and100 0.00	thr 146							
[DQN-Orig]	Ep 204 R	-0.87	avgR100	-40.20	eps 0.360	Qval	21.66	1
and100 0.00	thr 96							
[DQN-Orig]	Ep 205 R	-33.89	avgR100	-40.46	eps 0.358	Qval	22.53	1
and100 0.00	thr 880							
[DQN-Orig]	Ep 206 R	-2.22	avgR100	-39.81	eps 0.356	Qval	21.07	1
and100 0.00	thr 829							
[DQN-Orig]	Ep 207 R	-36.59	avgR100	-39.76	eps 0.354	Qval	20.28	1
and100 0.00	thr 881							
[DQN-Orig]	Ep 208 R	-20.54	avgR100	-39.53	eps 0.353	Qval	19.65	1
and100 0.00	thr 878							
[DQN-Orig]	Ep 209 R	-25.35	avgR100	-39.10	eps 0.351	Qval	18.41	1
and100 0.00	thr 883							
[DQN-Orig]	Ep 210 R	30.41	avgR100	-38.06	eps 0.349	Qval	18.63	1
and100 0.00	thr 893							

[DQN-Orig]	Ep 211 R	-10.44	avgR100	-37.79	eps 0.347	Qval 18.68	1
and100 0.00	thr 870						
[DQN-Orig]	Ep 212 R	7.16	avgR100	-36.63	eps 0.346	Qval 17.45	1
and100 0.00	thr 885						
[DQN-Orig]	Ep 213 R	0.68	avgR100	-36.28	eps 0.344	Qval 17.38	1
and100 0.00	thr 206						
[DQN-Orig]	Ep 214 R	-25.86	avgR100	-36.19	eps 0.342	Qval 16.49	1
and100 0.00	thr 878						
[DQN-Orig]	Ep 215 R	42.52	avgR100	-34.92	eps 0.340	Qval 17.15	1
and100 0.00	thr 868						
[DQN-Orig]	Ep 216 R	-9.61	avgR100	-34.33	eps 0.339	Qval 15.25	1
and100 0.00	thr 872						
[DQN-Orig]	Ep 217 R	4.22	avgR100	-33.07	eps 0.337	Qval 15.35	1
and100 0.00	thr 869						
[DQN-Orig]	Ep 218 R	-19.07	avgR100	-33.45	eps 0.335	Qval 14.52	1
and100 0.00	thr 875						
[DQN-Orig]	Ep 219 R	34.30	avgR100	-32.79	eps 0.334	Qval 13.31	1
and100 0.00	thr 840						
[DQN-Orig]	Ep 220 R	21.46	avgR100	-32.03	eps 0.332	Qval 13.52	1
and100 0.00	thr 848						
[DQN-Orig]	Ep 221 R	-50.53	avgR100	-32.04	eps 0.330	Qval 13.22	1
and100 0.00	thr 901						
[DQN-Orig]	Ep 222 R	39.06	avgR100	-31.87	eps 0.329	Qval 12.43	1
and100 0.00	thr 860						
[DQN-Orig]	Ep 223 R	9.85	avgR100	-31.59	eps 0.327	Qval 12.51	1
and100 0.00	thr 836						
[DQN-Orig]	Ep 224 R	17.25	avgR100	-31.60	eps 0.325	Qval 10.95	1
and100 0.00	thr 860						
[DQN-Orig]	Ep 225 R	39.61	avgR100	-30.94	eps 0.324	Qval 10.88	1
and100 0.00	thr 860						
[DQN-Orig]	Ep 226 R	20.86	avgR100	-30.07	eps 0.322	Qval 10.51	1
and100 0.00	thr 851						
[DQN-Orig]	Ep 227 R	-3.83	avgR100	-29.80	eps 0.321	Qval 10.77	1
and100 0.00	thr 802						
[DQN-Orig]	Ep 228 R	-28.35	avgR100	-28.15	eps 0.319	Qval 10.11	1
and100 0.00	thr 855						
[DQN-Orig]	Ep 229 R	63.16	avgR100	-27.27	eps 0.317	Qval 9.24	1
and100 0.00	thr 872						
[DQN-Orig]	Ep 230 R	3.50	avgR100	-27.08	eps 0.316	Qval 9.02	1
and100 0.00	thr 871						
[DQN-Orig]	Ep 231 R	-48.90	avgR100	-27.78	eps 0.314	Qval 9.29	1
and100 0.00	thr 882						
[DQN-Orig]	Ep 232 R	-20.76	avgR100	-27.60	eps 0.313	Qval 8.89	1
and100 0.00	thr 88						
[DQN-Orig]	Ep 233 R	35.87	avgR100	-27.37	eps 0.311	Qval 9.86	1
and100 0.00	thr 825						
[DQN-Orig]	Ep 234 R	53.24	avgR100	-26.49	eps 0.309	Qval 9.11	1
and100 0.00	thr 801						
[DQN-Orig]	Ep 235 R	12.77	avgR100	-26.65	eps 0.308	Qval 9.60	1
and100 0.00	thr 810						
[DQN-Orig]	Ep 236 R	-15.91	avgR100	-26.75	eps 0.306	Qval 9.94	1
and100 0.00	thr 848						
[DQN-Orig]	Ep 237 R	-1.67	avgR100	-26.90	eps 0.305	Qval 10.15	1
and100 0.00	thr 132						
[DQN-Orig]	Ep 238 R	5.67	avgR100	-26.41	eps 0.303	Qval 9.62	1
and100 0.00	thr 796						
[DQN-Orig]	Ep 239 R	-0.78	avgR100	-27.04	eps 0.302	Qval 9.24	1
and100 0.00	thr 120						
[DQN-Orig]	Ep 240 R	33.50	avgR100	-27.03	eps 0.300	Qval 8.89	1
and100 0.00	thr 114						

[DQN-Orig]	Ep 241 R	26.54 avgR100	-27.20 eps 0.299 Qval	9.10 1
and100 0.00	thr 838			
[DQN-Orig]	Ep 242 R	-4.85 avgR100	-27.08 eps 0.297 Qval	9.40 1
and100 0.00	thr 768			
[DQN-Orig]	Ep 243 R	-32.95 avgR100	-27.75 eps 0.296 Qval	10.62 1
and100 0.00	thr 147			
[DQN-Orig]	Ep 244 R	-4.74 avgR100	-27.92 eps 0.294 Qval	10.04 1
and100 0.00	thr 82			
[DQN-Orig]	Ep 245 R	6.41 avgR100	-24.54 eps 0.293 Qval	9.56 1
and100 0.00	thr 135			
[DQN-Orig]	Ep 246 R	85.21 avgR100	-22.92 eps 0.291 Qval	10.29 1
and100 0.00	thr 919			
[DQN-Orig]	Ep 247 R	-1.57 avgR100	-22.62 eps 0.290 Qval	8.98 1
and100 0.00	thr 798			
[DQN-Orig]	Ep 248 R	-43.42 avgR100	-23.05 eps 0.288 Qval	11.06 1
and100 0.00	thr 96			
[DQN-Orig]	Ep 249 R	-48.49 avgR100	-23.62 eps 0.287 Qval	9.09 1
and100 0.00	thr 776			
[DQN-Orig]	Ep 250 R	67.41 avgR100	-22.25 eps 0.286 Qval	9.75 1
and100 0.00	thr 818			
[DQN-Orig]	Ep 251 R	48.39 avgR100	-22.21 eps 0.284 Qval	9.12 1
and100 0.00	thr 845			
[DQN-Orig]	Ep 252 R	17.61 avgR100	-22.32 eps 0.283 Qval	9.50 1
and100 0.00	thr 856			
[DQN-Orig]	Ep 253 R	56.20 avgR100	-21.12 eps 0.281 Qval	9.59 1
and100 0.00	thr 731			
[DQN-Orig]	Ep 254 R	34.46 avgR100	-20.40 eps 0.280 Qval	9.84 1
and100 0.00	thr 773			
[DQN-Orig]	Ep 255 R	50.39 avgR100	-19.68 eps 0.279 Qval	9.17 1
and100 0.00	thr 812			
[DQN-Orig]	Ep 256 R	37.14 avgR100	-18.04 eps 0.277 Qval	9.51 1
and100 0.00	thr 774			
[DQN-Orig]	Ep 257 R	22.59 avgR100	-17.27 eps 0.276 Qval	9.31 1
and100 0.00	thr 796			
[DQN-Orig]	Ep 258 R	-24.56 avgR100	-17.17 eps 0.274 Qval	10.35 1
and100 0.00	thr 96			
[DQN-Orig]	Ep 259 R	8.35 avgR100	-16.86 eps 0.273 Qval	9.97 1
and100 0.00	thr 772			
[DQN-Orig]	Ep 260 R	30.13 avgR100	-16.19 eps 0.272 Qval	10.45 1
and100 0.00	thr 801			
[DQN-Orig]	Ep 261 R	73.97 avgR100	-15.03 eps 0.270 Qval	11.28 1
and100 0.00	thr 814			
[DQN-Orig]	Ep 262 R	187.44 avgR100	-12.81 eps 0.269 Qval	11.40 1
and100 0.01	thr 485			
[DQN-Orig]	Ep 263 R	2.17 avgR100	-12.87 eps 0.268 Qval	10.06 1
and100 0.01	thr 855			
[DQN-Orig]	Ep 264 R	-32.29 avgR100	-12.78 eps 0.266 Qval	9.66 1
and100 0.01	thr 858			
[DQN-Orig]	Ep 265 R	12.73 avgR100	-12.34 eps 0.265 Qval	10.36 1
and100 0.01	thr 802			
[DQN-Orig]	Ep 266 R	64.69 avgR100	-11.29 eps 0.264 Qval	10.28 1
and100 0.01	thr 672			
[DQN-Orig]	Ep 267 R	11.73 avgR100	-8.19 eps 0.262 Qval	9.84 1
and100 0.01	thr 841			
[DQN-Orig]	Ep 268 R	45.86 avgR100	-5.54 eps 0.261 Qval	11.65 1
and100 0.01	thr 110			
[DQN-Orig]	Ep 269 R	-12.61 avgR100	-5.37 eps 0.260 Qval	10.67 1
and100 0.01	thr 711			
[DQN-Orig]	Ep 270 R	60.12 avgR100	-3.17 eps 0.258 Qval	11.85 1
and100 0.01	thr 669			

[DQN-Orig]	Ep 271 R	92.52 avgR100	-2.23 eps 0.257 Qval	12.38 1
and100 0.01	thr 655			
[DQN-Orig]	Ep 272 R	73.12 avgR100	-1.43 eps 0.256 Qval	12.09 1
and100 0.01	thr 724			
[DQN-Orig]	Ep 273 R	28.65 avgR100	-1.11 eps 0.255 Qval	11.80 1
and100 0.01	thr 769			
[DQN-Orig]	Ep 274 R	18.87 avgR100	-0.84 eps 0.253 Qval	11.25 1
and100 0.01	thr 101			
[DQN-Orig]	Ep 275 R	40.97 avgR100	-0.18 eps 0.252 Qval	12.72 1
and100 0.01	thr 574			
[DQN-Orig]	Ep 276 R	74.35 avgR100	1.97 eps 0.251 Qval	12.76 1
and100 0.01	thr 740			
[DQN-Orig]	Ep 277 R	171.98 avgR100	3.78 eps 0.249 Qval	13.62 1
and100 0.01	thr 557			
[DQN-Orig]	Ep 278 R	30.06 avgR100	4.52 eps 0.248 Qval	13.29 1
and100 0.01	thr 751			
[DQN-Orig]	Ep 279 R	75.97 avgR100	5.61 eps 0.247 Qval	13.52 1
and100 0.01	thr 722			
[DQN-Orig]	Ep 280 R	27.60 avgR100	6.10 eps 0.246 Qval	12.96 1
and100 0.01	thr 803			
[DQN-Orig]	Ep 281 R	33.00 avgR100	6.47 eps 0.245 Qval	13.33 1
and100 0.01	thr 134			
[DQN-Orig]	Ep 282 R	67.14 avgR100	7.61 eps 0.243 Qval	14.05 1
and100 0.01	thr 678			
[DQN-Orig]	Ep 283 R	-8.81 avgR100	7.79 eps 0.242 Qval	15.36 1
and100 0.01	thr 770			
[DQN-Orig]	Ep 284 R	79.90 avgR100	9.16 eps 0.241 Qval	15.23 1
and100 0.01	thr 820			
[DQN-Orig]	Ep 285 R	109.03 avgR100	10.41 eps 0.240 Qval	15.10 1
and100 0.01	thr 676			
[DQN-Orig]	Ep 286 R	200.92 avgR100	12.58 eps 0.238 Qval	14.97 1
and100 0.02	thr 514			
[DQN-Orig]	Ep 287 R	197.04 avgR100	15.14 eps 0.237 Qval	15.26 1
and100 0.03	thr 486			
[DQN-Orig]	Ep 288 R	52.05 avgR100	15.50 eps 0.236 Qval	13.18 1
and100 0.03	thr 815			
[DQN-Orig]	Ep 289 R	4.02 avgR100	15.98 eps 0.235 Qval	15.51 1
and100 0.03	thr 818			
[DQN-Orig]	Ep 290 R	98.28 avgR100	18.14 eps 0.234 Qval	13.80 1
and100 0.03	thr 545			
[DQN-Orig]	Ep 291 R	-30.50 avgR100	18.04 eps 0.233 Qval	15.20 1
and100 0.03	thr 93			
[DQN-Orig]	Ep 292 R	-45.82 avgR100	17.53 eps 0.231 Qval	13.93 1
and100 0.03	thr 820			
[DQN-Orig]	Ep 293 R	10.58 avgR100	17.98 eps 0.230 Qval	15.49 1
and100 0.03	thr 873			
[DQN-Orig]	Ep 294 R	-61.31 avgR100	17.40 eps 0.229 Qval	14.80 1
and100 0.03	thr 884			
[DQN-Orig]	Ep 295 R	251.30 avgR100	19.93 eps 0.228 Qval	14.01 1
and100 0.04	thr 469			
[DQN-Orig]	Ep 296 R	223.36 avgR100	22.30 eps 0.227 Qval	14.56 1
and100 0.05	thr 440			
[DQN-Orig]	Ep 297 R	121.17 avgR100	24.21 eps 0.226 Qval	15.09 1
and100 0.05	thr 604			
[DQN-Orig]	Ep 298 R	296.73 avgR100	27.51 eps 0.225 Qval	16.75 1
and100 0.06	thr 279			
[DQN-Orig]	Ep 299 R	-10.73 avgR100	28.51 eps 0.223 Qval	16.80 1
and100 0.06	thr 890			
[DQN-Orig]	Ep 300 R	66.62 avgR100	29.46 eps 0.222 Qval	17.07 1
and100 0.06	thr 867			

[DQN-Orig]	Ep 301 R	0.76 avgR100	29.54 eps 0.221 Qval	16.28 1
and100 0.06	thr 100			
[DQN-Orig]	Ep 302 R	154.26 avgR100	31.67 eps 0.220 Qval	17.26 1
and100 0.06	thr 502			
[DQN-Orig]	Ep 303 R	20.38 avgR100	33.27 eps 0.219 Qval	17.32 1
and100 0.06	thr 890			
[DQN-Orig]	Ep 304 R	230.02 avgR100	35.58 eps 0.218 Qval	18.04 1
and100 0.07	thr 299			
[DQN-Orig]	Ep 305 R	110.53 avgR100	37.03 eps 0.217 Qval	18.33 1
and100 0.07	thr 517			
[DQN-Orig]	Ep 306 R	65.95 avgR100	37.71 eps 0.216 Qval	18.70 1
and100 0.07	thr 709			
[DQN-Orig]	Ep 307 R	234.80 avgR100	40.42 eps 0.215 Qval	19.26 1
and100 0.08	thr 393			
[DQN-Orig]	Ep 308 R	207.94 avgR100	42.71 eps 0.214 Qval	20.23 1
and100 0.09	thr 406			
[DQN-Orig]	Ep 309 R	5.68 avgR100	43.02 eps 0.212 Qval	18.96 1
and100 0.09	thr 75			
[DQN-Orig]	Ep 310 R	164.33 avgR100	44.36 eps 0.211 Qval	21.95 1
and100 0.09	thr 401			
[DQN-Orig]	Ep 311 R	25.49 avgR100	44.72 eps 0.210 Qval	21.36 1
and100 0.09	thr 120			
[DQN-Orig]	Ep 312 R	78.92 avgR100	45.43 eps 0.209 Qval	22.52 1
and100 0.09	thr 703			
[DQN-Orig]	Ep 313 R	103.22 avgR100	46.46 eps 0.208 Qval	23.33 1
and100 0.09	thr 447			
[DQN-Orig]	Ep 314 R	264.82 avgR100	49.37 eps 0.207 Qval	25.06 1
and100 0.10	thr 207			
[DQN-Orig]	Ep 315 R	256.74 avgR100	51.51 eps 0.206 Qval	25.29 1
and100 0.11	thr 435			
[DQN-Orig]	Ep 316 R	274.78 avgR100	54.35 eps 0.205 Qval	25.58 1
and100 0.12	thr 320			
[DQN-Orig]	Ep 317 R	110.36 avgR100	55.41 eps 0.204 Qval	26.50 1
and100 0.12	thr 695			
[DQN-Orig]	Ep 318 R	83.41 avgR100	56.44 eps 0.203 Qval	27.01 1
and100 0.12	thr 550			
[DQN-Orig]	Ep 319 R	147.23 avgR100	57.57 eps 0.202 Qval	26.89 1
and100 0.12	thr 675			
[DQN-Orig]	Ep 320 R	68.01 avgR100	58.03 eps 0.201 Qval	28.67 1
and100 0.12	thr 905			
[DQN-Orig]	Ep 321 R	108.76 avgR100	59.63 eps 0.200 Qval	29.63 1
and100 0.12	thr 407			
[DQN-Orig]	Ep 322 R	144.39 avgR100	60.68 eps 0.199 Qval	31.74 1
and100 0.12	thr 435			
[DQN-Orig]	Ep 323 R	223.51 avgR100	62.82 eps 0.198 Qval	29.08 1
and100 0.13	thr 266			
[DQN-Orig]	Ep 324 R	259.17 avgR100	65.24 eps 0.197 Qval	30.72 1
and100 0.14	thr 408			
[DQN-Orig]	Ep 325 R	210.16 avgR100	66.94 eps 0.196 Qval	31.61 1
and100 0.14	thr 352			
[DQN-Orig]	Ep 326 R	235.93 avgR100	69.09 eps 0.195 Qval	33.48 1
and100 0.15	thr 382			
[DQN-Orig]	Ep 327 R	258.44 avgR100	71.71 eps 0.194 Qval	32.51 1
and100 0.16	thr 336			
[DQN-Orig]	Ep 328 R	246.00 avgR100	74.46 eps 0.193 Qval	32.28 1
and100 0.17	thr 451			
[DQN-Orig]	Ep 329 R	136.96 avgR100	75.20 eps 0.192 Qval	33.68 1
and100 0.17	thr 425			
[DQN-Orig]	Ep 330 R	252.47 avgR100	77.69 eps 0.191 Qval	33.70 1
and100 0.18	thr 320			

[DQN-Orig]	Ep 331 R	262.23	avgR100	80.80	eps 0.190	Qval 33.18	1
and100 0.19	thr 409						
[DQN-Orig]	Ep 332 R	217.85	avgR100	83.18	eps 0.189	Qval 32.03	1
and100 0.20	thr 258						
[DQN-Orig]	Ep 333 R	172.80	avgR100	84.55	eps 0.188	Qval 35.36	1
and100 0.20	thr 422						
[DQN-Orig]	Ep 334 R	125.39	avgR100	85.27	eps 0.187	Qval 34.18	1
and100 0.20	thr 765						
[DQN-Orig]	Ep 335 R	-94.66	avgR100	84.20	eps 0.187	Qval 33.72	1
and100 0.20	thr 137						
[DQN-Orig]	Ep 336 R	232.36	avgR100	86.68	eps 0.186	Qval 34.03	1
and100 0.21	thr 487						
[DQN-Orig]	Ep 337 R	250.66	avgR100	89.21	eps 0.185	Qval 34.44	1
and100 0.22	thr 271						
[DQN-Orig]	Ep 338 R	108.20	avgR100	90.23	eps 0.184	Qval 35.44	1
and100 0.22	thr 424						
[DQN-Orig]	Ep 339 R	253.77	avgR100	92.78	eps 0.183	Qval 35.96	1
and100 0.23	thr 395						
[DQN-Orig]	Ep 340 R	23.51	avgR100	92.68	eps 0.182	Qval 34.79	1
and100 0.23	thr 257						
[DQN-Orig]	Ep 341 R	148.21	avgR100	93.89	eps 0.181	Qval 35.85	1
and100 0.23	thr 746						
[DQN-Orig]	Ep 342 R	210.77	avgR100	96.05	eps 0.180	Qval 34.89	1
and100 0.24	thr 339						
[DQN-Orig]	Ep 343 R	83.79	avgR100	97.22	eps 0.179	Qval 35.72	1
and100 0.24	thr 570						
[DQN-Orig]	Ep 344 R	107.07	avgR100	98.33	eps 0.178	Qval 36.71	1
and100 0.24	thr 654						
[DQN-Orig]	Ep 345 R	236.90	avgR100	100.64	eps 0.177	Qval 36.57	1
and100 0.25	thr 338						
[DQN-Orig]	Ep 346 R	-20.46	avgR100	99.58	eps 0.177	Qval 35.42	1
and100 0.25	thr 166						
[DQN-Orig]	Ep 347 R	283.90	avgR100	102.44	eps 0.176	Qval 36.79	1
and100 0.26	thr 208						
[DQN-Orig]	Ep 348 R	232.23	avgR100	105.19	eps 0.175	Qval 36.27	1
and100 0.27	thr 293						
[DQN-Orig]	Ep 349 R	260.96	avgR100	108.29	eps 0.174	Qval 35.12	1
and100 0.28	thr 421						
[DQN-Orig]	Ep 350 R	287.80	avgR100	110.49	eps 0.173	Qval 35.92	1
and100 0.29	thr 264						
[DQN-Orig]	Ep 351 R	153.14	avgR100	111.54	eps 0.172	Qval 36.56	1
and100 0.29	thr 491						
[DQN-Orig]	Ep 352 R	243.01	avgR100	113.79	eps 0.171	Qval 37.23	1
and100 0.29	thr 554						
[DQN-Orig]	Ep 353 R	243.51	avgR100	115.67	eps 0.170	Qval 36.72	1
and100 0.30	thr 378						
[DQN-Orig]	Ep 354 R	94.89	avgR100	116.27	eps 0.170	Qval 37.44	1
and100 0.30	thr 748						
[DQN-Orig]	Ep 355 R	237.18	avgR100	118.14	eps 0.169	Qval 37.22	1
and100 0.30	thr 420						
[DQN-Orig]	Ep 356 R	208.74	avgR100	119.86	eps 0.168	Qval 34.78	1
and100 0.31	thr 311						
[DQN-Orig]	Ep 357 R	243.56	avgR100	122.06	eps 0.167	Qval 36.52	1
and100 0.32	thr 273						
[DQN-Orig]	Ep 358 R	271.50	avgR100	125.03	eps 0.166	Qval 37.84	1
and100 0.33	thr 239						
[DQN-Orig]	Ep 359 R	92.81	avgR100	125.87	eps 0.165	Qval 36.67	1
and100 0.33	thr 765						
[DQN-Orig]	Ep 360 R	271.31	avgR100	128.28	eps 0.165	Qval 36.49	1
and100 0.34	thr 293						

[DQN-Orig]	Ep 361 R	103.07	avgR100	128.57	eps 0.164	Qval 35.67	1
and100 0.34	thr 548						
[DQN-Orig]	Ep 362 R	144.26	avgR100	128.14	eps 0.163	Qval 35.76	1
and100 0.33	thr 583						
[DQN-Orig]	Ep 363 R	239.66	avgR100	130.52	eps 0.162	Qval 36.23	1
and100 0.34	thr 498						
[DQN-Orig]	Ep 364 R	258.93	avgR100	133.43	eps 0.161	Qval 35.70	1
and100 0.34	thr 309						
[DQN-Orig]	Ep 365 R	249.63	avgR100	135.80	eps 0.160	Qval 36.27	1
and100 0.35	thr 193						
[DQN-Orig]	Ep 366 R	203.11	avgR100	137.18	eps 0.160	Qval 36.78	1
and100 0.35	thr 535						
[DQN-Orig]	Ep 367 R	285.20	avgR100	139.92	eps 0.159	Qval 34.66	1
and100 0.36	thr 241						
[DQN-Orig]	Ep 368 R	206.61	avgR100	141.52	eps 0.158	Qval 36.17	1
and100 0.36	thr 458						
[DQN-Orig]	Ep 369 R	245.99	avgR100	144.11	eps 0.157	Qval 34.08	1
and100 0.37	thr 471						
[DQN-Orig]	Ep 370 R	149.24	avgR100	145.00	eps 0.157	Qval 37.35	1
and100 0.37	thr 591						
[DQN-Orig]	Ep 371 R	234.04	avgR100	146.42	eps 0.156	Qval 35.82	1
and100 0.38	thr 215						
[DQN-Orig]	Ep 372 R	123.13	avgR100	146.92	eps 0.155	Qval 35.87	1
and100 0.38	thr 694						
[DQN-Orig]	Ep 373 R	238.51	avgR100	149.01	eps 0.154	Qval 36.02	1
and100 0.39	thr 149						
[DQN-Orig]	Ep 374 R	283.75	avgR100	151.66	eps 0.153	Qval 35.44	1
and100 0.40	thr 181						
[DQN-Orig]	Ep 375 R	239.19	avgR100	153.65	eps 0.153	Qval 34.98	1
and100 0.41	thr 263						
[DQN-Orig]	Ep 376 R	133.85	avgR100	154.24	eps 0.152	Qval 35.37	1
and100 0.41	thr 512						
[DQN-Orig]	Ep 377 R	240.48	avgR100	154.93	eps 0.151	Qval 36.47	1
and100 0.42	thr 474						
[DQN-Orig]	Ep 378 R	244.91	avgR100	157.07	eps 0.150	Qval 36.58	1
and100 0.43	thr 306						
[DQN-Orig]	Ep 379 R	210.54	avgR100	158.42	eps 0.150	Qval 37.07	1
and100 0.44	thr 476						
[DQN-Orig]	Ep 380 R	309.70	avgR100	161.24	eps 0.149	Qval 36.37	1
and100 0.45	thr 202						
[DQN-Orig]	Ep 381 R	242.19	avgR100	163.33	eps 0.148	Qval 36.32	1
and100 0.46	thr 260						
[DQN-Orig]	Ep 382 R	258.59	avgR100	165.25	eps 0.147	Qval 37.58	1
and100 0.47	thr 235						
[DQN-Orig]	Ep 383 R	266.23	avgR100	168.00	eps 0.147	Qval 37.64	1
and100 0.48	thr 214						
[DQN-Orig]	Ep 384 R	297.07	avgR100	170.17	eps 0.146	Qval 36.89	1
and100 0.49	thr 201						
[DQN-Orig]	Ep 385 R	259.85	avgR100	171.68	eps 0.145	Qval 38.67	1
and100 0.50	thr 251						
[DQN-Orig]	Ep 386 R	248.16	avgR100	172.15	eps 0.144	Qval 37.00	1
and100 0.50	thr 512						
[DQN-Orig]	Ep 387 R	277.53	avgR100	172.96	eps 0.144	Qval 37.87	1
and100 0.50	thr 269						
[DQN-Orig]	Ep 388 R	-8.26	avgR100	172.35	eps 0.143	Qval 37.43	1
and100 0.50	thr 248						
[DQN-Orig]	Ep 389 R	269.40	avgR100	175.01	eps 0.142	Qval 36.81	1
and100 0.51	thr 165						
[DQN-Orig]	Ep 390 R	275.30	avgR100	176.78	eps 0.142	Qval 37.55	1
and100 0.52	thr 262						

[DQN-Orig]	Ep 391 R	266.74 avgR100	179.75 eps 0.141 Qval	38.18 1
and100 0.53	thr 138			
[DQN-Orig]	Ep 392 R	278.98 avgR100	183.00 eps 0.140 Qval	37.48 1
and100 0.54	thr 235			
[DQN-Orig]	Ep 393 R	232.38 avgR100	185.21 eps 0.139 Qval	36.76 1
and100 0.55	thr 327			
[DQN-Orig]	Ep 394 R	226.64 avgR100	188.09 eps 0.139 Qval	37.13 1
and100 0.56	thr 356			
[DQN-Orig]	Ep 395 R	233.17 avgR100	187.91 eps 0.138 Qval	36.00 1
and100 0.56	thr 242			
[DQN-Orig]	Ep 396 R	185.32 avgR100	187.53 eps 0.137 Qval	39.27 1
and100 0.55	thr 590			
[DQN-Orig]	Ep 397 R	215.10 avgR100	188.47 eps 0.137 Qval	37.50 1
and100 0.56	thr 269			
[DQN-Orig]	Ep 398 R	250.68 avgR100	188.01 eps 0.136 Qval	37.55 1
and100 0.56	thr 345			
[DQN-Orig]	Ep 399 R	-0.10 avgR100	188.12 eps 0.135 Qval	37.22 1
and100 0.56	thr 215			
[DQN-Orig]	Ep 400 R	290.15 avgR100	190.35 eps 0.135 Qval	37.08 1
and100 0.57	thr 208			
[DQN-Orig]	Ep 401 R	233.50 avgR100	192.68 eps 0.134 Qval	37.79 1
and100 0.57	thr 325			
[DQN-Orig]	Ep 402 R	0.06 avgR100	191.14 eps 0.133 Qval	38.60 1
and100 0.57	thr 364			
[DQN-Orig]	Ep 403 R	237.28 avgR100	193.31 eps 0.133 Qval	38.02 1
and100 0.57	thr 244			
[DQN-Orig]	Ep 404 R	240.17 avgR100	193.41 eps 0.132 Qval	39.90 1
and100 0.57	thr 398			
[DQN-Orig]	Ep 405 R	211.59 avgR100	194.42 eps 0.131 Qval	39.59 1
and100 0.58	thr 358			
[DQN-Orig]	Ep 406 R	228.10 avgR100	196.04 eps 0.131 Qval	39.91 1
and100 0.59	thr 347			
[DQN-Orig]	Ep 407 R	296.72 avgR100	196.66 eps 0.130 Qval	40.60 1
and100 0.59	thr 171			
[DQN-Orig]	Ep 408 R	238.16 avgR100	196.96 eps 0.129 Qval	39.51 1
and100 0.59	thr 307			
[DQN-Orig]	Ep 409 R	244.22 avgR100	199.35 eps 0.129 Qval	40.29 1
and100 0.60	thr 347			
[DQN-Orig]	Ep 410 R	253.03 avgR100	200.23 eps 0.128 Qval	40.74 1
and100 0.61	thr 202			
[DQN-Orig]	Ep 411 R	240.40 avgR100	202.38 eps 0.127 Qval	40.18 1
and100 0.62	thr 187			
[DQN-Orig]	Ep 412 R	244.72 avgR100	204.04 eps 0.127 Qval	41.33 1
and100 0.62	thr 256			
[DQN-Orig]	Ep 413 R	276.11 avgR100	205.77 eps 0.126 Qval	40.65 1
and100 0.63	thr 168			
[DQN-Orig]	Ep 414 R	255.78 avgR100	205.68 eps 0.126 Qval	39.18 1
and100 0.63	thr 228			
[DQN-Orig]	Ep 415 R	294.47 avgR100	206.06 eps 0.125 Qval	40.61 1
and100 0.63	thr 216			
[DQN-Orig]	Ep 416 R	278.20 avgR100	206.09 eps 0.124 Qval	40.37 1
and100 0.63	thr 137			
[DQN-Orig]	Ep 417 R	105.37 avgR100	206.04 eps 0.124 Qval	39.34 1
and100 0.63	thr 626			
[DQN-Orig]	Ep 418 R	277.66 avgR100	207.98 eps 0.123 Qval	40.74 1
and100 0.64	thr 286			
[DQN-Orig]	Ep 419 R	233.87 avgR100	208.85 eps 0.122 Qval	41.95 1
and100 0.65	thr 181			
[DQN-Orig]	Ep 420 R	248.06 avgR100	210.65 eps 0.122 Qval	42.14 1
and100 0.66	thr 463			

[DQN-Orig]	Ep 421 R	282.48	avgR100	212.39	eps 0.121	Qval	42.41	1
and100 0.66	thr 249							
[DQN-Orig]	Ep 422 R	253.46	avgR100	213.48	eps 0.121	Qval	42.30	1
and100 0.67	thr 218							
[DQN-Orig]	Ep 423 R	223.87	avgR100	213.48	eps 0.120	Qval	43.21	1
and100 0.67	thr 313							
[DQN-Orig]	Ep 424 R	97.48	avgR100	211.87	eps 0.119	Qval	43.78	1
and100 0.66	thr 610							
[DQN-Orig]	Ep 425 R	288.77	avgR100	212.65	eps 0.119	Qval	43.81	1
and100 0.67	thr 226							
[DQN-Orig]	Ep 426 R	258.62	avgR100	212.88	eps 0.118	Qval	44.10	1
and100 0.67	thr 221							
[DQN-Orig]	Ep 427 R	274.05	avgR100	213.03	eps 0.118	Qval	43.69	1
and100 0.67	thr 183							
[DQN-Orig]	Ep 428 R	250.27	avgR100	213.08	eps 0.117	Qval	43.86	1
and100 0.67	thr 216							
[DQN-Orig]	Ep 429 R	232.91	avgR100	214.04	eps 0.116	Qval	44.10	1
and100 0.68	thr 258							
[DQN-Orig]	Ep 430 R	216.45	avgR100	213.68	eps 0.116	Qval	43.64	1
and100 0.67	thr 522							
[DQN-Orig]	Ep 431 R	259.84	avgR100	213.65	eps 0.115	Qval	42.62	1
and100 0.67	thr 191							
[DQN-Orig]	Ep 432 R	284.78	avgR100	214.32	eps 0.115	Qval	43.89	1
and100 0.67	thr 196							
[DQN-Orig]	Ep 433 R	246.63	avgR100	215.06	eps 0.114	Qval	45.05	1
and100 0.68	thr 356							
[DQN-Orig]	Ep 434 R	237.40	avgR100	216.18	eps 0.114	Qval	45.46	1
and100 0.69	thr 403							
[DQN-Orig]	Ep 435 R	289.16	avgR100	220.02	eps 0.113	Qval	43.87	1
and100 0.70	thr 181							
[DQN-Orig]	Ep 436 R	110.22	avgR100	218.80	eps 0.112	Qval	44.47	1
and100 0.69	thr 606							
[DQN-Orig]	Ep 437 R	256.46	avgR100	218.86	eps 0.112	Qval	44.75	1
and100 0.69	thr 237							
[DQN-Orig]	Ep 438 R	255.55	avgR100	220.33	eps 0.111	Qval	45.75	1
and100 0.69	thr 510							
[DQN-Orig]	Ep 439 R	244.52	avgR100	220.24	eps 0.111	Qval	45.34	1
and100 0.69	thr 267							
[DQN-Orig]	Ep 440 R	33.06	avgR100	220.33	eps 0.110	Qval	46.28	1
and100 0.69	thr 570							
[DQN-Orig]	Ep 441 R	232.14	avgR100	221.17	eps 0.110	Qval	45.60	1
and100 0.70	thr 187							
[DQN-Orig]	Ep 442 R	253.46	avgR100	221.60	eps 0.109	Qval	46.32	1
and100 0.70	thr 339							
[DQN-Orig]	Ep 443 R	224.20	avgR100	223.00	eps 0.109	Qval	44.92	1
and100 0.71	thr 267							
[DQN-Orig]	Ep 444 R	258.55	avgR100	224.52	eps 0.108	Qval	45.90	1
and100 0.72	thr 349							
[DQN-Orig]	Ep 445 R	262.57	avgR100	224.77	eps 0.107	Qval	46.25	1
and100 0.72	thr 182							
[DQN-Orig]	Ep 446 R	232.18	avgR100	227.30	eps 0.107	Qval	46.99	1
and100 0.73	thr 325							
[DQN-Orig]	Ep 447 R	256.42	avgR100	227.03	eps 0.106	Qval	47.03	1
and100 0.73	thr 189							
[DQN-Orig]	Ep 448 R	250.92	avgR100	227.21	eps 0.106	Qval	46.40	1
and100 0.73	thr 263							
[DQN-Orig]	Ep 449 R	228.68	avgR100	226.89	eps 0.105	Qval	47.54	1
and100 0.73	thr 512							
[DQN-Orig]	Ep 450 R	254.03	avgR100	226.55	eps 0.105	Qval	48.12	1
and100 0.73	thr 206							

[DQN-Orig]	Ep 451 R	173.39	avgR100	226.75	eps 0.104	Qval 47.26	1
and100 0.74	thr 622						
[DQN-Orig]	Ep 452 R	259.61	avgR100	226.92	eps 0.104	Qval 46.68	1
and100 0.75	thr 279						
[DQN-Orig]	Ep 453 R	253.49	avgR100	227.02	eps 0.103	Qval 47.82	1
and100 0.75	thr 268						
[DQN-Orig]	Ep 454 R	236.50	avgR100	228.44	eps 0.103	Qval 48.22	1
and100 0.76	thr 207						
[DQN-Orig]	Ep 455 R	250.91	avgR100	228.57	eps 0.102	Qval 46.82	1
and100 0.77	thr 208						
[DQN-Orig]	Ep 456 R	212.17	avgR100	228.61	eps 0.102	Qval 47.71	1
and100 0.76	thr 475						
[DQN-Orig]	Ep 457 R	246.84	avgR100	228.64	eps 0.101	Qval 48.58	1
and100 0.76	thr 276						
[DQN-Orig]	Ep 458 R	269.15	avgR100	228.62	eps 0.101	Qval 48.80	1
and100 0.76	thr 178						
[DQN-Orig]	Ep 459 R	213.77	avgR100	229.83	eps 0.100	Qval 48.40	1
and100 0.77	thr 615						
[DQN-Orig]	Ep 460 R	240.37	avgR100	229.52	eps 0.100	Qval 48.53	1
and100 0.77	thr 250						
[DQN-Orig]	Ep 461 R	241.10	avgR100	230.90	eps 0.099	Qval 48.73	1
and100 0.78	thr 146						
[DQN-Orig]	Ep 462 R	254.71	avgR100	232.00	eps 0.099	Qval 48.03	1
and100 0.79	thr 173						
[DQN-Orig]	Ep 463 R	237.38	avgR100	231.98	eps 0.098	Qval 48.60	1
and100 0.79	thr 291						
[DQN-Orig]	Ep 464 R	244.18	avgR100	231.83	eps 0.098	Qval 47.92	1
and100 0.80	thr 187						
[DQN-Orig]	Ep 465 R	249.60	avgR100	231.83	eps 0.097	Qval 48.92	1
and100 0.80	thr 198						
[DQN-Orig]	Ep 466 R	274.26	avgR100	232.54	eps 0.097	Qval 48.30	1
and100 0.81	thr 221						
[DQN-Orig]	Ep 467 R	280.33	avgR100	232.49	eps 0.096	Qval 48.83	1
and100 0.81	thr 143						
[DQN-Orig]	Ep 468 R	255.90	avgR100	232.99	eps 0.096	Qval 48.04	1
and100 0.82	thr 212						
[DQN-Orig]	Ep 469 R	219.47	avgR100	232.72	eps 0.095	Qval 47.60	1
and100 0.82	thr 227						
[DQN-Orig]	Ep 470 R	235.75	avgR100	233.59	eps 0.095	Qval 48.99	1
and100 0.83	thr 308						
[DQN-Orig]	Ep 471 R	146.28	avgR100	232.71	eps 0.094	Qval 48.76	1
and100 0.82	thr 712						
[DQN-Orig]	Ep 472 R	267.42	avgR100	234.15	eps 0.094	Qval 48.80	1
and100 0.83	thr 301						
[DQN-Orig]	Ep 473 R	298.66	avgR100	234.75	eps 0.093	Qval 50.66	1
and100 0.83	thr 200						
[DQN-Orig]	Ep 474 R	228.35	avgR100	234.20	eps 0.093	Qval 49.86	1
and100 0.83	thr 422						
[DQN-Orig]	Ep 475 R	255.13	avgR100	234.36	eps 0.092	Qval 51.41	1
and100 0.83	thr 299						
[DQN-Orig]	Ep 476 R	275.98	avgR100	235.78	eps 0.092	Qval 51.21	1
and100 0.84	thr 364						
[DQN-Orig]	Ep 477 R	231.38	avgR100	235.69	eps 0.092	Qval 50.35	1
and100 0.84	thr 325						
[DQN-Orig]	Ep 478 R	226.02	avgR100	235.50	eps 0.091	Qval 52.05	1
and100 0.84	thr 312						
[DQN-Orig]	Ep 479 R	281.20	avgR100	236.21	eps 0.091	Qval 50.28	1
and100 0.84	thr 137						
[DQN-Orig]	Ep 480 R	286.73	avgR100	235.98	eps 0.090	Qval 50.54	1
and100 0.84	thr 241						

[DQN-Orig]	Ep 481 R	216.30	avgR100	235.72	eps 0.090	Qval 48.48	1
and100 0.83	thr 436						
[DQN-Orig]	Ep 482 R	242.56	avgR100	235.56	eps 0.089	Qval 49.95	1
and100 0.83	thr 274						
[DQN-Orig]	Ep 483 R	16.88	avgR100	233.06	eps 0.089	Qval 49.07	1
and100 0.82	thr 766						
[DQN-Orig]	Ep 484 R	197.79	avgR100	232.07	eps 0.088	Qval 49.64	1
and100 0.82	thr 429						
[DQN-Orig]	Ep 485 R	244.26	avgR100	231.92	eps 0.088	Qval 49.57	1
and100 0.82	thr 232						
[DQN-Orig]	Ep 486 R	278.35	avgR100	232.22	eps 0.088	Qval 50.80	1
and100 0.81	thr 100						
[DQN-Orig]	Ep 487 R	298.63	avgR100	232.43	eps 0.087	Qval 50.43	1
and100 0.81	thr 187						
[DQN-Orig]	Ep 488 R	285.76	avgR100	235.37	eps 0.087	Qval 50.88	1
and100 0.82	thr 152						
[DQN-Orig]	Ep 489 R	277.06	avgR100	235.45	eps 0.086	Qval 49.80	1
and100 0.82	thr 215						
[DQN-Orig]	Ep 490 R	207.56	avgR100	234.77	eps 0.086	Qval 49.65	1
and100 0.82	thr 351						
[DQN-Orig]	Ep 491 R	277.40	avgR100	234.87	eps 0.085	Qval 48.18	1
and100 0.82	thr 190						
[DQN-Orig]	Ep 492 R	243.75	avgR100	234.52	eps 0.085	Qval 50.11	1
and100 0.82	thr 346						
[DQN-Orig]	Ep 493 R	282.11	avgR100	235.02	eps 0.084	Qval 48.26	1
and100 0.82	thr 139						
[DQN-Orig]	Ep 494 R	271.68	avgR100	235.47	eps 0.084	Qval 48.29	1
and100 0.82	thr 268						
[DQN-Orig]	Ep 495 R	229.16	avgR100	235.43	eps 0.084	Qval 48.22	1
and100 0.82	thr 183						
[DQN-Orig]	Ep 496 R	248.13	avgR100	236.06	eps 0.083	Qval 49.13	1
and100 0.83	thr 242						
[DQN-Orig]	Ep 497 R	280.60	avgR100	236.71	eps 0.083	Qval 48.38	1
and100 0.83	thr 166						
[DQN-Orig]	Ep 498 R	246.16	avgR100	236.67	eps 0.082	Qval 47.53	1
and100 0.83	thr 240						
[DQN-Orig]	Ep 499 R	255.63	avgR100	239.23	eps 0.082	Qval 47.35	1
and100 0.84	thr 182						
[DQN-Orig]	Ep 500 R	237.91	avgR100	238.70	eps 0.082	Qval 48.22	1
and100 0.84	thr 160						
[DQN-Orig]	Ep 501 R	220.22	avgR100	238.57	eps 0.081	Qval 46.52	1
and100 0.85	thr 271						
[DQN-Orig]	Ep 502 R	221.96	avgR100	240.79	eps 0.081	Qval 47.85	1
and100 0.86	thr 367						
[DQN-Orig]	Ep 503 R	25.70	avgR100	238.67	eps 0.080	Qval 47.43	1
and100 0.86	thr 796						
[DQN-Orig]	Ep 504 R	299.92	avgR100	239.27	eps 0.080	Qval 46.40	1
and100 0.86	thr 215						
[DQN-Orig]	Ep 505 R	255.84	avgR100	239.71	eps 0.080	Qval 46.89	1
and100 0.86	thr 143						
[DQN-Orig]	Ep 506 R	245.48	avgR100	239.89	eps 0.079	Qval 46.31	1
and100 0.86	thr 266						
[DQN-Orig]	Ep 507 R	259.27	avgR100	239.51	eps 0.079	Qval 45.63	1
and100 0.86	thr 213						
[DQN-Orig]	Ep 508 R	261.65	avgR100	239.75	eps 0.078	Qval 45.21	1
and100 0.86	thr 209						
[DQN-Orig]	Ep 509 R	262.11	avgR100	239.93	eps 0.078	Qval 44.04	1
and100 0.86	thr 288						
[DQN-Orig]	Ep 510 R	241.80	avgR100	239.81	eps 0.078	Qval 45.19	1
and100 0.86	thr 117						

[DQN-Orig]	Ep 511 R	261.51	avgR100	240.02	eps 0.077	Qval 44.74	1
and100 0.86	thr 198						
[DQN-Orig]	Ep 512 R	290.81	avgR100	240.49	eps 0.077	Qval 44.25	1
and100 0.87	thr 207						
[DQN-Orig]	Ep 513 R	21.92	avgR100	237.94	eps 0.076	Qval 44.59	1
and100 0.86	thr 829						
[DQN-Orig]	Ep 514 R	220.60	avgR100	237.59	eps 0.076	Qval 44.36	1
and100 0.86	thr 406						
[DQN-Orig]	Ep 515 R	246.58	avgR100	237.11	eps 0.076	Qval 44.14	1
and100 0.86	thr 215						
[DQN-Orig]	Ep 516 R	233.12	avgR100	236.66	eps 0.075	Qval 43.10	1
and100 0.86	thr 212						
[DQN-Orig]	Ep 517 R	223.06	avgR100	237.84	eps 0.075	Qval 44.11	1
and100 0.87	thr 196						
[DQN-Orig]	Ep 518 R	226.93	avgR100	237.33	eps 0.075	Qval 43.85	1
and100 0.87	thr 362						
[DQN-Orig]	Ep 519 R	271.11	avgR100	237.70	eps 0.074	Qval 42.63	1
and100 0.87	thr 252						
[DQN-Orig]	Ep 520 R	245.86	avgR100	237.68	eps 0.074	Qval 41.58	1
and100 0.87	thr 233						
[DQN-Orig]	Ep 521 R	221.84	avgR100	237.08	eps 0.073	Qval 42.01	1
and100 0.88	thr 241						
[DQN-Orig]	Ep 522 R	240.06	avgR100	236.94	eps 0.073	Qval 40.45	1
and100 0.88	thr 273						
[DQN-Orig]	Ep 523 R	252.79	avgR100	237.23	eps 0.073	Qval 42.46	1
and100 0.88	thr 302						
[DQN-Orig]	Ep 524 R	233.12	avgR100	238.59	eps 0.072	Qval 42.05	1
and100 0.89	thr 325						
[DQN-Orig]	Ep 525 R	252.55	avgR100	238.23	eps 0.072	Qval 39.72	1
and100 0.89	thr 337						
[DQN-Orig]	Ep 526 R	246.26	avgR100	238.10	eps 0.072	Qval 41.67	1
and100 0.89	thr 314						
[DQN-Orig]	Ep 527 R	252.54	avgR100	237.89	eps 0.071	Qval 40.69	1
and100 0.89	thr 225						
[DQN-Orig]	Ep 528 R	230.17	avgR100	237.69	eps 0.071	Qval 41.24	1
and100 0.89	thr 236						
[DQN-Orig]	Ep 529 R	299.93	avgR100	238.36	eps 0.071	Qval 41.62	1
and100 0.89	thr 184						
[DQN-Orig]	Ep 530 R	247.50	avgR100	238.67	eps 0.070	Qval 41.86	1
and100 0.90	thr 288						
[DQN-Orig]	Ep 531 R	283.55	avgR100	238.90	eps 0.070	Qval 40.33	1
and100 0.90	thr 186						
[DQN-Orig]	Ep 532 R	273.40	avgR100	238.79	eps 0.069	Qval 40.72	1
and100 0.90	thr 227						
[DQN-Orig]	Ep 533 R	309.44	avgR100	239.42	eps 0.069	Qval 41.31	1
and100 0.90	thr 175						
[DQN-Orig]	Ep 534 R	227.27	avgR100	239.32	eps 0.069	Qval 40.73	1
and100 0.90	thr 261						
[DQN-Orig]	Ep 535 R	258.00	avgR100	239.01	eps 0.068	Qval 40.28	1
and100 0.90	thr 270						
[DQN-Orig]	Ep 536 R	277.41	avgR100	240.68	eps 0.068	Qval 39.50	1
and100 0.91	thr 395						
[DQN-Orig]	Ep 537 R	300.43	avgR100	241.12	eps 0.068	Qval 39.57	1
and100 0.91	thr 175						
[DQN-Orig]	Ep 538 R	225.69	avgR100	240.82	eps 0.067	Qval 39.44	1
and100 0.92	thr 266						
[DQN-Orig]	Ep 539 R	246.27	avgR100	240.84	eps 0.067	Qval 41.00	1
and100 0.92	thr 350						
[DQN-Orig]	Ep 540 R	227.80	avgR100	242.78	eps 0.067	Qval 39.29	1
and100 0.93	thr 351						

[DQN-Orig]	Ep 541 R	259.61	avgR100	243.06	eps 0.066	Qval 39.92	1
and100 0.93	thr 191						
[DQN-Orig]	Ep 542 R	233.75	avgR100	242.86	eps 0.066	Qval 39.11	1
and100 0.93	thr 320						
[DQN-Orig]	Ep 543 R	262.37	avgR100	243.24	eps 0.066	Qval 37.88	1
and100 0.93	thr 246						
[DQN-Orig]	Ep 544 R	230.76	avgR100	242.96	eps 0.065	Qval 39.09	1
and100 0.93	thr 250						
[DQN-Orig]	Ep 545 R	247.30	avgR100	242.81	eps 0.065	Qval 40.29	1
and100 0.93	thr 383						
[DQN-Orig]	Ep 546 R	251.53	avgR100	243.01	eps 0.065	Qval 40.26	1
and100 0.93	thr 269						
[DQN-Orig]	Ep 547 R	262.50	avgR100	243.07	eps 0.064	Qval 38.75	1
and100 0.93	thr 162						
[DQN-Orig]	Ep 548 R	240.59	avgR100	242.96	eps 0.064	Qval 38.96	1
and100 0.93	thr 185						
[DQN-Orig]	Ep 549 R	251.73	avgR100	243.19	eps 0.064	Qval 39.92	1
and100 0.93	thr 234						
[DQN-Orig]	Ep 550 R	285.40	avgR100	243.51	eps 0.063	Qval 39.92	1
and100 0.93	thr 231						
[DQN-Orig]	Ep 551 R	244.63	avgR100	244.22	eps 0.063	Qval 40.01	1
and100 0.92	thr 227						
[DQN-Orig]	Ep 552 R	261.94	avgR100	244.24	eps 0.063	Qval 38.73	1
and100 0.92	thr 226						
[DQN-Orig]	Ep 553 R	262.00	avgR100	244.33	eps 0.063	Qval 38.83	1
and100 0.92	thr 186						
[DQN-Orig]	Ep 554 R	298.00	avgR100	244.94	eps 0.062	Qval 39.03	1
and100 0.92	thr 211						
[DQN-Orig]	Ep 555 R	255.94	avgR100	244.99	eps 0.062	Qval 38.15	1
and100 0.92	thr 183						
[DQN-Orig]	Ep 556 R	244.86	avgR100	245.32	eps 0.062	Qval 37.88	1
and100 0.93	thr 276						
[DQN-Orig]	Ep 557 R	239.80	avgR100	245.25	eps 0.061	Qval 38.25	1
and100 0.93	thr 135						
[DQN-Orig]	Ep 558 R	287.64	avgR100	245.43	eps 0.061	Qval 39.18	1
and100 0.93	thr 171						
[DQN-Orig]	Ep 559 R	264.32	avgR100	245.94	eps 0.061	Qval 38.59	1
and100 0.92	thr 247						
[DQN-Orig]	Ep 560 R	243.44	avgR100	245.97	eps 0.060	Qval 38.48	1
and100 0.92	thr 217						
[DQN-Orig]	Ep 561 R	301.44	avgR100	246.57	eps 0.060	Qval 38.34	1
and100 0.92	thr 164						
[DQN-Orig]	Ep 562 R	260.38	avgR100	246.63	eps 0.060	Qval 38.83	1
and100 0.92	thr 300						
[DQN-Orig]	Ep 563 R	273.00	avgR100	246.99	eps 0.059	Qval 38.72	1
and100 0.92	thr 184						
[DQN-Orig]	Ep 564 R	247.19	avgR100	247.02	eps 0.059	Qval 38.42	1
and100 0.92	thr 265						
[DQN-Orig]	Ep 565 R	253.80	avgR100	247.06	eps 0.059	Qval 38.37	1
and100 0.92	thr 199						
[DQN-Orig]	Ep 566 R	241.85	avgR100	246.74	eps 0.059	Qval 38.28	1
and100 0.92	thr 200						
[DQN-Orig]	Ep 567 R	237.53	avgR100	246.31	eps 0.058	Qval 39.37	1
and100 0.92	thr 267						
[DQN-Orig]	Ep 568 R	307.72	avgR100	246.83	eps 0.058	Qval 38.69	1
and100 0.92	thr 156						
[DQN-Orig]	Ep 569 R	227.61	avgR100	246.91	eps 0.058	Qval 38.29	1
and100 0.92	thr 221						
[DQN-Orig]	Ep 570 R	259.32	avgR100	247.14	eps 0.057	Qval 39.41	1
and100 0.92	thr 254						

[DQN-Orig]	Ep 571 R	219.19	avgR100	247.87	eps 0.057	Qval	38.59	1
and100 0.92	thr 210							
[DQN-Orig]	Ep 572 R	228.27	avgR100	247.48	eps 0.057	Qval	39.60	1
and100 0.92	thr 216							
[DQN-Orig]	Ep 573 R	258.56	avgR100	247.08	eps 0.057	Qval	38.31	1
and100 0.92	thr 153							
[DQN-Orig]	Ep 574 R	225.44	avgR100	247.05	eps 0.056	Qval	38.87	1
and100 0.92	thr 321							
[DQN-Orig]	Ep 575 R	250.88	avgR100	247.01	eps 0.056	Qval	37.35	1
and100 0.92	thr 279							
[DQN-Orig]	Ep 576 R	249.19	avgR100	246.74	eps 0.056	Qval	37.53	1
and100 0.92	thr 166							
[DQN-Orig]	Ep 577 R	235.07	avgR100	246.78	eps 0.055	Qval	37.06	1
and100 0.91	thr 371							
[DQN-Orig]	Ep 578 R	274.86	avgR100	247.27	eps 0.055	Qval	36.23	1
and100 0.91	thr 283							
[DQN-Orig]	Ep 579 R	244.90	avgR100	246.90	eps 0.055	Qval	38.02	1
and100 0.91	thr 260							
[DQN-Orig]	Ep 580 R	215.89	avgR100	246.19	eps 0.055	Qval	37.55	1
and100 0.91	thr 309							
[DQN-Orig]	Ep 581 R	268.52	avgR100	246.72	eps 0.054	Qval	36.96	1
and100 0.92	thr 141							
[DQN-Orig]	Ep 582 R	239.03	avgR100	246.68	eps 0.054	Qval	36.95	1
and100 0.92	thr 367							
[DQN-Orig]	Ep 583 R	236.41	avgR100	248.88	eps 0.054	Qval	38.03	1
and100 0.92	thr 252							
[DQN-Orig]	Ep 584 R	246.37	avgR100	249.36	eps 0.054	Qval	36.68	1
and100 0.92	thr 177							
[DQN-Orig]	Ep 585 R	223.58	avgR100	249.15	eps 0.053	Qval	36.51	1
and100 0.92	thr 429							
[DQN-Orig]	Ep 586 R	254.50	avgR100	248.92	eps 0.053	Qval	37.77	1
and100 0.93	thr 164							
[DQN-Orig]	Ep 587 R	231.52	avgR100	248.25	eps 0.053	Qval	37.58	1
and100 0.93	thr 167							
[DQN-Orig]	Ep 588 R	253.09	avgR100	247.92	eps 0.052	Qval	37.05	1
and100 0.93	thr 159							
[DQN-Orig]	Ep 589 R	225.75	avgR100	247.41	eps 0.052	Qval	34.84	1
and100 0.93	thr 332							
[DQN-Orig]	Ep 590 R	219.89	avgR100	247.53	eps 0.052	Qval	36.03	1
and100 0.93	thr 324							
[DQN-Orig]	Ep 591 R	278.59	avgR100	247.54	eps 0.052	Qval	35.95	1
and100 0.93	thr 199							
[DQN-Orig]	Ep 592 R	279.99	avgR100	247.90	eps 0.051	Qval	35.91	1
and100 0.92	thr 222							
[DQN-Orig]	Ep 593 R	266.96	avgR100	247.75	eps 0.051	Qval	35.17	1
and100 0.92	thr 362							
[DQN-Orig]	Ep 594 R	285.33	avgR100	247.89	eps 0.051	Qval	36.04	1
and100 0.91	thr 291							
[DQN-Orig]	Ep 595 R	261.63	avgR100	248.21	eps 0.051	Qval	34.84	1
and100 0.91	thr 315							
[DQN-Orig]	Ep 596 R	237.07	avgR100	248.10	eps 0.050	Qval	36.05	1
and100 0.90	thr 333							
[DQN-Orig]	Ep 597 R	286.43	avgR100	248.16	eps 0.050	Qval	35.89	1
and100 0.90	thr 138							
[DQN-Orig]	Ep 598 R	214.14	avgR100	247.84	eps 0.050	Qval	36.44	1
and100 0.90	thr 270							
[DQN-Orig]	Ep 599 R	254.82	avgR100	247.83	eps 0.050	Qval	35.24	1
and100 0.90	thr 299							
[DQN-Orig]	Ep 600 R	245.63	avgR100	247.91	eps 0.049	Qval	34.97	1
and100 0.89	thr 313							

[DQN-Orig]	Ep 601 R	263.10	avgR100	248.34	eps 0.049	Qval 34.85	1
and100 0.89	thr 282						
[DQN-Orig]	Ep 602 R	254.97	avgR100	248.67	eps 0.049	Qval 34.24	1
and100 0.88	thr 303						
[DQN-Orig]	Ep 603 R	258.82	avgR100	251.00	eps 0.049	Qval 34.55	1
and100 0.89	thr 186						
[DQN-Orig]	Ep 604 R	252.44	avgR100	250.52	eps 0.048	Qval 35.00	1
and100 0.89	thr 297						
[DQN-Orig]	Ep 605 R	252.98	avgR100	250.50	eps 0.048	Qval 35.08	1
and100 0.89	thr 237						
[DQN-Orig]	Ep 606 R	268.58	avgR100	250.73	eps 0.048	Qval 34.96	1
and100 0.89	thr 177						
[DQN-Orig]	Ep 607 R	203.93	avgR100	250.17	eps 0.048	Qval 34.69	1
and100 0.88	thr 367						
[DQN-Orig]	Ep 608 R	207.47	avgR100	249.63	eps 0.047	Qval 34.91	1
and100 0.87	thr 241						
[DQN-Orig]	Ep 609 R	264.85	avgR100	249.66	eps 0.047	Qval 35.11	1
and100 0.86	thr 303						
[DQN-Orig]	Ep 610 R	274.39	avgR100	249.98	eps 0.047	Qval 34.77	1
and100 0.86	thr 221						
[DQN-Orig]	Ep 611 R	222.12	avgR100	249.59	eps 0.047	Qval 34.08	1
and100 0.86	thr 323						
[DQN-Orig]	Ep 612 R	250.77	avgR100	249.19	eps 0.047	Qval 35.25	1
and100 0.86	thr 137						
[DQN-Orig]	Ep 613 R	222.78	avgR100	251.20	eps 0.046	Qval 35.53	1
and100 0.87	thr 165						
[DQN-Orig]	Ep 614 R	230.14	avgR100	251.29	eps 0.046	Qval 36.48	1
and100 0.86	thr 500						
[DQN-Orig]	Ep 615 R	259.89	avgR100	251.43	eps 0.046	Qval 36.42	1
and100 0.86	thr 157						
[DQN-Orig]	Ep 616 R	240.35	avgR100	251.50	eps 0.046	Qval 36.19	1
and100 0.86	thr 128						
[DQN-Orig]	Ep 617 R	282.15	avgR100	252.09	eps 0.045	Qval 36.37	1
and100 0.86	thr 170						
[DQN-Orig]	Ep 618 R	209.30	avgR100	251.91	eps 0.045	Qval 35.40	1
and100 0.85	thr 309						
[DQN-Orig]	Ep 619 R	277.74	avgR100	251.98	eps 0.045	Qval 35.59	1
and100 0.85	thr 130						
[DQN-Orig]	Ep 620 R	296.36	avgR100	252.49	eps 0.045	Qval 34.89	1
and100 0.85	thr 305						
[DQN-Orig]	Ep 621 R	237.56	avgR100	252.64	eps 0.044	Qval 34.32	1
and100 0.85	thr 155						
[DQN-Orig]	Ep 622 R	246.41	avgR100	252.71	eps 0.044	Qval 33.90	1
and100 0.85	thr 141						
[DQN-Orig]	Ep 623 R	243.76	avgR100	252.62	eps 0.044	Qval 35.21	1
and100 0.85	thr 240						
[DQN-Orig]	Ep 624 R	246.47	avgR100	252.75	eps 0.044	Qval 35.05	1
and100 0.85	thr 106						
[DQN-Orig]	Ep 625 R	247.70	avgR100	252.70	eps 0.044	Qval 34.37	1
and100 0.85	thr 153						
[DQN-Orig]	Ep 626 R	240.28	avgR100	252.64	eps 0.043	Qval 34.39	1
and100 0.85	thr 138						
[DQN-Orig]	Ep 627 R	273.02	avgR100	252.85	eps 0.043	Qval 34.46	1
and100 0.85	thr 201						
[DQN-Orig]	Ep 628 R	247.68	avgR100	253.02	eps 0.043	Qval 34.21	1
and100 0.85	thr 199						
[DQN-Orig]	Ep 629 R	277.48	avgR100	252.80	eps 0.043	Qval 32.83	1
and100 0.85	thr 227						
[DQN-Orig]	Ep 630 R	289.19	avgR100	253.21	eps 0.043	Qval 34.15	1
and100 0.85	thr 147						

[DQN-Orig] Ep 631 R and100 0.84 thr 271	240.97 avgR100	252.79 eps 0.042 Qval	32.53 1
[DQN-Orig] Ep 632 R and100 0.84 thr 205	301.03 avgR100	253.06 eps 0.042 Qval	34.17 1
[DQN-Orig] Ep 633 R and100 0.84 thr 517	249.55 avgR100	252.47 eps 0.042 Qval	34.38 1
[DQN-Orig] Ep 634 R and100 0.84 thr 139	282.11 avgR100	253.01 eps 0.042 Qval	34.11 1
[DQN-Orig] Ep 635 R and100 0.83 thr 620	152.72 avgR100	251.96 eps 0.041 Qval	34.55 1
[DQN-Orig] Ep 636 R and100 0.82 thr 433	167.75 avgR100	250.86 eps 0.041 Qval	33.11 1
[DQN-Orig] Ep 637 R and100 0.82 thr 181	291.05 avgR100	250.77 eps 0.041 Qval	32.43 1
[DQN-Orig] Ep 638 R and100 0.82 thr 384	302.68 avgR100	251.54 eps 0.041 Qval	32.31 1
[DQN-Orig] Ep 639 R and100 0.81 thr 163	253.14 avgR100	251.61 eps 0.041 Qval	32.61 1
[DQN-Orig] Ep 640 R and100 0.81 thr 218	305.87 avgR100	252.39 eps 0.040 Qval	32.14 1
[DQN-Orig] Ep 641 R and100 0.80 thr 347	236.32 avgR100	252.16 eps 0.040 Qval	32.44 1
[DQN-Orig] Ep 642 R and100 0.80 thr 155	254.74 avgR100	252.37 eps 0.040 Qval	32.46 1
[DQN-Orig] Ep 643 R and100 0.80 thr 251	265.11 avgR100	252.39 eps 0.040 Qval	32.67 1
[DQN-Orig] Ep 644 R and100 0.80 thr 128	238.95 avgR100	252.48 eps 0.040 Qval	34.26 1
[DQN-Orig] Ep 645 R and100 0.80 thr 227	259.74 avgR100	252.60 eps 0.039 Qval	33.86 1
[DQN-Orig] Ep 646 R and100 0.80 thr 164	277.37 avgR100	252.86 eps 0.039 Qval	33.82 1
[DQN-Orig] Ep 647 R and100 0.80 thr 137	236.79 avgR100	252.60 eps 0.039 Qval	33.32 1
[DQN-Orig] Ep 648 R and100 0.80 thr 389	215.40 avgR100	252.35 eps 0.039 Qval	33.01 1
[DQN-Orig] Ep 649 R and100 0.79 thr 488	202.14 avgR100	251.85 eps 0.039 Qval	32.61 1
[DQN-Orig] Ep 650 R and100 0.78 thr 193	275.65 avgR100	251.76 eps 0.038 Qval	32.39 1
[DQN-Orig] Ep 651 R and100 0.79 thr 181	250.77 avgR100	251.82 eps 0.038 Qval	32.21 1
[DQN-Orig] Ep 652 R and100 0.79 thr 183	301.07 avgR100	252.21 eps 0.038 Qval	32.54 1
[DQN-Orig] Ep 653 R and100 0.79 thr 145	273.08 avgR100	252.32 eps 0.038 Qval	31.65 1
[DQN-Orig] Ep 654 R and100 0.78 thr 241	279.21 avgR100	252.13 eps 0.038 Qval	32.15 1
[DQN-Orig] Ep 655 R and100 0.78 thr 183	303.97 avgR100	252.61 eps 0.038 Qval	32.34 1
[DQN-Orig] Ep 656 R and100 0.77 thr 338	201.16 avgR100	252.18 eps 0.037 Qval	32.57 1
[DQN-Orig] Ep 657 R and100 0.77 thr 169	242.84 avgR100	252.21 eps 0.037 Qval	33.06 1
[DQN-Orig] Ep 658 R and100 0.77 thr 148	229.85 avgR100	251.63 eps 0.037 Qval	33.03 1
[DQN-Orig] Ep 659 R and100 0.78 thr 277	283.27 avgR100	251.82 eps 0.037 Qval	32.69 1
[DQN-Orig] Ep 660 R and100 0.78 thr 157	264.59 avgR100	252.03 eps 0.037 Qval	33.36 1

[DQN-Orig] Ep 661 R and100 0.78 thr 195	250.52 avgR100	251.52 eps 0.036 Qval	32.84 1
[DQN-Orig] Ep 662 R and100 0.78 thr 435	218.47 avgR100	251.10 eps 0.036 Qval	33.32 1
[DQN-Orig] Ep 663 R and100 0.77 thr 322	241.05 avgR100	250.78 eps 0.036 Qval	32.00 1
[DQN-Orig] Ep 664 R and100 0.77 thr 225	270.21 avgR100	251.01 eps 0.036 Qval	33.30 1
[DQN-Orig] Ep 665 R and100 0.77 thr 345	251.19 avgR100	250.99 eps 0.036 Qval	33.84 1
[DQN-Orig] Ep 666 R and100 0.77 thr 285	254.09 avgR100	251.11 eps 0.035 Qval	32.48 1
[DQN-Orig] Ep 667 R and100 0.77 thr 491	247.62 avgR100	251.21 eps 0.035 Qval	31.83 1
[DQN-Orig] Ep 668 R and100 0.77 thr 141	230.77 avgR100	250.44 eps 0.035 Qval	32.31 1
[DQN-Orig] Ep 669 R and100 0.77 thr 234	244.19 avgR100	250.60 eps 0.035 Qval	31.56 1
[DQN-Orig] Ep 670 R and100 0.77 thr 159	277.65 avgR100	250.79 eps 0.035 Qval	33.86 1
[DQN-Orig] Ep 671 R and100 0.78 thr 236	245.60 avgR100	251.05 eps 0.035 Qval	32.03 1
[DQN-Orig] Ep 672 R and100 0.77 thr 134	2.31 avgR100	248.79 eps 0.034 Qval	31.68 1
[DQN-Orig] Ep 673 R and100 0.77 thr 406	227.81 avgR100	248.48 eps 0.034 Qval	31.44 1
[DQN-Orig] Ep 674 R and100 0.77 thr 193	277.05 avgR100	249.00 eps 0.034 Qval	33.07 1
[DQN-Orig] Ep 675 R and100 0.77 thr 178	243.80 avgR100	248.93 eps 0.034 Qval	31.66 1
[DQN-Orig] Ep 676 R and100 0.77 thr 303	249.22 avgR100	248.93 eps 0.034 Qval	31.58 1
[DQN-Orig] Ep 677 R and100 0.78 thr 182	276.36 avgR100	249.34 eps 0.034 Qval	31.29 1
[DQN-Orig] Ep 678 R and100 0.78 thr 252	277.35 avgR100	249.37 eps 0.033 Qval	31.53 1
[DQN-Orig] Ep 679 R and100 0.77 thr 316	266.24 avgR100	249.58 eps 0.033 Qval	32.48 1
[DQN-Orig] Ep 680 R and100 0.77 thr 432	252.71 avgR100	249.95 eps 0.033 Qval	32.20 1
[DQN-Orig] Ep 681 R and100 0.77 thr 185	243.86 avgR100	249.70 eps 0.033 Qval	31.80 1
[DQN-Orig] Ep 682 R and100 0.77 thr 130	277.02 avgR100	250.08 eps 0.033 Qval	31.55 1
[DQN-Orig] Ep 683 R and100 0.77 thr 233	233.18 avgR100	250.05 eps 0.033 Qval	30.90 1
[DQN-Orig] Ep 684 R and100 0.77 thr 173	251.23 avgR100	250.10 eps 0.032 Qval	32.40 1
[DQN-Orig] Ep 685 R and100 0.77 thr 173	265.48 avgR100	250.52 eps 0.032 Qval	33.14 1
[DQN-Orig] Ep 686 R and100 0.77 thr 122	241.79 avgR100	250.39 eps 0.032 Qval	31.78 1
[DQN-Orig] Ep 687 R and100 0.77 thr 570	212.04 avgR100	250.20 eps 0.032 Qval	32.41 1
[DQN-Orig] Ep 688 R and100 0.77 thr 238	257.78 avgR100	250.24 eps 0.032 Qval	32.47 1
[DQN-Orig] Ep 689 R and100 0.77 thr 352	265.88 avgR100	250.64 eps 0.032 Qval	32.44 1
[DQN-Orig] Ep 690 R and100 0.77 thr 162	240.60 avgR100	250.85 eps 0.031 Qval	32.44 1

[DQN-Orig]	Ep 691 R	272.32	avgR100	250.79	eps 0.031	Qval 33.27	1
and100 0.77	thr 136						
[DQN-Orig]	Ep 692 R	299.27	avgR100	250.98	eps 0.031	Qval 32.38	1
and100 0.78	thr 153						
[DQN-Orig]	Ep 693 R	235.70	avgR100	250.67	eps 0.031	Qval 32.30	1
and100 0.78	thr 280						
[DQN-Orig]	Ep 694 R	260.61	avgR100	250.42	eps 0.031	Qval 32.89	1
and100 0.79	thr 173						
[DQN-Orig]	Ep 695 R	261.40	avgR100	250.42	eps 0.031	Qval 32.35	1
and100 0.79	thr 170						
[DQN-Orig]	Ep 696 R	247.12	avgR100	250.52	eps 0.031	Qval 32.18	1
and100 0.80	thr 139						
[DQN-Orig]	Ep 697 R	282.15	avgR100	250.48	eps 0.030	Qval 31.99	1
and100 0.80	thr 130						
[DQN-Orig]	Ep 698 R	267.61	avgR100	251.01	eps 0.030	Qval 32.67	1
and100 0.80	thr 256						
[DQN-Orig]	Ep 699 R	262.30	avgR100	251.09	eps 0.030	Qval 32.01	1
and100 0.80	thr 122						
[DQN-Orig]	Ep 700 R	259.02	avgR100	251.22	eps 0.030	Qval 32.75	1
and100 0.81	thr 172						
[DQN-Orig]	Ep 701 R	203.48	avgR100	250.62	eps 0.030	Qval 31.87	1
and100 0.80	thr 443						
[DQN-Orig]	Ep 702 R	208.67	avgR100	250.16	eps 0.030	Qval 31.81	1
and100 0.81	thr 334						
[DQN-Orig]	Ep 703 R	268.05	avgR100	250.25	eps 0.029	Qval 32.55	1
and100 0.80	thr 185						
[DQN-Orig]	Ep 704 R	264.77	avgR100	250.38	eps 0.029	Qval 32.59	1
and100 0.80	thr 270						
[DQN-Orig]	Ep 705 R	247.60	avgR100	250.32	eps 0.029	Qval 32.89	1
and100 0.80	thr 155						
[DQN-Orig]	Ep 706 R	223.25	avgR100	249.87	eps 0.029	Qval 33.45	1
and100 0.79	thr 430						
[DQN-Orig]	Ep 707 R	275.49	avgR100	250.59	eps 0.029	Qval 33.46	1
and100 0.80	thr 262						
[DQN-Orig]	Ep 708 R	238.37	avgR100	250.89	eps 0.029	Qval 33.86	1
and100 0.81	thr 164						
[DQN-Orig]	Ep 709 R	245.68	avgR100	250.70	eps 0.029	Qval 32.48	1
and100 0.82	thr 234						
[DQN-Orig]	Ep 710 R	275.58	avgR100	250.72	eps 0.028	Qval 33.11	1
and100 0.82	thr 286						
[DQN-Orig]	Ep 711 R	262.93	avgR100	251.12	eps 0.028	Qval 33.88	1
and100 0.82	thr 160						
[DQN-Orig]	Ep 712 R	291.94	avgR100	251.54	eps 0.028	Qval 32.67	1
and100 0.82	thr 210						
[DQN-Orig]	Ep 713 R	270.49	avgR100	252.01	eps 0.028	Qval 33.38	1
and100 0.81	thr 230						
[DQN-Orig]	Ep 714 R	227.73	avgR100	251.99	eps 0.028	Qval 33.21	1
and100 0.82	thr 434						
[DQN-Orig]	Ep 715 R	263.44	avgR100	252.02	eps 0.028	Qval 33.36	1
and100 0.81	thr 445						
[DQN-Orig]	Ep 716 R	302.49	avgR100	252.65	eps 0.028	Qval 33.44	1
and100 0.81	thr 191						
[DQN-Orig]	Ep 717 R	254.79	avgR100	252.37	eps 0.027	Qval 33.04	1
and100 0.80	thr 222						
[DQN-Orig]	Ep 718 R	258.37	avgR100	252.86	eps 0.027	Qval 32.77	1
and100 0.81	thr 257						
[DQN-Orig]	Ep 719 R	282.84	avgR100	252.91	eps 0.027	Qval 32.87	1
and100 0.81	thr 232						
[DQN-Orig]	Ep 720 R	242.04	avgR100	252.37	eps 0.027	Qval 32.05	1
and100 0.81	thr 279						

[DQN-Orig]	Ep 721 R	294.63	avgR100	252.94	eps 0.027	Qval	32.13	1
and100 0.81	thr 174							
[DQN-Orig]	Ep 722 R	251.84	avgR100	252.99	eps 0.027	Qval	31.78	1
and100 0.81	thr 166							
[DQN-Orig]	Ep 723 R	260.85	avgR100	253.17	eps 0.027	Qval	33.01	1
and100 0.81	thr 270							
[DQN-Orig]	Ep 724 R	247.86	avgR100	253.18	eps 0.027	Qval	31.81	1
and100 0.81	thr 256							
[DQN-Orig]	Ep 725 R	272.40	avgR100	253.43	eps 0.026	Qval	31.82	1
and100 0.81	thr 314							
[DQN-Orig]	Ep 726 R	267.09	avgR100	253.69	eps 0.026	Qval	31.02	1
and100 0.80	thr 120							
[DQN-Orig]	Ep 727 R	262.04	avgR100	253.59	eps 0.026	Qval	31.16	1
and100 0.80	thr 186							
[DQN-Orig]	Ep 728 R	257.53	avgR100	253.68	eps 0.026	Qval	32.08	1
and100 0.80	thr 148							
[DQN-Orig]	Ep 729 R	284.99	avgR100	253.76	eps 0.026	Qval	31.67	1
and100 0.79	thr 201							
[DQN-Orig]	Ep 730 R	265.93	avgR100	253.53	eps 0.026	Qval	31.45	1
and100 0.79	thr 133							
[DQN-Orig]	Ep 731 R	241.91	avgR100	253.54	eps 0.026	Qval	32.18	1
and100 0.80	thr 194							
[DQN-Orig]	Ep 732 R	263.95	avgR100	253.16	eps 0.025	Qval	31.86	1
and100 0.80	thr 210							
[DQN-Orig]	Ep 733 R	280.47	avgR100	253.47	eps 0.025	Qval	33.37	1
and100 0.80	thr 301							
[DQN-Orig]	Ep 734 R	244.18	avgR100	253.09	eps 0.025	Qval	32.04	1
and100 0.80	thr 199							
[DQN-Orig]	Ep 735 R	255.87	avgR100	254.13	eps 0.025	Qval	32.87	1
and100 0.81	thr 240							
[DQN-Orig]	Ep 736 R	268.65	avgR100	255.14	eps 0.025	Qval	32.50	1
and100 0.82	thr 171							
[DQN-Orig]	Ep 737 R	249.82	avgR100	254.72	eps 0.025	Qval	32.90	1
and100 0.82	thr 323							
[DQN-Orig]	Ep 738 R	234.76	avgR100	254.04	eps 0.025	Qval	34.04	1
and100 0.82	thr 438							
[DQN-Orig]	Ep 739 R	281.77	avgR100	254.33	eps 0.025	Qval	33.52	1
and100 0.83	thr 194							
[DQN-Orig]	Ep 740 R	252.31	avgR100	253.79	eps 0.024	Qval	33.27	1
and100 0.83	thr 219							
[DQN-Orig]	Ep 741 R	125.94	avgR100	252.69	eps 0.024	Qval	32.89	1
and100 0.83	thr 701							
[DQN-Orig]	Ep 742 R	258.32	avgR100	252.73	eps 0.024	Qval	32.99	1
and100 0.83	thr 154							
[DQN-Orig]	Ep 743 R	228.91	avgR100	252.36	eps 0.024	Qval	32.54	1
and100 0.83	thr 281							
[DQN-Orig]	Ep 744 R	254.77	avgR100	252.52	eps 0.024	Qval	33.66	1
and100 0.83	thr 289							
[DQN-Orig]	Ep 745 R	273.29	avgR100	252.66	eps 0.024	Qval	34.09	1
and100 0.83	thr 165							
[DQN-Orig]	Ep 746 R	262.14	avgR100	252.51	eps 0.024	Qval	33.88	1
and100 0.83	thr 196							
[DQN-Orig]	Ep 747 R	239.09	avgR100	252.53	eps 0.024	Qval	33.59	1
and100 0.83	thr 185							
[DQN-Orig]	Ep 748 R	254.99	avgR100	252.92	eps 0.024	Qval	34.61	1
and100 0.83	thr 341							
[DQN-Orig]	Ep 749 R	258.30	avgR100	253.49	eps 0.023	Qval	33.72	1
and100 0.84	thr 277							
[DQN-Orig]	Ep 750 R	249.34	avgR100	253.22	eps 0.023	Qval	33.73	1
and100 0.85	thr 391							

[DQN-Orig]	Ep 751 R	258.74	avgR100	253.30	eps 0.023	Qval 33.64	1
and100 0.85	thr 185						
[DQN-Orig]	Ep 752 R	256.70	avgR100	252.86	eps 0.023	Qval 34.21	1
and100 0.85	thr 187						
[DQN-Orig]	Ep 753 R	295.19	avgR100	253.08	eps 0.023	Qval 33.39	1
and100 0.85	thr 216						
[DQN-Orig]	Ep 754 R	282.63	avgR100	253.11	eps 0.023	Qval 33.37	1
and100 0.86	thr 175						
[DQN-Orig]	Ep 755 R	260.33	avgR100	252.68	eps 0.023	Qval 33.09	1
and100 0.86	thr 282						
[DQN-Orig]	Ep 756 R	251.56	avgR100	253.18	eps 0.023	Qval 33.80	1
and100 0.87	thr 216						
[DQN-Orig]	Ep 757 R	207.64	avgR100	252.83	eps 0.022	Qval 34.22	1
and100 0.87	thr 309						
[DQN-Orig]	Ep 758 R	254.54	avgR100	253.08	eps 0.022	Qval 33.40	1
and100 0.87	thr 135						
[DQN-Orig]	Ep 759 R	247.98	avgR100	252.72	eps 0.022	Qval 34.22	1
and100 0.87	thr 247						
[DQN-Orig]	Ep 760 R	248.27	avgR100	252.56	eps 0.022	Qval 34.85	1
and100 0.87	thr 154						
[DQN-Orig]	Ep 761 R	277.77	avgR100	252.83	eps 0.022	Qval 33.06	1
and100 0.87	thr 171						
[DQN-Orig]	Ep 762 R	266.05	avgR100	253.31	eps 0.022	Qval 33.53	1
and100 0.87	thr 194						
[DQN-Orig]	Ep 763 R	273.35	avgR100	253.63	eps 0.022	Qval 34.16	1
and100 0.88	thr 205						
[DQN-Orig]	Ep 764 R	283.02	avgR100	253.76	eps 0.022	Qval 34.76	1
and100 0.88	thr 181						
[DQN-Orig]	Ep 765 R	240.16	avgR100	253.65	eps 0.022	Qval 33.15	1
and100 0.88	thr 420						
[DQN-Orig]	Ep 766 R	240.19	avgR100	253.51	eps 0.022	Qval 34.23	1
and100 0.88	thr 362						
[DQN-Orig]	Ep 767 R	240.58	avgR100	253.44	eps 0.021	Qval 33.34	1
and100 0.88	thr 517						
[DQN-Orig]	Ep 768 R	213.45	avgR100	253.27	eps 0.021	Qval 34.46	1
and100 0.87	thr 407						
[DQN-Orig]	Ep 769 R	314.00	avgR100	253.97	eps 0.021	Qval 34.18	1
and100 0.87	thr 183						
[DQN-Orig]	Ep 770 R	253.00	avgR100	253.72	eps 0.021	Qval 34.53	1
and100 0.87	thr 176						
[DQN-Orig]	Ep 771 R	262.11	avgR100	253.88	eps 0.021	Qval 35.67	1
and100 0.87	thr 144						
[DQN-Orig]	Ep 772 R	269.23	avgR100	256.55	eps 0.021	Qval 35.05	1
and100 0.88	thr 417						
[DQN-Orig]	Ep 773 R	264.33	avgR100	256.92	eps 0.021	Qval 35.51	1
and100 0.88	thr 183						
[DQN-Orig]	Ep 774 R	277.53	avgR100	256.92	eps 0.021	Qval 34.38	1
and100 0.88	thr 309						
[DQN-Orig]	Ep 775 R	277.50	avgR100	257.26	eps 0.021	Qval 35.08	1
and100 0.88	thr 133						
[DQN-Orig]	Ep 776 R	270.90	avgR100	257.48	eps 0.020	Qval 35.63	1
and100 0.88	thr 203						
[DQN-Orig]	Ep 777 R	280.92	avgR100	257.52	eps 0.020	Qval 36.59	1
and100 0.88	thr 154						
[DQN-Orig]	Ep 778 R	272.89	avgR100	257.48	eps 0.020	Qval 34.90	1
and100 0.88	thr 152						
[DQN-Orig]	Ep 779 R	298.91	avgR100	257.81	eps 0.020	Qval 35.94	1
and100 0.89	thr 167						
[DQN-Orig]	Ep 780 R	304.06	avgR100	258.32	eps 0.020	Qval 35.85	1
and100 0.89	thr 228						

[DQN-Orig]	Ep 781 R	272.44	avgR100	258.60	eps 0.020	Qval 35.39	1
and100 0.89	thr 137						
[DQN-Orig]	Ep 782 R	270.22	avgR100	258.54	eps 0.020	Qval 36.31	1
and100 0.89	thr 167						
[DQN-Orig]	Ep 783 R	261.36	avgR100	258.82	eps 0.020	Qval 36.43	1
and100 0.90	thr 157						
[DQN-Orig]	Ep 784 R	232.87	avgR100	258.63	eps 0.020	Qval 37.56	1
and100 0.90	thr 283						
[DQN-Orig]	Ep 785 R	278.76	avgR100	258.77	eps 0.020	Qval 35.68	1
and100 0.89	thr 300						
[DQN-Orig]	Ep 786 R	296.58	avgR100	259.32	eps 0.020	Qval 36.16	1
and100 0.89	thr 213						
[DQN-Orig]	Ep 787 R	255.22	avgR100	259.75	eps 0.020	Qval 36.28	1
and100 0.89	thr 220						
[DQN-Orig]	Ep 788 R	234.12	avgR100	259.51	eps 0.020	Qval 36.14	1
and100 0.89	thr 200						
[DQN-Orig]	Ep 789 R	295.59	avgR100	259.81	eps 0.020	Qval 36.61	1
and100 0.89	thr 173						
[DQN-Orig]	Ep 790 R	293.21	avgR100	260.33	eps 0.020	Qval 36.41	1
and100 0.89	thr 161						
[DQN-Orig]	Ep 791 R	268.68	avgR100	260.30	eps 0.020	Qval 37.60	1
and100 0.88	thr 386						
[DQN-Orig]	Ep 792 R	306.26	avgR100	260.37	eps 0.020	Qval 37.86	1
and100 0.88	thr 135						
[DQN-Orig]	Ep 793 R	286.27	avgR100	260.87	eps 0.020	Qval 37.20	1
and100 0.88	thr 277						
[DQN-Orig]	Ep 794 R	277.92	avgR100	261.05	eps 0.020	Qval 38.55	1
and100 0.88	thr 152						
[DQN-Orig]	Ep 795 R	299.38	avgR100	261.43	eps 0.020	Qval 38.12	1
and100 0.88	thr 153						
[DQN-Orig]	Ep 796 R	274.46	avgR100	261.70	eps 0.020	Qval 39.10	1
and100 0.88	thr 177						
[DQN-Orig]	Ep 797 R	257.58	avgR100	261.45	eps 0.020	Qval 38.99	1
and100 0.88	thr 356						
[DQN-Orig]	Ep 798 R	254.87	avgR100	261.33	eps 0.020	Qval 38.64	1
and100 0.88	thr 226						
[DQN-Orig]	Ep 799 R	259.24	avgR100	261.30	eps 0.020	Qval 39.05	1
and100 0.88	thr 167						
[DQN-Orig]	Ep 800 R	287.56	avgR100	261.58	eps 0.020	Qval 39.65	1
and100 0.88	thr 237						
[DQN-Orig]	Ep 801 R	266.52	avgR100	262.21	eps 0.020	Qval 38.48	1
and100 0.89	thr 222						
[DQN-Orig]	Ep 802 R	287.67	avgR100	263.00	eps 0.020	Qval 38.78	1
and100 0.89	thr 291						
[DQN-Orig]	Ep 803 R	307.98	avgR100	263.40	eps 0.020	Qval 38.66	1
and100 0.90	thr 132						
[DQN-Orig]	Ep 804 R	224.96	avgR100	263.00	eps 0.020	Qval 38.45	1
and100 0.90	thr 153						
[DQN-Orig]	Ep 805 R	232.82	avgR100	262.85	eps 0.020	Qval 39.04	1
and100 0.90	thr 200						
[DQN-Orig]	Ep 806 R	283.90	avgR100	263.46	eps 0.020	Qval 38.85	1
and100 0.91	thr 272						
[DQN-Orig]	Ep 807 R	308.33	avgR100	263.79	eps 0.020	Qval 39.32	1
and100 0.90	thr 175						
[DQN-Orig]	Ep 808 R	257.40	avgR100	263.98	eps 0.020	Qval 40.08	1
and100 0.90	thr 247						
[DQN-Orig]	Ep 809 R	236.78	avgR100	263.89	eps 0.020	Qval 39.77	1
and100 0.90	thr 139						
[DQN-Orig]	Ep 810 R	272.77	avgR100	263.86	eps 0.020	Qval 39.49	1
and100 0.90	thr 220						

[DQN-Orig]	Ep 811 R	241.81	avgR100	263.65	eps 0.020	Qval 39.95	1
and100 0.90	thr 184						
[DQN-Orig]	Ep 812 R	273.15	avgR100	263.46	eps 0.020	Qval 40.07	1
and100 0.90	thr 260						
[DQN-Orig]	Ep 813 R	244.44	avgR100	263.20	eps 0.020	Qval 40.25	1
and100 0.91	thr 393						
[DQN-Orig]	Ep 814 R	236.40	avgR100	263.29	eps 0.020	Qval 39.66	1
and100 0.91	thr 237						
[DQN-Orig]	Ep 815 R	269.27	avgR100	263.35	eps 0.020	Qval 39.59	1
and100 0.92	thr 204						
[DQN-Orig]	Ep 816 R	255.40	avgR100	262.88	eps 0.020	Qval 39.30	1
and100 0.92	thr 167						
[DQN-Orig]	Ep 817 R	266.11	avgR100	262.99	eps 0.020	Qval 40.56	1
and100 0.93	thr 129						
[DQN-Orig]	Ep 818 R	265.10	avgR100	263.06	eps 0.020	Qval 40.98	1
and100 0.93	thr 231						
[DQN-Orig]	Ep 819 R	267.92	avgR100	262.91	eps 0.020	Qval 40.41	1
and100 0.93	thr 372						
[DQN-Orig]	Ep 820 R	261.56	avgR100	263.10	eps 0.020	Qval 39.78	1
and100 0.93	thr 164						
[DQN-Orig]	Ep 821 R	233.78	avgR100	262.50	eps 0.020	Qval 41.08	1
and100 0.93	thr 173						
[DQN-Orig]	Ep 822 R	290.75	avgR100	262.88	eps 0.020	Qval 42.44	1
and100 0.93	thr 190						
[DQN-Orig]	Ep 823 R	295.47	avgR100	263.23	eps 0.020	Qval 42.76	1
and100 0.92	thr 255						
[DQN-Orig]	Ep 824 R	250.46	avgR100	263.26	eps 0.020	Qval 41.91	1
and100 0.92	thr 246						
[DQN-Orig]	Ep 825 R	220.36	avgR100	262.74	eps 0.020	Qval 42.33	1
and100 0.92	thr 247						
[DQN-Orig]	Ep 826 R	254.13	avgR100	262.61	eps 0.020	Qval 41.42	1
and100 0.93	thr 187						
[DQN-Orig]	Ep 827 R	264.98	avgR100	262.64	eps 0.020	Qval 41.73	1
and100 0.93	thr 172						
[DQN-Orig]	Ep 828 R	259.11	avgR100	262.65	eps 0.020	Qval 42.48	1
and100 0.93	thr 200						
[DQN-Orig]	Ep 829 R	280.53	avgR100	262.61	eps 0.020	Qval 42.74	1
and100 0.94	thr 268						
[DQN-Orig]	Ep 830 R	234.79	avgR100	262.30	eps 0.020	Qval 43.06	1
and100 0.94	thr 478						
[DQN-Orig]	Ep 831 R	251.14	avgR100	262.39	eps 0.020	Qval 43.31	1
and100 0.94	thr 188						
[DQN-Orig]	Ep 832 R	263.64	avgR100	262.39	eps 0.020	Qval 43.77	1
and100 0.94	thr 202						
[DQN-Orig]	Ep 833 R	262.23	avgR100	262.20	eps 0.020	Qval 43.32	1
and100 0.94	thr 367						
[DQN-Orig]	Ep 834 R	261.04	avgR100	262.37	eps 0.020	Qval 44.62	1
and100 0.94	thr 273						
[DQN-Orig]	Ep 835 R	250.83	avgR100	262.32	eps 0.020	Qval 44.07	1
and100 0.93	thr 224						
[DQN-Orig]	Ep 836 R	231.96	avgR100	261.95	eps 0.020	Qval 44.81	1
and100 0.93	thr 451						
[DQN-Orig]	Ep 837 R	249.06	avgR100	261.95	eps 0.020	Qval 45.22	1
and100 0.93	thr 338						
[DQN-Orig]	Ep 838 R	230.60	avgR100	261.90	eps 0.020	Qval 44.30	1
and100 0.93	thr 156						
[DQN-Orig]	Ep 839 R	262.38	avgR100	261.71	eps 0.020	Qval 45.45	1
and100 0.93	thr 419						
[DQN-Orig]	Ep 840 R	294.76	avgR100	262.14	eps 0.020	Qval 45.65	1
and100 0.93	thr 251						

[DQN-Orig]	Ep 841 R	258.72	avgR100	263.46	eps 0.020	Qval	46.37	1
and100 0.94	thr 172							
[DQN-Orig]	Ep 842 R	241.21	avgR100	263.29	eps 0.020	Qval	45.58	1
and100 0.94	thr 182							
[DQN-Orig]	Ep 843 R	288.64	avgR100	263.89	eps 0.020	Qval	46.03	1
and100 0.94	thr 168							
[DQN-Orig]	Ep 844 R	286.04	avgR100	264.20	eps 0.020	Qval	46.60	1
and100 0.94	thr 159							
[DQN-Orig]	Ep 845 R	307.75	avgR100	264.55	eps 0.020	Qval	46.12	1
and100 0.94	thr 163							
[DQN-Orig]	Ep 846 R	236.82	avgR100	264.29	eps 0.020	Qval	46.24	1
and100 0.94	thr 262							
[DQN-Orig]	Ep 847 R	268.55	avgR100	264.59	eps 0.020	Qval	45.89	1
and100 0.94	thr 187							
[DQN-Orig]	Ep 848 R	290.60	avgR100	264.94	eps 0.020	Qval	46.00	1
and100 0.94	thr 180							
[DQN-Orig]	Ep 849 R	288.34	avgR100	265.24	eps 0.020	Qval	46.88	1
and100 0.94	thr 217							
[DQN-Orig]	Ep 850 R	310.71	avgR100	265.86	eps 0.020	Qval	46.54	1
and100 0.94	thr 159							
[DQN-Orig]	Ep 851 R	236.29	avgR100	265.63	eps 0.020	Qval	46.46	1
and100 0.93	thr 328							
[DQN-Orig]	Ep 852 R	291.97	avgR100	265.99	eps 0.020	Qval	46.92	1
and100 0.93	thr 230							
[DQN-Orig]	Ep 853 R	274.16	avgR100	265.78	eps 0.020	Qval	46.13	1
and100 0.93	thr 145							
[DQN-Orig]	Ep 854 R	273.44	avgR100	265.68	eps 0.020	Qval	47.05	1
and100 0.93	thr 323							
[DQN-Orig]	Ep 855 R	262.70	avgR100	265.71	eps 0.020	Qval	46.72	1
and100 0.93	thr 356							
[DQN-Orig]	Ep 856 R	206.36	avgR100	265.26	eps 0.020	Qval	47.77	1
and100 0.93	thr 346							
[DQN-Orig]	Ep 857 R	249.50	avgR100	265.67	eps 0.020	Qval	46.99	1
and100 0.93	thr 171							
[DQN-Orig]	Ep 858 R	266.85	avgR100	265.80	eps 0.020	Qval	47.47	1
and100 0.93	thr 325							
[DQN-Orig]	Ep 859 R	298.80	avgR100	266.31	eps 0.020	Qval	47.48	1
and100 0.93	thr 199							
[DQN-Orig]	Ep 860 R	278.46	avgR100	266.61	eps 0.020	Qval	47.73	1
and100 0.93	thr 202							
[DQN-Orig]	Ep 861 R	264.10	avgR100	266.47	eps 0.020	Qval	47.01	1
and100 0.93	thr 165							
[DQN-Orig]	Ep 862 R	263.80	avgR100	266.45	eps 0.020	Qval	47.14	1
and100 0.93	thr 330							
[DQN-Orig]	Ep 863 R	271.71	avgR100	266.43	eps 0.020	Qval	47.34	1
and100 0.93	thr 382							
[DQN-Orig]	Ep 864 R	272.94	avgR100	266.33	eps 0.020	Qval	47.10	1
and100 0.93	thr 154							
[DQN-Orig]	Ep 865 R	252.10	avgR100	266.45	eps 0.020	Qval	47.73	1
and100 0.93	thr 136							
[DQN-Orig]	Ep 866 R	258.45	avgR100	266.63	eps 0.020	Qval	47.10	1
and100 0.93	thr 184							
[DQN-Orig]	Ep 867 R	259.49	avgR100	266.82	eps 0.020	Qval	47.15	1
and100 0.93	thr 184							
[DQN-Orig]	Ep 868 R	229.54	avgR100	266.98	eps 0.020	Qval	47.07	1
and100 0.94	thr 182							
[DQN-Orig]	Ep 869 R	268.38	avgR100	266.53	eps 0.020	Qval	46.97	1
and100 0.93	thr 273							
[DQN-Orig]	Ep 870 R	275.99	avgR100	266.76	eps 0.020	Qval	46.82	1
and100 0.93	thr 156							

[DQN-Orig]	Ep 871 R	265.17	avgR100	266.79	eps 0.020	Qval 47.39	1
and100 0.93	thr 172						
[DQN-Orig]	Ep 872 R	276.94	avgR100	266.86	eps 0.020	Qval 47.23	1
and100 0.93	thr 161						
[DQN-Orig]	Ep 873 R	226.81	avgR100	266.49	eps 0.020	Qval 46.89	1
and100 0.93	thr 275						
[DQN-Orig]	Ep 874 R	217.42	avgR100	265.89	eps 0.020	Qval 48.14	1
and100 0.93	thr 296						
[DQN-Orig]	Ep 875 R	251.87	avgR100	265.63	eps 0.020	Qval 47.58	1
and100 0.93	thr 150						
[DQN-Orig]	Ep 876 R	273.41	avgR100	265.66	eps 0.020	Qval 47.72	1
and100 0.93	thr 288						
[DQN-Orig]	Ep 877 R	268.86	avgR100	265.54	eps 0.020	Qval 47.35	1
and100 0.93	thr 217						
[DQN-Orig]	Ep 878 R	264.53	avgR100	265.45	eps 0.020	Qval 48.21	1
and100 0.93	thr 137						
[DQN-Orig]	Ep 879 R	289.41	avgR100	265.36	eps 0.020	Qval 47.22	1
and100 0.93	thr 168						
[DQN-Orig]	Ep 880 R	250.90	avgR100	264.83	eps 0.020	Qval 47.42	1
and100 0.92	thr 277						
[DQN-Orig]	Ep 881 R	243.84	avgR100	264.54	eps 0.020	Qval 47.68	1
and100 0.92	thr 225						
[DQN-Orig]	Ep 882 R	268.70	avgR100	264.53	eps 0.020	Qval 47.10	1
and100 0.92	thr 205						
[DQN-Orig]	Ep 883 R	241.90	avgR100	264.33	eps 0.020	Qval 47.55	1
and100 0.92	thr 152						
[DQN-Orig]	Ep 884 R	260.93	avgR100	264.61	eps 0.020	Qval 47.99	1
and100 0.92	thr 184						
[DQN-Orig]	Ep 885 R	233.67	avgR100	264.16	eps 0.020	Qval 48.95	1
and100 0.93	thr 332						
[DQN-Orig]	Ep 886 R	254.23	avgR100	263.74	eps 0.020	Qval 47.78	1
and100 0.93	thr 197						
[DQN-Orig]	Ep 887 R	233.98	avgR100	263.52	eps 0.020	Qval 47.07	1
and100 0.93	thr 145						
[DQN-Orig]	Ep 888 R	260.71	avgR100	263.79	eps 0.020	Qval 48.23	1
and100 0.93	thr 284						
[DQN-Orig]	Ep 889 R	246.98	avgR100	263.30	eps 0.020	Qval 48.61	1
and100 0.92	thr 188						
[DQN-Orig]	Ep 890 R	223.60	avgR100	262.61	eps 0.020	Qval 48.47	1
and100 0.92	thr 257						
[DQN-Orig]	Ep 891 R	270.25	avgR100	262.62	eps 0.020	Qval 49.27	1
and100 0.93	thr 175						
[DQN-Orig]	Ep 892 R	239.70	avgR100	261.96	eps 0.020	Qval 49.08	1
and100 0.93	thr 254						
[DQN-Orig]	Ep 893 R	293.63	avgR100	262.03	eps 0.020	Qval 48.91	1
and100 0.93	thr 290						
[DQN-Orig]	Ep 894 R	221.42	avgR100	261.47	eps 0.020	Qval 48.94	1
and100 0.93	thr 244						
[DQN-Orig]	Ep 895 R	285.45	avgR100	261.33	eps 0.020	Qval 48.73	1
and100 0.93	thr 195						
[DQN-Orig]	Ep 896 R	235.05	avgR100	260.93	eps 0.020	Qval 47.85	1
and100 0.93	thr 243						
[DQN-Orig]	Ep 897 R	267.98	avgR100	261.04	eps 0.020	Qval 48.87	1
and100 0.93	thr 142						
[DQN-Orig]	Ep 898 R	260.47	avgR100	261.09	eps 0.020	Qval 48.55	1
and100 0.92	thr 278						
[DQN-Orig]	Ep 899 R	259.72	avgR100	261.10	eps 0.020	Qval 48.38	1
and100 0.91	thr 405						
[DQN-Orig]	Ep 900 R	255.88	avgR100	260.78	eps 0.020	Qval 49.30	1
and100 0.91	thr 167						

[DQN-Orig]	Ep 901 R	251.08	avgR100	260.63	eps 0.020	Qval	48.30	1
and100 0.91	thr 174							
[DQN-Orig]	Ep 902 R	257.91	avgR100	260.33	eps 0.020	Qval	47.74	1
and100 0.91	thr 187							
[DQN-Orig]	Ep 903 R	239.81	avgR100	259.65	eps 0.020	Qval	48.78	1
and100 0.91	thr 281							
[DQN-Orig]	Ep 904 R	275.06	avgR100	260.15	eps 0.020	Qval	48.36	1
and100 0.91	thr 259							
[DQN-Orig]	Ep 905 R	249.26	avgR100	260.31	eps 0.020	Qval	49.60	1
and100 0.91	thr 160							
[DQN-Orig]	Ep 906 R	275.76	avgR100	260.23	eps 0.020	Qval	47.89	1
and100 0.91	thr 182							
[DQN-Orig]	Ep 907 R	282.42	avgR100	259.97	eps 0.020	Qval	47.66	1
and100 0.92	thr 161							
[DQN-Orig]	Ep 908 R	269.75	avgR100	260.10	eps 0.020	Qval	47.81	1
and100 0.92	thr 143							
[DQN-Orig]	Ep 909 R	249.11	avgR100	260.22	eps 0.020	Qval	48.01	1
and100 0.92	thr 295							
[DQN-Orig]	Ep 910 R	257.89	avgR100	260.07	eps 0.020	Qval	48.94	1
and100 0.92	thr 156							
[DQN-Orig]	Ep 911 R	286.54	avgR100	260.52	eps 0.020	Qval	48.14	1
and100 0.92	thr 264							
[DQN-Orig]	Ep 912 R	263.96	avgR100	260.43	eps 0.020	Qval	48.59	1
and100 0.92	thr 168							
[DQN-Orig]	Ep 913 R	252.42	avgR100	260.51	eps 0.020	Qval	48.40	1
and100 0.92	thr 131							
[DQN-Orig]	Ep 914 R	264.25	avgR100	260.78	eps 0.020	Qval	48.40	1
and100 0.92	thr 169							
[DQN-Orig]	Ep 915 R	274.06	avgR100	260.83	eps 0.020	Qval	48.15	1
and100 0.92	thr 142							
[DQN-Orig]	Ep 916 R	259.31	avgR100	260.87	eps 0.020	Qval	48.43	1
and100 0.92	thr 176							
[DQN-Orig]	Ep 917 R	221.89	avgR100	260.43	eps 0.020	Qval	47.95	1
and100 0.92	thr 206							
[DQN-Orig]	Ep 918 R	263.54	avgR100	260.41	eps 0.020	Qval	49.64	1
and100 0.92	thr 226							
[DQN-Orig]	Ep 919 R	248.40	avgR100	260.22	eps 0.020	Qval	48.33	1
and100 0.92	thr 172							
[DQN-Orig]	Ep 920 R	261.50	avgR100	260.22	eps 0.020	Qval	48.28	1
and100 0.92	thr 185							
[DQN-Orig]	Ep 921 R	241.27	avgR100	260.29	eps 0.020	Qval	47.99	1
and100 0.92	thr 247							
[DQN-Orig]	Ep 922 R	284.19	avgR100	260.23	eps 0.020	Qval	49.19	1
and100 0.92	thr 164							
[DQN-Orig]	Ep 923 R	280.76	avgR100	260.08	eps 0.020	Qval	48.59	1
and100 0.93	thr 133							
[DQN-Orig]	Ep 924 R	286.94	avgR100	260.44	eps 0.020	Qval	49.95	1
and100 0.93	thr 202							
[DQN-Orig]	Ep 925 R	70.96	avgR100	258.95	eps 0.020	Qval	48.90	1
and100 0.92	thr 126							
[DQN-Orig]	Ep 926 R	286.58	avgR100	259.28	eps 0.020	Qval	48.74	1
and100 0.92	thr 269							
[DQN-Orig]	Ep 927 R	246.12	avgR100	259.09	eps 0.020	Qval	48.77	1
and100 0.92	thr 134							
[DQN-Orig]	Ep 928 R	254.86	avgR100	259.04	eps 0.020	Qval	48.81	1
and100 0.92	thr 160							
[DQN-Orig]	Ep 929 R	275.65	avgR100	259.00	eps 0.020	Qval	48.76	1
and100 0.92	thr 208							
[DQN-Orig]	Ep 930 R	241.35	avgR100	259.06	eps 0.020	Qval	48.47	1
and100 0.92	thr 152							

[DQN-Orig] Ep 931 R and100 0.92 thr 201	298.51 avgR100	259.53 eps 0.020 Qval	48.31 1
[DQN-Orig] Ep 932 R and100 0.92 thr 186	263.73 avgR100	259.54 eps 0.020 Qval	48.53 1
[DQN-Orig] Ep 933 R and100 0.92 thr 168	245.55 avgR100	259.37 eps 0.020 Qval	49.76 1
[DQN-Orig] Ep 934 R and100 0.92 thr 132	262.98 avgR100	259.39 eps 0.020 Qval	49.66 1
[DQN-Orig] Ep 935 R and100 0.93 thr 313	257.82 avgR100	259.46 eps 0.020 Qval	49.83 1
[DQN-Orig] Ep 936 R and100 0.93 thr 170	254.52 avgR100	259.68 eps 0.020 Qval	49.66 1
[DQN-Orig] Ep 937 R and100 0.93 thr 135	303.54 avgR100	260.23 eps 0.020 Qval	49.43 1
[DQN-Orig] Ep 938 R and100 0.92 thr 165	246.97 avgR100	260.39 eps 0.020 Qval	49.55 1
[DQN-Orig] Ep 939 R and100 0.92 thr 170	279.58 avgR100	260.56 eps 0.020 Qval	50.48 1
[DQN-Orig] Ep 940 R and100 0.91 thr 149	265.67 avgR100	260.27 eps 0.020 Qval	50.05 1
[DQN-Orig] Ep 941 R and100 0.91 thr 163	235.74 avgR100	260.04 eps 0.020 Qval	50.26 1
[DQN-Orig] Ep 942 R and100 0.90 thr 153	252.19 avgR100	260.15 eps 0.020 Qval	49.77 1
[DQN-Orig] Ep 943 R and100 0.89 thr 168	276.49 avgR100	260.03 eps 0.020 Qval	50.16 1
[DQN-Orig] Ep 944 R and100 0.89 thr 164	299.77 avgR100	260.17 eps 0.020 Qval	50.24 1
[DQN-Orig] Ep 945 R and100 0.89 thr 201	274.54 avgR100	259.84 eps 0.020 Qval	49.76 1
[DQN-Orig] Ep 946 R and100 0.89 thr 371	265.29 avgR100	260.12 eps 0.020 Qval	50.10 1
[DQN-Orig] Ep 947 R and100 0.89 thr 162	301.27 avgR100	260.45 eps 0.020 Qval	50.66 1
[DQN-Orig] Ep 948 R and100 0.88 thr 286	265.88 avgR100	260.20 eps 0.020 Qval	49.78 1
[DQN-Orig] Ep 949 R and100 0.88 thr 166	274.54 avgR100	260.06 eps 0.020 Qval	50.18 1
[DQN-Orig] Ep 950 R and100 0.88 thr 204	287.00 avgR100	259.83 eps 0.020 Qval	50.90 1
[DQN-Orig] Ep 951 R and100 0.89 thr 207	273.56 avgR100	260.20 eps 0.020 Qval	50.77 1
[DQN-Orig] Ep 952 R and100 0.88 thr 323	242.75 avgR100	259.71 eps 0.020 Qval	50.22 1
[DQN-Orig] Ep 953 R and100 0.88 thr 222	289.13 avgR100	259.86 eps 0.020 Qval	50.09 1
[DQN-Orig] Ep 954 R and100 0.88 thr 243	271.30 avgR100	259.84 eps 0.020 Qval	50.56 1
[DQN-Orig] Ep 955 R and100 0.88 thr 238	247.36 avgR100	259.68 eps 0.020 Qval	50.03 1
[DQN-Orig] Ep 956 R and100 0.88 thr 282	278.67 avgR100	260.40 eps 0.020 Qval	49.74 1
[DQN-Orig] Ep 957 R and100 0.88 thr 143	250.47 avgR100	260.41 eps 0.020 Qval	50.60 1
[DQN-Orig] Ep 958 R and100 0.88 thr 181	282.23 avgR100	260.57 eps 0.020 Qval	49.89 1
[DQN-Orig] Ep 959 R and100 0.88 thr 165	254.93 avgR100	260.13 eps 0.020 Qval	49.67 1
[DQN-Orig] Ep 960 R and100 0.87 thr 180	291.18 avgR100	260.26 eps 0.020 Qval	50.68 1

[DQN-Orig]	Ep 961 R	242.87	avgR100	260.04	eps 0.020	Qval	49.85	1
and100 0.87	thr 190							
[DQN-Orig]	Ep 962 R	272.47	avgR100	260.13	eps 0.020	Qval	50.10	1
and100 0.87	thr 188							
[DQN-Orig]	Ep 963 R	292.53	avgR100	260.34	eps 0.020	Qval	49.15	1
and100 0.87	thr 227							
[DQN-Orig]	Ep 964 R	261.35	avgR100	260.22	eps 0.020	Qval	49.78	1
and100 0.87	thr 129							
[DQN-Orig]	Ep 965 R	308.95	avgR100	260.79	eps 0.020	Qval	49.23	1
and100 0.87	thr 186							
[DQN-Orig]	Ep 966 R	235.55	avgR100	260.56	eps 0.020	Qval	49.38	1
and100 0.87	thr 167							
[DQN-Orig]	Ep 967 R	254.18	avgR100	260.51	eps 0.020	Qval	49.52	1
and100 0.87	thr 279							
[DQN-Orig]	Ep 968 R	272.24	avgR100	260.94	eps 0.020	Qval	49.61	1
and100 0.87	thr 151							
[DQN-Orig]	Ep 969 R	263.54	avgR100	260.89	eps 0.020	Qval	50.69	1
and100 0.88	thr 318							
[DQN-Orig]	Ep 970 R	305.26	avgR100	261.18	eps 0.020	Qval	50.67	1
and100 0.88	thr 190							
[DQN-Orig]	Ep 971 R	281.39	avgR100	261.34	eps 0.020	Qval	50.26	1
and100 0.88	thr 204							
[DQN-Orig]	Ep 972 R	307.07	avgR100	261.64	eps 0.020	Qval	50.24	1
and100 0.88	thr 245							
[DQN-Orig]	Ep 973 R	272.77	avgR100	262.10	eps 0.020	Qval	50.56	1
and100 0.87	thr 195							
[DQN-Orig]	Ep 974 R	263.26	avgR100	262.56	eps 0.020	Qval	49.67	1
and100 0.87	thr 172							
[DQN-Orig]	Ep 975 R	268.07	avgR100	262.72	eps 0.020	Qval	50.16	1
and100 0.86	thr 190							
[DQN-Orig]	Ep 976 R	263.57	avgR100	262.63	eps 0.020	Qval	49.24	1
and100 0.86	thr 173							
[DQN-Orig]	Ep 977 R	255.95	avgR100	262.50	eps 0.020	Qval	49.76	1
and100 0.86	thr 148							
[DQN-Orig]	Ep 978 R	257.00	avgR100	262.42	eps 0.020	Qval	49.66	1
and100 0.86	thr 126							
[DQN-Orig]	Ep 979 R	285.13	avgR100	262.38	eps 0.020	Qval	48.82	1
and100 0.86	thr 236							
[DQN-Orig]	Ep 980 R	278.93	avgR100	262.66	eps 0.020	Qval	49.40	1
and100 0.87	thr 155							
[DQN-Orig]	Ep 981 R	246.39	avgR100	262.68	eps 0.020	Qval	48.79	1
and100 0.86	thr 197							
[DQN-Orig]	Ep 982 R	244.75	avgR100	262.45	eps 0.020	Qval	48.40	1
and100 0.86	thr 235							
[DQN-Orig]	Ep 983 R	262.56	avgR100	262.65	eps 0.020	Qval	48.41	1
and100 0.86	thr 203							
[DQN-Orig]	Ep 984 R	303.11	avgR100	263.07	eps 0.020	Qval	47.86	1
and100 0.86	thr 208							
[DQN-Orig]	Ep 985 R	290.48	avgR100	263.64	eps 0.020	Qval	48.74	1
and100 0.86	thr 221							
[DQN-Orig]	Ep 986 R	295.41	avgR100	264.05	eps 0.020	Qval	47.25	1
and100 0.86	thr 253							
[DQN-Orig]	Ep 987 R	265.31	avgR100	264.37	eps 0.020	Qval	47.90	1
and100 0.86	thr 154							
[DQN-Orig]	Ep 988 R	284.00	avgR100	264.60	eps 0.020	Qval	47.81	1
and100 0.86	thr 191							
[DQN-Orig]	Ep 989 R	275.09	avgR100	264.88	eps 0.020	Qval	47.24	1
and100 0.87	thr 131							
[DQN-Orig]	Ep 990 R	263.98	avgR100	265.28	eps 0.020	Qval	46.59	1
and100 0.87	thr 209							

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[DQN-Orig] Ep 991 | R 278.17 | avgR100 265.36 | eps 0.020 | Qval 47.33 | 1
and100 0.87 | thr 145
[DQN-Orig] Ep 992 | R 273.67 | avgR100 265.70 | eps 0.020 | Qval 47.64 | 1
and100 0.87 | thr 201
[DQN-Orig] Ep 993 | R 285.93 | avgR100 265.63 | eps 0.020 | Qval 46.23 | 1
and100 0.87 | thr 219
[DQN-Orig] Ep 994 | R 265.09 | avgR100 266.06 | eps 0.020 | Qval 47.35 | 1
and100 0.87 | thr 169
[DQN-Orig] Ep 995 | R 287.47 | avgR100 266.08 | eps 0.020 | Qval 46.92 | 1
and100 0.87 | thr 165
[DQN-Orig] Ep 996 | R 276.49 | avgR100 266.50 | eps 0.020 | Qval 46.95 | 1
and100 0.87 | thr 177
[DQN-Orig] Ep 997 | R 278.07 | avgR100 266.60 | eps 0.020 | Qval 47.62 | 1
and100 0.87 | thr 171
[DQN-Orig] Ep 998 | R 294.41 | avgR100 266.94 | eps 0.020 | Qval 46.26 | 1
and100 0.88 | thr 296
[DQN-Orig] Ep 999 | R 282.20 | avgR100 267.16 | eps 0.020 | Qval 46.48 | 1
and100 0.89 | thr 155
[DQN-Orig] Ep 1000 | R 280.34 | avgR100 267.41 | eps 0.020 | Qval 47.62 | 1
and100 0.89 | thr 164
[DQN-Orig] finished 1000 episodes in 3.7 min (final avgR100 = 267.41)

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In [28]: # ---- Configuration 2 of 4: DDQN on the ORIGINAL environment ----
RESULTS["DDQN-Orig"] = train_agent(modified=False, double=True, tag="DDQN-Orig")

```

[DDQN-Orig]	Ep	1		R	-76.76		avgR100	-76.76		eps	0.995		Qval	0.05	
land100 0.00		thr	43												
[DDQN-Orig]	Ep	2		R	-111.39		avgR100	-94.07		eps	0.990		Qval	0.05	
land100 0.00		thr	65												
[DDQN-Orig]	Ep	3		R	-248.17		avgR100	-145.44		eps	0.985		Qval	0.05	
land100 0.00		thr	64												
[DDQN-Orig]	Ep	4		R	-122.59		avgR100	-139.73		eps	0.980		Qval	0.05	
land100 0.00		thr	56												
[DDQN-Orig]	Ep	5		R	-292.21		avgR100	-170.22		eps	0.975		Qval	0.05	
land100 0.00		thr	73												
[DDQN-Orig]	Ep	6		R	-118.75		avgR100	-161.64		eps	0.970		Qval	0.05	
land100 0.00		thr	79												
[DDQN-Orig]	Ep	7		R	-500.86		avgR100	-210.10		eps	0.966		Qval	0.05	
land100 0.00		thr	74												
[DDQN-Orig]	Ep	8		R	-112.02		avgR100	-197.84		eps	0.961		Qval	0.05	
land100 0.00		thr	51												
[DDQN-Orig]	Ep	9		R	-95.80		avgR100	-186.51		eps	0.956		Qval	0.05	
land100 0.00		thr	59												
[DDQN-Orig]	Ep	10		R	-224.33		avgR100	-190.29		eps	0.951		Qval	0.05	
land100 0.00		thr	79												
[DDQN-Orig]	Ep	11		R	-289.42		avgR100	-199.30		eps	0.946		Qval	0.05	
land100 0.00		thr	81												
[DDQN-Orig]	Ep	12		R	-102.24		avgR100	-191.21		eps	0.942		Qval	0.09	
land100 0.00		thr	55												
[DDQN-Orig]	Ep	13		R	-48.74		avgR100	-180.25		eps	0.937		Qval	0.15	
land100 0.00		thr	63												
[DDQN-Orig]	Ep	14		R	-498.84		avgR100	-203.01		eps	0.932		Qval	0.25	
land100 0.00		thr	104												
[DDQN-Orig]	Ep	15		R	-93.01		avgR100	-195.67		eps	0.928		Qval	0.15	
land100 0.00		thr	56												
[DDQN-Orig]	Ep	16		R	-185.01		avgR100	-195.01		eps	0.923		Qval	0.18	
land100 0.00		thr	51												
[DDQN-Orig]	Ep	17		R	-183.11		avgR100	-194.31		eps	0.918		Qval	0.10	
land100 0.00		thr	106												
[DDQN-Orig]	Ep	18		R	-301.05		avgR100	-200.24		eps	0.914		Qval	-0.03	
land100 0.00		thr	94												
[DDQN-Orig]	Ep	19		R	-401.95		avgR100	-210.85		eps	0.909		Qval	-0.38	
land100 0.00		thr	97												
[DDQN-Orig]	Ep	20		R	-278.16		avgR100	-214.22		eps	0.905		Qval	-0.33	
land100 0.00		thr	75												
[DDQN-Orig]	Ep	21	</												

[illegible]

[illegible]

[DDQN-Orig]	Ep 91 R	-67.94	avgR100	-138.04	eps 0.634	Qval	6.70
land100 0.01	thr 127						
[DDQN-Orig]	Ep 92 R	15.58	avgR100	-136.37	eps 0.631	Qval	7.23
land100 0.01	thr 168						
[DDQN-Orig]	Ep 93 R	-229.43	avgR100	-137.38	eps 0.627	Qval	7.44
land100 0.01	thr 162						
[DDQN-Orig]	Ep 94 R	-59.70	avgR100	-136.55	eps 0.624	Qval	7.75
land100 0.01	thr 116						
[DDQN-Orig]	Ep 95 R	-185.71	avgR100	-137.07	eps 0.621	Qval	7.47
land100 0.01	thr 118						
[DDQN-Orig]	Ep 96 R	-84.11	avgR100	-136.51	eps 0.618	Qval	7.73
land100 0.01	thr 141						
[DDQN-Orig]	Ep 97 R	-171.69	avgR100	-136.88	eps 0.615	Qval	7.96
land100 0.01	thr 111						
[DDQN-Orig]	Ep 98 R	-125.47	avgR100	-136.76	eps 0.612	Qval	7.92
land100 0.01	thr 100						
[DDQN-Orig]	Ep 99 R	-77.58	avgR100	-136.16	eps 0.609	Qval	8.24
land100 0.01	thr 150						
[DDQN-Orig]	Ep 100 R	-47.63	avgR100	-135.28	eps 0.606	Qval	8.41
land100 0.01	thr 89						
[DDQN-Orig]	Ep 101 R	-78.44	avgR100	-135.29	eps 0.603	Qval	8.70
land100 0.01	thr 136						
[DDQN-Orig]	Ep 102 R	-137.34	avgR100	-135.55	eps 0.600	Qval	8.98
land100 0.01	thr 197						
[DDQN-Orig]	Ep 103 R	-54.27	avgR100	-133.62	eps 0.597	Qval	9.23
land100 0.01	thr 174						
[DDQN-Orig]	Ep 104 R	5.01	avgR100	-132.34	eps 0.594	Qval	9.20
land100 0.01	thr 96						
[DDQN-Orig]	Ep 105 R	-103.10	avgR100	-130.45	eps 0.591	Qval	9.53
land100 0.01	thr 188						
[DDQN-Orig]	Ep 106 R	-67.76	avgR100	-129.94	eps 0.588	Qval	9.87
land100 0.01	thr 77						
[DDQN-Orig]	Ep 107 R	-57.62	avgR100	-125.51	eps 0.585	Qval	9.47
land100 0.01	thr 122						
[DDQN-Orig]	Ep 108 R	-46.60	avgR100	-124.85	eps 0.582	Qval	10.34
land100 0.01	thr 175						
[DDQN-Orig]	Ep 109 R	-62.94	avgR100	-124.52	eps 0.579	Qval	10.12
land100 0.01	thr 80						
[DDQN-Orig]	Ep 110 R	-12.24	avgR100	-122.40	eps 0.576	Qval	10.57
land100 0.01	thr 115						
[DDQN-Orig]	Ep 111 R	-75.79	avgR100	-120.27	eps 0.573	Qval	10.71
land100 0.01	thr 148						
[DDQN-Orig]	Ep 112 R	-5.64	avgR100	-119.30	eps 0.570	Qval	10.82
land100 0.01	thr 132						
[DDQN-Orig]	Ep 113 R	-59.25	avgR100	-119.40	eps 0.568	Qval	10.85
land100 0.01	thr 146						
[DDQN-Orig]	Ep 114 R	-186.79	avgR100	-116.28	eps 0.565	Qval	11.19
land100 0.01	thr 98						
[DDQN-Orig]	Ep 115 R	-38.63	avgR100	-115.74	eps 0.562	Qval	11.34
land100 0.01	thr 156						
[DDQN-Orig]	Ep 116 R	-26.10	avgR100	-114.15	eps 0.559	Qval	12.01
land100 0.01	thr 135						
[DDQN-Orig]	Ep 117 R	-7.99	avgR100	-112.40	eps 0.556	Qval	11.48
land100 0.01	thr 91						
[DDQN-Orig]	Ep 118 R	-21.72	avgR100	-109.61	eps 0.554	Qval	12.26
land100 0.01	thr 143						
[DDQN-Orig]	Ep 119 R	-88.49	avgR100	-106.47	eps 0.551	Qval	12.70
land100 0.01	thr 170						
[DDQN-Orig]	Ep 120 R	-58.18	avgR100	-104.27	eps 0.548	Qval	12.81
land100 0.01	thr 179						

[DDQN-Orig]	Ep 121 R	-88.42 avgR100	-103.29 eps 0.545	Qval 12.51
land100 0.01 thr 104				
[DDQN-Orig]	Ep 122 R	-29.73 avgR100	-102.74 eps 0.543	Qval 14.56
land100 0.01 thr 862				
[DDQN-Orig]	Ep 123 R	-10.82 avgR100	-100.42 eps 0.540	Qval 14.60
land100 0.01 thr 109				
[DDQN-Orig]	Ep 124 R	-49.01 avgR100	-101.21 eps 0.537	Qval 14.73
land100 0.01 thr 88				
[DDQN-Orig]	Ep 125 R	-31.26 avgR100	-98.55 eps 0.534	Qval 14.99
land100 0.01 thr 66				
[DDQN-Orig]	Ep 126 R	18.92 avgR100	-97.15 eps 0.532	Qval 15.61
land100 0.01 thr 145				
[DDQN-Orig]	Ep 127 R	-82.20 avgR100	-96.93 eps 0.529	Qval 15.64
land100 0.01 thr 113				
[DDQN-Orig]	Ep 128 R	15.72 avgR100	-94.04 eps 0.526	Qval 16.74
land100 0.01 thr 852				
[DDQN-Orig]	Ep 129 R	20.17 avgR100	-92.52 eps 0.524	Qval 17.23
land100 0.01 thr 228				
[DDQN-Orig]	Ep 130 R	-1.81 avgR100	-91.50 eps 0.521	Qval 18.87
land100 0.01 thr 796				
[DDQN-Orig]	Ep 131 R	89.90 avgR100	-89.47 eps 0.519	Qval 20.41
land100 0.01 thr 701				
[DDQN-Orig]	Ep 132 R	49.33 avgR100	-88.02 eps 0.516	Qval 20.57
land100 0.01 thr 119				
[DDQN-Orig]	Ep 133 R	-85.24 avgR100	-88.41 eps 0.513	Qval 20.63
land100 0.01 thr 74				
[DDQN-Orig]	Ep 134 R	-64.86 avgR100	-87.04 eps 0.511	Qval 20.99
land100 0.01 thr 362				
[DDQN-Orig]	Ep 135 R	-48.90 avgR100	-85.86 eps 0.508	Qval 21.87
land100 0.01 thr 282				
[DDQN-Orig]	Ep 136 R	17.07 avgR100	-83.67 eps 0.506	Qval 22.58
land100 0.01 thr 256				
[DDQN-Orig]	Ep 137 R	-3.31 avgR100	-82.01 eps 0.503	Qval 22.61
land100 0.01 thr 175				
[DDQN-Orig]	Ep 138 R	-96.96 avgR100	-82.07 eps 0.501	Qval 22.77
land100 0.01 thr 138				
[DDQN-Orig]	Ep 139 R	-129.80 avgR100	-81.84 eps 0.498	Qval 23.31
land100 0.01 thr 181				
[DDQN-Orig]	Ep 140 R	-18.68 avgR100	-80.51 eps 0.496	Qval 23.39
land100 0.01 thr 201				
[DDQN-Orig]	Ep 141 R	-28.53 avgR100	-80.10 eps 0.493	Qval 23.79
land100 0.01 thr 123				
[DDQN-Orig]	Ep 142 R	-75.26 avgR100	-79.82 eps 0.491	Qval 23.79
land100 0.01 thr 135				
[DDQN-Orig]	Ep 143 R	30.20 avgR100	-77.13 eps 0.488	Qval 24.94
land100 0.01 thr 769				
[DDQN-Orig]	Ep 144 R	56.52 avgR100	-73.99 eps 0.486	Qval 26.24
land100 0.01 thr 791				
[DDQN-Orig]	Ep 145 R	2.83 avgR100	-73.15 eps 0.483	Qval 26.52
land100 0.01 thr 296				
[DDQN-Orig]	Ep 146 R	-67.38 avgR100	-72.74 eps 0.481	Qval 27.01
land100 0.01 thr 147				
[DDQN-Orig]	Ep 147 R	-12.35 avgR100	-71.07 eps 0.479	Qval 26.88
land100 0.01 thr 196				
[DDQN-Orig]	Ep 148 R	7.56 avgR100	-70.39 eps 0.476	Qval 26.94
land100 0.01 thr 125				
[DDQN-Orig]	Ep 149 R	-21.46 avgR100	-69.55 eps 0.474	Qval 28.74
land100 0.01 thr 808				
[DDQN-Orig]	Ep 150 R	36.57 avgR100	-68.22 eps 0.471	Qval 28.85
land100 0.01 thr 146				

[DDQN-Orig]	Ep 151 R	-36.13	avgR100	-66.25	eps 0.469	Qval	28.74
land100 0.01	thr 253						
[DDQN-Orig]	Ep 152 R	-20.47	avgR100	-65.19	eps 0.467	Qval	29.38
land100 0.01	thr 122						
[DDQN-Orig]	Ep 153 R	-22.62	avgR100	-64.47	eps 0.464	Qval	28.95
land100 0.01	thr 64						
[DDQN-Orig]	Ep 154 R	-91.96	avgR100	-64.06	eps 0.462	Qval	29.43
land100 0.01	thr 228						
[DDQN-Orig]	Ep 155 R	-44.88	avgR100	-63.07	eps 0.460	Qval	30.22
land100 0.01	thr 882						
[DDQN-Orig]	Ep 156 R	-53.62	avgR100	-62.46	eps 0.458	Qval	30.57
land100 0.01	thr 80						
[DDQN-Orig]	Ep 157 R	0.14	avgR100	-61.46	eps 0.455	Qval	30.23
land100 0.01	thr 99						
[DDQN-Orig]	Ep 158 R	-27.54	avgR100	-60.42	eps 0.453	Qval	31.71
land100 0.01	thr 802						
[DDQN-Orig]	Ep 159 R	-22.16	avgR100	-59.60	eps 0.451	Qval	31.98
land100 0.01	thr 849						
[DDQN-Orig]	Ep 160 R	29.45	avgR100	-58.66	eps 0.448	Qval	31.05
land100 0.01	thr 122						
[DDQN-Orig]	Ep 161 R	-24.68	avgR100	-58.55	eps 0.446	Qval	32.14
land100 0.01	thr 93						
[DDQN-Orig]	Ep 162 R	-10.00	avgR100	-57.35	eps 0.444	Qval	31.72
land100 0.01	thr 148						
[DDQN-Orig]	Ep 163 R	-68.04	avgR100	-55.99	eps 0.442	Qval	31.57
land100 0.01	thr 169						
[DDQN-Orig]	Ep 164 R	-22.46	avgR100	-54.07	eps 0.440	Qval	31.99
land100 0.01	thr 248						
[DDQN-Orig]	Ep 165 R	26.23	avgR100	-52.62	eps 0.437	Qval	31.57
land100 0.01	thr 137						
[DDQN-Orig]	Ep 166 R	-150.98	avgR100	-53.23	eps 0.435	Qval	32.38
land100 0.01	thr 146						
[DDQN-Orig]	Ep 167 R	-70.09	avgR100	-53.09	eps 0.433	Qval	32.27
land100 0.01	thr 73						
[DDQN-Orig]	Ep 168 R	-63.09	avgR100	-53.26	eps 0.431	Qval	32.61
land100 0.01	thr 113						
[DDQN-Orig]	Ep 169 R	-37.89	avgR100	-52.46	eps 0.429	Qval	32.17
land100 0.01	thr 206						
[DDQN-Orig]	Ep 170 R	1.27	avgR100	-51.06	eps 0.427	Qval	32.74
land100 0.01	thr 196						
[DDQN-Orig]	Ep 171 R	-26.61	avgR100	-50.24	eps 0.424	Qval	31.93
land100 0.01	thr 113						
[DDQN-Orig]	Ep 172 R	20.35	avgR100	-49.55	eps 0.422	Qval	32.72
land100 0.01	thr 848						
[DDQN-Orig]	Ep 173 R	-42.32	avgR100	-48.87	eps 0.420	Qval	32.35
land100 0.01	thr 72						
[DDQN-Orig]	Ep 174 R	-61.93	avgR100	-48.64	eps 0.418	Qval	33.18
land100 0.01	thr 871						
[DDQN-Orig]	Ep 175 R	15.38	avgR100	-48.09	eps 0.416	Qval	32.95
land100 0.00	thr 89						
[DDQN-Orig]	Ep 176 R	-24.43	avgR100	-47.06	eps 0.414	Qval	32.93
land100 0.00	thr 68						
[DDQN-Orig]	Ep 177 R	-12.91	avgR100	-46.05	eps 0.412	Qval	33.08
land100 0.00	thr 238						
[DDQN-Orig]	Ep 178 R	3.48	avgR100	-45.12	eps 0.410	Qval	33.51
land100 0.00	thr 844						
[DDQN-Orig]	Ep 179 R	7.44	avgR100	-43.89	eps 0.408	Qval	33.34
land100 0.01	thr 104						
[DDQN-Orig]	Ep 180 R	7.65	avgR100	-43.77	eps 0.406	Qval	32.23
land100 0.01	thr 103						

[DDQN-Orig]	Ep 181 R	-52.25	avgR100	-44.56	eps 0.404	Qval	33.41
land100 0.01	thr 876						
[DDQN-Orig]	Ep 182 R	-63.44	avgR100	-44.73	eps 0.402	Qval	32.66
land100 0.01	thr 216						
[DDQN-Orig]	Ep 183 R	-44.24	avgR100	-44.42	eps 0.400	Qval	32.62
land100 0.01	thr 73						
[DDQN-Orig]	Ep 184 R	6.57	avgR100	-43.45	eps 0.398	Qval	32.92
land100 0.01	thr 865						
[DDQN-Orig]	Ep 185 R	-15.23	avgR100	-42.89	eps 0.396	Qval	33.50
land100 0.01	thr 100						
[DDQN-Orig]	Ep 186 R	-102.54	avgR100	-42.72	eps 0.394	Qval	33.28
land100 0.01	thr 160						
[DDQN-Orig]	Ep 187 R	31.64	avgR100	-41.71	eps 0.392	Qval	32.85
land100 0.01	thr 846						
[DDQN-Orig]	Ep 188 R	-3.02	avgR100	-40.29	eps 0.390	Qval	32.69
land100 0.01	thr 130						
[DDQN-Orig]	Ep 189 R	-59.56	avgR100	-40.35	eps 0.388	Qval	32.70
land100 0.01	thr 63						
[DDQN-Orig]	Ep 190 R	-36.83	avgR100	-39.81	eps 0.386	Qval	32.28
land100 0.01	thr 853						
[DDQN-Orig]	Ep 191 R	10.24	avgR100	-39.03	eps 0.384	Qval	32.04
land100 0.01	thr 159						
[DDQN-Orig]	Ep 192 R	-33.21	avgR100	-39.52	eps 0.382	Qval	32.00
land100 0.01	thr 875						
[DDQN-Orig]	Ep 193 R	-116.87	avgR100	-38.39	eps 0.380	Qval	31.16
land100 0.01	thr 726						
[DDQN-Orig]	Ep 194 R	-44.50	avgR100	-38.24	eps 0.378	Qval	31.20
land100 0.01	thr 876						
[DDQN-Orig]	Ep 195 R	-93.50	avgR100	-37.32	eps 0.376	Qval	30.69
land100 0.01	thr 103						
[DDQN-Orig]	Ep 196 R	42.35	avgR100	-36.05	eps 0.374	Qval	31.02
land100 0.01	thr 118						
[DDQN-Orig]	Ep 197 R	1.70	avgR100	-34.32	eps 0.373	Qval	30.08
land100 0.01	thr 839						
[DDQN-Orig]	Ep 198 R	-25.36	avgR100	-33.32	eps 0.371	Qval	30.11
land100 0.01	thr 79						
[DDQN-Orig]	Ep 199 R	-80.46	avgR100	-33.35	eps 0.369	Qval	29.23
land100 0.01	thr 279						
[DDQN-Orig]	Ep 200 R	101.45	avgR100	-31.86	eps 0.367	Qval	29.22
land100 0.01	thr 870						
[DDQN-Orig]	Ep 201 R	-40.93	avgR100	-31.48	eps 0.365	Qval	29.23
land100 0.01	thr 860						
[DDQN-Orig]	Ep 202 R	-15.74	avgR100	-30.26	eps 0.363	Qval	28.11
land100 0.01	thr 124						
[DDQN-Orig]	Ep 203 R	-15.09	avgR100	-29.87	eps 0.361	Qval	27.90
land100 0.01	thr 828						
[DDQN-Orig]	Ep 204 R	-43.75	avgR100	-30.36	eps 0.360	Qval	27.46
land100 0.01	thr 123						
[DDQN-Orig]	Ep 205 R	-8.23	avgR100	-29.41	eps 0.358	Qval	28.48
land100 0.01	thr 80						
[DDQN-Orig]	Ep 206 R	-14.01	avgR100	-28.87	eps 0.356	Qval	26.72
land100 0.01	thr 833						
[DDQN-Orig]	Ep 207 R	-20.10	avgR100	-28.50	eps 0.354	Qval	26.88
land100 0.01	thr 89						
[DDQN-Orig]	Ep 208 R	-20.19	avgR100	-28.23	eps 0.353	Qval	25.89
land100 0.01	thr 873						
[DDQN-Orig]	Ep 209 R	-131.05	avgR100	-28.92	eps 0.351	Qval	26.72
land100 0.01	thr 129						
[DDQN-Orig]	Ep 210 R	-35.53	avgR100	-29.15	eps 0.349	Qval	25.38
land100 0.01	thr 918						

[DDQN-Orig]	Ep 211 R	-31.36	avgR100	-28.70	eps 0.347	Qval	24.70	
land100 0.01	thr 854							
[DDQN-Orig]	Ep 212 R	67.27	avgR100	-27.98	eps 0.346	Qval	23.19	
land100 0.01	thr 895							
[DDQN-Orig]	Ep 213 R	0.78	avgR100	-27.37	eps 0.344	Qval	23.15	
land100 0.01	thr 142							
[DDQN-Orig]	Ep 214 R	-49.23	avgR100	-26.00	eps 0.342	Qval	23.57	
land100 0.01	thr 87							
[DDQN-Orig]	Ep 215 R	20.91	avgR100	-25.40	eps 0.340	Qval	22.98	
land100 0.01	thr 109							
[DDQN-Orig]	Ep 216 R	-9.68	avgR100	-25.24	eps 0.339	Qval	22.67	
land100 0.01	thr 814							
[DDQN-Orig]	Ep 217 R	239.66	avgR100	-22.76	eps 0.337	Qval	22.17	
land100 0.01	thr 401							
[DDQN-Orig]	Ep 218 R	-26.37	avgR100	-22.81	eps 0.335	Qval	21.28	
land100 0.01	thr 858							
[DDQN-Orig]	Ep 219 R	-19.53	avgR100	-22.12	eps 0.334	Qval	19.66	
land100 0.01	thr 845							
[DDQN-Orig]	Ep 220 R	-60.78	avgR100	-22.15	eps 0.332	Qval	20.02	
land100 0.01	thr 71							
[DDQN-Orig]	Ep 221 R	-89.86	avgR100	-22.16	eps 0.330	Qval	20.12	
land100 0.01	thr 145							
[DDQN-Orig]	Ep 222 R	7.70	avgR100	-21.79	eps 0.329	Qval	18.64	
land100 0.01	thr 802							
[DDQN-Orig]	Ep 223 R	-36.03	avgR100	-22.04	eps 0.327	Qval	18.63	
land100 0.01	thr 838							
[DDQN-Orig]	Ep 224 R	-48.13	avgR100	-22.03	eps 0.325	Qval	16.91	
land100 0.01	thr 849							
[DDQN-Orig]	Ep 225 R	-11.76	avgR100	-21.83	eps 0.324	Qval	16.80	
land100 0.01	thr 834							
[DDQN-Orig]	Ep 226 R	-20.15	avgR100	-22.23	eps 0.322	Qval	16.19	
land100 0.01	thr 850							
[DDQN-Orig]	Ep 227 R	-20.50	avgR100	-21.61	eps 0.321	Qval	15.56	
land100 0.01	thr 855							
[DDQN-Orig]	Ep 228 R	-34.13	avgR100	-22.11	eps 0.319	Qval	13.87	
land100 0.01	thr 799							
[DDQN-Orig]	Ep 229 R	3.51	avgR100	-22.27	eps 0.317	Qval	13.61	
land100 0.01	thr 133							
[DDQN-Orig]	Ep 230 R	-1.65	avgR100	-22.27	eps 0.316	Qval	12.84	
land100 0.01	thr 817							
[DDQN-Orig]	Ep 231 R	-20.26	avgR100	-23.37	eps 0.314	Qval	12.65	
land100 0.01	thr 112							
[DDQN-Orig]	Ep 232 R	-14.24	avgR100	-24.01	eps 0.313	Qval	11.65	
land100 0.01	thr 794							
[DDQN-Orig]	Ep 233 R	-3.38	avgR100	-23.19	eps 0.311	Qval	12.08	
land100 0.01	thr 863							
[DDQN-Orig]	Ep 234 R	8.84	avgR100	-22.45	eps 0.309	Qval	9.48	
land100 0.01	thr 878							
[DDQN-Orig]	Ep 235 R	-15.00	avgR100	-22.11	eps 0.308	Qval	9.91	
land100 0.01	thr 853							
[DDQN-Orig]	Ep 236 R	-20.27	avgR100	-22.49	eps 0.306	Qval	9.74	
land100 0.01	thr 878							
[DDQN-Orig]	Ep 237 R	-9.83	avgR100	-22.55	eps 0.305	Qval	9.80	
land100 0.01	thr 821							
[DDQN-Orig]	Ep 238 R	25.22	avgR100	-21.33	eps 0.303	Qval	9.19	
land100 0.01	thr 167							
[DDQN-Orig]	Ep 239 R	57.34	avgR100	-19.46	eps 0.302	Qval	8.94	
land100 0.01	thr 742							
[DDQN-Orig]	Ep 240 R	50.42	avgR100	-18.77	eps 0.300	Qval	7.44	
land100 0.01	thr 800							

[DDQN-Orig] Ep 241 R	-3.93	avgR100	-18.52	eps 0.299	Qval	6.88
land100 0.01 thr 840						
[DDQN-Orig] Ep 242 R	-1.04	avgR100	-17.78	eps 0.297	Qval	6.54
land100 0.01 thr 741						
[DDQN-Orig] Ep 243 R	253.74	avgR100	-15.55	eps 0.296	Qval	7.37
land100 0.02 thr 297						
[DDQN-Orig] Ep 244 R	4.05	avgR100	-16.07	eps 0.294	Qval	7.36
land100 0.02 thr 109						
[DDQN-Orig] Ep 245 R	26.08	avgR100	-15.84	eps 0.293	Qval	5.27
land100 0.02 thr 764						
[DDQN-Orig] Ep 246 R	18.26	avgR100	-14.98	eps 0.291	Qval	5.88
land100 0.02 thr 884						
[DDQN-Orig] Ep 247 R	46.51	avgR100	-14.39	eps 0.290	Qval	5.76
land100 0.02 thr 138						
[DDQN-Orig] Ep 248 R	-38.67	avgR100	-14.85	eps 0.288	Qval	5.58
land100 0.02 thr 112						
[DDQN-Orig] Ep 249 R	-7.45	avgR100	-14.71	eps 0.287	Qval	5.99
land100 0.02 thr 833						
[DDQN-Orig] Ep 250 R	6.10	avgR100	-15.02	eps 0.286	Qval	5.13
land100 0.02 thr 821						
[DDQN-Orig] Ep 251 R	52.07	avgR100	-14.14	eps 0.284	Qval	5.79
land100 0.02 thr 740						
[DDQN-Orig] Ep 252 R	30.45	avgR100	-13.63	eps 0.283	Qval	5.46
land100 0.02 thr 790						
[DDQN-Orig] Ep 253 R	25.94	avgR100	-13.14	eps 0.281	Qval	7.69
land100 0.02 thr 726						
[DDQN-Orig] Ep 254 R	81.38	avgR100	-11.41	eps 0.280	Qval	5.03
land100 0.02 thr 863						
[DDQN-Orig] Ep 255 R	-55.70	avgR100	-11.52	eps 0.279	Qval	5.08
land100 0.02 thr 131						
[DDQN-Orig] Ep 256 R	25.18	avgR100	-10.73	eps 0.277	Qval	5.64
land100 0.02 thr 717						
[DDQN-Orig] Ep 257 R	37.55	avgR100	-10.36	eps 0.276	Qval	4.35
land100 0.02 thr 744						
[DDQN-Orig] Ep 258 R	180.52	avgR100	-8.27	eps 0.274	Qval	4.06
land100 0.03 thr 624						
[DDQN-Orig] Ep 259 R	59.87	avgR100	-7.45	eps 0.273	Qval	4.98
land100 0.03 thr 782						
[DDQN-Orig] Ep 260 R	13.23	avgR100	-7.62	eps 0.272	Qval	4.95
land100 0.03 thr 159						
[DDQN-Orig] Ep 261 R	71.54	avgR100	-6.65	eps 0.270	Qval	5.00
land100 0.03 thr 168						
[DDQN-Orig] Ep 262 R	-30.56	avgR100	-6.86	eps 0.269	Qval	4.82
land100 0.03 thr 118						
[DDQN-Orig] Ep 263 R	51.21	avgR100	-5.67	eps 0.268	Qval	5.20
land100 0.03 thr 731						
[DDQN-Orig] Ep 264 R	-4.57	avgR100	-5.49	eps 0.266	Qval	4.02
land100 0.03 thr 91						
[DDQN-Orig] Ep 265 R	55.63	avgR100	-5.19	eps 0.265	Qval	4.08
land100 0.03 thr 815						
[DDQN-Orig] Ep 266 R	40.37	avgR100	-3.28	eps 0.264	Qval	3.82
land100 0.03 thr 768						
[DDQN-Orig] Ep 267 R	111.13	avgR100	-1.47	eps 0.262	Qval	2.65
land100 0.03 thr 612						
[DDQN-Orig] Ep 268 R	109.22	avgR100	0.25	eps 0.261	Qval	1.84
land100 0.03 thr 586						
[DDQN-Orig] Ep 269 R	90.99	avgR100	1.54	eps 0.260	Qval	2.19
land100 0.03 thr 727						
[DDQN-Orig] Ep 270 R	-13.43	avgR100	1.40	eps 0.258	Qval	2.42
land100 0.03 thr 94						

[DDQN-Orig]	Ep 271 R	58.67	avgR100	2.25	eps 0.257	Qval	1.70
land100 0.03	thr 736						
[DDQN-Orig]	Ep 272 R	34.50	avgR100	2.39	eps 0.256	Qval	2.66
land100 0.03	thr 690						
[DDQN-Orig]	Ep 273 R	-1.41	avgR100	2.80	eps 0.255	Qval	1.14
land100 0.03	thr 105						
[DDQN-Orig]	Ep 274 R	64.60	avgR100	4.06	eps 0.253	Qval	2.39
land100 0.03	thr 777						
[DDQN-Orig]	Ep 275 R	42.32	avgR100	4.33	eps 0.252	Qval	1.82
land100 0.03	thr 720						
[DDQN-Orig]	Ep 276 R	12.34	avgR100	4.70	eps 0.251	Qval	2.46
land100 0.03	thr 834						
[DDQN-Orig]	Ep 277 R	71.13	avgR100	5.54	eps 0.249	Qval	2.19
land100 0.03	thr 749						
[DDQN-Orig]	Ep 278 R	25.04	avgR100	5.76	eps 0.248	Qval	2.03
land100 0.03	thr 146						
[DDQN-Orig]	Ep 279 R	91.16	avgR100	6.59	eps 0.247	Qval	2.07
land100 0.02	thr 618						
[DDQN-Orig]	Ep 280 R	5.96	avgR100	6.58	eps 0.246	Qval	2.14
land100 0.02	thr 156						
[DDQN-Orig]	Ep 281 R	61.26	avgR100	7.71	eps 0.245	Qval	2.07
land100 0.02	thr 737						
[DDQN-Orig]	Ep 282 R	-34.58	avgR100	8.00	eps 0.243	Qval	1.78
land100 0.02	thr 850						
[DDQN-Orig]	Ep 283 R	55.89	avgR100	9.00	eps 0.242	Qval	1.99
land100 0.02	thr 657						
[DDQN-Orig]	Ep 284 R	124.09	avgR100	10.18	eps 0.241	Qval	2.13
land100 0.02	thr 620						
[DDQN-Orig]	Ep 285 R	78.10	avgR100	11.11	eps 0.240	Qval	4.12
land100 0.02	thr 647						
[DDQN-Orig]	Ep 286 R	254.57	avgR100	14.68	eps 0.238	Qval	5.04
land100 0.03	thr 422						
[DDQN-Orig]	Ep 287 R	23.97	avgR100	14.61	eps 0.237	Qval	5.39
land100 0.03	thr 873						
[DDQN-Orig]	Ep 288 R	21.45	avgR100	14.85	eps 0.236	Qval	4.45
land100 0.03	thr 140						
[DDQN-Orig]	Ep 289 R	125.71	avgR100	16.70	eps 0.235	Qval	6.06
land100 0.03	thr 639						
[DDQN-Orig]	Ep 290 R	93.93	avgR100	18.01	eps 0.234	Qval	6.63
land100 0.03	thr 643						
[DDQN-Orig]	Ep 291 R	198.27	avgR100	19.89	eps 0.233	Qval	6.74
land100 0.04	thr 471						
[DDQN-Orig]	Ep 292 R	202.38	avgR100	22.25	eps 0.231	Qval	6.79
land100 0.05	thr 587						
[DDQN-Orig]	Ep 293 R	99.29	avgR100	24.41	eps 0.230	Qval	9.45
land100 0.05	thr 609						
[DDQN-Orig]	Ep 294 R	278.71	avgR100	27.64	eps 0.229	Qval	8.29
land100 0.06	thr 168						
[DDQN-Orig]	Ep 295 R	143.82	avgR100	30.01	eps 0.228	Qval	9.28
land100 0.06	thr 564						
[DDQN-Orig]	Ep 296 R	273.83	avgR100	32.33	eps 0.227	Qval	9.26
land100 0.07	thr 266						
[DDQN-Orig]	Ep 297 R	-8.96	avgR100	32.22	eps 0.226	Qval	9.72
land100 0.07	thr 95						
[DDQN-Orig]	Ep 298 R	81.54	avgR100	33.29	eps 0.225	Qval	9.54
land100 0.07	thr 597						
[DDQN-Orig]	Ep 299 R	16.75	avgR100	34.26	eps 0.223	Qval	9.88
land100 0.07	thr 137						
[DDQN-Orig]	Ep 300 R	216.18	avgR100	35.41	eps 0.222	Qval	9.55
land100 0.07	thr 253						

[DDQN-Orig]	Ep 301 R	283.45 avgR100	38.65 eps 0.221 Qval	9.04
land100 0.08 thr	349			
[DDQN-Orig]	Ep 302 R	130.92 avgR100	40.12 eps 0.220 Qval	9.86
land100 0.08 thr	653			
[DDQN-Orig]	Ep 303 R	208.82 avgR100	42.36 eps 0.219 Qval	10.00
land100 0.09 thr	396			
[DDQN-Orig]	Ep 304 R	122.10 avgR100	44.02 eps 0.218 Qval	10.70
land100 0.09 thr	552			
[DDQN-Orig]	Ep 305 R	110.29 avgR100	45.20 eps 0.217 Qval	10.20
land100 0.09 thr	651			
[DDQN-Orig]	Ep 306 R	183.46 avgR100	47.18 eps 0.216 Qval	11.64
land100 0.10 thr	541			
[DDQN-Orig]	Ep 307 R	43.38 avgR100	47.81 eps 0.215 Qval	10.90
land100 0.10 thr	717			
[DDQN-Orig]	Ep 308 R	198.84 avgR100	50.00 eps 0.214 Qval	11.43
land100 0.11 thr	442			
[DDQN-Orig]	Ep 309 R	282.72 avgR100	54.14 eps 0.212 Qval	11.93
land100 0.12 thr	227			
[DDQN-Orig]	Ep 310 R	8.52 avgR100	54.58 eps 0.211 Qval	12.03
land100 0.12 thr	171			
[DDQN-Orig]	Ep 311 R	58.77 avgR100	55.48 eps 0.210 Qval	11.45
land100 0.12 thr	723			
[DDQN-Orig]	Ep 312 R	-18.24 avgR100	54.63 eps 0.209 Qval	12.52
land100 0.12 thr	65			
[DDQN-Orig]	Ep 313 R	199.22 avgR100	56.61 eps 0.208 Qval	12.04
land100 0.12 thr	704			
[DDQN-Orig]	Ep 314 R	264.06 avgR100	59.75 eps 0.207 Qval	13.01
land100 0.13 thr	286			
[DDQN-Orig]	Ep 315 R	73.45 avgR100	60.27 eps 0.206 Qval	12.53
land100 0.13 thr	602			
[DDQN-Orig]	Ep 316 R	260.36 avgR100	62.97 eps 0.205 Qval	13.95
land100 0.14 thr	308			
[DDQN-Orig]	Ep 317 R	145.12 avgR100	62.03 eps 0.204 Qval	14.46
land100 0.14 thr	477			
[DDQN-Orig]	Ep 318 R	149.86 avgR100	63.79 eps 0.203 Qval	14.75
land100 0.15 thr	783			
[DDQN-Orig]	Ep 319 R	237.50 avgR100	66.36 eps 0.202 Qval	14.20
land100 0.16 thr	258			
[DDQN-Orig]	Ep 320 R	253.06 avgR100	69.50 eps 0.201 Qval	14.79
land100 0.17 thr	282			
[DDQN-Orig]	Ep 321 R	238.64 avgR100	72.78 eps 0.200 Qval	14.90
land100 0.18 thr	213			
[DDQN-Orig]	Ep 322 R	160.45 avgR100	74.31 eps 0.199 Qval	15.25
land100 0.18 thr	708			
[DDQN-Orig]	Ep 323 R	252.85 avgR100	77.20 eps 0.198 Qval	14.89
land100 0.19 thr	251			
[DDQN-Orig]	Ep 324 R	106.60 avgR100	78.75 eps 0.197 Qval	13.98
land100 0.19 thr	575			
[DDQN-Orig]	Ep 325 R	199.92 avgR100	80.86 eps 0.196 Qval	15.47
land100 0.20 thr	669			
[DDQN-Orig]	Ep 326 R	11.89 avgR100	81.18 eps 0.195 Qval	15.03
land100 0.20 thr	111			
[DDQN-Orig]	Ep 327 R	277.11 avgR100	84.16 eps 0.194 Qval	15.13
land100 0.21 thr	450			
[DDQN-Orig]	Ep 328 R	248.19 avgR100	86.98 eps 0.193 Qval	15.41
land100 0.22 thr	447			
[DDQN-Orig]	Ep 329 R	127.79 avgR100	88.22 eps 0.192 Qval	15.54
land100 0.22 thr	418			
[DDQN-Orig]	Ep 330 R	229.73 avgR100	90.54 eps 0.191 Qval	14.70
land100 0.23 thr	320			

[DDQN-Orig]	Ep 331 R	275.09 avgR100	93.49 eps 0.190 Qval	16.81
land100 0.24 thr 437				
[DDQN-Orig]	Ep 332 R	264.11 avgR100	96.28 eps 0.189 Qval	16.81
land100 0.25 thr 296				
[DDQN-Orig]	Ep 333 R	-45.52 avgR100	95.85 eps 0.188 Qval	16.38
land100 0.25 thr 117				
[DDQN-Orig]	Ep 334 R	185.87 avgR100	97.62 eps 0.187 Qval	17.64
land100 0.26 thr 582				
[DDQN-Orig]	Ep 335 R	233.20 avgR100	100.11 eps 0.187 Qval	18.59
land100 0.26 thr 274				
[DDQN-Orig]	Ep 336 R	204.03 avgR100	102.35 eps 0.186 Qval	17.79
land100 0.26 thr 664				
[DDQN-Orig]	Ep 337 R	-53.39 avgR100	101.91 eps 0.185 Qval	17.78
land100 0.26 thr 915				
[DDQN-Orig]	Ep 338 R	107.32 avgR100	102.73 eps 0.184 Qval	17.53
land100 0.26 thr 712				
[DDQN-Orig]	Ep 339 R	271.56 avgR100	104.88 eps 0.183 Qval	18.05
land100 0.27 thr 299				
[DDQN-Orig]	Ep 340 R	235.86 avgR100	106.73 eps 0.182 Qval	16.59
land100 0.28 thr 260				
[DDQN-Orig]	Ep 341 R	57.89 avgR100	107.35 eps 0.181 Qval	17.09
land100 0.28 thr 190				
[DDQN-Orig]	Ep 342 R	271.70 avgR100	110.08 eps 0.180 Qval	16.36
land100 0.29 thr 329				
[DDQN-Orig]	Ep 343 R	243.29 avgR100	109.97 eps 0.179 Qval	16.59
land100 0.29 thr 377				
[DDQN-Orig]	Ep 344 R	107.35 avgR100	111.01 eps 0.178 Qval	17.28
land100 0.29 thr 705				
[DDQN-Orig]	Ep 345 R	43.57 avgR100	111.18 eps 0.177 Qval	18.28
land100 0.29 thr 80				
[DDQN-Orig]	Ep 346 R	264.99 avgR100	113.65 eps 0.177 Qval	18.08
land100 0.29 thr 268				
[DDQN-Orig]	Ep 347 R	220.36 avgR100	115.39 eps 0.176 Qval	16.96
land100 0.30 thr 366				
[DDQN-Orig]	Ep 348 R	176.38 avgR100	117.54 eps 0.175 Qval	18.55
land100 0.31 thr 584				
[DDQN-Orig]	Ep 349 R	300.75 avgR100	120.62 eps 0.174 Qval	19.22
land100 0.32 thr 335				
[DDQN-Orig]	Ep 350 R	27.30 avgR100	120.83 eps 0.173 Qval	18.33
land100 0.32 thr 84				
[DDQN-Orig]	Ep 351 R	247.06 avgR100	122.78 eps 0.172 Qval	17.64
land100 0.33 thr 465				
[DDQN-Orig]	Ep 352 R	206.13 avgR100	124.54 eps 0.171 Qval	19.77
land100 0.34 thr 474				
[DDQN-Orig]	Ep 353 R	231.38 avgR100	126.59 eps 0.170 Qval	19.65
land100 0.35 thr 243				
[DDQN-Orig]	Ep 354 R	241.02 avgR100	128.19 eps 0.170 Qval	19.23
land100 0.36 thr 282				
[DDQN-Orig]	Ep 355 R	132.34 avgR100	130.07 eps 0.169 Qval	20.73
land100 0.36 thr 858				
[DDQN-Orig]	Ep 356 R	233.80 avgR100	132.15 eps 0.168 Qval	21.14
land100 0.37 thr 457				
[DDQN-Orig]	Ep 357 R	264.92 avgR100	134.43 eps 0.167 Qval	19.98
land100 0.38 thr 206				
[DDQN-Orig]	Ep 358 R	114.06 avgR100	133.76 eps 0.166 Qval	20.53
land100 0.37 thr 407				
[DDQN-Orig]	Ep 359 R	236.53 avgR100	135.53 eps 0.165 Qval	21.11
land100 0.38 thr 572				
[DDQN-Orig]	Ep 360 R	136.58 avgR100	136.76 eps 0.165 Qval	19.84
land100 0.38 thr 629				

[DDQN-Orig]	Ep 361 R	212.85 avgR100	138.18 eps 0.164 Qval	21.24
land100 0.39 thr 459				
[DDQN-Orig]	Ep 362 R	248.30 avgR100	140.97 eps 0.163 Qval	22.06
land100 0.40 thr 253				
[DDQN-Orig]	Ep 363 R	187.95 avgR100	142.33 eps 0.162 Qval	20.96
land100 0.41 thr 446				
[DDQN-Orig]	Ep 364 R	274.87 avgR100	145.13 eps 0.161 Qval	20.66
land100 0.42 thr 157				
[DDQN-Orig]	Ep 365 R	238.26 avgR100	146.95 eps 0.160 Qval	20.31
land100 0.43 thr 181				
[DDQN-Orig]	Ep 366 R	76.84 avgR100	147.32 eps 0.160 Qval	20.53
land100 0.43 thr 731				
[DDQN-Orig]	Ep 367 R	232.24 avgR100	148.53 eps 0.159 Qval	21.49
land100 0.43 thr 386				
[DDQN-Orig]	Ep 368 R	273.87 avgR100	150.18 eps 0.158 Qval	20.39
land100 0.44 thr 183				
[DDQN-Orig]	Ep 369 R	263.95 avgR100	151.91 eps 0.157 Qval	21.46
land100 0.45 thr 211				
[DDQN-Orig]	Ep 370 R	235.39 avgR100	154.39 eps 0.157 Qval	21.42
land100 0.45 thr 368				
[DDQN-Orig]	Ep 371 R	-3.16 avgR100	153.78 eps 0.156 Qval	21.44
land100 0.45 thr 83				
[DDQN-Orig]	Ep 372 R	0.80 avgR100	153.44 eps 0.155 Qval	20.41
land100 0.45 thr 111				
[DDQN-Orig]	Ep 373 R	226.85 avgR100	155.72 eps 0.154 Qval	22.41
land100 0.45 thr 194				
[DDQN-Orig]	Ep 374 R	-6.06 avgR100	155.01 eps 0.153 Qval	21.01
land100 0.45 thr 173				
[DDQN-Orig]	Ep 375 R	245.52 avgR100	157.05 eps 0.153 Qval	22.03
land100 0.46 thr 621				
[DDQN-Orig]	Ep 376 R	224.39 avgR100	159.17 eps 0.152 Qval	22.48
land100 0.47 thr 412				
[DDQN-Orig]	Ep 377 R	161.02 avgR100	160.07 eps 0.151 Qval	22.54
land100 0.47 thr 591				
[DDQN-Orig]	Ep 378 R	221.41 avgR100	162.03 eps 0.150 Qval	22.68
land100 0.47 thr 724				
[DDQN-Orig]	Ep 379 R	184.10 avgR100	162.96 eps 0.150 Qval	24.06
land100 0.48 thr 539				
[DDQN-Orig]	Ep 380 R	217.67 avgR100	165.08 eps 0.149 Qval	24.78
land100 0.48 thr 512				
[DDQN-Orig]	Ep 381 R	216.56 avgR100	166.63 eps 0.148 Qval	25.20
land100 0.49 thr 601				
[DDQN-Orig]	Ep 382 R	24.83 avgR100	167.22 eps 0.147 Qval	24.30
land100 0.49 thr 235				
[DDQN-Orig]	Ep 383 R	184.52 avgR100	168.51 eps 0.147 Qval	26.40
land100 0.50 thr 503				
[DDQN-Orig]	Ep 384 R	221.29 avgR100	169.48 eps 0.146 Qval	25.01
land100 0.50 thr 255				
[DDQN-Orig]	Ep 385 R	177.40 avgR100	170.47 eps 0.145 Qval	25.27
land100 0.50 thr 560				
[DDQN-Orig]	Ep 386 R	228.40 avgR100	170.21 eps 0.144 Qval	25.91
land100 0.50 thr 363				
[DDQN-Orig]	Ep 387 R	251.13 avgR100	172.48 eps 0.144 Qval	26.04
land100 0.51 thr 200				
[DDQN-Orig]	Ep 388 R	241.57 avgR100	174.69 eps 0.143 Qval	25.27
land100 0.52 thr 175				
[DDQN-Orig]	Ep 389 R	118.79 avgR100	174.62 eps 0.142 Qval	25.75
land100 0.52 thr 613				
[DDQN-Orig]	Ep 390 R	190.17 avgR100	175.58 eps 0.142 Qval	26.11
land100 0.52 thr 489				

[DDQN-Orig]	Ep 391 R	67.69 avgR100	174.27 eps 0.141 Qval	25.78
land100 0.51 thr 899				
[DDQN-Orig]	Ep 392 R	122.77 avgR100	173.48 eps 0.140 Qval	25.64
land100 0.50 thr 470				
[DDQN-Orig]	Ep 393 R	228.35 avgR100	174.77 eps 0.139 Qval	25.95
land100 0.51 thr 234				
[DDQN-Orig]	Ep 394 R	290.29 avgR100	174.88 eps 0.139 Qval	28.16
land100 0.51 thr 179				
[DDQN-Orig]	Ep 395 R	242.22 avgR100	175.87 eps 0.138 Qval	27.29
land100 0.52 thr 384				
[DDQN-Orig]	Ep 396 R	260.70 avgR100	175.74 eps 0.137 Qval	27.40
land100 0.52 thr 209				
[DDQN-Orig]	Ep 397 R	222.80 avgR100	178.05 eps 0.137 Qval	28.67
land100 0.52 thr 264				
[DDQN-Orig]	Ep 398 R	267.73 avgR100	179.92 eps 0.136 Qval	27.38
land100 0.53 thr 153				
[DDQN-Orig]	Ep 399 R	181.65 avgR100	181.56 eps 0.135 Qval	27.28
land100 0.53 thr 403				
[DDQN-Orig]	Ep 400 R	266.28 avgR100	182.07 eps 0.135 Qval	27.18
land100 0.54 thr 151				
[DDQN-Orig]	Ep 401 R	238.48 avgR100	181.62 eps 0.134 Qval	26.13
land100 0.54 thr 328				
[DDQN-Orig]	Ep 402 R	248.75 avgR100	182.79 eps 0.133 Qval	27.52
land100 0.55 thr 201				
[DDQN-Orig]	Ep 403 R	250.32 avgR100	183.21 eps 0.133 Qval	25.68
land100 0.55 thr 205				
[DDQN-Orig]	Ep 404 R	281.88 avgR100	184.81 eps 0.132 Qval	26.05
land100 0.56 thr 179				
[DDQN-Orig]	Ep 405 R	260.48 avgR100	186.31 eps 0.131 Qval	26.78
land100 0.57 thr 268				
[DDQN-Orig]	Ep 406 R	243.43 avgR100	186.91 eps 0.131 Qval	27.40
land100 0.57 thr 159				
[DDQN-Orig]	Ep 407 R	256.02 avgR100	189.03 eps 0.130 Qval	26.94
land100 0.58 thr 139				
[DDQN-Orig]	Ep 408 R	240.76 avgR100	189.45 eps 0.129 Qval	26.85
land100 0.58 thr 161				
[DDQN-Orig]	Ep 409 R	254.33 avgR100	189.17 eps 0.129 Qval	27.78
land100 0.57 thr 240				
[DDQN-Orig]	Ep 410 R	70.98 avgR100	189.79 eps 0.128 Qval	28.44
land100 0.57 thr 781				
[DDQN-Orig]	Ep 411 R	279.94 avgR100	192.01 eps 0.127 Qval	29.15
land100 0.58 thr 331				
[DDQN-Orig]	Ep 412 R	117.55 avgR100	193.36 eps 0.127 Qval	29.28
land100 0.58 thr 645				
[DDQN-Orig]	Ep 413 R	63.68 avgR100	192.01 eps 0.126 Qval	29.73
land100 0.58 thr 898				
[DDQN-Orig]	Ep 414 R	264.45 avgR100	192.01 eps 0.126 Qval	30.47
land100 0.58 thr 232				
[DDQN-Orig]	Ep 415 R	231.03 avgR100	193.59 eps 0.125 Qval	31.48
land100 0.59 thr 472				
[DDQN-Orig]	Ep 416 R	270.37 avgR100	193.69 eps 0.124 Qval	30.66
land100 0.59 thr 307				
[DDQN-Orig]	Ep 417 R	293.09 avgR100	195.17 eps 0.124 Qval	31.77
land100 0.60 thr 242				
[DDQN-Orig]	Ep 418 R	256.17 avgR100	196.23 eps 0.123 Qval	29.30
land100 0.60 thr 307				
[DDQN-Orig]	Ep 419 R	262.26 avgR100	196.48 eps 0.122 Qval	29.61
land100 0.60 thr 265				
[DDQN-Orig]	Ep 420 R	262.28 avgR100	196.57 eps 0.122 Qval	30.62
land100 0.60 thr 155				

[DDQN-Orig]	Ep 421 R	286.90 avgR100	197.05 eps 0.121 Qval	29.96
land100 0.60 thr 187				
[DDQN-Orig]	Ep 422 R	143.49 avgR100	196.88 eps 0.121 Qval	30.54
land100 0.60 thr 401				
[DDQN-Orig]	Ep 423 R	192.33 avgR100	196.28 eps 0.120 Qval	30.35
land100 0.60 thr 710				
[DDQN-Orig]	Ep 424 R	243.07 avgR100	197.64 eps 0.119 Qval	30.91
land100 0.61 thr 149				
[DDQN-Orig]	Ep 425 R	254.95 avgR100	198.19 eps 0.119 Qval	29.50
land100 0.61 thr 356				
[DDQN-Orig]	Ep 426 R	227.05 avgR100	200.35 eps 0.118 Qval	28.57
land100 0.62 thr 485				
[DDQN-Orig]	Ep 427 R	228.24 avgR100	199.86 eps 0.118 Qval	28.68
land100 0.61 thr 267				
[DDQN-Orig]	Ep 428 R	261.05 avgR100	199.99 eps 0.117 Qval	29.82
land100 0.61 thr 241				
[DDQN-Orig]	Ep 429 R	69.99 avgR100	199.41 eps 0.116 Qval	29.37
land100 0.61 thr 813				
[DDQN-Orig]	Ep 430 R	247.01 avgR100	199.58 eps 0.116 Qval	29.31
land100 0.61 thr 327				
[DDQN-Orig]	Ep 431 R	230.01 avgR100	199.13 eps 0.115 Qval	30.11
land100 0.61 thr 192				
[DDQN-Orig]	Ep 432 R	259.54 avgR100	199.08 eps 0.115 Qval	29.40
land100 0.61 thr 195				
[DDQN-Orig]	Ep 433 R	239.34 avgR100	201.93 eps 0.114 Qval	28.27
land100 0.61 thr 300				
[DDQN-Orig]	Ep 434 R	236.58 avgR100	202.44 eps 0.114 Qval	28.59
land100 0.61 thr 288				
[DDQN-Orig]	Ep 435 R	239.91 avgR100	202.51 eps 0.113 Qval	30.21
land100 0.61 thr 302				
[DDQN-Orig]	Ep 436 R	241.57 avgR100	202.88 eps 0.112 Qval	29.16
land100 0.62 thr 438				
[DDQN-Orig]	Ep 437 R	230.09 avgR100	205.72 eps 0.112 Qval	29.56
land100 0.63 thr 341				
[DDQN-Orig]	Ep 438 R	232.99 avgR100	206.97 eps 0.111 Qval	28.74
land100 0.64 thr 278				
[DDQN-Orig]	Ep 439 R	265.64 avgR100	206.91 eps 0.111 Qval	28.50
land100 0.64 thr 276				
[DDQN-Orig]	Ep 440 R	254.41 avgR100	207.10 eps 0.110 Qval	30.14
land100 0.64 thr 258				
[DDQN-Orig]	Ep 441 R	269.19 avgR100	209.21 eps 0.110 Qval	27.92
land100 0.65 thr 296				
[DDQN-Orig]	Ep 442 R	251.92 avgR100	209.02 eps 0.109 Qval	28.81
land100 0.64 thr 193				
[DDQN-Orig]	Ep 443 R	252.73 avgR100	209.11 eps 0.109 Qval	29.85
land100 0.64 thr 347				
[DDQN-Orig]	Ep 444 R	227.81 avgR100	210.31 eps 0.108 Qval	29.97
land100 0.65 thr 362				
[DDQN-Orig]	Ep 445 R	243.97 avgR100	212.32 eps 0.107 Qval	29.54
land100 0.66 thr 301				
[DDQN-Orig]	Ep 446 R	260.47 avgR100	212.27 eps 0.107 Qval	31.15
land100 0.67 thr 299				
[DDQN-Orig]	Ep 447 R	219.71 avgR100	212.27 eps 0.106 Qval	29.47
land100 0.67 thr 351				
[DDQN-Orig]	Ep 448 R	268.41 avgR100	213.19 eps 0.106 Qval	29.65
land100 0.67 thr 243				
[DDQN-Orig]	Ep 449 R	254.25 avgR100	212.72 eps 0.105 Qval	29.15
land100 0.67 thr 225				
[DDQN-Orig]	Ep 450 R	249.25 avgR100	214.94 eps 0.105 Qval	29.71
land100 0.68 thr 365				

[DDQN-Orig]	Ep 451 R	251.23	avgR100	214.98	eps 0.104	Qval 28.94
land100 0.68	thr 186					
[DDQN-Orig]	Ep 452 R	212.13	avgR100	215.04	eps 0.104	Qval 29.58
land100 0.67	thr 359					
[DDQN-Orig]	Ep 453 R	205.07	avgR100	214.78	eps 0.103	Qval 30.21
land100 0.67	thr 517					
[DDQN-Orig]	Ep 454 R	259.94	avgR100	214.97	eps 0.103	Qval 29.33
land100 0.67	thr 192					
[DDQN-Orig]	Ep 455 R	236.90	avgR100	216.01	eps 0.102	Qval 30.06
land100 0.67	thr 381					
[DDQN-Orig]	Ep 456 R	231.41	avgR100	215.99	eps 0.102	Qval 30.70
land100 0.67	thr 155					
[DDQN-Orig]	Ep 457 R	252.46	avgR100	215.87	eps 0.101	Qval 29.98
land100 0.67	thr 352					
[DDQN-Orig]	Ep 458 R	254.86	avgR100	217.27	eps 0.101	Qval 29.73
land100 0.68	thr 304					
[DDQN-Orig]	Ep 459 R	282.99	avgR100	217.74	eps 0.100	Qval 29.90
land100 0.68	thr 136					
[DDQN-Orig]	Ep 460 R	230.78	avgR100	218.68	eps 0.100	Qval 29.89
land100 0.69	thr 265					
[DDQN-Orig]	Ep 461 R	-45.61	avgR100	216.10	eps 0.099	Qval 31.90
land100 0.68	thr 95					
[DDQN-Orig]	Ep 462 R	169.80	avgR100	215.31	eps 0.099	Qval 31.71
land100 0.67	thr 702					
[DDQN-Orig]	Ep 463 R	284.14	avgR100	216.27	eps 0.098	Qval 31.59
land100 0.67	thr 217					
[DDQN-Orig]	Ep 464 R	246.21	avgR100	215.99	eps 0.098	Qval 31.28
land100 0.66	thr 216					
[DDQN-Orig]	Ep 465 R	272.67	avgR100	216.33	eps 0.097	Qval 29.79
land100 0.66	thr 214					
[DDQN-Orig]	Ep 466 R	249.96	avgR100	218.06	eps 0.097	Qval 30.78
land100 0.67	thr 243					
[DDQN-Orig]	Ep 467 R	217.22	avgR100	217.91	eps 0.096	Qval 30.26
land100 0.68	thr 499					
[DDQN-Orig]	Ep 468 R	197.51	avgR100	217.15	eps 0.096	Qval 30.78
land100 0.68	thr 378					
[DDQN-Orig]	Ep 469 R	248.12	avgR100	216.99	eps 0.095	Qval 29.33
land100 0.67	thr 234					
[DDQN-Orig]	Ep 470 R	253.00	avgR100	217.17	eps 0.095	Qval 31.56
land100 0.68	thr 268					
[DDQN-Orig]	Ep 471 R	234.96	avgR100	219.55	eps 0.094	Qval 28.77
land100 0.69	thr 189					
[DDQN-Orig]	Ep 472 R	208.36	avgR100	221.62	eps 0.094	Qval 32.57
land100 0.70	thr 451					
[DDQN-Orig]	Ep 473 R	269.81	avgR100	222.05	eps 0.093	Qval 30.96
land100 0.71	thr 261					
[DDQN-Orig]	Ep 474 R	32.04	avgR100	222.43	eps 0.093	Qval 31.28
land100 0.71	thr 83					
[DDQN-Orig]	Ep 475 R	172.07	avgR100	221.70	eps 0.092	Qval 32.29
land100 0.70	thr 686					
[DDQN-Orig]	Ep 476 R	128.35	avgR100	220.74	eps 0.092	Qval 31.42
land100 0.69	thr 412					
[DDQN-Orig]	Ep 477 R	271.68	avgR100	221.84	eps 0.092	Qval 32.85
land100 0.70	thr 189					
[DDQN-Orig]	Ep 478 R	-59.10	avgR100	219.04	eps 0.091	Qval 32.68
land100 0.70	thr 930					
[DDQN-Orig]	Ep 479 R	210.95	avgR100	219.31	eps 0.091	Qval 33.21
land100 0.70	thr 360					
[DDQN-Orig]	Ep 480 R	272.66	avgR100	219.86	eps 0.090	Qval 32.92
land100 0.71	thr 263					

[DDQN-Orig]	Ep 481 R	282.77 avgR100	220.52 eps 0.090	Qval 33.36
land100 0.71 thr 209				
[DDQN-Orig]	Ep 482 R	276.06 avgR100	223.03 eps 0.089	Qval 32.79
land100 0.72 thr 330				
[DDQN-Orig]	Ep 483 R	17.41 avgR100	221.36 eps 0.089	Qval 34.19
land100 0.71 thr 88				
[DDQN-Orig]	Ep 484 R	205.92 avgR100	221.21 eps 0.088	Qval 34.12
land100 0.71 thr 234				
[DDQN-Orig]	Ep 485 R	110.68 avgR100	220.54 eps 0.088	Qval 33.58
land100 0.71 thr 515				
[DDQN-Orig]	Ep 486 R	254.23 avgR100	220.80 eps 0.088	Qval 34.36
land100 0.71 thr 279				
[DDQN-Orig]	Ep 487 R	265.57 avgR100	220.94 eps 0.087	Qval 34.37
land100 0.71 thr 297				
[DDQN-Orig]	Ep 488 R	42.98 avgR100	218.96 eps 0.087	Qval 33.53
land100 0.70 thr 862				
[DDQN-Orig]	Ep 489 R	201.69 avgR100	219.79 eps 0.086	Qval 33.47
land100 0.71 thr 438				
[DDQN-Orig]	Ep 490 R	150.99 avgR100	219.39 eps 0.086	Qval 33.80
land100 0.72 thr 781				
[DDQN-Orig]	Ep 491 R	263.87 avgR100	221.36 eps 0.085	Qval 33.00
land100 0.73 thr 208				
[DDQN-Orig]	Ep 492 R	265.34 avgR100	222.78 eps 0.085	Qval 32.90
land100 0.74 thr 258				
[DDQN-Orig]	Ep 493 R	250.72 avgR100	223.01 eps 0.084	Qval 32.78
land100 0.74 thr 383				
[DDQN-Orig]	Ep 494 R	268.22 avgR100	222.79 eps 0.084	Qval 33.17
land100 0.74 thr 184				
[DDQN-Orig]	Ep 495 R	287.45 avgR100	223.24 eps 0.084	Qval 32.77
land100 0.74 thr 262				
[DDQN-Orig]	Ep 496 R	248.25 avgR100	223.11 eps 0.083	Qval 32.60
land100 0.73 thr 507				
[DDQN-Orig]	Ep 497 R	226.64 avgR100	223.15 eps 0.083	Qval 32.29
land100 0.74 thr 211				
[DDQN-Orig]	Ep 498 R	224.88 avgR100	222.72 eps 0.082	Qval 34.06
land100 0.74 thr 285				
[DDQN-Orig]	Ep 499 R	242.22 avgR100	223.33 eps 0.082	Qval 34.12
land100 0.75 thr 179				
[DDQN-Orig]	Ep 500 R	270.42 avgR100	223.37 eps 0.082	Qval 33.44
land100 0.75 thr 314				
[DDQN-Orig]	Ep 501 R	257.51 avgR100	223.56 eps 0.081	Qval 34.88
land100 0.75 thr 188				
[DDQN-Orig]	Ep 502 R	231.47 avgR100	223.39 eps 0.081	Qval 34.54
land100 0.75 thr 307				
[DDQN-Orig]	Ep 503 R	277.27 avgR100	223.66 eps 0.080	Qval 33.84
land100 0.75 thr 251				
[DDQN-Orig]	Ep 504 R	212.66 avgR100	222.96 eps 0.080	Qval 33.50
land100 0.74 thr 694				
[DDQN-Orig]	Ep 505 R	24.48 avgR100	220.60 eps 0.080	Qval 33.06
land100 0.73 thr 163				
[DDQN-Orig]	Ep 506 R	157.51 avgR100	219.75 eps 0.079	Qval 33.01
land100 0.73 thr 769				
[DDQN-Orig]	Ep 507 R	206.91 avgR100	219.25 eps 0.079	Qval 32.88
land100 0.73 thr 256				
[DDQN-Orig]	Ep 508 R	90.07 avgR100	217.75 eps 0.078	Qval 33.22
land100 0.72 thr 667				
[DDQN-Orig]	Ep 509 R	217.09 avgR100	217.38 eps 0.078	Qval 32.56
land100 0.72 thr 426				
[DDQN-Orig]	Ep 510 R	239.03 avgR100	219.06 eps 0.078	Qval 33.71
land100 0.73 thr 179				

[DDQN-Orig]	Ep 511 R	229.35	avgR100	218.55	eps 0.077	Qval 32.75
land100 0.73	thr 215					
[DDQN-Orig]	Ep 512 R	202.06	avgR100	219.39	eps 0.077	Qval 33.74
land100 0.74	thr 609					
[DDQN-Orig]	Ep 513 R	229.02	avgR100	221.05	eps 0.076	Qval 33.93
land100 0.75	thr 320					
[DDQN-Orig]	Ep 514 R	233.67	avgR100	220.74	eps 0.076	Qval 32.37
land100 0.75	thr 191					
[DDQN-Orig]	Ep 515 R	250.40	avgR100	220.93	eps 0.076	Qval 33.79
land100 0.75	thr 268					
[DDQN-Orig]	Ep 516 R	213.05	avgR100	220.36	eps 0.075	Qval 31.81
land100 0.75	thr 561					
[DDQN-Orig]	Ep 517 R	288.77	avgR100	220.32	eps 0.075	Qval 31.99
land100 0.74	thr 195					
[DDQN-Orig]	Ep 518 R	250.02	avgR100	220.26	eps 0.075	Qval 33.46
land100 0.74	thr 310					
[DDQN-Orig]	Ep 519 R	69.32	avgR100	218.33	eps 0.074	Qval 33.11
land100 0.73	thr 788					
[DDQN-Orig]	Ep 520 R	190.05	avgR100	217.60	eps 0.074	Qval 31.58
land100 0.73	thr 509					
[DDQN-Orig]	Ep 521 R	264.69	avgR100	217.38	eps 0.073	Qval 32.88
land100 0.73	thr 408					
[DDQN-Orig]	Ep 522 R	261.25	avgR100	218.56	eps 0.073	Qval 33.20
land100 0.74	thr 173					
[DDQN-Orig]	Ep 523 R	240.42	avgR100	219.04	eps 0.073	Qval 33.13
land100 0.74	thr 228					
[DDQN-Orig]	Ep 524 R	228.80	avgR100	218.90	eps 0.072	Qval 32.97
land100 0.74	thr 373					
[DDQN-Orig]	Ep 525 R	191.38	avgR100	218.26	eps 0.072	Qval 33.96
land100 0.74	thr 595					
[DDQN-Orig]	Ep 526 R	244.78	avgR100	218.44	eps 0.072	Qval 34.38
land100 0.74	thr 170					
[DDQN-Orig]	Ep 527 R	227.50	avgR100	218.43	eps 0.071	Qval 34.43
land100 0.75	thr 266					
[DDQN-Orig]	Ep 528 R	214.60	avgR100	217.97	eps 0.071	Qval 35.96
land100 0.75	thr 288					
[DDQN-Orig]	Ep 529 R	267.60	avgR100	219.94	eps 0.071	Qval 36.75
land100 0.76	thr 214					
[DDQN-Orig]	Ep 530 R	230.66	avgR100	219.78	eps 0.070	Qval 36.06
land100 0.76	thr 263					
[DDQN-Orig]	Ep 531 R	195.25	avgR100	219.43	eps 0.070	Qval 35.55
land100 0.75	thr 587					
[DDQN-Orig]	Ep 532 R	194.27	avgR100	218.78	eps 0.069	Qval 35.50
land100 0.74	thr 329					
[DDQN-Orig]	Ep 533 R	236.79	avgR100	218.75	eps 0.069	Qval 35.67
land100 0.74	thr 322					
[DDQN-Orig]	Ep 534 R	218.70	avgR100	218.58	eps 0.069	Qval 36.59
land100 0.74	thr 226					
[DDQN-Orig]	Ep 535 R	169.46	avgR100	217.87	eps 0.068	Qval 36.98
land100 0.75	thr 656					
[DDQN-Orig]	Ep 536 R	272.21	avgR100	218.18	eps 0.068	Qval 35.71
land100 0.75	thr 348					
[DDQN-Orig]	Ep 537 R	249.72	avgR100	218.37	eps 0.068	Qval 36.71
land100 0.74	thr 350					
[DDQN-Orig]	Ep 538 R	246.66	avgR100	218.51	eps 0.067	Qval 35.92
land100 0.74	thr 382					
[DDQN-Orig]	Ep 539 R	249.68	avgR100	218.35	eps 0.067	Qval 37.09
land100 0.74	thr 354					
[DDQN-Orig]	Ep 540 R	277.05	avgR100	218.58	eps 0.067	Qval 36.27
land100 0.74	thr 155					

[DDQN-Orig]	Ep 541 R	250.51 avgR100	218.39 eps 0.066 Qval	36.42
land100 0.73 thr 213				
[DDQN-Orig]	Ep 542 R	239.25 avgR100	218.26 eps 0.066 Qval	36.17
land100 0.74 thr 230				
[DDQN-Orig]	Ep 543 R	256.92 avgR100	218.31 eps 0.066 Qval	36.60
land100 0.74 thr 196				
[DDQN-Orig]	Ep 544 R	220.68 avgR100	218.23 eps 0.065 Qval	37.55
land100 0.74 thr 430				
[DDQN-Orig]	Ep 545 R	271.46 avgR100	218.51 eps 0.065 Qval	36.40
land100 0.74 thr 267				
[DDQN-Orig]	Ep 546 R	246.16 avgR100	218.37 eps 0.065 Qval	36.36
land100 0.74 thr 258				
[DDQN-Orig]	Ep 547 R	197.67 avgR100	218.15 eps 0.064 Qval	36.85
land100 0.73 thr 538				
[DDQN-Orig]	Ep 548 R	286.48 avgR100	218.33 eps 0.064 Qval	37.06
land100 0.73 thr 205				
[DDQN-Orig]	Ep 549 R	234.15 avgR100	218.13 eps 0.064 Qval	36.55
land100 0.73 thr 301				
[DDQN-Orig]	Ep 550 R	293.95 avgR100	218.57 eps 0.063 Qval	36.67
land100 0.73 thr 213				
[DDQN-Orig]	Ep 551 R	228.71 avgR100	218.35 eps 0.063 Qval	37.58
land100 0.72 thr 239				
[DDQN-Orig]	Ep 552 R	49.77 avgR100	216.72 eps 0.063 Qval	34.90
land100 0.72 thr 760				
[DDQN-Orig]	Ep 553 R	219.47 avgR100	216.87 eps 0.063 Qval	36.43
land100 0.71 thr 528				
[DDQN-Orig]	Ep 554 R	245.32 avgR100	216.72 eps 0.062 Qval	36.31
land100 0.71 thr 301				
[DDQN-Orig]	Ep 555 R	193.66 avgR100	216.29 eps 0.062 Qval	36.03
land100 0.71 thr 342				
[DDQN-Orig]	Ep 556 R	228.83 avgR100	216.26 eps 0.062 Qval	35.26
land100 0.70 thr 335				
[DDQN-Orig]	Ep 557 R	236.87 avgR100	216.11 eps 0.061 Qval	36.40
land100 0.70 thr 263				
[DDQN-Orig]	Ep 558 R	239.73 avgR100	215.96 eps 0.061 Qval	36.08
land100 0.70 thr 244				
[DDQN-Orig]	Ep 559 R	235.94 avgR100	215.49 eps 0.061 Qval	36.23
land100 0.70 thr 224				
[DDQN-Orig]	Ep 560 R	191.99 avgR100	215.10 eps 0.060 Qval	34.45
land100 0.70 thr 567				
[DDQN-Orig]	Ep 561 R	108.09 avgR100	216.64 eps 0.060 Qval	35.14
land100 0.71 thr 810				
[DDQN-Orig]	Ep 562 R	226.67 avgR100	217.20 eps 0.060 Qval	34.91
land100 0.71 thr 327				
[DDQN-Orig]	Ep 563 R	272.90 avgR100	217.09 eps 0.059 Qval	34.65
land100 0.71 thr 414				
[DDQN-Orig]	Ep 564 R	230.80 avgR100	216.94 eps 0.059 Qval	35.27
land100 0.72 thr 255				
[DDQN-Orig]	Ep 565 R	272.51 avgR100	216.94 eps 0.059 Qval	35.48
land100 0.72 thr 240				
[DDQN-Orig]	Ep 566 R	210.03 avgR100	216.54 eps 0.059 Qval	35.06
land100 0.72 thr 375				
[DDQN-Orig]	Ep 567 R	254.75 avgR100	216.91 eps 0.058 Qval	35.49
land100 0.72 thr 351				
[DDQN-Orig]	Ep 568 R	251.45 avgR100	217.45 eps 0.058 Qval	36.25
land100 0.72 thr 316				
[DDQN-Orig]	Ep 569 R	276.34 avgR100	217.73 eps 0.058 Qval	36.83
land100 0.73 thr 242				
[DDQN-Orig]	Ep 570 R	240.88 avgR100	217.61 eps 0.057 Qval	35.76
land100 0.72 thr 383				

[DDQN-Orig]	Ep 571 R	242.80 avgR100	217.69 eps 0.057 Qval	36.05
land100 0.72 thr 357				
[DDQN-Orig]	Ep 572 R	227.02 avgR100	217.88 eps 0.057 Qval	36.61
land100 0.71 thr 307				
[DDQN-Orig]	Ep 573 R	241.26 avgR100	217.59 eps 0.057 Qval	37.23
land100 0.71 thr 321				
[DDQN-Orig]	Ep 574 R	160.84 avgR100	218.88 eps 0.056 Qval	37.24
land100 0.72 thr 664				
[DDQN-Orig]	Ep 575 R	39.11 avgR100	217.55 eps 0.056 Qval	38.62
land100 0.72 thr 762				
[DDQN-Orig]	Ep 576 R	171.07 avgR100	217.98 eps 0.056 Qval	38.29
land100 0.73 thr 498				
[DDQN-Orig]	Ep 577 R	242.95 avgR100	217.69 eps 0.055 Qval	38.89
land100 0.73 thr 184				
[DDQN-Orig]	Ep 578 R	256.52 avgR100	220.85 eps 0.055 Qval	38.08
land100 0.74 thr 173				
[DDQN-Orig]	Ep 579 R	241.46 avgR100	221.15 eps 0.055 Qval	39.11
land100 0.73 thr 394				
[DDQN-Orig]	Ep 580 R	224.46 avgR100	220.67 eps 0.055 Qval	39.02
land100 0.73 thr 464				
[DDQN-Orig]	Ep 581 R	219.19 avgR100	220.03 eps 0.054 Qval	37.97
land100 0.73 thr 389				
[DDQN-Orig]	Ep 582 R	285.01 avgR100	220.12 eps 0.054 Qval	38.59
land100 0.73 thr 167				
[DDQN-Orig]	Ep 583 R	244.94 avgR100	222.40 eps 0.054 Qval	38.72
land100 0.74 thr 163				
[DDQN-Orig]	Ep 584 R	236.18 avgR100	222.70 eps 0.054 Qval	38.90
land100 0.74 thr 312				
[DDQN-Orig]	Ep 585 R	190.49 avgR100	223.50 eps 0.053 Qval	38.37
land100 0.74 thr 241				
[DDQN-Orig]	Ep 586 R	258.21 avgR100	223.54 eps 0.053 Qval	38.63
land100 0.74 thr 224				
[DDQN-Orig]	Ep 587 R	192.26 avgR100	222.81 eps 0.053 Qval	39.34
land100 0.73 thr 419				
[DDQN-Orig]	Ep 588 R	210.20 avgR100	224.48 eps 0.052 Qval	40.15
land100 0.74 thr 432				
[DDQN-Orig]	Ep 589 R	280.49 avgR100	225.27 eps 0.052 Qval	39.18
land100 0.74 thr 557				
[DDQN-Orig]	Ep 590 R	242.33 avgR100	226.18 eps 0.052 Qval	38.65
land100 0.74 thr 187				
[DDQN-Orig]	Ep 591 R	117.65 avgR100	224.72 eps 0.052 Qval	40.57
land100 0.73 thr 538				
[DDQN-Orig]	Ep 592 R	227.57 avgR100	224.34 eps 0.051 Qval	39.17
land100 0.72 thr 380				
[DDQN-Orig]	Ep 593 R	232.06 avgR100	224.15 eps 0.051 Qval	40.00
land100 0.71 thr 257				
[DDQN-Orig]	Ep 594 R	264.90 avgR100	224.12 eps 0.051 Qval	40.01
land100 0.71 thr 214				
[DDQN-Orig]	Ep 595 R	234.30 avgR100	223.59 eps 0.051 Qval	39.54
land100 0.70 thr 232				
[DDQN-Orig]	Ep 596 R	-21.09 avgR100	220.89 eps 0.050 Qval	38.63
land100 0.70 thr 135				
[DDQN-Orig]	Ep 597 R	268.49 avgR100	221.31 eps 0.050 Qval	39.14
land100 0.70 thr 195				
[DDQN-Orig]	Ep 598 R	233.37 avgR100	221.40 eps 0.050 Qval	40.46
land100 0.70 thr 526				
[DDQN-Orig]	Ep 599 R	227.91 avgR100	221.26 eps 0.050 Qval	39.66
land100 0.69 thr 533				
[DDQN-Orig]	Ep 600 R	287.24 avgR100	221.42 eps 0.049 Qval	40.73
land100 0.69 thr 166				

[DDQN-Orig]	Ep 601 R	-59.82 avgR100	218.25 eps 0.049	Qval 39.63
land100 0.68 thr 874				
[DDQN-Orig]	Ep 602 R	144.32 avgR100	217.38 eps 0.049	Qval 40.40
land100 0.67 thr 689				
[DDQN-Orig]	Ep 603 R	242.59 avgR100	217.03 eps 0.049	Qval 41.55
land100 0.67 thr 166				
[DDQN-Orig]	Ep 604 R	206.97 avgR100	216.97 eps 0.048	Qval 42.00
land100 0.68 thr 453				
[DDQN-Orig]	Ep 605 R	260.45 avgR100	219.33 eps 0.048	Qval 41.43
land100 0.69 thr 191				
[DDQN-Orig]	Ep 606 R	225.63 avgR100	220.02 eps 0.048	Qval 40.98
land100 0.69 thr 228				
[DDQN-Orig]	Ep 607 R	20.49 avgR100	218.15 eps 0.048	Qval 41.13
land100 0.68 thr 721				
[DDQN-Orig]	Ep 608 R	211.80 avgR100	219.37 eps 0.047	Qval 40.67
land100 0.68 thr 423				
[DDQN-Orig]	Ep 609 R	257.95 avgR100	219.78 eps 0.047	Qval 40.55
land100 0.69 thr 209				
[DDQN-Orig]	Ep 610 R	197.86 avgR100	219.37 eps 0.047	Qval 42.15
land100 0.69 thr 613				
[DDQN-Orig]	Ep 611 R	216.43 avgR100	219.24 eps 0.047	Qval 42.51
land100 0.69 thr 290				
[DDQN-Orig]	Ep 612 R	241.59 avgR100	219.63 eps 0.047	Qval 41.93
land100 0.68 thr 451				
[DDQN-Orig]	Ep 613 R	271.31 avgR100	220.05 eps 0.046	Qval 43.66
land100 0.68 thr 350				
[DDQN-Orig]	Ep 614 R	231.86 avgR100	220.04 eps 0.046	Qval 42.76
land100 0.68 thr 303				
[DDQN-Orig]	Ep 615 R	71.51 avgR100	218.25 eps 0.046	Qval 42.48
land100 0.67 thr 845				
[DDQN-Orig]	Ep 616 R	220.53 avgR100	218.32 eps 0.046	Qval 43.11
land100 0.67 thr 404				
[DDQN-Orig]	Ep 617 R	178.50 avgR100	217.22 eps 0.045	Qval 41.83
land100 0.68 thr 415				
[DDQN-Orig]	Ep 618 R	205.15 avgR100	216.77 eps 0.045	Qval 43.22
land100 0.68 thr 382				
[DDQN-Orig]	Ep 619 R	238.24 avgR100	218.46 eps 0.045	Qval 43.84
land100 0.69 thr 265				
[DDQN-Orig]	Ep 620 R	294.31 avgR100	219.50 eps 0.045	Qval 42.64
land100 0.69 thr 198				
[DDQN-Orig]	Ep 621 R	261.87 avgR100	219.47 eps 0.044	Qval 43.49
land100 0.69 thr 174				
[DDQN-Orig]	Ep 622 R	250.47 avgR100	219.37 eps 0.044	Qval 43.61
land100 0.69 thr 302				
[DDQN-Orig]	Ep 623 R	214.07 avgR100	219.10 eps 0.044	Qval 42.88
land100 0.69 thr 320				
[DDQN-Orig]	Ep 624 R	8.79 avgR100	216.90 eps 0.044	Qval 42.64
land100 0.68 thr 92				
[DDQN-Orig]	Ep 625 R	232.84 avgR100	217.32 eps 0.044	Qval 43.64
land100 0.67 thr 210				
[DDQN-Orig]	Ep 626 R	262.12 avgR100	217.49 eps 0.043	Qval 43.46
land100 0.67 thr 168				
[DDQN-Orig]	Ep 627 R	247.81 avgR100	217.69 eps 0.043	Qval 43.11
land100 0.67 thr 258				
[DDQN-Orig]	Ep 628 R	290.03 avgR100	218.45 eps 0.043	Qval 42.80
land100 0.67 thr 246				
[DDQN-Orig]	Ep 629 R	141.55 avgR100	217.19 eps 0.043	Qval 42.55
land100 0.67 thr 693				
[DDQN-Orig]	Ep 630 R	174.75 avgR100	216.63 eps 0.043	Qval 42.03
land100 0.67 thr 490				

[DDQN-Orig]	Ep 631 R	263.97	avgR100	217.32	eps 0.042	Qval	42.32
land100 0.68	thr 171						
[DDQN-Orig]	Ep 632 R	236.15	avgR100	217.74	eps 0.042	Qval	42.18
land100 0.69	thr 275						
[DDQN-Orig]	Ep 633 R	200.55	avgR100	217.37	eps 0.042	Qval	42.65
land100 0.70	thr 500						
[DDQN-Orig]	Ep 634 R	250.12	avgR100	217.69	eps 0.042	Qval	43.05
land100 0.70	thr 114						
[DDQN-Orig]	Ep 635 R	258.84	avgR100	218.58	eps 0.041	Qval	43.33
land100 0.70	thr 194						
[DDQN-Orig]	Ep 636 R	198.92	avgR100	217.85	eps 0.041	Qval	43.12
land100 0.70	thr 343						
[DDQN-Orig]	Ep 637 R	221.57	avgR100	217.57	eps 0.041	Qval	43.23
land100 0.70	thr 311						
[DDQN-Orig]	Ep 638 R	251.79	avgR100	217.62	eps 0.041	Qval	43.71
land100 0.70	thr 228						
[DDQN-Orig]	Ep 639 R	311.99	avgR100	218.24	eps 0.041	Qval	43.57
land100 0.70	thr 214						
[DDQN-Orig]	Ep 640 R	272.58	avgR100	218.20	eps 0.040	Qval	44.56
land100 0.70	thr 284						
[DDQN-Orig]	Ep 641 R	291.13	avgR100	218.60	eps 0.040	Qval	45.98
land100 0.71	thr 187						
[DDQN-Orig]	Ep 642 R	271.18	avgR100	218.92	eps 0.040	Qval	45.01
land100 0.71	thr 214						
[DDQN-Orig]	Ep 643 R	230.74	avgR100	218.66	eps 0.040	Qval	44.81
land100 0.71	thr 458						
[DDQN-Orig]	Ep 644 R	184.54	avgR100	218.30	eps 0.040	Qval	44.08
land100 0.71	thr 465						
[DDQN-Orig]	Ep 645 R	171.57	avgR100	217.30	eps 0.039	Qval	45.23
land100 0.71	thr 696						
[DDQN-Orig]	Ep 646 R	248.56	avgR100	217.32	eps 0.039	Qval	44.40
land100 0.71	thr 337						
[DDQN-Orig]	Ep 647 R	200.34	avgR100	217.35	eps 0.039	Qval	45.38
land100 0.72	thr 520						
[DDQN-Orig]	Ep 648 R	238.17	avgR100	216.87	eps 0.039	Qval	44.95
land100 0.72	thr 310						
[DDQN-Orig]	Ep 649 R	302.35	avgR100	217.55	eps 0.039	Qval	45.12
land100 0.72	thr 255						
[DDQN-Orig]	Ep 650 R	190.04	avgR100	216.51	eps 0.038	Qval	44.75
land100 0.72	thr 477						
[DDQN-Orig]	Ep 651 R	222.93	avgR100	216.45	eps 0.038	Qval	45.50
land100 0.72	thr 378						
[DDQN-Orig]	Ep 652 R	256.81	avgR100	218.52	eps 0.038	Qval	46.12
land100 0.73	thr 420						
[DDQN-Orig]	Ep 653 R	188.95	avgR100	218.22	eps 0.038	Qval	47.40
land100 0.74	thr 365						
[DDQN-Orig]	Ep 654 R	-30.22	avgR100	215.46	eps 0.038	Qval	47.95
land100 0.73	thr 787						
[DDQN-Orig]	Ep 655 R	291.46	avgR100	216.44	eps 0.038	Qval	45.79
land100 0.73	thr 183						
[DDQN-Orig]	Ep 656 R	223.97	avgR100	216.39	eps 0.037	Qval	47.13
land100 0.74	thr 385						
[DDQN-Orig]	Ep 657 R	266.38	avgR100	216.69	eps 0.037	Qval	46.24
land100 0.74	thr 267						
[DDQN-Orig]	Ep 658 R	261.46	avgR100	216.90	eps 0.037	Qval	46.00
land100 0.74	thr 188						
[DDQN-Orig]	Ep 659 R	256.16	avgR100	217.11	eps 0.037	Qval	47.91
land100 0.74	thr 181						
[DDQN-Orig]	Ep 660 R	237.28	avgR100	217.56	eps 0.037	Qval	47.17
land100 0.74	thr 314						

[DDQN-Orig]	Ep 661 R	261.95 avgR100	219.10 eps 0.036 Qval	46.87
land100 0.73 thr 262				
[DDQN-Orig]	Ep 662 R	266.94 avgR100	219.50 eps 0.036 Qval	46.43
land100 0.74 thr 285				
[DDQN-Orig]	Ep 663 R	256.02 avgR100	219.33 eps 0.036 Qval	47.13
land100 0.74 thr 351				
[DDQN-Orig]	Ep 664 R	252.91 avgR100	219.55 eps 0.036 Qval	47.78
land100 0.74 thr 197				
[DDQN-Orig]	Ep 665 R	299.79 avgR100	219.82 eps 0.036 Qval	46.75
land100 0.73 thr 201				
[DDQN-Orig]	Ep 666 R	236.83 avgR100	220.09 eps 0.035 Qval	47.45
land100 0.73 thr 368				
[DDQN-Orig]	Ep 667 R	230.95 avgR100	219.85 eps 0.035 Qval	46.57
land100 0.73 thr 254				
[DDQN-Orig]	Ep 668 R	253.83 avgR100	219.88 eps 0.035 Qval	46.36
land100 0.73 thr 200				
[DDQN-Orig]	Ep 669 R	295.12 avgR100	220.07 eps 0.035 Qval	47.97
land100 0.73 thr 337				
[DDQN-Orig]	Ep 670 R	53.18 avgR100	218.19 eps 0.035 Qval	48.15
land100 0.73 thr 732				
[DDQN-Orig]	Ep 671 R	272.77 avgR100	218.49 eps 0.035 Qval	47.87
land100 0.73 thr 301				
[DDQN-Orig]	Ep 672 R	252.60 avgR100	218.75 eps 0.034 Qval	47.76
land100 0.74 thr 164				
[DDQN-Orig]	Ep 673 R	168.00 avgR100	218.01 eps 0.034 Qval	47.98
land100 0.74 thr 684				
[DDQN-Orig]	Ep 674 R	220.10 avgR100	218.61 eps 0.034 Qval	48.63
land100 0.74 thr 344				
[DDQN-Orig]	Ep 675 R	267.41 avgR100	220.89 eps 0.034 Qval	48.31
land100 0.75 thr 143				
[DDQN-Orig]	Ep 676 R	274.56 avgR100	221.92 eps 0.034 Qval	47.50
land100 0.74 thr 343				
[DDQN-Orig]	Ep 677 R	269.10 avgR100	222.18 eps 0.034 Qval	47.54
land100 0.74 thr 327				
[DDQN-Orig]	Ep 678 R	228.59 avgR100	221.91 eps 0.033 Qval	49.08
land100 0.74 thr 424				
[DDQN-Orig]	Ep 679 R	240.93 avgR100	221.90 eps 0.033 Qval	48.04
land100 0.74 thr 199				
[DDQN-Orig]	Ep 680 R	238.44 avgR100	222.04 eps 0.033 Qval	47.51
land100 0.74 thr 311				
[DDQN-Orig]	Ep 681 R	245.61 avgR100	222.30 eps 0.033 Qval	48.32
land100 0.74 thr 240				
[DDQN-Orig]	Ep 682 R	239.53 avgR100	221.85 eps 0.033 Qval	49.06
land100 0.74 thr 424				
[DDQN-Orig]	Ep 683 R	230.48 avgR100	221.70 eps 0.033 Qval	50.13
land100 0.74 thr 328				
[DDQN-Orig]	Ep 684 R	206.95 avgR100	221.41 eps 0.032 Qval	49.47
land100 0.75 thr 464				
[DDQN-Orig]	Ep 685 R	214.32 avgR100	221.65 eps 0.032 Qval	48.04
land100 0.76 thr 482				
[DDQN-Orig]	Ep 686 R	238.18 avgR100	221.45 eps 0.032 Qval	48.43
land100 0.76 thr 173				
[DDQN-Orig]	Ep 687 R	246.45 avgR100	221.99 eps 0.032 Qval	49.74
land100 0.77 thr 326				
[DDQN-Orig]	Ep 688 R	266.96 avgR100	222.56 eps 0.032 Qval	49.80
land100 0.77 thr 212				
[DDQN-Orig]	Ep 689 R	246.29 avgR100	222.22 eps 0.032 Qval	50.04
land100 0.77 thr 225				
[DDQN-Orig]	Ep 690 R	239.94 avgR100	222.19 eps 0.031 Qval	49.46
land100 0.77 thr 310				

[DDQN-Orig]	Ep 691 R	233.95 avgR100	223.36 eps 0.031 Qval	49.53
land100 0.78 thr 399				
[DDQN-Orig]	Ep 692 R	270.34 avgR100	223.78 eps 0.031 Qval	50.38
land100 0.79 thr 206				
[DDQN-Orig]	Ep 693 R	256.33 avgR100	224.03 eps 0.031 Qval	50.14
land100 0.80 thr 213				
[DDQN-Orig]	Ep 694 R	205.24 avgR100	223.43 eps 0.031 Qval	49.71
land100 0.80 thr 388				
[DDQN-Orig]	Ep 695 R	213.63 avgR100	223.22 eps 0.031 Qval	49.96
land100 0.81 thr 414				
[DDQN-Orig]	Ep 696 R	226.20 avgR100	225.70 eps 0.031 Qval	52.09
land100 0.82 thr 262				
[DDQN-Orig]	Ep 697 R	197.95 avgR100	224.99 eps 0.030 Qval	52.38
land100 0.82 thr 426				
[DDQN-Orig]	Ep 698 R	254.01 avgR100	225.20 eps 0.030 Qval	52.69
land100 0.82 thr 206				
[DDQN-Orig]	Ep 699 R	247.41 avgR100	225.39 eps 0.030 Qval	52.02
land100 0.83 thr 214				
[DDQN-Orig]	Ep 700 R	254.24 avgR100	225.06 eps 0.030 Qval	52.26
land100 0.83 thr 226				
[DDQN-Orig]	Ep 701 R	234.21 avgR100	228.00 eps 0.030 Qval	52.40
land100 0.84 thr 348				
[DDQN-Orig]	Ep 702 R	272.30 avgR100	229.28 eps 0.030 Qval	52.90
land100 0.85 thr 318				
[DDQN-Orig]	Ep 703 R	225.90 avgR100	229.12 eps 0.029 Qval	53.02
land100 0.85 thr 327				
[DDQN-Orig]	Ep 704 R	230.23 avgR100	229.35 eps 0.029 Qval	54.20
land100 0.85 thr 456				
[DDQN-Orig]	Ep 705 R	242.04 avgR100	229.16 eps 0.029 Qval	55.30
land100 0.85 thr 262				
[DDQN-Orig]	Ep 706 R	142.40 avgR100	228.33 eps 0.029 Qval	55.43
land100 0.85 thr 686				
[DDQN-Orig]	Ep 707 R	258.96 avgR100	230.72 eps 0.029 Qval	55.18
land100 0.86 thr 337				
[DDQN-Orig]	Ep 708 R	237.10 avgR100	230.97 eps 0.029 Qval	55.27
land100 0.87 thr 191				
[DDQN-Orig]	Ep 709 R	224.75 avgR100	230.64 eps 0.029 Qval	55.66
land100 0.87 thr 199				
[DDQN-Orig]	Ep 710 R	233.02 avgR100	230.99 eps 0.028 Qval	56.32
land100 0.87 thr 265				
[DDQN-Orig]	Ep 711 R	276.31 avgR100	231.59 eps 0.028 Qval	57.01
land100 0.87 thr 188				
[DDQN-Orig]	Ep 712 R	248.43 avgR100	231.66 eps 0.028 Qval	57.82
land100 0.88 thr 443				
[DDQN-Orig]	Ep 713 R	228.19 avgR100	231.23 eps 0.028 Qval	57.44
land100 0.88 thr 212				
[DDQN-Orig]	Ep 714 R	198.22 avgR100	230.89 eps 0.028 Qval	58.57
land100 0.87 thr 250				
[DDQN-Orig]	Ep 715 R	252.08 avgR100	232.69 eps 0.028 Qval	58.17
land100 0.88 thr 289				
[DDQN-Orig]	Ep 716 R	253.32 avgR100	233.02 eps 0.028 Qval	58.31
land100 0.87 thr 421				
[DDQN-Orig]	Ep 717 R	256.15 avgR100	233.80 eps 0.027 Qval	58.89
land100 0.87 thr 276				
[DDQN-Orig]	Ep 718 R	236.88 avgR100	234.12 eps 0.027 Qval	60.64
land100 0.87 thr 321				
[DDQN-Orig]	Ep 719 R	213.06 avgR100	233.86 eps 0.027 Qval	60.21
land100 0.87 thr 235				
[DDQN-Orig]	Ep 720 R	257.72 avgR100	233.50 eps 0.027 Qval	59.63
land100 0.87 thr 158				

[DDQN-Orig]	Ep 721 R	228.13	avgR100	233.16	eps 0.027	Qval	59.76
land100 0.87	thr 326						
[DDQN-Orig]	Ep 722 R	259.69	avgR100	233.25	eps 0.027	Qval	59.89
land100 0.87	thr 155						
[DDQN-Orig]	Ep 723 R	132.96	avgR100	232.44	eps 0.027	Qval	60.45
land100 0.87	thr 633						
[DDQN-Orig]	Ep 724 R	219.63	avgR100	234.55	eps 0.027	Qval	60.44
land100 0.87	thr 473						
[DDQN-Orig]	Ep 725 R	242.06	avgR100	234.64	eps 0.026	Qval	59.72
land100 0.88	thr 196						
[DDQN-Orig]	Ep 726 R	270.43	avgR100	234.73	eps 0.026	Qval	59.85
land100 0.88	thr 279						
[DDQN-Orig]	Ep 727 R	168.93	avgR100	233.94	eps 0.026	Qval	61.67
land100 0.88	thr 571						
[DDQN-Orig]	Ep 728 R	257.26	avgR100	233.61	eps 0.026	Qval	60.40
land100 0.88	thr 196						
[DDQN-Orig]	Ep 729 R	260.10	avgR100	234.80	eps 0.026	Qval	61.96
land100 0.88	thr 437						
[DDQN-Orig]	Ep 730 R	257.52	avgR100	235.62	eps 0.026	Qval	61.37
land100 0.88	thr 179						
[DDQN-Orig]	Ep 731 R	260.81	avgR100	235.59	eps 0.026	Qval	61.49
land100 0.88	thr 176						
[DDQN-Orig]	Ep 732 R	229.04	avgR100	235.52	eps 0.025	Qval	62.43
land100 0.88	thr 365						
[DDQN-Orig]	Ep 733 R	243.72	avgR100	235.95	eps 0.025	Qval	62.51
land100 0.88	thr 289						
[DDQN-Orig]	Ep 734 R	266.93	avgR100	236.12	eps 0.025	Qval	62.96
land100 0.88	thr 183						
[DDQN-Orig]	Ep 735 R	242.73	avgR100	235.96	eps 0.025	Qval	63.40
land100 0.87	thr 261						
[DDQN-Orig]	Ep 736 R	174.89	avgR100	235.72	eps 0.025	Qval	64.30
land100 0.87	thr 653						
[DDQN-Orig]	Ep 737 R	251.95	avgR100	236.02	eps 0.025	Qval	65.01
land100 0.88	thr 200						
[DDQN-Orig]	Ep 738 R	223.93	avgR100	235.74	eps 0.025	Qval	64.63
land100 0.88	thr 295						
[DDQN-Orig]	Ep 739 R	199.16	avgR100	234.62	eps 0.025	Qval	65.42
land100 0.88	thr 347						
[DDQN-Orig]	Ep 740 R	258.11	avgR100	234.47	eps 0.024	Qval	65.94
land100 0.87	thr 222						
[DDQN-Orig]	Ep 741 R	255.39	avgR100	234.11	eps 0.024	Qval	66.44
land100 0.87	thr 515						
[DDQN-Orig]	Ep 742 R	230.34	avgR100	233.71	eps 0.024	Qval	65.91
land100 0.87	thr 460						
[DDQN-Orig]	Ep 743 R	251.68	avgR100	233.91	eps 0.024	Qval	66.31
land100 0.87	thr 267						
[DDQN-Orig]	Ep 744 R	270.15	avgR100	234.77	eps 0.024	Qval	66.10
land100 0.87	thr 199						
[DDQN-Orig]	Ep 745 R	267.85	avgR100	235.73	eps 0.024	Qval	66.41
land100 0.87	thr 243						
[DDQN-Orig]	Ep 746 R	253.06	avgR100	235.78	eps 0.024	Qval	66.57
land100 0.86	thr 349						
[DDQN-Orig]	Ep 747 R	262.52	avgR100	236.40	eps 0.024	Qval	66.59
land100 0.86	thr 292						
[DDQN-Orig]	Ep 748 R	262.41	avgR100	236.64	eps 0.024	Qval	66.46
land100 0.86	thr 205						
[DDQN-Orig]	Ep 749 R	296.97	avgR100	236.59	eps 0.023	Qval	66.06
land100 0.86	thr 186						
[DDQN-Orig]	Ep 750 R	204.09	avgR100	236.73	eps 0.023	Qval	65.83
land100 0.86	thr 477						

[DDQN-Orig]	Ep 751 R	42.33 avgR100	234.92 eps 0.023 Qval	65.58
land100 0.86 thr 97				
[DDQN-Orig]	Ep 752 R	251.53 avgR100	234.87 eps 0.023 Qval	65.29
land100 0.86 thr 307				
[DDQN-Orig]	Ep 753 R	40.09 avgR100	233.38 eps 0.023 Qval	65.64
land100 0.85 thr 52				
[DDQN-Orig]	Ep 754 R	244.82 avgR100	236.13 eps 0.023 Qval	66.82
land100 0.85 thr 454				
[DDQN-Orig]	Ep 755 R	192.33 avgR100	235.14 eps 0.023 Qval	66.88
land100 0.86 thr 463				
[DDQN-Orig]	Ep 756 R	243.67 avgR100	235.34 eps 0.023 Qval	67.41
land100 0.85 thr 115				
[DDQN-Orig]	Ep 757 R	207.83 avgR100	234.75 eps 0.022 Qval	67.39
land100 0.85 thr 429				
[DDQN-Orig]	Ep 758 R	180.48 avgR100	233.94 eps 0.022 Qval	66.92
land100 0.85 thr 486				
[DDQN-Orig]	Ep 759 R	263.04 avgR100	234.01 eps 0.022 Qval	67.47
land100 0.84 thr 263				
[DDQN-Orig]	Ep 760 R	233.21 avgR100	233.97 eps 0.022 Qval	67.13
land100 0.84 thr 291				
[DDQN-Orig]	Ep 761 R	233.80 avgR100	233.69 eps 0.022 Qval	66.48
land100 0.85 thr 215				
[DDQN-Orig]	Ep 762 R	220.18 avgR100	233.22 eps 0.022 Qval	67.40
land100 0.85 thr 374				
[DDQN-Orig]	Ep 763 R	215.39 avgR100	232.82 eps 0.022 Qval	67.29
land100 0.85 thr 292				
[DDQN-Orig]	Ep 764 R	259.16 avgR100	232.88 eps 0.022 Qval	67.78
land100 0.84 thr 188				
[DDQN-Orig]	Ep 765 R	298.75 avgR100	232.87 eps 0.022 Qval	67.21
land100 0.85 thr 174				
[DDQN-Orig]	Ep 766 R	237.13 avgR100	232.87 eps 0.022 Qval	66.51
land100 0.85 thr 181				
[DDQN-Orig]	Ep 767 R	261.18 avgR100	233.17 eps 0.021 Qval	67.17
land100 0.85 thr 100				
[DDQN-Orig]	Ep 768 R	185.70 avgR100	232.49 eps 0.021 Qval	67.88
land100 0.85 thr 526				
[DDQN-Orig]	Ep 769 R	261.27 avgR100	232.15 eps 0.021 Qval	66.60
land100 0.85 thr 282				
[DDQN-Orig]	Ep 770 R	274.39 avgR100	234.37 eps 0.021 Qval	66.74
land100 0.86 thr 246				
[DDQN-Orig]	Ep 771 R	264.05 avgR100	234.28 eps 0.021 Qval	67.42
land100 0.86 thr 193				
[DDQN-Orig]	Ep 772 R	-4.92 avgR100	231.70 eps 0.021 Qval	67.77
land100 0.85 thr 66				
[DDQN-Orig]	Ep 773 R	273.67 avgR100	232.76 eps 0.021 Qval	67.09
land100 0.85 thr 337				
[DDQN-Orig]	Ep 774 R	243.44 avgR100	232.99 eps 0.021 Qval	68.05
land100 0.84 thr 278				
[DDQN-Orig]	Ep 775 R	233.24 avgR100	232.65 eps 0.021 Qval	68.62
land100 0.84 thr 228				
[DDQN-Orig]	Ep 776 R	217.27 avgR100	232.08 eps 0.020 Qval	68.13
land100 0.85 thr 299				
[DDQN-Orig]	Ep 777 R	211.44 avgR100	231.50 eps 0.020 Qval	68.48
land100 0.85 thr 479				
[DDQN-Orig]	Ep 778 R	177.23 avgR100	230.99 eps 0.020 Qval	69.99
land100 0.85 thr 538				
[DDQN-Orig]	Ep 779 R	285.72 avgR100	231.44 eps 0.020 Qval	69.34
land100 0.86 thr 164				
[DDQN-Orig]	Ep 780 R	253.48 avgR100	231.59 eps 0.020 Qval	69.53
land100 0.86 thr 207				

[DDQN-Orig]	Ep 781 R	209.56 avgR100	231.23 eps 0.020 Qval	69.32
land100 0.86 thr 590				
[DDQN-Orig]	Ep 782 R	201.34 avgR100	230.84 eps 0.020 Qval	70.05
land100 0.86 thr 460				
[DDQN-Orig]	Ep 783 R	192.96 avgR100	230.47 eps 0.020 Qval	68.80
land100 0.86 thr 368				
[DDQN-Orig]	Ep 784 R	182.01 avgR100	230.22 eps 0.020 Qval	68.82
land100 0.86 thr 520				
[DDQN-Orig]	Ep 785 R	255.85 avgR100	230.63 eps 0.020 Qval	69.47
land100 0.86 thr 207				
[DDQN-Orig]	Ep 786 R	221.27 avgR100	230.47 eps 0.020 Qval	69.76
land100 0.86 thr 436				
[DDQN-Orig]	Ep 787 R	243.13 avgR100	230.43 eps 0.020 Qval	69.94
land100 0.86 thr 201				
[DDQN-Orig]	Ep 788 R	265.42 avgR100	230.42 eps 0.020 Qval	70.31
land100 0.85 thr 200				
[DDQN-Orig]	Ep 789 R	252.03 avgR100	230.47 eps 0.020 Qval	70.03
land100 0.84 thr 176				
[DDQN-Orig]	Ep 790 R	266.33 avgR100	230.74 eps 0.020 Qval	69.32
land100 0.84 thr 233				
[DDQN-Orig]	Ep 791 R	254.28 avgR100	230.94 eps 0.020 Qval	70.69
land100 0.84 thr 176				
[DDQN-Orig]	Ep 792 R	241.65 avgR100	230.65 eps 0.020 Qval	68.94
land100 0.84 thr 224				
[DDQN-Orig]	Ep 793 R	217.26 avgR100	230.26 eps 0.020 Qval	69.54
land100 0.84 thr 315				
[DDQN-Orig]	Ep 794 R	214.47 avgR100	230.36 eps 0.020 Qval	69.49
land100 0.84 thr 252				
[DDQN-Orig]	Ep 795 R	274.60 avgR100	230.97 eps 0.020 Qval	68.85
land100 0.84 thr 290				
[DDQN-Orig]	Ep 796 R	218.79 avgR100	230.89 eps 0.020 Qval	69.80
land100 0.84 thr 388				
[DDQN-Orig]	Ep 797 R	232.00 avgR100	231.23 eps 0.020 Qval	70.13
land100 0.84 thr 366				
[DDQN-Orig]	Ep 798 R	225.57 avgR100	230.95 eps 0.020 Qval	70.35
land100 0.84 thr 364				
[DDQN-Orig]	Ep 799 R	209.73 avgR100	230.57 eps 0.020 Qval	69.91
land100 0.84 thr 354				
[DDQN-Orig]	Ep 800 R	221.47 avgR100	230.24 eps 0.020 Qval	70.55
land100 0.83 thr 535				
[DDQN-Orig]	Ep 801 R	226.89 avgR100	230.17 eps 0.020 Qval	70.55
land100 0.83 thr 312				
[DDQN-Orig]	Ep 802 R	262.20 avgR100	230.07 eps 0.020 Qval	70.46
land100 0.82 thr 154				
[DDQN-Orig]	Ep 803 R	237.88 avgR100	230.19 eps 0.020 Qval	69.51
land100 0.82 thr 363				
[DDQN-Orig]	Ep 804 R	237.57 avgR100	230.26 eps 0.020 Qval	70.10
land100 0.81 thr 309				
[DDQN-Orig]	Ep 805 R	235.75 avgR100	230.20 eps 0.020 Qval	70.84
land100 0.81 thr 252				
[DDQN-Orig]	Ep 806 R	235.72 avgR100	231.13 eps 0.020 Qval	70.63
land100 0.80 thr 340				
[DDQN-Orig]	Ep 807 R	242.27 avgR100	230.97 eps 0.020 Qval	70.85
land100 0.80 thr 175				
[DDQN-Orig]	Ep 808 R	222.97 avgR100	230.82 eps 0.020 Qval	71.04
land100 0.80 thr 276				
[DDQN-Orig]	Ep 809 R	263.34 avgR100	231.21 eps 0.020 Qval	70.88
land100 0.80 thr 223				
[DDQN-Orig]	Ep 810 R	260.39 avgR100	231.48 eps 0.020 Qval	71.71
land100 0.80 thr 290				

[DDQN-Orig]	Ep 811 R	256.43	avgR100	231.29	eps 0.020	Qval	70.27	
land100 0.80	thr 224							
[DDQN-Orig]	Ep 812 R	248.52	avgR100	231.29	eps 0.020	Qval	71.12	
land100 0.80	thr 274							
[DDQN-Orig]	Ep 813 R	281.05	avgR100	231.81	eps 0.020	Qval	71.19	
land100 0.80	thr 285							
[DDQN-Orig]	Ep 814 R	260.96	avgR100	232.44	eps 0.020	Qval	70.98	
land100 0.81	thr 311							
[DDQN-Orig]	Ep 815 R	234.21	avgR100	232.26	eps 0.020	Qval	71.10	
land100 0.81	thr 362							
[DDQN-Orig]	Ep 816 R	245.30	avgR100	232.18	eps 0.020	Qval	71.44	
land100 0.81	thr 399							
[DDQN-Orig]	Ep 817 R	276.57	avgR100	232.39	eps 0.020	Qval	72.23	
land100 0.81	thr 272							
[DDQN-Orig]	Ep 818 R	198.23	avgR100	232.00	eps 0.020	Qval	71.31	
land100 0.81	thr 374							
[DDQN-Orig]	Ep 819 R	231.25	avgR100	232.18	eps 0.020	Qval	71.92	
land100 0.81	thr 269							
[DDQN-Orig]	Ep 820 R	229.86	avgR100	231.90	eps 0.020	Qval	72.63	
land100 0.81	thr 306							
[DDQN-Orig]	Ep 821 R	260.94	avgR100	232.23	eps 0.020	Qval	72.11	
land100 0.81	thr 162							
[DDQN-Orig]	Ep 822 R	166.56	avgR100	231.30	eps 0.020	Qval	72.02	
land100 0.81	thr 562							
[DDQN-Orig]	Ep 823 R	253.73	avgR100	232.51	eps 0.020	Qval	73.10	
land100 0.81	thr 299							
[DDQN-Orig]	Ep 824 R	268.56	avgR100	233.00	eps 0.020	Qval	73.21	
land100 0.82	thr 143							
[DDQN-Orig]	Ep 825 R	253.05	avgR100	233.11	eps 0.020	Qval	73.67	
land100 0.82	thr 382							
[DDQN-Orig]	Ep 826 R	269.52	avgR100	233.10	eps 0.020	Qval	73.32	
land100 0.82	thr 223							
[DDQN-Orig]	Ep 827 R	216.53	avgR100	233.57	eps 0.020	Qval	72.03	
land100 0.82	thr 356							
[DDQN-Orig]	Ep 828 R	286.51	avgR100	233.87	eps 0.020	Qval	73.67	
land100 0.82	thr 257							
[DDQN-Orig]	Ep 829 R	274.81	avgR100	234.01	eps 0.020	Qval	72.86	
land100 0.82	thr 295							
[DDQN-Orig]	Ep 830 R	227.17	avgR100	233.71	eps 0.020	Qval	72.72	
land100 0.82	thr 503							
[DDQN-Orig]	Ep 831 R	268.89	avgR100	233.79	eps 0.020	Qval	71.37	
land100 0.82	thr 231							
[DDQN-Orig]	Ep 832 R	213.47	avgR100	233.64	eps 0.020	Qval	72.74	
land100 0.82	thr 374							
[DDQN-Orig]	Ep 833 R	231.85	avgR100	233.52	eps 0.020	Qval	73.61	
land100 0.82	thr 425							
[DDQN-Orig]	Ep 834 R	226.64	avgR100	233.11	eps 0.020	Qval	74.04	
land100 0.82	thr 180							
[DDQN-Orig]	Ep 835 R	230.82	avgR100	233.00	eps 0.020	Qval	72.49	
land100 0.83	thr 248							
[DDQN-Orig]	Ep 836 R	226.24	avgR100	233.51	eps 0.020	Qval	71.98	
land100 0.83	thr 193							
[DDQN-Orig]	Ep 837 R	261.31	avgR100	233.60	eps 0.020	Qval	72.76	
land100 0.83	thr 192							
[DDQN-Orig]	Ep 838 R	245.78	avgR100	233.82	eps 0.020	Qval	72.49	
land100 0.83	thr 328							
[DDQN-Orig]	Ep 839 R	251.95	avgR100	234.35	eps 0.020	Qval	73.02	
land100 0.83	thr 210							
[DDQN-Orig]	Ep 840 R	261.52	avgR100	234.38	eps 0.020	Qval	72.65	
land100 0.84	thr 177							

[DDQN-Orig]	Ep 841 R	275.22 avgR100	234.58 eps 0.020	Qval 73.16
land100 0.84 thr 201				
[DDQN-Orig]	Ep 842 R	265.63 avgR100	234.93 eps 0.020	Qval 73.03
land100 0.84 thr 194				
[DDQN-Orig]	Ep 843 R	272.51 avgR100	235.14 eps 0.020	Qval 73.28
land100 0.84 thr 273				
[DDQN-Orig]	Ep 844 R	259.94 avgR100	235.04 eps 0.020	Qval 73.31
land100 0.84 thr 214				
[DDQN-Orig]	Ep 845 R	261.32 avgR100	234.97 eps 0.020	Qval 73.88
land100 0.84 thr 229				
[DDQN-Orig]	Ep 846 R	249.50 avgR100	234.94 eps 0.020	Qval 74.59
land100 0.85 thr 202				
[DDQN-Orig]	Ep 847 R	254.83 avgR100	234.86 eps 0.020	Qval 74.68
land100 0.85 thr 290				
[DDQN-Orig]	Ep 848 R	255.87 avgR100	234.80 eps 0.020	Qval 74.18
land100 0.85 thr 338				
[DDQN-Orig]	Ep 849 R	247.91 avgR100	234.31 eps 0.020	Qval 74.01
land100 0.85 thr 228				
[DDQN-Orig]	Ep 850 R	224.52 avgR100	234.51 eps 0.020	Qval 74.56
land100 0.85 thr 256				
[DDQN-Orig]	Ep 851 R	181.27 avgR100	235.90 eps 0.020	Qval 74.33
land100 0.86 thr 467				
[DDQN-Orig]	Ep 852 R	256.45 avgR100	235.95 eps 0.020	Qval 73.59
land100 0.86 thr 280				
[DDQN-Orig]	Ep 853 R	273.16 avgR100	238.28 eps 0.020	Qval 74.39
land100 0.87 thr 195				
[DDQN-Orig]	Ep 854 R	224.73 avgR100	238.08 eps 0.020	Qval 74.08
land100 0.88 thr 361				
[DDQN-Orig]	Ep 855 R	279.19 avgR100	238.95 eps 0.020	Qval 74.30
land100 0.88 thr 283				
[DDQN-Orig]	Ep 856 R	262.19 avgR100	239.13 eps 0.020	Qval 74.34
land100 0.89 thr 318				
[DDQN-Orig]	Ep 857 R	236.64 avgR100	239.42 eps 0.020	Qval 73.37
land100 0.89 thr 196				
[DDQN-Orig]	Ep 858 R	269.54 avgR100	240.31 eps 0.020	Qval 72.98
land100 0.89 thr 221				
[DDQN-Orig]	Ep 859 R	245.85 avgR100	240.14 eps 0.020	Qval 73.91
land100 0.90 thr 148				
[DDQN-Orig]	Ep 860 R	233.01 avgR100	240.14 eps 0.020	Qval 74.55
land100 0.90 thr 179				
[DDQN-Orig]	Ep 861 R	280.37 avgR100	240.60 eps 0.020	Qval 74.92
land100 0.90 thr 241				
[DDQN-Orig]	Ep 862 R	280.92 avgR100	241.21 eps 0.020	Qval 74.47
land100 0.90 thr 283				
[DDQN-Orig]	Ep 863 R	232.12 avgR100	241.38 eps 0.020	Qval 74.31
land100 0.90 thr 486				
[DDQN-Orig]	Ep 864 R	293.48 avgR100	241.72 eps 0.020	Qval 74.68
land100 0.91 thr 217				
[DDQN-Orig]	Ep 865 R	268.46 avgR100	241.42 eps 0.020	Qval 74.47
land100 0.91 thr 237				
[DDQN-Orig]	Ep 866 R	260.87 avgR100	241.66 eps 0.020	Qval 74.86
land100 0.91 thr 194				
[DDQN-Orig]	Ep 867 R	279.54 avgR100	241.84 eps 0.020	Qval 75.16
land100 0.91 thr 208				
[DDQN-Orig]	Ep 868 R	218.88 avgR100	242.17 eps 0.020	Qval 75.32
land100 0.91 thr 302				
[DDQN-Orig]	Ep 869 R	225.71 avgR100	241.82 eps 0.020	Qval 75.86
land100 0.91 thr 494				
[DDQN-Orig]	Ep 870 R	250.52 avgR100	241.58 eps 0.020	Qval 76.11
land100 0.91 thr 189				

[DDQN-Orig]	Ep 871 R	265.69 avgR100	241.59 eps 0.020	Qval 75.55
land100 0.91	thr 171			
[DDQN-Orig]	Ep 872 R	208.65 avgR100	243.73 eps 0.020	Qval 75.63
land100 0.92	thr 188			
[DDQN-Orig]	Ep 873 R	260.46 avgR100	243.60 eps 0.020	Qval 74.92
land100 0.92	thr 253			
[DDQN-Orig]	Ep 874 R	280.03 avgR100	243.96 eps 0.020	Qval 75.41
land100 0.93	thr 344			
[DDQN-Orig]	Ep 875 R	209.18 avgR100	243.72 eps 0.020	Qval 75.00
land100 0.93	thr 430			
[DDQN-Orig]	Ep 876 R	257.20 avgR100	244.12 eps 0.020	Qval 75.79
land100 0.93	thr 236			
[DDQN-Orig]	Ep 877 R	252.80 avgR100	244.53 eps 0.020	Qval 76.02
land100 0.93	thr 349			
[DDQN-Orig]	Ep 878 R	233.63 avgR100	245.10 eps 0.020	Qval 75.64
land100 0.93	thr 218			
[DDQN-Orig]	Ep 879 R	253.04 avgR100	244.77 eps 0.020	Qval 76.14
land100 0.93	thr 186			
[DDQN-Orig]	Ep 880 R	275.08 avgR100	244.99 eps 0.020	Qval 76.11
land100 0.93	thr 151			
[DDQN-Orig]	Ep 881 R	249.93 avgR100	245.39 eps 0.020	Qval 76.29
land100 0.93	thr 354			
[DDQN-Orig]	Ep 882 R	265.95 avgR100	246.04 eps 0.020	Qval 75.88
land100 0.93	thr 290			
[DDQN-Orig]	Ep 883 R	37.64 avgR100	244.48 eps 0.020	Qval 75.71
land100 0.92	thr 53			
[DDQN-Orig]	Ep 884 R	261.16 avgR100	245.28 eps 0.020	Qval 76.57
land100 0.92	thr 175			
[DDQN-Orig]	Ep 885 R	250.97 avgR100	245.23 eps 0.020	Qval 76.36
land100 0.92	thr 408			
[DDQN-Orig]	Ep 886 R	275.55 avgR100	245.77 eps 0.020	Qval 76.96
land100 0.92	thr 302			
[DDQN-Orig]	Ep 887 R	272.67 avgR100	246.07 eps 0.020	Qval 77.41
land100 0.91	thr 140			
[DDQN-Orig]	Ep 888 R	262.23 avgR100	246.03 eps 0.020	Qval 76.68
land100 0.92	thr 215			
[DDQN-Orig]	Ep 889 R	249.57 avgR100	246.01 eps 0.020	Qval 77.86
land100 0.93	thr 348			
[DDQN-Orig]	Ep 890 R	241.01 avgR100	245.76 eps 0.020	Qval 77.87
land100 0.93	thr 131			
[DDQN-Orig]	Ep 891 R	219.10 avgR100	245.40 eps 0.020	Qval 78.24
land100 0.93	thr 249			
[DDQN-Orig]	Ep 892 R	262.06 avgR100	245.61 eps 0.020	Qval 77.33
land100 0.93	thr 160			
[DDQN-Orig]	Ep 893 R	210.45 avgR100	245.54 eps 0.020	Qval 77.41
land100 0.93	thr 264			
[DDQN-Orig]	Ep 894 R	249.05 avgR100	245.89 eps 0.020	Qval 78.11
land100 0.93	thr 187			
[DDQN-Orig]	Ep 895 R	254.37 avgR100	245.68 eps 0.020	Qval 77.92
land100 0.93	thr 229			
[DDQN-Orig]	Ep 896 R	197.45 avgR100	245.47 eps 0.020	Qval 77.75
land100 0.93	thr 280			
[DDQN-Orig]	Ep 897 R	12.04 avgR100	243.27 eps 0.020	Qval 77.76
land100 0.92	thr 43			
[DDQN-Orig]	Ep 898 R	-30.89 avgR100	240.71 eps 0.020	Qval 78.00
land100 0.91	thr 39			
[DDQN-Orig]	Ep 899 R	-18.91 avgR100	238.42 eps 0.020	Qval 78.50
land100 0.90	thr 29			
[DDQN-Orig]	Ep 900 R	276.71 avgR100	238.97 eps 0.020	Qval 77.13
land100 0.91	thr 213			

[DDQN-Orig]	Ep 901 R	301.48	avgR100	239.72	eps 0.020	Qval	78.24	
land100 0.91	thr 209							
[DDQN-Orig]	Ep 902 R	252.11	avgR100	239.62	eps 0.020	Qval	77.46	
land100 0.92	thr 229							
[DDQN-Orig]	Ep 903 R	259.21	avgR100	239.83	eps 0.020	Qval	76.63	
land100 0.92	thr 255							
[DDQN-Orig]	Ep 904 R	279.74	avgR100	240.25	eps 0.020	Qval	76.28	
land100 0.93	thr 239							
[DDQN-Orig]	Ep 905 R	268.41	avgR100	240.58	eps 0.020	Qval	76.49	
land100 0.93	thr 344							
[DDQN-Orig]	Ep 906 R	268.38	avgR100	240.91	eps 0.020	Qval	76.59	
land100 0.94	thr 249							
[DDQN-Orig]	Ep 907 R	212.36	avgR100	240.61	eps 0.020	Qval	76.04	
land100 0.94	thr 376							
[DDQN-Orig]	Ep 908 R	62.42	avgR100	239.00	eps 0.020	Qval	76.33	
land100 0.93	thr 124							
[DDQN-Orig]	Ep 909 R	255.44	avgR100	238.92	eps 0.020	Qval	77.07	
land100 0.93	thr 192							
[DDQN-Orig]	Ep 910 R	242.88	avgR100	238.75	eps 0.020	Qval	76.75	
land100 0.93	thr 171							
[DDQN-Orig]	Ep 911 R	231.06	avgR100	238.49	eps 0.020	Qval	76.39	
land100 0.93	thr 320							
[DDQN-Orig]	Ep 912 R	44.85	avgR100	236.46	eps 0.020	Qval	76.05	
land100 0.92	thr 97							
[DDQN-Orig]	Ep 913 R	-11.28	avgR100	233.53	eps 0.020	Qval	76.23	
land100 0.91	thr 51							
[DDQN-Orig]	Ep 914 R	247.84	avgR100	233.40	eps 0.020	Qval	77.33	
land100 0.91	thr 281							
[DDQN-Orig]	Ep 915 R	264.45	avgR100	233.70	eps 0.020	Qval	75.99	
land100 0.91	thr 191							
[DDQN-Orig]	Ep 916 R	32.44	avgR100	231.58	eps 0.020	Qval	75.96	
land100 0.91	thr 92							
[DDQN-Orig]	Ep 917 R	244.44	avgR100	231.25	eps 0.020	Qval	75.67	
land100 0.91	thr 248							
[DDQN-Orig]	Ep 918 R	244.64	avgR100	231.72	eps 0.020	Qval	75.38	
land100 0.91	thr 203							
[DDQN-Orig]	Ep 919 R	250.58	avgR100	231.91	eps 0.020	Qval	76.84	
land100 0.91	thr 182							
[DDQN-Orig]	Ep 920 R	240.66	avgR100	232.02	eps 0.020	Qval	76.44	
land100 0.90	thr 283							
[DDQN-Orig]	Ep 921 R	273.63	avgR100	232.15	eps 0.020	Qval	75.02	
land100 0.90	thr 198							
[DDQN-Orig]	Ep 922 R	245.86	avgR100	232.94	eps 0.020	Qval	75.55	
land100 0.90	thr 228							
[DDQN-Orig]	Ep 923 R	197.89	avgR100	232.38	eps 0.020	Qval	75.25	
land100 0.90	thr 355							
[DDQN-Orig]	Ep 924 R	258.81	avgR100	232.28	eps 0.020	Qval	74.98	
land100 0.90	thr 182							
[DDQN-Orig]	Ep 925 R	277.48	avgR100	232.53	eps 0.020	Qval	74.58	
land100 0.90	thr 225							
[DDQN-Orig]	Ep 926 R	250.34	avgR100	232.34	eps 0.020	Qval	75.78	
land100 0.89	thr 162							
[DDQN-Orig]	Ep 927 R	225.09	avgR100	232.42	eps 0.020	Qval	76.05	
land100 0.89	thr 265							
[DDQN-Orig]	Ep 928 R	213.14	avgR100	231.69	eps 0.020	Qval	74.28	
land100 0.89	thr 241							
[DDQN-Orig]	Ep 929 R	264.96	avgR100	231.59	eps 0.020	Qval	73.42	
land100 0.89	thr 165							
[DDQN-Orig]	Ep 930 R	214.02	avgR100	231.46	eps 0.020	Qval	73.57	
land100 0.89	thr 284							

[DDQN-Orig]	Ep 931 R	289.41	avgR100	231.66	eps 0.020	Qval	73.44
land100 0.89	thr 151						
[DDQN-Orig]	Ep 932 R	284.29	avgR100	232.37	eps 0.020	Qval	74.00
land100 0.89	thr 256						
[DDQN-Orig]	Ep 933 R	226.42	avgR100	232.32	eps 0.020	Qval	74.72
land100 0.89	thr 317						
[DDQN-Orig]	Ep 934 R	253.55	avgR100	232.59	eps 0.020	Qval	74.39
land100 0.88	thr 222						
[DDQN-Orig]	Ep 935 R	268.72	avgR100	232.97	eps 0.020	Qval	74.51
land100 0.88	thr 278						
[DDQN-Orig]	Ep 936 R	265.52	avgR100	233.36	eps 0.020	Qval	73.89
land100 0.88	thr 234						
[DDQN-Orig]	Ep 937 R	256.99	avgR100	233.32	eps 0.020	Qval	72.02
land100 0.88	thr 416						
[DDQN-Orig]	Ep 938 R	279.80	avgR100	233.66	eps 0.020	Qval	72.62
land100 0.88	thr 205						
[DDQN-Orig]	Ep 939 R	277.70	avgR100	233.91	eps 0.020	Qval	73.64
land100 0.88	thr 217						
[DDQN-Orig]	Ep 940 R	275.17	avgR100	234.05	eps 0.020	Qval	73.98
land100 0.88	thr 191						
[DDQN-Orig]	Ep 941 R	262.57	avgR100	233.92	eps 0.020	Qval	72.60
land100 0.88	thr 249						
[DDQN-Orig]	Ep 942 R	242.49	avgR100	233.69	eps 0.020	Qval	73.43
land100 0.88	thr 305						
[DDQN-Orig]	Ep 943 R	240.15	avgR100	233.37	eps 0.020	Qval	73.27
land100 0.88	thr 167						
[DDQN-Orig]	Ep 944 R	273.84	avgR100	233.51	eps 0.020	Qval	73.29
land100 0.88	thr 281						
[DDQN-Orig]	Ep 945 R	243.40	avgR100	233.33	eps 0.020	Qval	72.26
land100 0.88	thr 249						
[DDQN-Orig]	Ep 946 R	245.85	avgR100	233.29	eps 0.020	Qval	73.10
land100 0.88	thr 332						
[DDQN-Orig]	Ep 947 R	285.25	avgR100	233.60	eps 0.020	Qval	72.47
land100 0.88	thr 219						
[DDQN-Orig]	Ep 948 R	233.55	avgR100	233.37	eps 0.020	Qval	72.72
land100 0.88	thr 447						
[DDQN-Orig]	Ep 949 R	245.88	avgR100	233.35	eps 0.020	Qval	73.21
land100 0.88	thr 259						
[DDQN-Orig]	Ep 950 R	257.55	avgR100	233.68	eps 0.020	Qval	74.05
land100 0.87	thr 198						
[DDQN-Orig]	Ep 951 R	246.62	avgR100	234.34	eps 0.020	Qval	73.02
land100 0.86	thr 245						
[DDQN-Orig]	Ep 952 R	247.28	avgR100	234.24	eps 0.020	Qval	73.32
land100 0.86	thr 324						
[DDQN-Orig]	Ep 953 R	294.39	avgR100	234.46	eps 0.020	Qval	72.85
land100 0.86	thr 196						
[DDQN-Orig]	Ep 954 R	277.46	avgR100	234.98	eps 0.020	Qval	72.50
land100 0.86	thr 245						
[DDQN-Orig]	Ep 955 R	238.98	avgR100	234.58	eps 0.020	Qval	73.01
land100 0.86	thr 357						
[DDQN-Orig]	Ep 956 R	274.92	avgR100	234.71	eps 0.020	Qval	72.94
land100 0.86	thr 251						
[DDQN-Orig]	Ep 957 R	268.04	avgR100	235.02	eps 0.020	Qval	73.01
land100 0.86	thr 223						
[DDQN-Orig]	Ep 958 R	263.07	avgR100	234.96	eps 0.020	Qval	72.15
land100 0.86	thr 230						
[DDQN-Orig]	Ep 959 R	267.88	avgR100	235.18	eps 0.020	Qval	71.85
land100 0.86	thr 146						
[DDQN-Orig]	Ep 960 R	252.91	avgR100	235.38	eps 0.020	Qval	71.42
land100 0.86	thr 205						

[DDQN-Orig]	Ep 961 R	283.77 avgR100	235.41 eps 0.020	Qval 71.06
land100 0.86 thr 239				
[DDQN-Orig]	Ep 962 R	248.69 avgR100	235.09 eps 0.020	Qval 70.40
land100 0.86 thr 234				
[DDQN-Orig]	Ep 963 R	232.54 avgR100	235.09 eps 0.020	Qval 71.69
land100 0.85 thr 217				
[DDQN-Orig]	Ep 964 R	252.44 avgR100	234.68 eps 0.020	Qval 71.22
land100 0.85 thr 248				
[DDQN-Orig]	Ep 965 R	281.97 avgR100	234.82 eps 0.020	Qval 71.02
land100 0.85 thr 168				
[DDQN-Orig]	Ep 966 R	289.88 avgR100	235.11 eps 0.020	Qval 70.62
land100 0.85 thr 268				
[DDQN-Orig]	Ep 967 R	-0.65 avgR100	232.31 eps 0.020	Qval 70.50
land100 0.84 thr 80				
[DDQN-Orig]	Ep 968 R	287.69 avgR100	232.99 eps 0.020	Qval 70.12
land100 0.84 thr 117				
[DDQN-Orig]	Ep 969 R	284.81 avgR100	233.59 eps 0.020	Qval 70.38
land100 0.84 thr 202				
[DDQN-Orig]	Ep 970 R	256.80 avgR100	233.65 eps 0.020	Qval 70.18
land100 0.84 thr 166				
[DDQN-Orig]	Ep 971 R	-98.74 avgR100	230.00 eps 0.020	Qval 70.87
land100 0.83 thr 137				
[DDQN-Orig]	Ep 972 R	274.52 avgR100	230.66 eps 0.020	Qval 69.90
land100 0.83 thr 181				
[DDQN-Orig]	Ep 973 R	243.52 avgR100	230.49 eps 0.020	Qval 70.59
land100 0.83 thr 196				
[DDQN-Orig]	Ep 974 R	217.06 avgR100	229.86 eps 0.020	Qval 70.32
land100 0.83 thr 289				
[DDQN-Orig]	Ep 975 R	252.56 avgR100	230.30 eps 0.020	Qval 69.81
land100 0.83 thr 237				
[DDQN-Orig]	Ep 976 R	310.35 avgR100	230.83 eps 0.020	Qval 70.03
land100 0.83 thr 147				
[DDQN-Orig]	Ep 977 R	302.42 avgR100	231.33 eps 0.020	Qval 69.34
land100 0.83 thr 193				
[DDQN-Orig]	Ep 978 R	276.66 avgR100	231.76 eps 0.020	Qval 70.89
land100 0.83 thr 209				
[DDQN-Orig]	Ep 979 R	279.16 avgR100	232.02 eps 0.020	Qval 71.04
land100 0.83 thr 232				
[DDQN-Orig]	Ep 980 R	261.14 avgR100	231.88 eps 0.020	Qval 69.83
land100 0.83 thr 311				
[DDQN-Orig]	Ep 981 R	213.24 avgR100	231.51 eps 0.020	Qval 69.89
land100 0.83 thr 408				
[DDQN-Orig]	Ep 982 R	230.61 avgR100	231.16 eps 0.020	Qval 69.13
land100 0.83 thr 248				
[DDQN-Orig]	Ep 983 R	227.91 avgR100	233.06 eps 0.020	Qval 69.36
land100 0.84 thr 221				
[DDQN-Orig]	Ep 984 R	267.72 avgR100	233.13 eps 0.020	Qval 68.96
land100 0.84 thr 336				
[DDQN-Orig]	Ep 985 R	224.75 avgR100	232.86 eps 0.020	Qval 69.40
land100 0.84 thr 558				
[DDQN-Orig]	Ep 986 R	256.36 avgR100	232.67 eps 0.020	Qval 70.04
land100 0.84 thr 173				
[DDQN-Orig]	Ep 987 R	272.08 avgR100	232.67 eps 0.020	Qval 69.79
land100 0.85 thr 140				
[DDQN-Orig]	Ep 988 R	224.07 avgR100	232.28 eps 0.020	Qval 69.59
land100 0.85 thr 259				
[DDQN-Orig]	Ep 989 R	264.69 avgR100	232.44 eps 0.020	Qval 68.59
land100 0.85 thr 282				
[DDQN-Orig]	Ep 990 R	237.81 avgR100	232.40 eps 0.020	Qval 68.69
land100 0.85 thr 267				

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[DDQN-Orig] Ep 991 | R 264.00 | avgR100 232.85 | eps 0.020 | Qval 68.86 |
land100 0.84 | thr 295
[DDQN-Orig] Ep 992 | R 257.90 | avgR100 232.81 | eps 0.020 | Qval 68.50 |
land100 0.84 | thr 156
[DDQN-Orig] Ep 993 | R 270.37 | avgR100 233.41 | eps 0.020 | Qval 68.34 |
land100 0.84 | thr 145
[DDQN-Orig] Ep 994 | R 272.63 | avgR100 233.65 | eps 0.020 | Qval 68.54 |
land100 0.84 | thr 169
[DDQN-Orig] Ep 995 | R 270.21 | avgR100 233.80 | eps 0.020 | Qval 68.72 |
land100 0.84 | thr 220
[DDQN-Orig] Ep 996 | R 280.67 | avgR100 234.64 | eps 0.020 | Qval 68.53 |
land100 0.84 | thr 236
[DDQN-Orig] Ep 997 | R 271.18 | avgR100 237.23 | eps 0.020 | Qval 68.97 |
land100 0.84 | thr 544
[DDQN-Orig] Ep 998 | R 250.33 | avgR100 240.04 | eps 0.020 | Qval 67.99 |
land100 0.85 | thr 275
[DDQN-Orig] Ep 999 | R 270.16 | avgR100 242.93 | eps 0.020 | Qval 68.24 |
land100 0.86 | thr 182
[DDQN-Orig] Ep 1000 | R 11.65 | avgR100 240.28 | eps 0.020 | Qval 67.92 |
land100 0.85 | thr 77
[DDQN-Orig] finished 1000 episodes in 4.2 min (final avgR100 = 240.28)

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In [29]: # ---- Configuration 3 of 4: DQN on the MODIFIED environment ----
RESULTS["DQN-Mod"] = train_agent(modified=True, double=False, tag="DQN-Mod")

```

[DQN-Mod]	Ep	1		R	-102.54		avgR100	-102.54		eps	0.995		Qval	0.05		la
nd100 0.00		thr	41													
[DQN-Mod]	Ep	2		R	-98.27		avgR100	-100.40		eps	0.990		Qval	0.05		la
nd100 0.00		thr	50													
[DQN-Mod]	Ep	3		R	-307.70		avgR100	-169.50		eps	0.985		Qval	0.05		la
nd100 0.00		thr	87													
[DQN-Mod]	Ep	4		R	-166.45		avgR100	-168.74		eps	0.980		Qval	0.05		la
nd100 0.00		thr	75													
[DQN-Mod]	Ep	5		R	-122.17		avgR100	-159.43		eps	0.975		Qval	0.05		la
nd100 0.00		thr	54													
[DQN-Mod]	Ep	6		R	-281.18		avgR100	-179.72		eps	0.970		Qval	0.05		la
nd100 0.00		thr	69													
[DQN-Mod]	Ep	7		R	-391.00		avgR100	-209.90		eps	0.966		Qval	0.05		la
nd100 0.00		thr	58													
[DQN-Mod]	Ep	8		R	-64.07		avgR100	-191.67		eps	0.961		Qval	0.05		la
nd100 0.12		thr	64													
[DQN-Mod]	Ep	9		R	-240.66		avgR100	-197.12		eps	0.956		Qval	0.05		la
nd100 0.11		thr	67													
[DQN-Mod]	Ep	10		R	-122.26		avgR100	-189.63		eps	0.951		Qval	0.05		la
nd100 0.10		thr	80													
[DQN-Mod]	Ep	11		R	-137.70		avgR100	-184.91		eps	0.946		Qval	0.05		la
nd100 0.09		thr	78													
[DQN-Mod]	Ep	12		R	-336.48		avgR100	-197.54		eps	0.942		Qval	0.14		la
nd100 0.08		thr	71													
[DQN-Mod]	Ep	13		R	-243.07		avgR100	-201.04		eps	0.937		Qval	-0.18		la
nd100 0.08		thr	93													
[DQN-Mod]	Ep	14		R	-195.35		avgR100	-200.64		eps	0.932		Qval	-0.48		la
nd100 0.07		thr	121													
[DQN-Mod]	Ep	15		R	-172.56		avgR100	-198.76		eps	0.928		Qval	-0.57		la
nd100 0.07		thr	56													
[DQN-Mod]	Ep	16		R	-491.52		avgR100	-217.06		eps	0.923		Qval	-0.43		la
nd100 0.06		thr	80													
[DQN-Mod]	Ep	17		R	-324.77		avgR100	-223.40		eps	0.918		Qval	-0.52		la
nd100 0.06		thr	73													
[DQN-Mod]	Ep	18		R	-275.27		avgR100	-226.28		eps	0.914		Qval	-0.62		la
nd100 0.06		thr	106													
[DQN-Mod]	Ep	19		R	-348.87		avgR100	-232.73		eps	0.909		Qval	-0.57		la
nd100 0.05		thr	87													
[DQN-Mod]	Ep	20		R	-4											

[DQN-Mod] Ep	31	R	-93.22	avgR100	-196.56	eps	0.856	Qval	-0.01	la
nd100 0.10	thr	69								
[DQN-Mod] Ep	32	R	-224.29	avgR100	-197.43	eps	0.852	Qval	0.24	la
nd100 0.09	thr	89								
[DQN-Mod] Ep	33	R	-250.08	avgR100	-199.03	eps	0.848	Qval	-0.05	la
nd100 0.09	thr	60								
[DQN-Mod] Ep	34	R	-62.95	avgR100	-195.02	eps	0.843	Qval	0.05	la
nd100 0.09	thr	80								
[DQN-Mod] Ep	35	R	-146.18	avgR100	-193.63	eps	0.839	Qval	0.04	la
nd100 0.09	thr	64								
[DQN-Mod] Ep	36	R	-101.66	avgR100	-191.07	eps	0.835	Qval	-0.21	la
nd100 0.08	thr	94								
[DQN-Mod] Ep	37	R	-153.46	avgR100	-190.06	eps	0.831	Qval	0.07	la
nd100 0.08	thr	77								
[DQN-Mod] Ep	38	R	-102.53	avgR100	-187.75	eps	0.827	Qval	-0.15	la
nd100 0.08	thr	67								
[DQN-Mod] Ep	39	R	-49.74	avgR100	-184.21	eps	0.822	Qval	0.20	la
nd100 0.08	thr	75								
[DQN-Mod] Ep	40	R	-185.58	avgR100	-184.25	eps	0.818	Qval	0.14	la
nd100 0.07	thr	71								
[DQN-Mod] Ep	41	R	-173.25	avgR100	-183.98	eps	0.814	Qval	0.07	la
nd100 0.07	thr	50								
[DQN-Mod] Ep	42	R	-136.69	avgR100	-182.85	eps	0.810	Qval	-0.02	la
nd100 0.07	thr	78								
[DQN-Mod] Ep	43	R	-282.54	avgR100	-185.17	eps	0.806	Qval	0.34	la
nd100 0.07	thr	75								
[DQN-Mod] Ep	44	R	-300.69	avgR100	-187.80	eps	0.802	Qval	0.08	la
nd100 0.07	thr	76								
[DQN-Mod] Ep	45	R	-107.06	avgR100	-186.00	eps	0.798	Qval	0.25	la
nd100 0.07	thr	69								
[DQN-Mod] Ep	46	R	-143.83	avgR100	-185.09	eps	0.794	Qval	0.13	la
nd100 0.07	thr	124								
[DQN-Mod] Ep	47	R	-157.12	avgR100	-184.49	eps	0.790	Qval	0.13	la
nd100 0.06	thr	47								
[DQN-Mod] Ep	48	R	-90.15	avgR100	-182.53	eps	0.786	Qval	0.03	la
nd100 0.06	thr	58								
[DQN-Mod] Ep	49	R	-122.04	avgR100	-181.29	eps	0.782	Qval	0.23	la
nd100 0.06	thr	82								
[DQN-Mod] Ep	50	R	-195.94	avgR100	-181.59	eps	0.778	Qval	0.15	la
nd100 0.06	thr	96								
[DQN-Mod] Ep	51	R	-118.71	avgR100	-180.35	eps	0.774	Qval	0.07	la
nd100 0.06	thr	60								
[DQN-Mod] Ep	52	R	-208.88	avgR100	-180.90	eps	0.771	Qval	0.04	la
nd100 0.06	thr	73								
[DQN-Mod] Ep	53	R	-347.54	avgR100	-184.05	eps	0.767	Qval	-0.15	la
nd100 0.06	thr	115								
[DQN-Mod] Ep	54	R	-98.50	avgR100	-182.46	eps	0.763	Qval	0.47	la
nd100 0.06	thr	57								
[DQN-Mod] Ep	55	R	-151.01	avgR100	-181.89	eps	0.759	Qval	0.09	la
nd100 0.05	thr	91								
[DQN-Mod] Ep	56	R	-126.28	avgR100	-180.90	eps	0.755	Qval	-0.01	la
nd100 0.05	thr	117								
[DQN-Mod] Ep	57	R	-132.40	avgR100	-180.05	eps	0.751	Qval	0.21	la
nd100 0.05	thr	86								
[DQN-Mod] Ep	58	R	-126.18	avgR100	-179.12	eps	0.748	Qval	0.16	la
nd100 0.05	thr	94								
[DQN-Mod] Ep	59	R	-100.44	avgR100	-177.78	eps	0.744	Qval	0.02	la
nd100 0.05	thr	72								
[DQN-Mod] Ep	60	R	-173.67	avgR100	-177.72	eps	0.740	Qval	-0.22	la
nd100 0.05	thr	89								

[DQN-Mod] Ep	61	R	-166.16	avgR100	-177.53	eps	0.737	Qval	-0.02	la
nd100 0.05	thr	93								
[DQN-Mod] Ep	62	R	-192.67	avgR100	-177.77	eps	0.733	Qval	-0.03	la
nd100 0.05	thr	85								
[DQN-Mod] Ep	63	R	-104.57	avgR100	-176.61	eps	0.729	Qval	-0.02	la
nd100 0.05	thr	49								
[DQN-Mod] Ep	64	R	-13.11	avgR100	-174.05	eps	0.726	Qval	0.02	la
nd100 0.06	thr	118								
[DQN-Mod] Ep	65	R	-81.36	avgR100	-172.63	eps	0.722	Qval	0.42	la
nd100 0.06	thr	102								
[DQN-Mod] Ep	66	R	-112.86	avgR100	-171.72	eps	0.718	Qval	0.23	la
nd100 0.06	thr	74								
[DQN-Mod] Ep	67	R	-167.11	avgR100	-171.65	eps	0.715	Qval	0.37	la
nd100 0.06	thr	54								
[DQN-Mod] Ep	68	R	-166.29	avgR100	-171.57	eps	0.711	Qval	0.23	la
nd100 0.06	thr	114								
[DQN-Mod] Ep	69	R	-97.99	avgR100	-170.51	eps	0.708	Qval	-0.07	la
nd100 0.06	thr	47								
[DQN-Mod] Ep	70	R	-162.97	avgR100	-170.40	eps	0.704	Qval	0.14	la
nd100 0.06	thr	73								
[DQN-Mod] Ep	71	R	-61.30	avgR100	-168.86	eps	0.701	Qval	0.15	la
nd100 0.06	thr	98								
[DQN-Mod] Ep	72	R	-95.25	avgR100	-167.84	eps	0.697	Qval	0.08	la
nd100 0.06	thr	80								
[DQN-Mod] Ep	73	R	-125.22	avgR100	-167.26	eps	0.694	Qval	-0.18	la
nd100 0.05	thr	91								
[DQN-Mod] Ep	74	R	-114.86	avgR100	-166.55	eps	0.690	Qval	0.24	la
nd100 0.05	thr	69								
[DQN-Mod] Ep	75	R	-291.95	avgR100	-168.22	eps	0.687	Qval	-0.05	la
nd100 0.05	thr	88								
[DQN-Mod] Ep	76	R	-125.71	avgR100	-167.66	eps	0.683	Qval	0.17	la
nd100 0.05	thr	78								
[DQN-Mod] Ep	77	R	-35.22	avgR100	-165.94	eps	0.680	Qval	0.10	la
nd100 0.06	thr	72								
[DQN-Mod] Ep	78	R	-84.15	avgR100	-164.89	eps	0.676	Qval	-0.01	la
nd100 0.06	thr	95								
[DQN-Mod] Ep	79	R	-126.53	avgR100	-164.41	eps	0.673	Qval	0.29	la
nd100 0.06	thr	73								
[DQN-Mod] Ep	80	R	-117.76	avgR100	-163.82	eps	0.670	Qval	0.21	la
nd100 0.06	thr	75								
[DQN-Mod] Ep	81	R	-119.26	avgR100	-163.27	eps	0.666	Qval	0.28	la
nd100 0.06	thr	94								
[DQN-Mod] Ep	82	R	-195.52	avgR100	-163.67	eps	0.663	Qval	0.57	la
nd100 0.06	thr	86								
[DQN-Mod] Ep	83	R	-120.55	avgR100	-163.15	eps	0.660	Qval	0.20	la
nd100 0.06	thr	129								
[DQN-Mod] Ep	84	R	-188.20	avgR100	-163.45	eps	0.656	Qval	0.18	la
nd100 0.06	thr	108								
[DQN-Mod] Ep	85	R	-171.66	avgR100	-163.54	eps	0.653	Qval	0.41	la
nd100 0.06	thr	78								
[DQN-Mod] Ep	86	R	-57.02	avgR100	-162.30	eps	0.650	Qval	0.26	la
nd100 0.06	thr	79								
[DQN-Mod] Ep	87	R	-56.17	avgR100	-161.08	eps	0.647	Qval	0.80	la
nd100 0.06	thr	110								
[DQN-Mod] Ep	88	R	-161.26	avgR100	-161.09	eps	0.643	Qval	0.79	la
nd100 0.06	thr	84								
[DQN-Mod] Ep	89	R	-92.95	avgR100	-160.32	eps	0.640	Qval	0.23	la
nd100 0.06	thr	69								
[DQN-Mod] Ep	90	R	-122.08	avgR100	-159.90	eps	0.637	Qval	0.51	la
nd100 0.06	thr	65								

[DQN-Mod] Ep nd100 0.05	91 R thr 69	-14.85	avgR100	-158.30	eps 0.634	Qval	0.62	la
[DQN-Mod] Ep nd100 0.05	92 R thr 126	-118.43	avgR100	-157.87	eps 0.631	Qval	0.48	la
[DQN-Mod] Ep nd100 0.06	93 R thr 98	-59.22	avgR100	-156.81	eps 0.627	Qval	1.03	la
[DQN-Mod] Ep nd100 0.06	94 R thr 116	-110.86	avgR100	-156.32	eps 0.624	Qval	1.06	la
[DQN-Mod] Ep nd100 0.06	95 R thr 64	-141.16	avgR100	-156.16	eps 0.621	Qval	0.86	la
[DQN-Mod] Ep nd100 0.06	96 R thr 81	-136.19	avgR100	-155.95	eps 0.618	Qval	0.82	la
[DQN-Mod] Ep nd100 0.06	97 R thr 133	-103.65	avgR100	-155.41	eps 0.615	Qval	0.95	la
[DQN-Mod] Ep nd100 0.06	98 R thr 153	-16.73	avgR100	-154.00	eps 0.612	Qval	1.08	la
[DQN-Mod] Ep nd100 0.06	99 R thr 105	-68.83	avgR100	-153.14	eps 0.609	Qval	0.73	la
[DQN-Mod] Ep nd100 0.07	100 R thr 77	18.55	avgR100	-151.42	eps 0.606	Qval	0.90	la
[DQN-Mod] Ep nd100 0.07	101 R thr 65	-79.74	avgR100	-151.19	eps 0.603	Qval	1.21	la
[DQN-Mod] Ep nd100 0.07	102 R thr 121	-125.60	avgR100	-151.47	eps 0.600	Qval	0.99	la
[DQN-Mod] Ep nd100 0.07	103 R thr 107	-138.30	avgR100	-149.77	eps 0.597	Qval	1.12	la
[DQN-Mod] Ep nd100 0.07	104 R thr 81	-81.40	avgR100	-148.92	eps 0.594	Qval	1.18	la
[DQN-Mod] Ep nd100 0.07	105 R thr 105	-93.66	avgR100	-148.64	eps 0.591	Qval	1.47	la
[DQN-Mod] Ep nd100 0.07	106 R thr 56	-63.63	avgR100	-146.46	eps 0.588	Qval	1.14	la
[DQN-Mod] Ep nd100 0.07	107 R thr 78	-98.86	avgR100	-143.54	eps 0.585	Qval	1.63	la
[DQN-Mod] Ep nd100 0.06	108 R thr 61	-90.16	avgR100	-143.80	eps 0.582	Qval	1.45	la
[DQN-Mod] Ep nd100 0.06	109 R thr 143	-131.79	avgR100	-142.71	eps 0.579	Qval	1.74	la
[DQN-Mod] Ep nd100 0.07	110 R thr 68	-1.37	avgR100	-141.50	eps 0.576	Qval	1.68	la
[DQN-Mod] Ep nd100 0.07	111 R thr 100	-105.30	avgR100	-141.18	eps 0.573	Qval	1.93	la
[DQN-Mod] Ep nd100 0.07	112 R thr 140	-37.10	avgR100	-138.18	eps 0.570	Qval	1.72	la
[DQN-Mod] Ep nd100 0.07	113 R thr 177	-116.45	avgR100	-136.92	eps 0.568	Qval	1.52	la
[DQN-Mod] Ep nd100 0.07	114 R thr 114	-115.50	avgR100	-136.12	eps 0.565	Qval	1.88	la
[DQN-Mod] Ep nd100 0.07	115 R thr 162	-73.08	avgR100	-135.12	eps 0.562	Qval	2.38	la
[DQN-Mod] Ep nd100 0.07	116 R thr 128	-137.16	avgR100	-131.58	eps 0.559	Qval	2.56	la
[DQN-Mod] Ep nd100 0.07	117 R thr 74	-84.21	avgR100	-129.18	eps 0.556	Qval	2.19	la
[DQN-Mod] Ep nd100 0.07	118 R thr 73	-41.73	avgR100	-126.84	eps 0.554	Qval	2.36	la
[DQN-Mod] Ep nd100 0.07	119 R thr 70	-37.76	avgR100	-123.73	eps 0.551	Qval	2.55	la
[DQN-Mod] Ep nd100 0.06	120 R thr 117	-133.50	avgR100	-124.63	eps 0.548	Qval	2.54	la

[DQN-Mod] Ep nd100 0.06	121 R thr 114	-78.83	avgR100	-124.06	eps 0.545	Qval	2.54	1a
[DQN-Mod] Ep nd100 0.06	122 R thr 74	-94.27	avgR100	-123.85	eps 0.543	Qval	2.89	1a
[DQN-Mod] Ep nd100 0.06	123 R thr 74	-91.29	avgR100	-121.71	eps 0.540	Qval	2.59	1a
[DQN-Mod] Ep nd100 0.06	124 R thr 90	-160.88	avgR100	-121.47	eps 0.537	Qval	2.99	1a
[DQN-Mod] Ep nd100 0.06	125 R thr 112	-49.45	avgR100	-121.02	eps 0.534	Qval	2.90	1a
[DQN-Mod] Ep nd100 0.06	126 R thr 110	-133.82	avgR100	-120.37	eps 0.532	Qval	2.61	1a
[DQN-Mod] Ep nd100 0.06	127 R thr 109	-132.14	avgR100	-118.85	eps 0.529	Qval	2.68	1a
[DQN-Mod] Ep nd100 0.06	128 R thr 117	-52.03	avgR100	-117.55	eps 0.526	Qval	2.77	1a
[DQN-Mod] Ep nd100 0.06	129 R thr 135	-67.15	avgR100	-118.08	eps 0.524	Qval	3.26	1a
[DQN-Mod] Ep nd100 0.05	130 R thr 63	-91.05	avgR100	-118.79	eps 0.521	Qval	3.05	1a
[DQN-Mod] Ep nd100 0.05	131 R thr 152	-68.67	avgR100	-118.54	eps 0.519	Qval	3.05	1a
[DQN-Mod] Ep nd100 0.05	132 R thr 120	-61.64	avgR100	-116.92	eps 0.516	Qval	3.53	1a
[DQN-Mod] Ep nd100 0.05	133 R thr 152	-84.70	avgR100	-115.26	eps 0.513	Qval	3.64	1a
[DQN-Mod] Ep nd100 0.05	134 R thr 168	-72.29	avgR100	-115.36	eps 0.511	Qval	3.33	1a
[DQN-Mod] Ep nd100 0.05	135 R thr 145	-0.70	avgR100	-113.90	eps 0.508	Qval	3.55	1a
[DQN-Mod] Ep nd100 0.05	136 R thr 143	-17.32	avgR100	-113.06	eps 0.506	Qval	3.77	1a
[DQN-Mod] Ep nd100 0.05	137 R thr 122	-92.47	avgR100	-112.45	eps 0.503	Qval	3.82	1a
[DQN-Mod] Ep nd100 0.05	138 R thr 137	-146.33	avgR100	-112.89	eps 0.501	Qval	3.90	1a
[DQN-Mod] Ep nd100 0.05	139 R thr 108	-56.04	avgR100	-112.95	eps 0.498	Qval	3.74	1a
[DQN-Mod] Ep nd100 0.05	140 R thr 119	-113.32	avgR100	-112.23	eps 0.496	Qval	4.19	1a
[DQN-Mod] Ep nd100 0.05	141 R thr 121	-33.95	avgR100	-110.83	eps 0.493	Qval	4.43	1a
[DQN-Mod] Ep nd100 0.05	142 R thr 182	-86.18	avgR100	-110.33	eps 0.491	Qval	3.99	1a
[DQN-Mod] Ep nd100 0.05	143 R thr 130	-90.52	avgR100	-108.41	eps 0.488	Qval	4.48	1a
[DQN-Mod] Ep nd100 0.05	144 R thr 170	-113.47	avgR100	-106.54	eps 0.486	Qval	4.95	1a
[DQN-Mod] Ep nd100 0.05	145 R thr 108	-67.31	avgR100	-106.14	eps 0.483	Qval	5.48	1a
[DQN-Mod] Ep nd100 0.05	146 R thr 102	-63.89	avgR100	-105.34	eps 0.481	Qval	5.31	1a
[DQN-Mod] Ep nd100 0.05	147 R thr 119	-92.75	avgR100	-104.70	eps 0.479	Qval	5.24	1a
[DQN-Mod] Ep nd100 0.05	148 R thr 94	-121.33	avgR100	-105.01	eps 0.476	Qval	5.08	1a
[DQN-Mod] Ep nd100 0.05	149 R thr 168	-61.75	avgR100	-104.40	eps 0.474	Qval	5.58	1a
[DQN-Mod] Ep nd100 0.05	150 R thr 128	-93.86	avgR100	-103.38	eps 0.471	Qval	5.63	1a

[DQN-Mod] Ep 151 R	-58.85	avgR100	-102.79	eps 0.469	Qval	5.62	1a
nd100 0.05 thr 143							
[DQN-Mod] Ep 152 R	-67.50	avgR100	-101.37	eps 0.467	Qval	5.80	1a
nd100 0.05 thr 74							
[DQN-Mod] Ep 153 R	-41.09	avgR100	-98.31	eps 0.464	Qval	6.15	1a
nd100 0.05 thr 129							
[DQN-Mod] Ep 154 R	-87.76	avgR100	-98.20	eps 0.462	Qval	6.17	1a
nd100 0.05 thr 191							
[DQN-Mod] Ep 155 R	-37.63	avgR100	-97.07	eps 0.460	Qval	6.46	1a
nd100 0.05 thr 116							
[DQN-Mod] Ep 156 R	-82.53	avgR100	-96.63	eps 0.458	Qval	7.08	1a
nd100 0.05 thr 172							
[DQN-Mod] Ep 157 R	-101.68	avgR100	-96.32	eps 0.455	Qval	7.36	1a
nd100 0.05 thr 139							
[DQN-Mod] Ep 158 R	-60.81	avgR100	-95.67	eps 0.453	Qval	6.76	1a
nd100 0.05 thr 82							
[DQN-Mod] Ep 159 R	-94.27	avgR100	-95.61	eps 0.451	Qval	7.25	1a
nd100 0.05 thr 138							
[DQN-Mod] Ep 160 R	-48.56	avgR100	-94.35	eps 0.448	Qval	7.29	1a
nd100 0.05 thr 77							
[DQN-Mod] Ep 161 R	-322.07	avgR100	-95.91	eps 0.446	Qval	7.82	1a
nd100 0.05 thr 883							
[DQN-Mod] Ep 162 R	-43.42	avgR100	-94.42	eps 0.444	Qval	8.00	1a
nd100 0.05 thr 102							
[DQN-Mod] Ep 163 R	-55.68	avgR100	-93.93	eps 0.442	Qval	8.20	1a
nd100 0.05 thr 94							
[DQN-Mod] Ep 164 R	-40.83	avgR100	-94.21	eps 0.440	Qval	8.82	1a
nd100 0.04 thr 167							
[DQN-Mod] Ep 165 R	-63.18	avgR100	-94.03	eps 0.437	Qval	8.45	1a
nd100 0.04 thr 124							
[DQN-Mod] Ep 166 R	-18.76	avgR100	-93.09	eps 0.435	Qval	8.72	1a
nd100 0.04 thr 82							
[DQN-Mod] Ep 167 R	-234.44	avgR100	-93.76	eps 0.433	Qval	8.63	1a
nd100 0.04 thr 719							
[DQN-Mod] Ep 168 R	-54.84	avgR100	-92.65	eps 0.431	Qval	8.72	1a
nd100 0.04 thr 104							
[DQN-Mod] Ep 169 R	-40.69	avgR100	-92.07	eps 0.429	Qval	9.50	1a
nd100 0.04 thr 94							
[DQN-Mod] Ep 170 R	-56.81	avgR100	-91.01	eps 0.427	Qval	9.41	1a
nd100 0.04 thr 77							
[DQN-Mod] Ep 171 R	-12.82	avgR100	-90.53	eps 0.424	Qval	9.76	1a
nd100 0.04 thr 190							
[DQN-Mod] Ep 172 R	-115.60	avgR100	-90.73	eps 0.422	Qval	10.12	1a
nd100 0.04 thr 112							
[DQN-Mod] Ep 173 R	-24.78	avgR100	-89.72	eps 0.420	Qval	9.65	1a
nd100 0.04 thr 105							
[DQN-Mod] Ep 174 R	-35.11	avgR100	-88.93	eps 0.418	Qval	9.96	1a
nd100 0.04 thr 133							
[DQN-Mod] Ep 175 R	-45.93	avgR100	-86.47	eps 0.416	Qval	10.01	1a
nd100 0.04 thr 112							
[DQN-Mod] Ep 176 R	26.62	avgR100	-84.94	eps 0.414	Qval	10.02	1a
nd100 0.05 thr 85							
[DQN-Mod] Ep 177 R	-51.67	avgR100	-85.11	eps 0.412	Qval	9.99	1a
nd100 0.04 thr 116							
[DQN-Mod] Ep 178 R	-57.95	avgR100	-84.85	eps 0.410	Qval	10.52	1a
nd100 0.04 thr 86							
[DQN-Mod] Ep 179 R	-51.77	avgR100	-84.10	eps 0.408	Qval	10.05	1a
nd100 0.04 thr 155							
[DQN-Mod] Ep 180 R	-24.76	avgR100	-83.17	eps 0.406	Qval	10.80	1a
nd100 0.04 thr 121							

[DQN-Mod] Ep nd100 0.04	181 R thr 103	-62.28 avgR100	-82.60 eps 0.404 Qval	10.97 la
[DQN-Mod] Ep nd100 0.04	182 R thr 129	-40.00 avgR100	-81.04 eps 0.402 Qval	11.39 la
[DQN-Mod] Ep nd100 0.04	183 R thr 77	-54.58 avgR100	-80.38 eps 0.400 Qval	10.57 la
[DQN-Mod] Ep nd100 0.04	184 R thr 720	-79.92 avgR100	-79.30 eps 0.398 Qval	12.03 la
[DQN-Mod] Ep nd100 0.04	185 R thr 191	-101.88 avgR100	-78.60 eps 0.396 Qval	11.25 la
[DQN-Mod] Ep nd100 0.04	186 R thr 122	-57.16 avgR100	-78.61 eps 0.394 Qval	12.56 la
[DQN-Mod] Ep nd100 0.04	187 R thr 148	-24.05 avgR100	-78.28 eps 0.392 Qval	12.41 la
[DQN-Mod] Ep nd100 0.04	188 R thr 136	-69.93 avgR100	-77.37 eps 0.390 Qval	12.45 la
[DQN-Mod] Ep nd100 0.04	189 R thr 99	-74.11 avgR100	-77.18 eps 0.388 Qval	12.65 la
[DQN-Mod] Ep nd100 0.04	190 R thr 91	-33.67 avgR100	-76.30 eps 0.386 Qval	12.52 la
[DQN-Mod] Ep nd100 0.04	191 R thr 99	-67.82 avgR100	-76.83 eps 0.384 Qval	12.83 la
[DQN-Mod] Ep nd100 0.04	192 R thr 399	-116.49 avgR100	-76.81 eps 0.382 Qval	12.97 la
[DQN-Mod] Ep nd100 0.03	193 R thr 191	-52.76 avgR100	-76.74 eps 0.380 Qval	13.39 la
[DQN-Mod] Ep nd100 0.03	194 R thr 102	-70.67 avgR100	-76.34 eps 0.378 Qval	13.11 la
[DQN-Mod] Ep nd100 0.03	195 R thr 108	-98.43 avgR100	-75.91 eps 0.376 Qval	13.20 la
[DQN-Mod] Ep nd100 0.03	196 R thr 131	-79.11 avgR100	-75.34 eps 0.374 Qval	13.38 la
[DQN-Mod] Ep nd100 0.03	197 R thr 142	-36.00 avgR100	-74.67 eps 0.373 Qval	13.99 la
[DQN-Mod] Ep nd100 0.03	198 R thr 101	5.29 avgR100	-74.45 eps 0.371 Qval	13.28 la
[DQN-Mod] Ep nd100 0.03	199 R thr 161	-105.45 avgR100	-74.81 eps 0.369 Qval	13.51 la
[DQN-Mod] Ep nd100 0.02	200 R thr 92	-79.78 avgR100	-75.80 eps 0.367 Qval	13.63 la
[DQN-Mod] Ep nd100 0.02	201 R thr 119	-62.55 avgR100	-75.62 eps 0.365 Qval	13.95 la
[DQN-Mod] Ep nd100 0.02	202 R thr 146	-112.76 avgR100	-75.50 eps 0.363 Qval	13.47 la
[DQN-Mod] Ep nd100 0.02	203 R thr 125	-66.73 avgR100	-74.78 eps 0.361 Qval	13.93 la
[DQN-Mod] Ep nd100 0.02	204 R thr 89	-42.93 avgR100	-74.40 eps 0.360 Qval	14.07 la
[DQN-Mod] Ep nd100 0.02	205 R thr 118	-91.84 avgR100	-74.38 eps 0.358 Qval	13.57 la
[DQN-Mod] Ep nd100 0.02	206 R thr 132	-66.57 avgR100	-74.41 eps 0.356 Qval	13.84 la
[DQN-Mod] Ep nd100 0.02	207 R thr 181	-19.13 avgR100	-73.61 eps 0.354 Qval	14.72 la
[DQN-Mod] Ep nd100 0.02	208 R thr 175	-78.72 avgR100	-73.50 eps 0.353 Qval	14.06 la
[DQN-Mod] Ep nd100 0.02	209 R thr 137	-20.88 avgR100	-72.39 eps 0.351 Qval	14.06 la
[DQN-Mod] Ep nd100 0.01	210 R thr 75	-64.22 avgR100	-73.01 eps 0.349 Qval	14.69 la

[DQN-Mod] Ep nd100 0.01	211 R thr 154	-83.68 avgR100	-72.80 eps 0.347 Qval	14.72 la
[DQN-Mod] Ep nd100 0.01	212 R thr 114	-60.13 avgR100	-73.03 eps 0.346 Qval	14.64 la
[DQN-Mod] Ep nd100 0.01	213 R thr 83	-30.52 avgR100	-72.17 eps 0.344 Qval	14.58 la
[DQN-Mod] Ep nd100 0.01	214 R thr 631	-440.32 avgR100	-75.42 eps 0.342 Qval	15.46 la
[DQN-Mod] Ep nd100 0.01	215 R thr 94	-72.61 avgR100	-75.41 eps 0.340 Qval	14.90 la
[DQN-Mod] Ep nd100 0.01	216 R thr 160	-67.42 avgR100	-74.72 eps 0.339 Qval	15.29 la
[DQN-Mod] Ep nd100 0.01	217 R thr 71	-59.74 avgR100	-74.47 eps 0.337 Qval	15.13 la
[DQN-Mod] Ep nd100 0.01	218 R thr 591	-111.44 avgR100	-75.17 eps 0.335 Qval	15.37 la
[DQN-Mod] Ep nd100 0.02	219 R thr 218	-23.24 avgR100	-75.02 eps 0.334 Qval	15.58 la
[DQN-Mod] Ep nd100 0.02	220 R thr 692	-167.09 avgR100	-75.36 eps 0.332 Qval	15.86 la
[DQN-Mod] Ep nd100 0.02	221 R thr 95	-129.32 avgR100	-75.86 eps 0.330 Qval	15.96 la
[DQN-Mod] Ep nd100 0.02	222 R thr 56	-50.60 avgR100	-75.43 eps 0.329 Qval	15.50 la
[DQN-Mod] Ep nd100 0.02	223 R thr 134	-42.85 avgR100	-74.94 eps 0.327 Qval	15.45 la
[DQN-Mod] Ep nd100 0.02	224 R thr 123	-46.57 avgR100	-73.80 eps 0.325 Qval	15.98 la
[DQN-Mod] Ep nd100 0.03	225 R thr 64	14.83 avgR100	-73.16 eps 0.324 Qval	15.12 la
[DQN-Mod] Ep nd100 0.03	226 R thr 86	-52.80 avgR100	-72.35 eps 0.322 Qval	15.67 la
[DQN-Mod] Ep nd100 0.03	227 R thr 86	-102.99 avgR100	-72.05 eps 0.321 Qval	15.98 la
[DQN-Mod] Ep nd100 0.03	228 R thr 100	-64.09 avgR100	-72.18 eps 0.319 Qval	16.05 la
[DQN-Mod] Ep nd100 0.03	229 R thr 68	-54.37 avgR100	-72.05 eps 0.317 Qval	15.95 la
[DQN-Mod] Ep nd100 0.03	230 R thr 91	-25.36 avgR100	-71.39 eps 0.316 Qval	16.38 la
[DQN-Mod] Ep nd100 0.03	231 R thr 891	-270.40 avgR100	-73.41 eps 0.314 Qval	15.89 la
[DQN-Mod] Ep nd100 0.03	232 R thr 121	-45.53 avgR100	-73.25 eps 0.313 Qval	15.69 la
[DQN-Mod] Ep nd100 0.03	233 R thr 113	-202.05 avgR100	-74.42 eps 0.311 Qval	16.22 la
[DQN-Mod] Ep nd100 0.03	234 R thr 153	-64.70 avgR100	-74.34 eps 0.309 Qval	15.02 la
[DQN-Mod] Ep nd100 0.03	235 R thr 95	-60.65 avgR100	-74.94 eps 0.308 Qval	15.24 la
[DQN-Mod] Ep nd100 0.03	236 R thr 353	-151.51 avgR100	-76.29 eps 0.306 Qval	15.72 la
[DQN-Mod] Ep nd100 0.03	237 R thr 126	-24.17 avgR100	-75.60 eps 0.305 Qval	15.09 la
[DQN-Mod] Ep nd100 0.03	238 R thr 133	-25.39 avgR100	-74.39 eps 0.303 Qval	15.27 la
[DQN-Mod] Ep nd100 0.03	239 R thr 443	-7.06 avgR100	-73.90 eps 0.302 Qval	15.03 la
[DQN-Mod] Ep nd100 0.03	240 R thr 73	-97.61 avgR100	-73.75 eps 0.300 Qval	14.96 la

[DQN-Mod] Ep nd100 0.03	241 R thr 98	-75.98 avgR100	-74.17 eps 0.299 Qval	15.61 la
[DQN-Mod] Ep nd100 0.03	242 R thr 107	-62.51 avgR100	-73.93 eps 0.297 Qval	15.05 la
[DQN-Mod] Ep nd100 0.03	243 R thr 755	-210.26 avgR100	-75.13 eps 0.296 Qval	14.53 la
[DQN-Mod] Ep nd100 0.03	244 R thr 67	-77.62 avgR100	-74.77 eps 0.294 Qval	14.33 la
[DQN-Mod] Ep nd100 0.03	245 R thr 87	-42.55 avgR100	-74.52 eps 0.293 Qval	14.57 la
[DQN-Mod] Ep nd100 0.03	246 R thr 123	-63.42 avgR100	-74.52 eps 0.291 Qval	14.26 la
[DQN-Mod] Ep nd100 0.03	247 R thr 55	-42.46 avgR100	-74.01 eps 0.290 Qval	14.16 la
[DQN-Mod] Ep nd100 0.03	248 R thr 135	-47.74 avgR100	-73.28 eps 0.288 Qval	14.21 la
[DQN-Mod] Ep nd100 0.03	249 R thr 96	-74.54 avgR100	-73.41 eps 0.287 Qval	14.18 la
[DQN-Mod] Ep nd100 0.03	250 R thr 64	-50.82 avgR100	-72.98 eps 0.286 Qval	14.00 la
[DQN-Mod] Ep nd100 0.03	251 R thr 81	-47.37 avgR100	-72.86 eps 0.284 Qval	13.90 la
[DQN-Mod] Ep nd100 0.03	252 R thr 74	-59.62 avgR100	-72.78 eps 0.283 Qval	14.38 la
[DQN-Mod] Ep nd100 0.03	253 R thr 612	-526.79 avgR100	-77.64 eps 0.281 Qval	13.90 la
[DQN-Mod] Ep nd100 0.03	254 R thr 64	-53.04 avgR100	-77.29 eps 0.280 Qval	13.54 la
[DQN-Mod] Ep nd100 0.03	255 R thr 115	-49.06 avgR100	-77.41 eps 0.279 Qval	13.69 la
[DQN-Mod] Ep nd100 0.03	256 R thr 86	7.21 avgR100	-76.51 eps 0.277 Qval	13.55 la
[DQN-Mod] Ep nd100 0.03	257 R thr 61	-83.66 avgR100	-76.33 eps 0.276 Qval	13.79 la
[DQN-Mod] Ep nd100 0.03	258 R thr 149	-22.61 avgR100	-75.95 eps 0.274 Qval	13.41 la
[DQN-Mod] Ep nd100 0.03	259 R thr 95	-18.77 avgR100	-75.19 eps 0.273 Qval	13.58 la
[DQN-Mod] Ep nd100 0.03	260 R thr 78	-54.99 avgR100	-75.26 eps 0.272 Qval	12.90 la
[DQN-Mod] Ep nd100 0.03	261 R thr 91	-25.59 avgR100	-72.29 eps 0.270 Qval	13.20 la
[DQN-Mod] Ep nd100 0.04	262 R thr 210	219.37 avgR100	-69.66 eps 0.269 Qval	12.09 la
[DQN-Mod] Ep nd100 0.04	263 R thr 843	-223.87 avgR100	-71.35 eps 0.268 Qval	12.52 la
[DQN-Mod] Ep nd100 0.04	264 R thr 110	-37.15 avgR100	-71.31 eps 0.266 Qval	12.41 la
[DQN-Mod] Ep nd100 0.04	265 R thr 408	-153.34 avgR100	-72.21 eps 0.265 Qval	12.12 la
[DQN-Mod] Ep nd100 0.04	266 R thr 99	-3.90 avgR100	-72.06 eps 0.264 Qval	11.54 la
[DQN-Mod] Ep nd100 0.04	267 R thr 97	-27.07 avgR100	-69.99 eps 0.262 Qval	11.61 la
[DQN-Mod] Ep nd100 0.04	268 R thr 76	-34.79 avgR100	-69.79 eps 0.261 Qval	11.46 la
[DQN-Mod] Ep nd100 0.04	269 R thr 863	-319.80 avgR100	-72.58 eps 0.260 Qval	10.94 la
[DQN-Mod] Ep nd100 0.04	270 R thr 88	-118.83 avgR100	-73.20 eps 0.258 Qval	10.76 la

[DQN-Mod] Ep nd100 0.04	271 R thr 219	-121.21	avgR100	-74.28	eps 0.257	Qval	9.97	1a
[DQN-Mod] Ep nd100 0.04	272 R thr 79	-13.55	avgR100	-73.26	eps 0.256	Qval	10.71	1a
[DQN-Mod] Ep nd100 0.04	273 R thr 156	-80.49	avgR100	-73.82	eps 0.255	Qval	9.80	1a
[DQN-Mod] Ep nd100 0.04	274 R thr 890	-355.08	avgR100	-77.02	eps 0.253	Qval	8.67	1a
[DQN-Mod] Ep nd100 0.04	275 R thr 88	-12.69	avgR100	-76.69	eps 0.252	Qval	9.03	1a
[DQN-Mod] Ep nd100 0.03	276 R thr 285	-94.88	avgR100	-77.90	eps 0.251	Qval	8.34	1a
[DQN-Mod] Ep nd100 0.03	277 R thr 115	-24.10	avgR100	-77.63	eps 0.249	Qval	8.08	1a
[DQN-Mod] Ep nd100 0.03	278 R thr 132	-52.60	avgR100	-77.57	eps 0.248	Qval	8.24	1a
[DQN-Mod] Ep nd100 0.03	279 R thr 245	-70.89	avgR100	-77.76	eps 0.247	Qval	7.71	1a
[DQN-Mod] Ep nd100 0.03	280 R thr 835	-232.14	avgR100	-79.84	eps 0.246	Qval	7.18	1a
[DQN-Mod] Ep nd100 0.03	281 R thr 92	-71.47	avgR100	-79.93	eps 0.245	Qval	6.42	1a
[DQN-Mod] Ep nd100 0.03	282 R thr 91	-29.58	avgR100	-79.82	eps 0.243	Qval	7.16	1a
[DQN-Mod] Ep nd100 0.03	283 R thr 97	-46.21	avgR100	-79.74	eps 0.242	Qval	6.71	1a
[DQN-Mod] Ep nd100 0.03	284 R thr 74	-63.33	avgR100	-79.58	eps 0.241	Qval	6.66	1a
[DQN-Mod] Ep nd100 0.03	285 R thr 870	-294.09	avgR100	-81.50	eps 0.240	Qval	5.22	1a
[DQN-Mod] Ep nd100 0.03	286 R thr 357	-130.94	avgR100	-82.24	eps 0.238	Qval	4.34	1a
[DQN-Mod] Ep nd100 0.03	287 R thr 415	-152.18	avgR100	-83.52	eps 0.237	Qval	4.42	1a
[DQN-Mod] Ep nd100 0.03	288 R thr 78	-15.43	avgR100	-82.97	eps 0.236	Qval	3.81	1a
[DQN-Mod] Ep nd100 0.03	289 R thr 818	-259.40	avgR100	-84.82	eps 0.235	Qval	2.35	1a
[DQN-Mod] Ep nd100 0.03	290 R thr 81	-80.62	avgR100	-85.29	eps 0.234	Qval	2.38	1a
[DQN-Mod] Ep nd100 0.03	291 R thr 91	-26.86	avgR100	-84.88	eps 0.233	Qval	2.17	1a
[DQN-Mod] Ep nd100 0.03	292 R thr 127	-58.82	avgR100	-84.31	eps 0.231	Qval	1.98	1a
[DQN-Mod] Ep nd100 0.03	293 R thr 103	-64.10	avgR100	-84.42	eps 0.230	Qval	1.53	1a
[DQN-Mod] Ep nd100 0.03	294 R thr 121	-22.51	avgR100	-83.94	eps 0.229	Qval	1.41	1a
[DQN-Mod] Ep nd100 0.03	295 R thr 176	-94.92	avgR100	-83.90	eps 0.228	Qval	1.59	1a
[DQN-Mod] Ep nd100 0.03	296 R thr 57	-69.29	avgR100	-83.81	eps 0.227	Qval	1.64	1a
[DQN-Mod] Ep nd100 0.03	297 R thr 87	-26.97	avgR100	-83.72	eps 0.226	Qval	1.34	1a
[DQN-Mod] Ep nd100 0.04	298 R thr 526	105.44	avgR100	-82.71	eps 0.225	Qval	0.11	1a
[DQN-Mod] Ep nd100 0.04	299 R thr 54	-66.85	avgR100	-82.33	eps 0.223	Qval	0.23	1a
[DQN-Mod] Ep nd100 0.04	300 R thr 835	-233.67	avgR100	-83.87	eps 0.222	Qval	-1.11	1a

[DQN-Mod] Ep nd100 0.04	301 R thr 128	-194.98 avgR100	-85.19 eps 0.221 Qval	-1.13 la
[DQN-Mod] Ep nd100 0.05	302 R thr 404	132.68 avgR100	-82.74 eps 0.220 Qval	-2.25 la
[DQN-Mod] Ep nd100 0.05	303 R thr 86	-24.46 avgR100	-82.31 eps 0.219 Qval	-2.63 la
[DQN-Mod] Ep nd100 0.05	304 R thr 809	-223.37 avgR100	-84.12 eps 0.218 Qval	-4.19 la
[DQN-Mod] Ep nd100 0.05	305 R thr 887	-281.72 avgR100	-86.02 eps 0.217 Qval	-4.97 la
[DQN-Mod] Ep nd100 0.05	306 R thr 74	-60.83 avgR100	-85.96 eps 0.216 Qval	-6.16 la
[DQN-Mod] Ep nd100 0.05	307 R thr 141	-43.77 avgR100	-86.21 eps 0.215 Qval	-5.69 la
[DQN-Mod] Ep nd100 0.05	308 R thr 710	-163.06 avgR100	-87.05 eps 0.214 Qval	-6.74 la
[DQN-Mod] Ep nd100 0.05	309 R thr 806	-239.87 avgR100	-89.24 eps 0.212 Qval	-8.63 la
[DQN-Mod] Ep nd100 0.05	310 R thr 120	-21.22 avgR100	-88.81 eps 0.211 Qval	-9.04 la
[DQN-Mod] Ep nd100 0.05	311 R thr 744	-242.35 avgR100	-90.40 eps 0.210 Qval	-9.63 la
[DQN-Mod] Ep nd100 0.05	312 R thr 116	-6.79 avgR100	-89.86 eps 0.209 Qval	-10.13 la
[DQN-Mod] Ep nd100 0.05	313 R thr 776	-232.85 avgR100	-91.89 eps 0.208 Qval	-11.46 la
[DQN-Mod] Ep nd100 0.05	314 R thr 106	-19.28 avgR100	-87.68 eps 0.207 Qval	-11.49 la
[DQN-Mod] Ep nd100 0.05	315 R thr 820	-265.99 avgR100	-89.61 eps 0.206 Qval	-13.33 la
[DQN-Mod] Ep nd100 0.05	316 R thr 798	-240.26 avgR100	-91.34 eps 0.205 Qval	-13.98 la
[DQN-Mod] Ep nd100 0.05	317 R thr 401	-155.35 avgR100	-92.29 eps 0.204 Qval	-15.19 la
[DQN-Mod] Ep nd100 0.05	318 R thr 685	-169.27 avgR100	-92.87 eps 0.203 Qval	-16.36 la
[DQN-Mod] Ep nd100 0.04	319 R thr 847	-294.87 avgR100	-95.59 eps 0.202 Qval	-17.71 la
[DQN-Mod] Ep nd100 0.04	320 R thr 216	171.93 avgR100	-92.20 eps 0.201 Qval	-17.89 la
[DQN-Mod] Ep nd100 0.05	321 R thr 701	46.79 avgR100	-90.44 eps 0.200 Qval	-18.88 la
[DQN-Mod] Ep nd100 0.05	322 R thr 828	-238.44 avgR100	-92.32 eps 0.199 Qval	-19.62 la
[DQN-Mod] Ep nd100 0.05	323 R thr 111	-73.58 avgR100	-92.62 eps 0.198 Qval	-20.54 la
[DQN-Mod] Ep nd100 0.06	324 R thr 499	129.04 avgR100	-90.87 eps 0.197 Qval	-20.68 la
[DQN-Mod] Ep nd100 0.05	325 R thr 141	-53.33 avgR100	-91.55 eps 0.196 Qval	-20.43 la
[DQN-Mod] Ep nd100 0.05	326 R thr 671	-131.68 avgR100	-92.34 eps 0.195 Qval	-23.07 la
[DQN-Mod] Ep nd100 0.05	327 R thr 853	-262.37 avgR100	-93.93 eps 0.194 Qval	-23.66 la
[DQN-Mod] Ep nd100 0.05	328 R thr 841	-231.42 avgR100	-95.60 eps 0.193 Qval	-24.07 la
[DQN-Mod] Ep nd100 0.05	329 R thr 821	-234.60 avgR100	-97.41 eps 0.192 Qval	-26.83 la
[DQN-Mod] Ep nd100 0.05	330 R thr 730	-184.59 avgR100	-99.00 eps 0.191 Qval	-27.23 la

[DQN-Mod] Ep nd100 0.05	331 R thr 814	-297.96	avgR100	-99.27	eps 0.190	Qval -29.24	la
[DQN-Mod] Ep nd100 0.06	332 R thr 297	202.38	avgR100	-96.80	eps 0.189	Qval -28.82	la
[DQN-Mod] Ep nd100 0.06	333 R thr 639	-275.98	avgR100	-97.54	eps 0.188	Qval -29.72	la
[DQN-Mod] Ep nd100 0.06	334 R thr 852	-280.66	avgR100	-99.69	eps 0.187	Qval -30.97	la
[DQN-Mod] Ep nd100 0.06	335 R thr 746	-222.57	avgR100	-101.31	eps 0.187	Qval -31.44	la
[DQN-Mod] Ep nd100 0.07	336 R thr 609	49.86	avgR100	-99.30	eps 0.186	Qval -34.77	la
[DQN-Mod] Ep nd100 0.07	337 R thr 593	-81.49	avgR100	-99.87	eps 0.185	Qval -35.00	la
[DQN-Mod] Ep nd100 0.08	338 R thr 584	96.39	avgR100	-98.66	eps 0.184	Qval -35.79	la
[DQN-Mod] Ep nd100 0.08	339 R thr 745	-195.33	avgR100	-100.54	eps 0.183	Qval -36.77	la
[DQN-Mod] Ep nd100 0.08	340 R thr 821	-257.50	avgR100	-102.14	eps 0.182	Qval -37.63	la
[DQN-Mod] Ep nd100 0.08	341 R thr 859	-253.42	avgR100	-103.91	eps 0.181	Qval -38.26	la
[DQN-Mod] Ep nd100 0.08	342 R thr 121	-70.50	avgR100	-103.99	eps 0.180	Qval -39.06	la
[DQN-Mod] Ep nd100 0.08	343 R thr 853	-267.91	avgR100	-104.57	eps 0.179	Qval -38.52	la
[DQN-Mod] Ep nd100 0.09	344 R thr 713	-16.25	avgR100	-103.95	eps 0.178	Qval -40.75	la
[DQN-Mod] Ep nd100 0.10	345 R thr 217	253.38	avgR100	-101.00	eps 0.177	Qval -40.36	la
[DQN-Mod] Ep nd100 0.10	346 R thr 770	-214.35	avgR100	-102.50	eps 0.177	Qval -42.65	la
[DQN-Mod] Ep nd100 0.10	347 R thr 813	-240.29	avgR100	-104.48	eps 0.176	Qval -43.61	la
[DQN-Mod] Ep nd100 0.10	348 R thr 787	-256.41	avgR100	-106.57	eps 0.175	Qval -44.16	la
[DQN-Mod] Ep nd100 0.10	349 R thr 93	-87.63	avgR100	-106.70	eps 0.174	Qval -44.70	la
[DQN-Mod] Ep nd100 0.10	350 R thr 119	-57.07	avgR100	-106.76	eps 0.173	Qval -45.13	la
[DQN-Mod] Ep nd100 0.11	351 R thr 276	274.36	avgR100	-103.55	eps 0.172	Qval -44.22	la
[DQN-Mod] Ep nd100 0.11	352 R thr 821	-299.60	avgR100	-105.95	eps 0.171	Qval -43.72	la
[DQN-Mod] Ep nd100 0.11	353 R thr 778	-230.53	avgR100	-102.98	eps 0.170	Qval -45.78	la
[DQN-Mod] Ep nd100 0.11	354 R thr 124	-24.33	avgR100	-102.70	eps 0.170	Qval -45.14	la
[DQN-Mod] Ep nd100 0.11	355 R thr 180	-41.76	avgR100	-102.62	eps 0.169	Qval -45.01	la
[DQN-Mod] Ep nd100 0.11	356 R thr 800	-281.59	avgR100	-105.51	eps 0.168	Qval -44.92	la
[DQN-Mod] Ep nd100 0.11	357 R thr 771	-197.66	avgR100	-106.65	eps 0.167	Qval -46.57	la
[DQN-Mod] Ep nd100 0.11	358 R thr 100	-80.35	avgR100	-107.23	eps 0.166	Qval -46.29	la
[DQN-Mod] Ep nd100 0.11	359 R thr 824	-245.65	avgR100	-109.50	eps 0.165	Qval -45.60	la
[DQN-Mod] Ep nd100 0.11	360 R thr 107	-74.83	avgR100	-109.70	eps 0.165	Qval -46.62	la

[DQN-Mod] Ep nd100 0.11	361 R thr 782	-147.59	avgR100	-110.92	eps 0.164	Qval -46.99	la
[DQN-Mod] Ep nd100 0.11	362 R thr 419	147.39	avgR100	-111.63	eps 0.163	Qval -47.44	la
[DQN-Mod] Ep nd100 0.12	363 R thr 114	72.93	avgR100	-108.67	eps 0.162	Qval -47.14	la
[DQN-Mod] Ep nd100 0.12	364 R thr 748	-205.96	avgR100	-110.36	eps 0.161	Qval -48.78	la
[DQN-Mod] Ep nd100 0.12	365 R thr 809	-167.89	avgR100	-110.50	eps 0.160	Qval -49.00	la
[DQN-Mod] Ep nd100 0.12	366 R thr 797	-240.46	avgR100	-112.87	eps 0.160	Qval -50.23	la
[DQN-Mod] Ep nd100 0.13	367 R thr 665	-28.97	avgR100	-112.89	eps 0.159	Qval -48.27	la
[DQN-Mod] Ep nd100 0.13	368 R thr 165	-32.70	avgR100	-112.86	eps 0.158	Qval -48.89	la
[DQN-Mod] Ep nd100 0.14	369 R thr 663	30.31	avgR100	-109.36	eps 0.157	Qval -49.88	la
[DQN-Mod] Ep nd100 0.15	370 R thr 558	69.03	avgR100	-107.48	eps 0.157	Qval -49.67	la
[DQN-Mod] Ep nd100 0.15	371 R thr 764	-197.33	avgR100	-108.25	eps 0.156	Qval -49.94	la
[DQN-Mod] Ep nd100 0.16	372 R thr 479	88.44	avgR100	-107.23	eps 0.155	Qval -50.81	la
[DQN-Mod] Ep nd100 0.17	373 R thr 471	123.92	avgR100	-105.18	eps 0.154	Qval -51.15	la
[DQN-Mod] Ep nd100 0.17	374 R thr 691	-173.29	avgR100	-103.36	eps 0.153	Qval -52.11	la
[DQN-Mod] Ep nd100 0.18	375 R thr 601	35.68	avgR100	-102.88	eps 0.153	Qval -49.50	la
[DQN-Mod] Ep nd100 0.19	376 R thr 384	177.68	avgR100	-100.15	eps 0.152	Qval -50.18	la
[DQN-Mod] Ep nd100 0.19	377 R thr 773	-225.37	avgR100	-102.17	eps 0.151	Qval -49.47	la
[DQN-Mod] Ep nd100 0.19	378 R thr 786	-232.98	avgR100	-103.97	eps 0.150	Qval -50.63	la
[DQN-Mod] Ep nd100 0.19	379 R thr 774	-229.06	avgR100	-105.55	eps 0.150	Qval -50.35	la
[DQN-Mod] Ep nd100 0.20	380 R thr 533	107.35	avgR100	-102.16	eps 0.149	Qval -50.79	la
[DQN-Mod] Ep nd100 0.20	381 R thr 129	-36.26	avgR100	-101.81	eps 0.148	Qval -51.47	la
[DQN-Mod] Ep nd100 0.21	382 R thr 397	156.81	avgR100	-99.94	eps 0.147	Qval -50.53	la
[DQN-Mod] Ep nd100 0.21	383 R thr 734	-215.56	avgR100	-101.64	eps 0.147	Qval -49.66	la
[DQN-Mod] Ep nd100 0.22	384 R thr 460	99.46	avgR100	-100.01	eps 0.146	Qval -49.83	la
[DQN-Mod] Ep nd100 0.23	385 R thr 203	215.79	avgR100	-94.91	eps 0.145	Qval -50.13	la
[DQN-Mod] Ep nd100 0.23	386 R thr 817	-262.21	avgR100	-96.22	eps 0.144	Qval -51.66	la
[DQN-Mod] Ep nd100 0.24	387 R thr 269	225.65	avgR100	-92.44	eps 0.144	Qval -51.09	la
[DQN-Mod] Ep nd100 0.24	388 R thr 115	-8.33	avgR100	-92.37	eps 0.143	Qval -50.91	la
[DQN-Mod] Ep nd100 0.25	389 R thr 479	128.51	avgR100	-88.49	eps 0.142	Qval -51.04	la
[DQN-Mod] Ep nd100 0.25	390 R thr 189	-81.34	avgR100	-88.50	eps 0.142	Qval -50.53	la

[DQN-Mod] Ep nd100 0.25	391 R thr 99	-94.20	avgR100	-89.17	eps 0.141	Qval -51.66	la
[DQN-Mod] Ep nd100 0.25	392 R thr 832	-239.08	avgR100	-90.98	eps 0.140	Qval -50.86	la
[DQN-Mod] Ep nd100 0.25	393 R thr 667	-154.16	avgR100	-91.88	eps 0.139	Qval -51.93	la
[DQN-Mod] Ep nd100 0.25	394 R thr 106	0.54	avgR100	-91.65	eps 0.139	Qval -52.32	la
[DQN-Mod] Ep nd100 0.25	395 R thr 217	-33.89	avgR100	-91.04	eps 0.138	Qval -51.40	la
[DQN-Mod] Ep nd100 0.25	396 R thr 860	-235.01	avgR100	-92.69	eps 0.137	Qval -52.77	la
[DQN-Mod] Ep nd100 0.25	397 R thr 739	-240.48	avgR100	-94.83	eps 0.137	Qval -52.76	la
[DQN-Mod] Ep nd100 0.24	398 R thr 794	-198.16	avgR100	-97.86	eps 0.136	Qval -50.86	la
[DQN-Mod] Ep nd100 0.24	399 R thr 808	-200.76	avgR100	-99.20	eps 0.135	Qval -52.39	la
[DQN-Mod] Ep nd100 0.24	400 R thr 83	-94.09	avgR100	-97.81	eps 0.135	Qval -52.95	la
[DQN-Mod] Ep nd100 0.24	401 R thr 811	-247.67	avgR100	-98.33	eps 0.134	Qval -51.84	la
[DQN-Mod] Ep nd100 0.23	402 R thr 791	-254.54	avgR100	-102.21	eps 0.133	Qval -52.94	la
[DQN-Mod] Ep nd100 0.23	403 R thr 837	-260.43	avgR100	-104.57	eps 0.133	Qval -54.61	la
[DQN-Mod] Ep nd100 0.23	404 R thr 804	-253.52	avgR100	-104.87	eps 0.132	Qval -57.59	la
[DQN-Mod] Ep nd100 0.23	405 R thr 394	-113.52	avgR100	-103.19	eps 0.131	Qval -56.42	la
[DQN-Mod] Ep nd100 0.23	406 R thr 531	19.68	avgR100	-102.38	eps 0.131	Qval -57.36	la
[DQN-Mod] Ep nd100 0.23	407 R thr 168	-38.60	avgR100	-102.33	eps 0.130	Qval -57.27	la
[DQN-Mod] Ep nd100 0.23	408 R thr 122	-64.93	avgR100	-101.35	eps 0.129	Qval -56.57	la
[DQN-Mod] Ep nd100 0.23	409 R thr 801	-249.98	avgR100	-101.45	eps 0.129	Qval -55.79	la
[DQN-Mod] Ep nd100 0.23	410 R thr 815	-288.44	avgR100	-104.12	eps 0.128	Qval -58.73	la
[DQN-Mod] Ep nd100 0.23	411 R thr 309	-385.54	avgR100	-105.55	eps 0.127	Qval -58.95	la
[DQN-Mod] Ep nd100 0.23	412 R thr 568	-264.42	avgR100	-108.13	eps 0.127	Qval -58.25	la
[DQN-Mod] Ep nd100 0.23	413 R thr 815	-300.19	avgR100	-108.80	eps 0.126	Qval -58.15	la
[DQN-Mod] Ep nd100 0.23	414 R thr 819	-277.33	avgR100	-111.38	eps 0.126	Qval -58.66	la
[DQN-Mod] Ep nd100 0.23	415 R thr 784	-290.74	avgR100	-111.63	eps 0.125	Qval -60.45	la
[DQN-Mod] Ep nd100 0.24	416 R thr 514	114.91	avgR100	-108.08	eps 0.124	Qval -59.69	la
[DQN-Mod] Ep nd100 0.24	417 R thr 737	-178.54	avgR100	-108.31	eps 0.124	Qval -60.63	la
[DQN-Mod] Ep nd100 0.24	418 R thr 128	-44.45	avgR100	-107.06	eps 0.123	Qval -61.01	la
[DQN-Mod] Ep nd100 0.24	419 R thr 113	-71.05	avgR100	-104.82	eps 0.122	Qval -61.92	la
[DQN-Mod] Ep nd100 0.24	420 R thr 763	-249.76	avgR100	-109.04	eps 0.122	Qval -62.81	la

[DQN-Mod] Ep nd100 0.24	421 R thr 669	7.62	avgR100	-109.43	eps 0.121	Qval -59.92	la
[DQN-Mod] Ep nd100 0.24	422 R thr 790	-243.72	avgR100	-109.49	eps 0.121	Qval -61.10	la
[DQN-Mod] Ep nd100 0.24	423 R thr 786	-258.11	avgR100	-111.33	eps 0.120	Qval -62.82	la
[DQN-Mod] Ep nd100 0.23	424 R thr 111	-36.43	avgR100	-112.99	eps 0.119	Qval -62.52	la
[DQN-Mod] Ep nd100 0.23	425 R thr 777	-284.81	avgR100	-115.30	eps 0.119	Qval -63.95	la
[DQN-Mod] Ep nd100 0.24	426 R thr 456	122.93	avgR100	-112.75	eps 0.118	Qval -63.86	la
[DQN-Mod] Ep nd100 0.24	427 R thr 793	-260.57	avgR100	-112.74	eps 0.118	Qval -63.89	la
[DQN-Mod] Ep nd100 0.24	428 R thr 769	-287.37	avgR100	-113.30	eps 0.117	Qval -66.76	la
[DQN-Mod] Ep nd100 0.24	429 R thr 795	-227.61	avgR100	-113.23	eps 0.116	Qval -65.77	la
[DQN-Mod] Ep nd100 0.24	430 R thr 797	-260.88	avgR100	-113.99	eps 0.116	Qval -68.07	la
[DQN-Mod] Ep nd100 0.24	431 R thr 781	-287.04	avgR100	-113.88	eps 0.115	Qval -68.88	la
[DQN-Mod] Ep nd100 0.24	432 R thr 617	81.38	avgR100	-115.09	eps 0.115	Qval -70.50	la
[DQN-Mod] Ep nd100 0.25	433 R thr 152	276.15	avgR100	-109.57	eps 0.114	Qval -69.65	la
[DQN-Mod] Ep nd100 0.25	434 R thr 822	-300.39	avgR100	-109.77	eps 0.114	Qval -72.49	la
[DQN-Mod] Ep nd100 0.25	435 R thr 134	-22.31	avgR100	-107.76	eps 0.113	Qval -72.34	la
[DQN-Mod] Ep nd100 0.24	436 R thr 857	-300.29	avgR100	-111.26	eps 0.112	Qval -73.62	la
[DQN-Mod] Ep nd100 0.24	437 R thr 795	-274.03	avgR100	-113.19	eps 0.112	Qval -72.33	la
[DQN-Mod] Ep nd100 0.24	438 R thr 607	9.67	avgR100	-114.06	eps 0.111	Qval -73.11	la
[DQN-Mod] Ep nd100 0.24	439 R thr 75	-3.08	avgR100	-112.14	eps 0.111	Qval -73.49	la
[DQN-Mod] Ep nd100 0.24	440 R thr 812	-233.72	avgR100	-111.90	eps 0.110	Qval -74.15	la
[DQN-Mod] Ep nd100 0.25	441 R thr 569	72.59	avgR100	-108.64	eps 0.110	Qval -74.02	la
[DQN-Mod] Ep nd100 0.25	442 R thr 115	-19.27	avgR100	-108.12	eps 0.109	Qval -74.04	la
[DQN-Mod] Ep nd100 0.25	443 R thr 643	-128.17	avgR100	-106.73	eps 0.109	Qval -74.30	la
[DQN-Mod] Ep nd100 0.24	444 R thr 803	-256.92	avgR100	-109.13	eps 0.108	Qval -75.16	la
[DQN-Mod] Ep nd100 0.23	445 R thr 175	11.35	avgR100	-111.55	eps 0.107	Qval -73.33	la
[DQN-Mod] Ep nd100 0.23	446 R thr 826	-240.29	avgR100	-111.81	eps 0.107	Qval -74.41	la
[DQN-Mod] Ep nd100 0.23	447 R thr 826	-271.97	avgR100	-112.13	eps 0.106	Qval -77.04	la
[DQN-Mod] Ep nd100 0.23	448 R thr 103	-11.80	avgR100	-109.68	eps 0.106	Qval -75.19	la
[DQN-Mod] Ep nd100 0.23	449 R thr 785	-251.17	avgR100	-111.32	eps 0.105	Qval -76.23	la
[DQN-Mod] Ep nd100 0.23	450 R thr 761	-163.69	avgR100	-112.39	eps 0.105	Qval -76.01	la

[DQN-Mod] Ep nd100 0.22	451 R thr 830	-241.85	avgR100	-117.55	eps 0.104	Qval -75.79	la
[DQN-Mod] Ep nd100 0.22	452 R thr 804	-223.24	avgR100	-116.78	eps 0.104	Qval -75.36	la
[DQN-Mod] Ep nd100 0.22	453 R thr 787	-190.08	avgR100	-116.38	eps 0.103	Qval -78.23	la
[DQN-Mod] Ep nd100 0.22	454 R thr 796	-273.02	avgR100	-118.87	eps 0.103	Qval -78.32	la
[DQN-Mod] Ep nd100 0.22	455 R thr 799	-259.78	avgR100	-121.05	eps 0.102	Qval -77.84	la
[DQN-Mod] Ep nd100 0.22	456 R thr 190	-37.15	avgR100	-118.60	eps 0.102	Qval -77.74	la
[DQN-Mod] Ep nd100 0.22	457 R thr 789	-211.25	avgR100	-118.74	eps 0.101	Qval -78.71	la
[DQN-Mod] Ep nd100 0.22	458 R thr 768	-246.47	avgR100	-120.40	eps 0.101	Qval -77.67	la
[DQN-Mod] Ep nd100 0.23	459 R thr 346	202.24	avgR100	-115.92	eps 0.100	Qval -76.78	la
[DQN-Mod] Ep nd100 0.23	460 R thr 811	-240.97	avgR100	-117.58	eps 0.100	Qval -77.17	la
[DQN-Mod] Ep nd100 0.24	461 R thr 273	228.28	avgR100	-113.82	eps 0.099	Qval -78.87	la
[DQN-Mod] Ep nd100 0.23	462 R thr 803	-235.86	avgR100	-117.66	eps 0.099	Qval -78.75	la
[DQN-Mod] Ep nd100 0.22	463 R thr 131	-29.67	avgR100	-118.68	eps 0.098	Qval -77.51	la
[DQN-Mod] Ep nd100 0.22	464 R thr 786	-219.47	avgR100	-118.82	eps 0.098	Qval -76.00	la
[DQN-Mod] Ep nd100 0.22	465 R thr 754	-212.94	avgR100	-119.27	eps 0.097	Qval -78.22	la
[DQN-Mod] Ep nd100 0.23	466 R thr 611	74.51	avgR100	-116.12	eps 0.097	Qval -75.76	la
[DQN-Mod] Ep nd100 0.23	467 R thr 552	64.06	avgR100	-115.19	eps 0.096	Qval -75.65	la
[DQN-Mod] Ep nd100 0.23	468 R thr 792	-216.22	avgR100	-117.02	eps 0.096	Qval -76.15	la
[DQN-Mod] Ep nd100 0.22	469 R thr 138	-84.95	avgR100	-118.18	eps 0.095	Qval -76.19	la
[DQN-Mod] Ep nd100 0.21	470 R thr 798	-239.05	avgR100	-121.26	eps 0.095	Qval -75.68	la
[DQN-Mod] Ep nd100 0.21	471 R thr 754	-209.88	avgR100	-121.38	eps 0.094	Qval -76.82	la
[DQN-Mod] Ep nd100 0.20	472 R thr 803	-264.33	avgR100	-124.91	eps 0.094	Qval -77.24	la
[DQN-Mod] Ep nd100 0.19	473 R thr 770	-251.34	avgR100	-128.66	eps 0.093	Qval -76.46	la
[DQN-Mod] Ep nd100 0.20	474 R thr 552	27.17	avgR100	-126.66	eps 0.093	Qval -77.54	la
[DQN-Mod] Ep nd100 0.20	475 R thr 643	68.35	avgR100	-126.33	eps 0.092	Qval -76.19	la
[DQN-Mod] Ep nd100 0.19	476 R thr 758	-170.99	avgR100	-129.82	eps 0.092	Qval -76.55	la
[DQN-Mod] Ep nd100 0.20	477 R thr 293	217.01	avgR100	-125.39	eps 0.092	Qval -75.04	la
[DQN-Mod] Ep nd100 0.21	478 R thr 449	161.55	avgR100	-121.45	eps 0.091	Qval -77.18	la
[DQN-Mod] Ep nd100 0.21	479 R thr 114	-15.88	avgR100	-119.32	eps 0.091	Qval -77.89	la
[DQN-Mod] Ep nd100 0.21	480 R thr 598	62.43	avgR100	-119.77	eps 0.090	Qval -77.66	la

[DQN-Mod] Ep nd100 0.22	481 R thr 642	12.52	avgR100	-119.28	eps 0.090	Qval -77.08	la
[DQN-Mod] Ep nd100 0.22	482 R thr 261	223.22	avgR100	-118.61	eps 0.089	Qval -77.34	la
[DQN-Mod] Ep nd100 0.23	483 R thr 406	167.70	avgR100	-114.78	eps 0.089	Qval -76.12	la
[DQN-Mod] Ep nd100 0.23	484 R thr 273	228.86	avgR100	-113.49	eps 0.088	Qval -76.74	la
[DQN-Mod] Ep nd100 0.22	485 R thr 773	-223.09	avgR100	-117.88	eps 0.088	Qval -74.37	la
[DQN-Mod] Ep nd100 0.22	486 R thr 763	-224.10	avgR100	-117.50	eps 0.088	Qval -73.98	la
[DQN-Mod] Ep nd100 0.22	487 R thr 447	121.94	avgR100	-118.53	eps 0.087	Qval -76.08	la
[DQN-Mod] Ep nd100 0.23	488 R thr 287	187.93	avgR100	-116.57	eps 0.087	Qval -75.35	la
[DQN-Mod] Ep nd100 0.23	489 R thr 656	41.34	avgR100	-117.44	eps 0.086	Qval -75.26	la
[DQN-Mod] Ep nd100 0.24	490 R thr 448	95.33	avgR100	-115.67	eps 0.086	Qval -73.46	la
[DQN-Mod] Ep nd100 0.25	491 R thr 379	162.84	avgR100	-113.10	eps 0.085	Qval -76.02	la
[DQN-Mod] Ep nd100 0.26	492 R thr 489	101.02	avgR100	-109.70	eps 0.085	Qval -74.33	la
[DQN-Mod] Ep nd100 0.26	493 R thr 95	-53.42	avgR100	-108.70	eps 0.084	Qval -75.21	la
[DQN-Mod] Ep nd100 0.27	494 R thr 516	76.81	avgR100	-107.93	eps 0.084	Qval -74.27	la
[DQN-Mod] Ep nd100 0.28	495 R thr 644	55.58	avgR100	-107.04	eps 0.084	Qval -72.89	la
[DQN-Mod] Ep nd100 0.28	496 R thr 791	-272.48	avgR100	-107.41	eps 0.083	Qval -72.58	la
[DQN-Mod] Ep nd100 0.29	497 R thr 505	71.36	avgR100	-104.29	eps 0.083	Qval -72.13	la
[DQN-Mod] Ep nd100 0.30	498 R thr 583	68.22	avgR100	-101.63	eps 0.082	Qval -71.13	la
[DQN-Mod] Ep nd100 0.30	499 R thr 789	-227.23	avgR100	-101.90	eps 0.082	Qval -73.22	la
[DQN-Mod] Ep nd100 0.31	500 R thr 337	168.61	avgR100	-99.27	eps 0.082	Qval -72.73	la
[DQN-Mod] Ep nd100 0.32	501 R thr 202	213.19	avgR100	-94.66	eps 0.081	Qval -71.93	la
[DQN-Mod] Ep nd100 0.32	502 R thr 768	-218.04	avgR100	-94.29	eps 0.081	Qval -71.46	la
[DQN-Mod] Ep nd100 0.33	503 R thr 407	179.87	avgR100	-89.89	eps 0.080	Qval -70.19	la
[DQN-Mod] Ep nd100 0.34	504 R thr 447	123.36	avgR100	-86.12	eps 0.080	Qval -68.62	la
[DQN-Mod] Ep nd100 0.35	505 R thr 647	38.06	avgR100	-84.61	eps 0.080	Qval -67.03	la
[DQN-Mod] Ep nd100 0.36	506 R thr 299	201.99	avgR100	-82.78	eps 0.079	Qval -65.95	la
[DQN-Mod] Ep nd100 0.37	507 R thr 309	185.89	avgR100	-80.54	eps 0.079	Qval -66.07	la
[DQN-Mod] Ep nd100 0.38	508 R thr 346	132.06	avgR100	-78.57	eps 0.078	Qval -65.00	la
[DQN-Mod] Ep nd100 0.38	509 R thr 355	-286.86	avgR100	-78.94	eps 0.078	Qval -64.83	la
[DQN-Mod] Ep nd100 0.39	510 R thr 396	142.33	avgR100	-74.63	eps 0.078	Qval -64.23	la

[DQN-Mod] Ep 511 R	77.16	avgR100	-70.00	eps 0.077	Qval	-62.77	la
nd100 0.40 thr 545							
[DQN-Mod] Ep 512 R	99.37	avgR100	-66.37	eps 0.077	Qval	-60.93	la
nd100 0.41 thr 548							
[DQN-Mod] Ep 513 R	222.71	avgR100	-61.14	eps 0.076	Qval	-59.80	la
nd100 0.42 thr 306							
[DQN-Mod] Ep 514 R	209.67	avgR100	-56.27	eps 0.076	Qval	-59.19	la
nd100 0.43 thr 336							
[DQN-Mod] Ep 515 R	162.45	avgR100	-51.73	eps 0.076	Qval	-58.17	la
nd100 0.44 thr 386							
[DQN-Mod] Ep 516 R	101.64	avgR100	-51.87	eps 0.075	Qval	-57.07	la
nd100 0.44 thr 446							
[DQN-Mod] Ep 517 R	208.34	avgR100	-48.00	eps 0.075	Qval	-57.29	la
nd100 0.45 thr 340							
[DQN-Mod] Ep 518 R	163.60	avgR100	-45.92	eps 0.075	Qval	-57.02	la
nd100 0.46 thr 341							
[DQN-Mod] Ep 519 R	239.92	avgR100	-42.81	eps 0.074	Qval	-57.37	la
nd100 0.47 thr 234							
[DQN-Mod] Ep 520 R	86.33	avgR100	-39.45	eps 0.074	Qval	-56.61	la
nd100 0.48 thr 456							
[DQN-Mod] Ep 521 R	234.60	avgR100	-37.18	eps 0.073	Qval	-56.60	la
nd100 0.48 thr 286							
[DQN-Mod] Ep 522 R	247.15	avgR100	-32.27	eps 0.073	Qval	-55.71	la
nd100 0.49 thr 201							
[DQN-Mod] Ep 523 R	208.80	avgR100	-27.60	eps 0.073	Qval	-54.91	la
nd100 0.50 thr 278							
[DQN-Mod] Ep 524 R	30.21	avgR100	-26.93	eps 0.072	Qval	-53.97	la
nd100 0.51 thr 599							
[DQN-Mod] Ep 525 R	158.48	avgR100	-22.50	eps 0.072	Qval	-52.90	la
nd100 0.52 thr 421							
[DQN-Mod] Ep 526 R	223.29	avgR100	-21.50	eps 0.072	Qval	-51.88	la
nd100 0.52 thr 262							
[DQN-Mod] Ep 527 R	236.91	avgR100	-16.52	eps 0.071	Qval	-52.54	la
nd100 0.53 thr 241							
[DQN-Mod] Ep 528 R	151.67	avgR100	-12.13	eps 0.071	Qval	-51.62	la
nd100 0.54 thr 342							
[DQN-Mod] Ep 529 R	185.15	avgR100	-8.00	eps 0.071	Qval	-48.88	la
nd100 0.55 thr 309							
[DQN-Mod] Ep 530 R	150.21	avgR100	-3.89	eps 0.070	Qval	-50.61	la
nd100 0.56 thr 440							
[DQN-Mod] Ep 531 R	154.61	avgR100	0.52	eps 0.070	Qval	-48.97	la
nd100 0.57 thr 390							
[DQN-Mod] Ep 532 R	197.55	avgR100	1.69	eps 0.069	Qval	-47.43	la
nd100 0.57 thr 262							
[DQN-Mod] Ep 533 R	207.36	avgR100	1.00	eps 0.069	Qval	-48.84	la
nd100 0.57 thr 331							
[DQN-Mod] Ep 534 R	247.89	avgR100	6.48	eps 0.069	Qval	-47.96	la
nd100 0.58 thr 177							
[DQN-Mod] Ep 535 R	171.58	avgR100	8.42	eps 0.068	Qval	-46.57	la
nd100 0.59 thr 371							
[DQN-Mod] Ep 536 R	137.23	avgR100	12.79	eps 0.068	Qval	-47.08	la
nd100 0.60 thr 422							
[DQN-Mod] Ep 537 R	187.11	avgR100	17.41	eps 0.068	Qval	-46.36	la
nd100 0.61 thr 328							
[DQN-Mod] Ep 538 R	116.58	avgR100	18.47	eps 0.067	Qval	-46.42	la
nd100 0.61 thr 465							
[DQN-Mod] Ep 539 R	140.56	avgR100	19.91	eps 0.067	Qval	-45.91	la
nd100 0.62 thr 359							
[DQN-Mod] Ep 540 R	208.67	avgR100	24.34	eps 0.067	Qval	-44.40	la
nd100 0.63 thr 214							

[DQN-Mod] Ep 541 R 186.67 avgR100 25.48 eps 0.066 Qval -44.41 la
nd100 0.63 thr 235
[DQN-Mod] Ep 542 R 134.22 avgR100 27.01 eps 0.066 Qval -43.33 la
nd100 0.64 thr 386
[DQN-Mod] Ep 543 R 54.15 avgR100 28.83 eps 0.066 Qval -44.43 la
nd100 0.64 thr 563
[DQN-Mod] Ep 544 R 276.71 avgR100 34.17 eps 0.065 Qval -43.15 la
nd100 0.65 thr 200
[DQN-Mod] Ep 545 R 139.86 avgR100 35.46 eps 0.065 Qval -42.58 la
nd100 0.66 thr 340
[DQN-Mod] Ep 546 R 239.85 avgR100 40.26 eps 0.065 Qval -41.70 la
nd100 0.67 thr 249
[DQN-Mod] Ep 547 R 171.64 avgR100 44.69 eps 0.064 Qval -41.11 la
nd100 0.68 thr 403
[DQN-Mod] Ep 548 R 225.21 avgR100 47.06 eps 0.064 Qval -41.89 la
nd100 0.69 thr 294
[DQN-Mod] Ep 549 R 233.90 avgR100 51.91 eps 0.064 Qval -39.91 la
nd100 0.70 thr 235
[DQN-Mod] Ep 550 R 228.41 avgR100 55.83 eps 0.063 Qval -39.06 la
nd100 0.71 thr 293
[DQN-Mod] Ep 551 R 233.90 avgR100 60.59 eps 0.063 Qval -37.50 la
nd100 0.72 thr 190
[DQN-Mod] Ep 552 R -47.69 avgR100 62.35 eps 0.063 Qval -39.62 la
nd100 0.72 thr 288
[DQN-Mod] Ep 553 R 240.95 avgR100 66.66 eps 0.063 Qval -38.98 la
nd100 0.73 thr 251
[DQN-Mod] Ep 554 R 237.51 avgR100 71.76 eps 0.062 Qval -38.82 la
nd100 0.74 thr 177
[DQN-Mod] Ep 555 R 150.25 avgR100 75.86 eps 0.062 Qval -38.00 la
nd100 0.75 thr 365
[DQN-Mod] Ep 556 R 148.90 avgR100 77.72 eps 0.062 Qval -36.74 la
nd100 0.76 thr 369
[DQN-Mod] Ep 557 R 198.28 avgR100 81.82 eps 0.061 Qval -37.10 la
nd100 0.77 thr 247
[DQN-Mod] Ep 558 R 186.64 avgR100 86.15 eps 0.061 Qval -35.74 la
nd100 0.78 thr 248
[DQN-Mod] Ep 559 R 203.20 avgR100 86.16 eps 0.061 Qval -36.48 la
nd100 0.78 thr 301
[DQN-Mod] Ep 560 R 105.14 avgR100 89.62 eps 0.060 Qval -35.57 la
nd100 0.79 thr 532
[DQN-Mod] Ep 561 R 198.60 avgR100 89.32 eps 0.060 Qval -34.09 la
nd100 0.79 thr 324
[DQN-Mod] Ep 562 R 13.66 avgR100 91.82 eps 0.060 Qval -35.46 la
nd100 0.79 thr 104
[DQN-Mod] Ep 563 R 258.53 avgR100 94.70 eps 0.059 Qval -32.18 la
nd100 0.80 thr 181
[DQN-Mod] Ep 564 R 213.34 avgR100 99.03 eps 0.059 Qval -31.96 la
nd100 0.81 thr 333
[DQN-Mod] Ep 565 R 103.63 avgR100 102.20 eps 0.059 Qval -30.79 la
nd100 0.82 thr 514
[DQN-Mod] Ep 566 R 220.32 avgR100 103.65 eps 0.059 Qval -30.88 la
nd100 0.82 thr 320
[DQN-Mod] Ep 567 R 207.15 avgR100 105.08 eps 0.058 Qval -29.84 la
nd100 0.82 thr 327
[DQN-Mod] Ep 568 R -49.17 avgR100 106.75 eps 0.058 Qval -30.09 la
nd100 0.82 thr 108
[DQN-Mod] Ep 569 R 287.90 avgR100 110.48 eps 0.058 Qval -30.31 la
nd100 0.83 thr 148
[DQN-Mod] Ep 570 R 168.82 avgR100 114.56 eps 0.057 Qval -29.10 la
nd100 0.84 thr 425

[DQN-Mod] Ep 571 R 183.05 avgR100 118.49 eps 0.057 Qval -28.18 la
nd100 0.85 thr 347
[DQN-Mod] Ep 572 R 204.81 avgR100 123.18 eps 0.057 Qval -26.65 la
nd100 0.85 thr 256
[DQN-Mod] Ep 573 R 219.13 avgR100 127.89 eps 0.057 Qval -27.46 la
nd100 0.86 thr 323
[DQN-Mod] Ep 574 R 266.94 avgR100 130.28 eps 0.056 Qval -26.45 la
nd100 0.86 thr 210
[DQN-Mod] Ep 575 R 135.26 avgR100 130.95 eps 0.056 Qval -25.26 la
nd100 0.86 thr 382
[DQN-Mod] Ep 576 R 225.47 avgR100 134.92 eps 0.056 Qval -23.79 la
nd100 0.87 thr 177
[DQN-Mod] Ep 577 R 70.37 avgR100 133.45 eps 0.055 Qval -23.65 la
nd100 0.87 thr 585
[DQN-Mod] Ep 578 R 52.42 avgR100 132.36 eps 0.055 Qval -22.79 la
nd100 0.87 thr 625
[DQN-Mod] Ep 579 R 232.88 avgR100 134.85 eps 0.055 Qval -22.59 la
nd100 0.88 thr 272
[DQN-Mod] Ep 580 R 207.75 avgR100 136.30 eps 0.055 Qval -22.32 la
nd100 0.88 thr 361
[DQN-Mod] Ep 581 R 289.91 avgR100 139.08 eps 0.054 Qval -22.43 la
nd100 0.88 thr 126
[DQN-Mod] Ep 582 R -41.78 avgR100 136.43 eps 0.054 Qval -20.30 la
nd100 0.87 thr 151
[DQN-Mod] Ep 583 R 217.84 avgR100 136.93 eps 0.054 Qval -22.38 la
nd100 0.87 thr 228
[DQN-Mod] Ep 584 R -37.44 avgR100 134.26 eps 0.054 Qval -21.90 la
nd100 0.86 thr 119
[DQN-Mod] Ep 585 R 174.32 avgR100 138.24 eps 0.053 Qval -21.52 la
nd100 0.86 thr 211
[DQN-Mod] Ep 586 R -12.02 avgR100 140.36 eps 0.053 Qval -21.93 la
nd100 0.86 thr 132
[DQN-Mod] Ep 587 R 207.33 avgR100 141.21 eps 0.053 Qval -21.98 la
nd100 0.86 thr 200
[DQN-Mod] Ep 588 R 221.91 avgR100 141.55 eps 0.052 Qval -20.86 la
nd100 0.85 thr 201
[DQN-Mod] Ep 589 R 214.51 avgR100 143.28 eps 0.052 Qval -21.53 la
nd100 0.85 thr 233
[DQN-Mod] Ep 590 R 132.67 avgR100 143.66 eps 0.052 Qval -20.92 la
nd100 0.85 thr 396
[DQN-Mod] Ep 591 R 262.23 avgR100 144.65 eps 0.052 Qval -20.09 la
nd100 0.85 thr 233
[DQN-Mod] Ep 592 R 222.00 avgR100 145.86 eps 0.051 Qval -21.75 la
nd100 0.85 thr 185
[DQN-Mod] Ep 593 R 245.39 avgR100 148.85 eps 0.051 Qval -19.98 la
nd100 0.86 thr 249
[DQN-Mod] Ep 594 R 73.92 avgR100 148.82 eps 0.051 Qval -20.32 la
nd100 0.85 thr 370
[DQN-Mod] Ep 595 R 209.99 avgR100 150.36 eps 0.051 Qval -19.38 la
nd100 0.85 thr 270
[DQN-Mod] Ep 596 R 282.06 avgR100 155.91 eps 0.050 Qval -19.23 la
nd100 0.86 thr 150
[DQN-Mod] Ep 597 R -17.52 avgR100 155.02 eps 0.050 Qval -18.35 la
nd100 0.85 thr 157
[DQN-Mod] Ep 598 R 263.46 avgR100 156.97 eps 0.050 Qval -18.11 la
nd100 0.85 thr 187
[DQN-Mod] Ep 599 R -92.72 avgR100 158.32 eps 0.050 Qval -18.11 la
nd100 0.85 thr 646
[DQN-Mod] Ep 600 R 179.13 avgR100 158.42 eps 0.049 Qval -16.55 la
nd100 0.85 thr 315

[DQN-Mod] Ep 601 R	175.77	avgR100	158.05	eps 0.049	Qval -16.61	la
nd100 0.85 thr 289						
[DQN-Mod] Ep 602 R	126.16	avgR100	161.49	eps 0.049	Qval -15.05	la
nd100 0.86 thr 451						
[DQN-Mod] Ep 603 R	204.57	avgR100	161.74	eps 0.049	Qval -15.91	la
nd100 0.85 thr 215						
[DQN-Mod] Ep 604 R	244.94	avgR100	162.95	eps 0.048	Qval -15.07	la
nd100 0.85 thr 240						
[DQN-Mod] Ep 605 R	232.40	avgR100	164.90	eps 0.048	Qval -16.04	la
nd100 0.85 thr 193						
[DQN-Mod] Ep 606 R	6.97	avgR100	162.95	eps 0.048	Qval -14.94	la
nd100 0.84 thr 148						
[DQN-Mod] Ep 607 R	224.99	avgR100	163.34	eps 0.048	Qval -16.12	la
nd100 0.84 thr 260						
[DQN-Mod] Ep 608 R	286.40	avgR100	164.88	eps 0.047	Qval -15.64	la
nd100 0.84 thr 161						
[DQN-Mod] Ep 609 R	-6.61	avgR100	167.68	eps 0.047	Qval -15.08	la
nd100 0.84 thr 126						
[DQN-Mod] Ep 610 R	240.09	avgR100	168.66	eps 0.047	Qval -14.72	la
nd100 0.84 thr 210						
[DQN-Mod] Ep 611 R	267.55	avgR100	170.57	eps 0.047	Qval -13.52	la
nd100 0.84 thr 195						
[DQN-Mod] Ep 612 R	149.46	avgR100	171.07	eps 0.047	Qval -14.43	la
nd100 0.84 thr 331						
[DQN-Mod] Ep 613 R	211.79	avgR100	170.96	eps 0.046	Qval -12.41	la
nd100 0.84 thr 283						
[DQN-Mod] Ep 614 R	-177.11	avgR100	167.09	eps 0.046	Qval -13.35	la
nd100 0.83 thr 830						
[DQN-Mod] Ep 615 R	199.91	avgR100	167.46	eps 0.046	Qval -12.39	la
nd100 0.83 thr 329						
[DQN-Mod] Ep 616 R	298.61	avgR100	169.43	eps 0.046	Qval -14.14	la
nd100 0.83 thr 195						
[DQN-Mod] Ep 617 R	199.34	avgR100	169.34	eps 0.045	Qval -12.33	la
nd100 0.83 thr 261						
[DQN-Mod] Ep 618 R	240.14	avgR100	170.11	eps 0.045	Qval -12.98	la
nd100 0.83 thr 184						
[DQN-Mod] Ep 619 R	215.89	avgR100	169.87	eps 0.045	Qval -11.64	la
nd100 0.83 thr 275						
[DQN-Mod] Ep 620 R	212.98	avgR100	171.14	eps 0.045	Qval -12.40	la
nd100 0.83 thr 255						
[DQN-Mod] Ep 621 R	202.44	avgR100	170.81	eps 0.044	Qval -12.94	la
nd100 0.83 thr 317						
[DQN-Mod] Ep 622 R	251.62	avgR100	170.86	eps 0.044	Qval -12.49	la
nd100 0.83 thr 216						
[DQN-Mod] Ep 623 R	243.91	avgR100	171.21	eps 0.044	Qval -11.57	la
nd100 0.83 thr 240						
[DQN-Mod] Ep 624 R	217.12	avgR100	173.08	eps 0.044	Qval -12.23	la
nd100 0.83 thr 191						
[DQN-Mod] Ep 625 R	167.30	avgR100	173.17	eps 0.044	Qval -11.18	la
nd100 0.83 thr 339						
[DQN-Mod] Ep 626 R	251.36	avgR100	173.45	eps 0.043	Qval -10.99	la
nd100 0.83 thr 233						
[DQN-Mod] Ep 627 R	216.26	avgR100	173.24	eps 0.043	Qval -12.13	la
nd100 0.83 thr 293						
[DQN-Mod] Ep 628 R	73.27	avgR100	172.46	eps 0.043	Qval -10.78	la
nd100 0.83 thr 569						
[DQN-Mod] Ep 629 R	243.12	avgR100	173.04	eps 0.043	Qval -11.09	la
nd100 0.83 thr 185						
[DQN-Mod] Ep 630 R	50.46	avgR100	172.04	eps 0.043	Qval -10.10	la
nd100 0.83 thr 591						

[DQN-Mod] Ep 631 R 212.73 avgR100 172.62 eps 0.042 Qval -9.11 la
nd100 0.83 thr 195
[DQN-Mod] Ep 632 R 213.77 avgR100 172.78 eps 0.042 Qval -9.17 la
nd100 0.83 thr 257
[DQN-Mod] Ep 633 R 282.12 avgR100 173.53 eps 0.042 Qval -9.24 la
nd100 0.83 thr 176
[DQN-Mod] Ep 634 R 223.35 avgR100 173.29 eps 0.042 Qval -8.51 la
nd100 0.83 thr 241
[DQN-Mod] Ep 635 R 241.90 avgR100 173.99 eps 0.041 Qval -9.38 la
nd100 0.83 thr 281
[DQN-Mod] Ep 636 R 230.24 avgR100 174.92 eps 0.041 Qval -8.89 la
nd100 0.83 thr 262
[DQN-Mod] Ep 637 R 131.66 avgR100 174.36 eps 0.041 Qval -7.93 la
nd100 0.83 thr 353
[DQN-Mod] Ep 638 R 117.28 avgR100 174.37 eps 0.041 Qval -8.84 la
nd100 0.83 thr 503
[DQN-Mod] Ep 639 R 238.74 avgR100 175.35 eps 0.041 Qval -9.44 la
nd100 0.83 thr 227
[DQN-Mod] Ep 640 R 254.06 avgR100 175.81 eps 0.040 Qval -8.83 la
nd100 0.83 thr 218
[DQN-Mod] Ep 641 R 181.57 avgR100 175.76 eps 0.040 Qval -8.24 la
nd100 0.83 thr 252
[DQN-Mod] Ep 642 R 240.16 avgR100 176.81 eps 0.040 Qval -8.70 la
nd100 0.83 thr 290
[DQN-Mod] Ep 643 R 235.10 avgR100 178.62 eps 0.040 Qval -8.65 la
nd100 0.84 thr 286
[DQN-Mod] Ep 644 R 244.78 avgR100 178.31 eps 0.040 Qval -7.17 la
nd100 0.84 thr 195
[DQN-Mod] Ep 645 R 252.40 avgR100 179.43 eps 0.039 Qval -9.03 la
nd100 0.84 thr 161
[DQN-Mod] Ep 646 R 227.33 avgR100 179.31 eps 0.039 Qval -8.52 la
nd100 0.84 thr 175
[DQN-Mod] Ep 647 R 0.72 avgR100 177.60 eps 0.039 Qval -8.26 la
nd100 0.83 thr 135
[DQN-Mod] Ep 648 R 183.44 avgR100 177.18 eps 0.039 Qval -8.05 la
nd100 0.83 thr 294
[DQN-Mod] Ep 649 R -173.59 avgR100 173.10 eps 0.039 Qval -8.69 la
nd100 0.82 thr 915
[DQN-Mod] Ep 650 R 172.98 avgR100 172.55 eps 0.038 Qval -8.90 la
nd100 0.81 thr 243
[DQN-Mod] Ep 651 R -1.50 avgR100 170.20 eps 0.038 Qval -9.98 la
nd100 0.81 thr 723
[DQN-Mod] Ep 652 R 144.39 avgR100 172.12 eps 0.038 Qval -8.28 la
nd100 0.82 thr 432
[DQN-Mod] Ep 653 R 225.46 avgR100 171.96 eps 0.038 Qval -8.86 la
nd100 0.82 thr 229
[DQN-Mod] Ep 654 R 245.70 avgR100 172.04 eps 0.038 Qval -8.77 la
nd100 0.82 thr 241
[DQN-Mod] Ep 655 R 267.89 avgR100 173.22 eps 0.038 Qval -9.96 la
nd100 0.82 thr 152
[DQN-Mod] Ep 656 R 288.53 avgR100 174.62 eps 0.037 Qval -8.99 la
nd100 0.82 thr 213
[DQN-Mod] Ep 657 R 247.37 avgR100 175.11 eps 0.037 Qval -9.18 la
nd100 0.82 thr 241
[DQN-Mod] Ep 658 R 188.94 avgR100 175.13 eps 0.037 Qval -9.01 la
nd100 0.82 thr 318
[DQN-Mod] Ep 659 R -162.82 avgR100 171.47 eps 0.037 Qval -6.93 la
nd100 0.81 thr 746
[DQN-Mod] Ep 660 R 308.61 avgR100 173.50 eps 0.037 Qval -7.27 la
nd100 0.81 thr 134

[DQN-Mod] Ep 661 R 255.61 avgR100 174.07 eps 0.036 Qval -7.15 la
nd100 0.81 thr 191
[DQN-Mod] Ep 662 R 200.75 avgR100 175.95 eps 0.036 Qval -6.05 la
nd100 0.82 thr 230
[DQN-Mod] Ep 663 R 214.07 avgR100 175.50 eps 0.036 Qval -6.90 la
nd100 0.82 thr 214
[DQN-Mod] Ep 664 R 116.74 avgR100 174.53 eps 0.036 Qval -7.91 la
nd100 0.82 thr 400
[DQN-Mod] Ep 665 R -240.20 avgR100 171.10 eps 0.036 Qval -5.97 la
nd100 0.81 thr 752
[DQN-Mod] Ep 666 R 213.92 avgR100 171.03 eps 0.035 Qval -5.78 la
nd100 0.81 thr 280
[DQN-Mod] Ep 667 R 178.33 avgR100 170.74 eps 0.035 Qval -5.46 la
nd100 0.81 thr 389
[DQN-Mod] Ep 668 R 126.32 avgR100 172.50 eps 0.035 Qval -4.84 la
nd100 0.82 thr 495
[DQN-Mod] Ep 669 R 200.54 avgR100 171.63 eps 0.035 Qval -5.48 la
nd100 0.82 thr 309
[DQN-Mod] Ep 670 R 68.91 avgR100 170.63 eps 0.035 Qval -4.82 la
nd100 0.82 thr 563
[DQN-Mod] Ep 671 R 178.36 avgR100 170.58 eps 0.035 Qval -4.61 la
nd100 0.82 thr 311
[DQN-Mod] Ep 672 R 159.38 avgR100 170.13 eps 0.034 Qval -4.19 la
nd100 0.83 thr 367
[DQN-Mod] Ep 673 R 168.90 avgR100 169.62 eps 0.034 Qval -5.03 la
nd100 0.83 thr 369
[DQN-Mod] Ep 674 R 276.03 avgR100 169.71 eps 0.034 Qval -4.31 la
nd100 0.83 thr 170
[DQN-Mod] Ep 675 R 251.21 avgR100 170.87 eps 0.034 Qval -4.68 la
nd100 0.83 thr 205
[DQN-Mod] Ep 676 R 245.56 avgR100 171.07 eps 0.034 Qval -4.63 la
nd100 0.83 thr 196
[DQN-Mod] Ep 677 R 139.35 avgR100 171.76 eps 0.034 Qval -4.03 la
nd100 0.83 thr 473
[DQN-Mod] Ep 678 R 232.77 avgR100 173.57 eps 0.033 Qval -4.05 la
nd100 0.82 thr 199
[DQN-Mod] Ep 679 R 163.04 avgR100 172.87 eps 0.033 Qval -4.57 la
nd100 0.82 thr 322
[DQN-Mod] Ep 680 R 223.49 avgR100 173.03 eps 0.033 Qval -3.24 la
nd100 0.82 thr 239
[DQN-Mod] Ep 681 R 151.55 avgR100 171.64 eps 0.033 Qval -3.76 la
nd100 0.82 thr 381
[DQN-Mod] Ep 682 R 204.84 avgR100 174.11 eps 0.033 Qval -4.17 la
nd100 0.83 thr 244
[DQN-Mod] Ep 683 R 135.58 avgR100 173.29 eps 0.033 Qval -4.13 la
nd100 0.83 thr 455
[DQN-Mod] Ep 684 R 229.85 avgR100 175.96 eps 0.032 Qval -4.19 la
nd100 0.84 thr 246
[DQN-Mod] Ep 685 R 1.21 avgR100 174.23 eps 0.032 Qval -3.55 la
nd100 0.84 thr 686
[DQN-Mod] Ep 686 R -41.23 avgR100 173.94 eps 0.032 Qval -4.66 la
nd100 0.84 thr 154
[DQN-Mod] Ep 687 R -108.15 avgR100 170.78 eps 0.032 Qval -4.53 la
nd100 0.83 thr 580
[DQN-Mod] Ep 688 R 196.28 avgR100 170.53 eps 0.032 Qval -3.80 la
nd100 0.84 thr 338
[DQN-Mod] Ep 689 R 250.52 avgR100 170.89 eps 0.032 Qval -3.28 la
nd100 0.84 thr 237
[DQN-Mod] Ep 690 R 239.02 avgR100 171.95 eps 0.031 Qval -3.88 la
nd100 0.84 thr 192

[DQN-Mod] Ep 691 R 253.35 avgR100 171.86 eps 0.031 Qval -5.28 la
nd100 0.84 thr 133
[DQN-Mod] Ep 692 R 227.88 avgR100 171.92 eps 0.031 Qval -4.13 la
nd100 0.84 thr 181
[DQN-Mod] Ep 693 R 260.25 avgR100 172.07 eps 0.031 Qval -4.54 la
nd100 0.84 thr 177
[DQN-Mod] Ep 694 R 219.85 avgR100 173.53 eps 0.031 Qval -3.53 la
nd100 0.85 thr 199
[DQN-Mod] Ep 695 R 188.00 avgR100 173.31 eps 0.031 Qval -5.75 la
nd100 0.84 thr 130
[DQN-Mod] Ep 696 R 101.54 avgR100 171.50 eps 0.031 Qval -5.49 la
nd100 0.84 thr 514
[DQN-Mod] Ep 697 R -42.84 avgR100 171.25 eps 0.030 Qval -4.86 la
nd100 0.84 thr 145
[DQN-Mod] Ep 698 R 266.43 avgR100 171.28 eps 0.030 Qval -3.21 la
nd100 0.84 thr 254
[DQN-Mod] Ep 699 R 9.18 avgR100 172.30 eps 0.030 Qval -3.64 la
nd100 0.84 thr 490
[DQN-Mod] Ep 700 R 263.75 avgR100 173.14 eps 0.030 Qval -2.91 la
nd100 0.84 thr 171
[DQN-Mod] Ep 701 R 170.33 avgR100 173.09 eps 0.030 Qval -3.67 la
nd100 0.84 thr 293
[DQN-Mod] Ep 702 R 23.74 avgR100 172.06 eps 0.030 Qval -5.03 la
nd100 0.84 thr 672
[DQN-Mod] Ep 703 R 226.54 avgR100 172.28 eps 0.029 Qval -4.09 la
nd100 0.85 thr 319
[DQN-Mod] Ep 704 R 227.98 avgR100 172.12 eps 0.029 Qval -3.98 la
nd100 0.85 thr 255
[DQN-Mod] Ep 705 R 220.35 avgR100 171.99 eps 0.029 Qval -3.55 la
nd100 0.84 thr 171
[DQN-Mod] Ep 706 R 70.93 avgR100 172.63 eps 0.029 Qval -3.55 la
nd100 0.84 thr 543
[DQN-Mod] Ep 707 R 7.00 avgR100 170.45 eps 0.029 Qval -3.44 la
nd100 0.83 thr 150
[DQN-Mod] Ep 708 R 267.46 avgR100 170.26 eps 0.029 Qval -2.82 la
nd100 0.83 thr 164
[DQN-Mod] Ep 709 R 253.69 avgR100 172.87 eps 0.029 Qval -3.31 la
nd100 0.84 thr 159
[DQN-Mod] Ep 710 R 244.60 avgR100 172.91 eps 0.028 Qval -4.38 la
nd100 0.84 thr 267
[DQN-Mod] Ep 711 R 161.35 avgR100 171.85 eps 0.028 Qval -4.08 la
nd100 0.84 thr 388
[DQN-Mod] Ep 712 R 101.27 avgR100 171.37 eps 0.028 Qval -4.44 la
nd100 0.84 thr 136
[DQN-Mod] Ep 713 R 202.97 avgR100 171.28 eps 0.028 Qval -4.56 la
nd100 0.84 thr 248
[DQN-Mod] Ep 714 R 235.30 avgR100 175.41 eps 0.028 Qval -4.15 la
nd100 0.85 thr 258
[DQN-Mod] Ep 715 R 176.05 avgR100 175.17 eps 0.028 Qval -4.21 la
nd100 0.85 thr 355
[DQN-Mod] Ep 716 R 276.77 avgR100 174.95 eps 0.028 Qval -4.04 la
nd100 0.85 thr 191
[DQN-Mod] Ep 717 R 231.79 avgR100 175.27 eps 0.027 Qval -4.01 la
nd100 0.85 thr 237
[DQN-Mod] Ep 718 R 254.94 avgR100 175.42 eps 0.027 Qval -4.53 la
nd100 0.85 thr 216
[DQN-Mod] Ep 719 R 231.57 avgR100 175.58 eps 0.027 Qval -4.53 la
nd100 0.85 thr 198
[DQN-Mod] Ep 720 R 218.32 avgR100 175.63 eps 0.027 Qval -5.34 la
nd100 0.85 thr 252

[DQN-Mod] Ep nd100 0.85	721 R thr 355	172.93	avgR100	175.34	eps 0.027	Qval	-4.15	la
[DQN-Mod] Ep nd100 0.85	722 R thr 242	232.47	avgR100	175.14	eps 0.027	Qval	-3.35	la
[DQN-Mod] Ep nd100 0.85	723 R thr 523	92.30	avgR100	173.63	eps 0.027	Qval	-4.78	la
[DQN-Mod] Ep nd100 0.85	724 R thr 273	209.22	avgR100	173.55	eps 0.027	Qval	-3.97	la
[DQN-Mod] Ep nd100 0.85	725 R thr 172	249.28	avgR100	174.37	eps 0.026	Qval	-2.90	la
[DQN-Mod] Ep nd100 0.84	726 R thr 279	185.30	avgR100	173.71	eps 0.026	Qval	-3.54	la
[DQN-Mod] Ep nd100 0.84	727 R thr 104	286.75	avgR100	174.41	eps 0.026	Qval	-3.92	la
[DQN-Mod] Ep nd100 0.84	728 R thr 117	252.53	avgR100	176.21	eps 0.026	Qval	-3.66	la
[DQN-Mod] Ep nd100 0.84	729 R thr 287	193.81	avgR100	175.71	eps 0.026	Qval	-2.55	la
[DQN-Mod] Ep nd100 0.84	730 R thr 189	246.38	avgR100	177.67	eps 0.026	Qval	-3.96	la
[DQN-Mod] Ep nd100 0.84	731 R thr 195	245.81	avgR100	178.00	eps 0.026	Qval	-4.05	la
[DQN-Mod] Ep nd100 0.83	732 R thr 568	-70.70	avgR100	175.16	eps 0.025	Qval	-3.53	la
[DQN-Mod] Ep nd100 0.83	733 R thr 193	258.30	avgR100	174.92	eps 0.025	Qval	-4.17	la
[DQN-Mod] Ep nd100 0.83	734 R thr 294	170.72	avgR100	174.39	eps 0.025	Qval	-2.53	la
[DQN-Mod] Ep nd100 0.83	735 R thr 453	84.60	avgR100	172.82	eps 0.025	Qval	-4.14	la
[DQN-Mod] Ep nd100 0.83	736 R thr 227	213.16	avgR100	172.65	eps 0.025	Qval	-4.23	la
[DQN-Mod] Ep nd100 0.82	737 R thr 146	-43.95	avgR100	170.89	eps 0.025	Qval	-3.87	la
[DQN-Mod] Ep nd100 0.81	738 R thr 139	-8.04	avgR100	169.64	eps 0.025	Qval	-3.76	la
[DQN-Mod] Ep nd100 0.81	739 R thr 249	263.64	avgR100	169.89	eps 0.025	Qval	-4.78	la
[DQN-Mod] Ep nd100 0.81	740 R thr 177	256.14	avgR100	169.91	eps 0.024	Qval	-3.56	la
[DQN-Mod] Ep nd100 0.80	741 R thr 149	-23.63	avgR100	167.86	eps 0.024	Qval	-4.35	la
[DQN-Mod] Ep nd100 0.80	742 R thr 227	207.15	avgR100	167.53	eps 0.024	Qval	-4.96	la
[DQN-Mod] Ep nd100 0.80	743 R thr 271	223.98	avgR100	167.42	eps 0.024	Qval	-4.52	la
[DQN-Mod] Ep nd100 0.79	744 R thr 150	-28.68	avgR100	164.68	eps 0.024	Qval	-4.96	la
[DQN-Mod] Ep nd100 0.78	745 R thr 166	-27.19	avgR100	161.89	eps 0.024	Qval	-5.74	la
[DQN-Mod] Ep nd100 0.78	746 R thr 218	218.45	avgR100	161.80	eps 0.024	Qval	-4.52	la
[DQN-Mod] Ep nd100 0.78	747 R thr 119	-36.34	avgR100	161.43	eps 0.024	Qval	-5.18	la
[DQN-Mod] Ep nd100 0.78	748 R thr 231	212.16	avgR100	161.71	eps 0.024	Qval	-4.86	la
[DQN-Mod] Ep nd100 0.79	749 R thr 181	242.05	avgR100	165.87	eps 0.023	Qval	-5.14	la
[DQN-Mod] Ep nd100 0.80	750 R thr 221	272.31	avgR100	166.86	eps 0.023	Qval	-5.75	la

[DQN-Mod] Ep nd100 0.80	751 R thr 112	253.90	avgR100	169.42	eps 0.023	Qval	-5.15	la
[DQN-Mod] Ep nd100 0.80	752 R thr 265	226.95	avgR100	170.24	eps 0.023	Qval	-4.12	la
[DQN-Mod] Ep nd100 0.80	753 R thr 206	243.33	avgR100	170.42	eps 0.023	Qval	-5.33	la
[DQN-Mod] Ep nd100 0.80	754 R thr 168	273.63	avgR100	170.70	eps 0.023	Qval	-6.15	la
[DQN-Mod] Ep nd100 0.79	755 R thr 103	238.17	avgR100	170.40	eps 0.023	Qval	-5.37	la
[DQN-Mod] Ep nd100 0.79	756 R thr 164	256.23	avgR100	170.08	eps 0.023	Qval	-5.09	la
[DQN-Mod] Ep nd100 0.79	757 R thr 173	285.87	avgR100	170.47	eps 0.022	Qval	-4.01	la
[DQN-Mod] Ep nd100 0.79	758 R thr 224	244.05	avgR100	171.02	eps 0.022	Qval	-4.79	la
[DQN-Mod] Ep nd100 0.79	759 R thr 125	-41.10	avgR100	172.23	eps 0.022	Qval	-4.76	la
[DQN-Mod] Ep nd100 0.79	760 R thr 276	204.71	avgR100	171.20	eps 0.022	Qval	-4.80	la
[DQN-Mod] Ep nd100 0.79	761 R thr 242	214.87	avgR100	170.79	eps 0.022	Qval	-4.47	la
[DQN-Mod] Ep nd100 0.79	762 R thr 345	219.37	avgR100	170.97	eps 0.022	Qval	-4.92	la
[DQN-Mod] Ep nd100 0.79	763 R thr 168	262.57	avgR100	171.46	eps 0.022	Qval	-4.97	la
[DQN-Mod] Ep nd100 0.79	764 R thr 340	210.49	avgR100	172.40	eps 0.022	Qval	-3.04	la
[DQN-Mod] Ep nd100 0.80	765 R thr 127	278.75	avgR100	177.59	eps 0.022	Qval	-4.59	la
[DQN-Mod] Ep nd100 0.80	766 R thr 211	228.26	avgR100	177.73	eps 0.022	Qval	-3.72	la
[DQN-Mod] Ep nd100 0.80	767 R thr 188	259.04	avgR100	178.54	eps 0.021	Qval	-4.07	la
[DQN-Mod] Ep nd100 0.80	768 R thr 407	129.57	avgR100	178.57	eps 0.021	Qval	-4.61	la
[DQN-Mod] Ep nd100 0.80	769 R thr 169	268.25	avgR100	179.25	eps 0.021	Qval	-5.03	la
[DQN-Mod] Ep nd100 0.80	770 R thr 116	295.29	avgR100	181.51	eps 0.021	Qval	-3.46	la
[DQN-Mod] Ep nd100 0.79	771 R thr 151	-22.27	avgR100	179.50	eps 0.021	Qval	-4.27	la
[DQN-Mod] Ep nd100 0.78	772 R thr 803	-133.77	avgR100	176.57	eps 0.021	Qval	-2.80	la
[DQN-Mod] Ep nd100 0.78	773 R thr 167	247.85	avgR100	177.36	eps 0.021	Qval	-2.41	la
[DQN-Mod] Ep nd100 0.77	774 R thr 171	-65.46	avgR100	173.95	eps 0.021	Qval	-3.27	la
[DQN-Mod] Ep nd100 0.77	775 R thr 137	254.27	avgR100	173.98	eps 0.021	Qval	-3.26	la
[DQN-Mod] Ep nd100 0.77	776 R thr 208	264.25	avgR100	174.16	eps 0.020	Qval	-4.47	la
[DQN-Mod] Ep nd100 0.77	777 R thr 391	169.91	avgR100	174.47	eps 0.020	Qval	-4.11	la
[DQN-Mod] Ep nd100 0.78	778 R thr 156	271.79	avgR100	174.86	eps 0.020	Qval	-4.44	la
[DQN-Mod] Ep nd100 0.78	779 R thr 223	242.04	avgR100	175.65	eps 0.020	Qval	-3.91	la
[DQN-Mod] Ep nd100 0.78	780 R thr 157	254.23	avgR100	175.96	eps 0.020	Qval	-3.79	la

[DQN-Mod] Ep 781 R	-4.44	avgR100	174.40	eps 0.020	Qval	-3.73	1a
nd100 0.77 thr 122							
[DQN-Mod] Ep 782 R	7.99	avgR100	172.43	eps 0.020	Qval	-2.02	1a
nd100 0.76 thr 123							
[DQN-Mod] Ep 783 R	-56.44	avgR100	170.51	eps 0.020	Qval	-4.20	1a
nd100 0.75 thr 128							
[DQN-Mod] Ep 784 R	239.07	avgR100	170.60	eps 0.020	Qval	-3.91	1a
nd100 0.75 thr 195							
[DQN-Mod] Ep 785 R	297.91	avgR100	173.57	eps 0.020	Qval	-3.82	1a
nd100 0.76 thr 117							
[DQN-Mod] Ep 786 R	250.20	avgR100	176.48	eps 0.020	Qval	-4.23	1a
nd100 0.77 thr 225							
[DQN-Mod] Ep 787 R	-49.18	avgR100	177.07	eps 0.020	Qval	-3.71	1a
nd100 0.77 thr 146							
[DQN-Mod] Ep 788 R	228.95	avgR100	177.40	eps 0.020	Qval	-3.67	1a
nd100 0.77 thr 145							
[DQN-Mod] Ep 789 R	-31.13	avgR100	174.58	eps 0.020	Qval	-5.59	1a
nd100 0.76 thr 122							
[DQN-Mod] Ep 790 R	-44.76	avgR100	171.74	eps 0.020	Qval	-5.16	1a
nd100 0.75 thr 188							
[DQN-Mod] Ep 791 R	207.48	avgR100	171.29	eps 0.020	Qval	-5.65	1a
nd100 0.75 thr 296							
[DQN-Mod] Ep 792 R	263.68	avgR100	171.64	eps 0.020	Qval	-4.60	1a
nd100 0.75 thr 136							
[DQN-Mod] Ep 793 R	-60.09	avgR100	168.44	eps 0.020	Qval	-4.31	1a
nd100 0.74 thr 137							
[DQN-Mod] Ep 794 R	-40.33	avgR100	165.84	eps 0.020	Qval	-5.21	1a
nd100 0.73 thr 144							
[DQN-Mod] Ep 795 R	278.36	avgR100	166.74	eps 0.020	Qval	-4.67	1a
nd100 0.74 thr 148							
[DQN-Mod] Ep 796 R	40.99	avgR100	166.14	eps 0.020	Qval	-4.47	1a
nd100 0.73 thr 358							
[DQN-Mod] Ep 797 R	297.20	avgR100	169.54	eps 0.020	Qval	-4.15	1a
nd100 0.74 thr 112							
[DQN-Mod] Ep 798 R	111.30	avgR100	167.99	eps 0.020	Qval	-4.78	1a
nd100 0.74 thr 547							
[DQN-Mod] Ep 799 R	159.24	avgR100	169.49	eps 0.020	Qval	-5.99	1a
nd100 0.74 thr 239							
[DQN-Mod] Ep 800 R	-1.05	avgR100	166.84	eps 0.020	Qval	-5.25	1a
nd100 0.73 thr 107							
[DQN-Mod] Ep 801 R	273.72	avgR100	167.87	eps 0.020	Qval	-6.26	1a
nd100 0.73 thr 167							
[DQN-Mod] Ep 802 R	244.61	avgR100	170.08	eps 0.020	Qval	-6.43	1a
nd100 0.73 thr 117							
[DQN-Mod] Ep 803 R	250.06	avgR100	170.32	eps 0.020	Qval	-5.09	1a
nd100 0.73 thr 148							
[DQN-Mod] Ep 804 R	245.68	avgR100	170.49	eps 0.020	Qval	-3.99	1a
nd100 0.73 thr 171							
[DQN-Mod] Ep 805 R	-30.66	avgR100	167.98	eps 0.020	Qval	-4.70	1a
nd100 0.73 thr 125							
[DQN-Mod] Ep 806 R	245.88	avgR100	169.73	eps 0.020	Qval	-4.03	1a
nd100 0.74 thr 204							
[DQN-Mod] Ep 807 R	220.92	avgR100	171.87	eps 0.020	Qval	-5.31	1a
nd100 0.75 thr 245							
[DQN-Mod] Ep 808 R	243.80	avgR100	171.64	eps 0.020	Qval	-4.14	1a
nd100 0.75 thr 181							
[DQN-Mod] Ep 809 R	132.53	avgR100	170.42	eps 0.020	Qval	-6.43	1a
nd100 0.75 thr 358							
[DQN-Mod] Ep 810 R	271.78	avgR100	170.70	eps 0.020	Qval	-5.08	1a
nd100 0.75 thr 252							

[DQN-Mod] Ep nd100 0.75	811 R thr 192	290.04	avgR100	171.98	eps 0.020	Qval	-4.32	1a
[DQN-Mod] Ep nd100 0.75	812 R thr 269	209.52	avgR100	173.06	eps 0.020	Qval	-4.79	1a
[DQN-Mod] Ep nd100 0.75	813 R thr 306	187.23	avgR100	172.91	eps 0.020	Qval	-4.38	1a
[DQN-Mod] Ep nd100 0.74	814 R thr 72	-2.68	avgR100	170.53	eps 0.020	Qval	-4.53	1a
[DQN-Mod] Ep nd100 0.74	815 R thr 291	198.76	avgR100	170.75	eps 0.020	Qval	-3.75	1a
[DQN-Mod] Ep nd100 0.73	816 R thr 232	199.78	avgR100	169.98	eps 0.020	Qval	-4.63	1a
[DQN-Mod] Ep nd100 0.73	817 R thr 185	232.75	avgR100	169.99	eps 0.020	Qval	-4.25	1a
[DQN-Mod] Ep nd100 0.73	818 R thr 259	182.36	avgR100	169.27	eps 0.020	Qval	-4.35	1a
[DQN-Mod] Ep nd100 0.72	819 R thr 134	12.00	avgR100	167.07	eps 0.020	Qval	-4.32	1a
[DQN-Mod] Ep nd100 0.71	820 R thr 148	-8.48	avgR100	164.80	eps 0.020	Qval	-5.79	1a
[DQN-Mod] Ep nd100 0.71	821 R thr 208	217.40	avgR100	165.25	eps 0.020	Qval	-3.75	1a
[DQN-Mod] Ep nd100 0.70	822 R thr 97	25.76	avgR100	163.18	eps 0.020	Qval	-4.76	1a
[DQN-Mod] Ep nd100 0.70	823 R thr 382	147.84	avgR100	163.74	eps 0.020	Qval	-3.51	1a
[DQN-Mod] Ep nd100 0.69	824 R thr 235	168.22	avgR100	163.33	eps 0.020	Qval	-4.77	1a
[DQN-Mod] Ep nd100 0.68	825 R thr 142	-22.11	avgR100	160.61	eps 0.020	Qval	-5.57	1a
[DQN-Mod] Ep nd100 0.69	826 R thr 164	249.25	avgR100	161.25	eps 0.020	Qval	-5.57	1a
[DQN-Mod] Ep nd100 0.69	827 R thr 223	263.88	avgR100	161.02	eps 0.020	Qval	-4.35	1a
[DQN-Mod] Ep nd100 0.69	828 R thr 343	200.42	avgR100	160.50	eps 0.020	Qval	-3.77	1a
[DQN-Mod] Ep nd100 0.69	829 R thr 192	221.17	avgR100	160.78	eps 0.020	Qval	-3.34	1a
[DQN-Mod] Ep nd100 0.69	830 R thr 198	264.05	avgR100	160.95	eps 0.020	Qval	-3.91	1a
[DQN-Mod] Ep nd100 0.68	831 R thr 89	0.59	avgR100	158.50	eps 0.020	Qval	-3.37	1a
[DQN-Mod] Ep nd100 0.69	832 R thr 309	184.46	avgR100	161.05	eps 0.020	Qval	-2.65	1a
[DQN-Mod] Ep nd100 0.69	833 R thr 455	96.10	avgR100	159.43	eps 0.020	Qval	-3.58	1a
[DQN-Mod] Ep nd100 0.69	834 R thr 254	222.75	avgR100	159.95	eps 0.020	Qval	-2.81	1a
[DQN-Mod] Ep nd100 0.69	835 R thr 171	259.17	avgR100	161.70	eps 0.020	Qval	-3.91	1a
[DQN-Mod] Ep nd100 0.69	836 R thr 159	240.17	avgR100	161.97	eps 0.020	Qval	-2.63	1a
[DQN-Mod] Ep nd100 0.70	837 R thr 200	265.09	avgR100	165.06	eps 0.020	Qval	-2.87	1a
[DQN-Mod] Ep nd100 0.71	838 R thr 176	302.19	avgR100	168.16	eps 0.020	Qval	-3.81	1a
[DQN-Mod] Ep nd100 0.70	839 R thr 138	20.29	avgR100	165.73	eps 0.020	Qval	-2.87	1a
[DQN-Mod] Ep nd100 0.70	840 R thr 274	215.84	avgR100	165.32	eps 0.020	Qval	-2.12	1a

[DQN-Mod] Ep 841 R	-15.86	avgR100	165.40	eps 0.020	Qval	-2.29	la
nd100 0.70 thr 173							
[DQN-Mod] Ep 842 R	276.65	avgR100	166.10	eps 0.020	Qval	-3.06	la
nd100 0.70 thr 147							
[DQN-Mod] Ep 843 R	221.87	avgR100	166.08	eps 0.020	Qval	-4.17	la
nd100 0.70 thr 269							
[DQN-Mod] Ep 844 R	244.39	avgR100	168.81	eps 0.020	Qval	-3.37	la
nd100 0.71 thr 166							
[DQN-Mod] Ep 845 R	275.95	avgR100	171.84	eps 0.020	Qval	-3.84	la
nd100 0.72 thr 169							
[DQN-Mod] Ep 846 R	201.03	avgR100	171.66	eps 0.020	Qval	-4.67	la
nd100 0.72 thr 331							
[DQN-Mod] Ep 847 R	158.87	avgR100	173.62	eps 0.020	Qval	-5.60	la
nd100 0.73 thr 415							
[DQN-Mod] Ep 848 R	-27.13	avgR100	171.22	eps 0.020	Qval	-5.14	la
nd100 0.72 thr 127							
[DQN-Mod] Ep 849 R	294.13	avgR100	171.74	eps 0.020	Qval	-5.03	la
nd100 0.72 thr 156							
[DQN-Mod] Ep 850 R	232.49	avgR100	171.34	eps 0.020	Qval	-3.21	la
nd100 0.72 thr 297							
[DQN-Mod] Ep 851 R	230.63	avgR100	171.11	eps 0.020	Qval	-1.64	la
nd100 0.72 thr 200							
[DQN-Mod] Ep 852 R	259.71	avgR100	171.44	eps 0.020	Qval	-1.29	la
nd100 0.72 thr 133							
[DQN-Mod] Ep 853 R	231.26	avgR100	171.32	eps 0.020	Qval	-1.27	la
nd100 0.72 thr 254							
[DQN-Mod] Ep 854 R	265.40	avgR100	171.24	eps 0.020	Qval	-1.66	la
nd100 0.72 thr 147							
[DQN-Mod] Ep 855 R	213.10	avgR100	170.99	eps 0.020	Qval	-1.94	la
nd100 0.73 thr 263							
[DQN-Mod] Ep 856 R	253.42	avgR100	170.96	eps 0.020	Qval	-2.27	la
nd100 0.73 thr 130							
[DQN-Mod] Ep 857 R	4.94	avgR100	168.15	eps 0.020	Qval	-2.30	la
nd100 0.72 thr 124							
[DQN-Mod] Ep 858 R	211.33	avgR100	167.82	eps 0.020	Qval	-1.77	la
nd100 0.72 thr 273							
[DQN-Mod] Ep 859 R	251.63	avgR100	170.75	eps 0.020	Qval	-2.02	la
nd100 0.73 thr 134							
[DQN-Mod] Ep 860 R	181.70	avgR100	170.52	eps 0.020	Qval	-1.15	la
nd100 0.73 thr 281							
[DQN-Mod] Ep 861 R	263.89	avgR100	171.01	eps 0.020	Qval	-1.89	la
nd100 0.73 thr 184							
[DQN-Mod] Ep 862 R	223.34	avgR100	171.05	eps 0.020	Qval	-2.48	la
nd100 0.73 thr 186							
[DQN-Mod] Ep 863 R	4.84	avgR100	168.47	eps 0.020	Qval	-2.35	la
nd100 0.72 thr 143							
[DQN-Mod] Ep 864 R	235.56	avgR100	168.72	eps 0.020	Qval	-1.39	la
nd100 0.71 thr 134							
[DQN-Mod] Ep 865 R	226.49	avgR100	168.20	eps 0.020	Qval	-2.31	la
nd100 0.71 thr 167							
[DQN-Mod] Ep 866 R	272.54	avgR100	168.64	eps 0.020	Qval	-2.44	la
nd100 0.71 thr 166							
[DQN-Mod] Ep 867 R	262.31	avgR100	168.67	eps 0.020	Qval	-0.80	la
nd100 0.71 thr 196							
[DQN-Mod] Ep 868 R	193.46	avgR100	169.31	eps 0.020	Qval	-1.62	la
nd100 0.70 thr 205							
[DQN-Mod] Ep 869 R	284.82	avgR100	169.48	eps 0.020	Qval	-1.67	la
nd100 0.70 thr 167							
[DQN-Mod] Ep 870 R	265.55	avgR100	169.18	eps 0.020	Qval	-0.95	la
nd100 0.70 thr 163							

[DQN-Mod] Ep nd100 0.70	871 R thr 284	128.21	avgR100	170.69	eps 0.020	Qval	-0.70	la
[DQN-Mod] Ep nd100 0.71	872 R thr 146	244.05	avgR100	174.47	eps 0.020	Qval	-1.47	la
[DQN-Mod] Ep nd100 0.70	873 R thr 122	-38.16	avgR100	171.61	eps 0.020	Qval	-1.06	la
[DQN-Mod] Ep nd100 0.70	874 R thr 124	-36.79	avgR100	171.89	eps 0.020	Qval	-1.88	la
[DQN-Mod] Ep nd100 0.70	875 R thr 159	255.88	avgR100	171.91	eps 0.020	Qval	-2.27	la
[DQN-Mod] Ep nd100 0.70	876 R thr 270	198.26	avgR100	171.25	eps 0.020	Qval	-2.82	la
[DQN-Mod] Ep nd100 0.70	877 R thr 124	295.93	avgR100	172.51	eps 0.020	Qval	-3.37	la
[DQN-Mod] Ep nd100 0.70	878 R thr 190	260.74	avgR100	172.40	eps 0.020	Qval	-0.74	la
[DQN-Mod] Ep nd100 0.70	879 R thr 279	229.60	avgR100	172.27	eps 0.020	Qval	-2.78	la
[DQN-Mod] Ep nd100 0.70	880 R thr 211	245.63	avgR100	172.19	eps 0.020	Qval	-2.40	la
[DQN-Mod] Ep nd100 0.71	881 R thr 138	307.81	avgR100	175.31	eps 0.020	Qval	-2.36	la
[DQN-Mod] Ep nd100 0.72	882 R thr 106	292.44	avgR100	178.15	eps 0.020	Qval	-1.30	la
[DQN-Mod] Ep nd100 0.72	883 R thr 168	245.07	avgR100	181.17	eps 0.020	Qval	-2.35	la
[DQN-Mod] Ep nd100 0.72	884 R thr 135	243.80	avgR100	181.22	eps 0.020	Qval	-2.37	la
[DQN-Mod] Ep nd100 0.72	885 R thr 133	253.61	avgR100	180.77	eps 0.020	Qval	-0.96	la
[DQN-Mod] Ep nd100 0.71	886 R thr 111	-15.18	avgR100	178.12	eps 0.020	Qval	-0.75	la
[DQN-Mod] Ep nd100 0.71	887 R thr 299	167.52	avgR100	180.29	eps 0.020	Qval	0.19	la
[DQN-Mod] Ep nd100 0.71	888 R thr 221	253.71	avgR100	180.53	eps 0.020	Qval	-1.38	la
[DQN-Mod] Ep nd100 0.72	889 R thr 240	224.20	avgR100	183.09	eps 0.020	Qval	-1.50	la
[DQN-Mod] Ep nd100 0.72	890 R thr 653	-126.00	avgR100	182.28	eps 0.020	Qval	-3.58	la
[DQN-Mod] Ep nd100 0.71	891 R thr 142	186.65	avgR100	182.07	eps 0.020	Qval	-2.39	la
[DQN-Mod] Ep nd100 0.70	892 R thr 534	-70.47	avgR100	178.73	eps 0.020	Qval	-2.14	la
[DQN-Mod] Ep nd100 0.71	893 R thr 214	267.64	avgR100	182.00	eps 0.020	Qval	-1.28	la
[DQN-Mod] Ep nd100 0.72	894 R thr 398	122.88	avgR100	183.63	eps 0.020	Qval	-1.25	la
[DQN-Mod] Ep nd100 0.72	895 R thr 497	83.27	avgR100	181.68	eps 0.020	Qval	-0.99	la
[DQN-Mod] Ep nd100 0.72	896 R thr 124	-5.59	avgR100	181.22	eps 0.020	Qval	-0.72	la
[DQN-Mod] Ep nd100 0.72	897 R thr 254	178.23	avgR100	180.03	eps 0.020	Qval	-2.61	la
[DQN-Mod] Ep nd100 0.72	898 R thr 180	235.35	avgR100	181.27	eps 0.020	Qval	-1.96	la
[DQN-Mod] Ep nd100 0.73	899 R thr 278	208.42	avgR100	181.76	eps 0.020	Qval	-1.85	la
[DQN-Mod] Ep nd100 0.74	900 R thr 312	177.24	avgR100	183.54	eps 0.020	Qval	-1.06	la

[DQN-Mod] Ep 901 R	231.67	avgR100	183.12	eps 0.020	Qval	-1.71	la
nd100 0.74 thr 220							
[DQN-Mod] Ep 902 R	233.32	avgR100	183.01	eps 0.020	Qval	-1.46	la
nd100 0.74 thr 199							
[DQN-Mod] Ep 903 R	240.52	avgR100	182.91	eps 0.020	Qval	-1.89	la
nd100 0.74 thr 215							
[DQN-Mod] Ep 904 R	276.06	avgR100	183.22	eps 0.020	Qval	-1.41	la
nd100 0.74 thr 174							
[DQN-Mod] Ep 905 R	240.85	avgR100	185.93	eps 0.020	Qval	-1.96	la
nd100 0.75 thr 235							
[DQN-Mod] Ep 906 R	217.18	avgR100	185.65	eps 0.020	Qval	-2.46	la
nd100 0.75 thr 196							
[DQN-Mod] Ep 907 R	8.63	avgR100	183.52	eps 0.020	Qval	-1.44	la
nd100 0.74 thr 130							
[DQN-Mod] Ep 908 R	8.76	avgR100	181.17	eps 0.020	Qval	-2.24	la
nd100 0.73 thr 110							
[DQN-Mod] Ep 909 R	212.16	avgR100	181.97	eps 0.020	Qval	-1.18	la
nd100 0.73 thr 268							
[DQN-Mod] Ep 910 R	244.58	avgR100	181.70	eps 0.020	Qval	0.47	la
nd100 0.73 thr 213							
[DQN-Mod] Ep 911 R	247.27	avgR100	181.27	eps 0.020	Qval	0.02	la
nd100 0.73 thr 265							
[DQN-Mod] Ep 912 R	255.90	avgR100	181.73	eps 0.020	Qval	-0.03	la
nd100 0.73 thr 191							
[DQN-Mod] Ep 913 R	143.43	avgR100	181.30	eps 0.020	Qval	0.90	la
nd100 0.73 thr 327							
[DQN-Mod] Ep 914 R	241.35	avgR100	183.74	eps 0.020	Qval	0.60	la
nd100 0.74 thr 165							
[DQN-Mod] Ep 915 R	231.38	avgR100	184.06	eps 0.020	Qval	1.51	la
nd100 0.74 thr 278							
[DQN-Mod] Ep 916 R	4.47	avgR100	182.11	eps 0.020	Qval	0.88	la
nd100 0.74 thr 93							
[DQN-Mod] Ep 917 R	243.78	avgR100	182.22	eps 0.020	Qval	0.65	la
nd100 0.74 thr 190							
[DQN-Mod] Ep 918 R	-25.45	avgR100	180.14	eps 0.020	Qval	0.90	la
nd100 0.73 thr 119							
[DQN-Mod] Ep 919 R	254.96	avgR100	182.57	eps 0.020	Qval	1.51	la
nd100 0.74 thr 243							
[DQN-Mod] Ep 920 R	163.99	avgR100	184.30	eps 0.020	Qval	1.39	la
nd100 0.75 thr 417							
[DQN-Mod] Ep 921 R	233.84	avgR100	184.46	eps 0.020	Qval	0.46	la
nd100 0.75 thr 260							
[DQN-Mod] Ep 922 R	268.97	avgR100	186.89	eps 0.020	Qval	1.68	la
nd100 0.76 thr 222							
[DQN-Mod] Ep 923 R	253.54	avgR100	187.95	eps 0.020	Qval	2.83	la
nd100 0.76 thr 218							
[DQN-Mod] Ep 924 R	236.00	avgR100	188.63	eps 0.020	Qval	2.00	la
nd100 0.77 thr 170							
[DQN-Mod] Ep 925 R	250.57	avgR100	191.35	eps 0.020	Qval	2.25	la
nd100 0.78 thr 235							
[DQN-Mod] Ep 926 R	-43.02	avgR100	188.43	eps 0.020	Qval	1.77	la
nd100 0.77 thr 91							
[DQN-Mod] Ep 927 R	242.58	avgR100	188.22	eps 0.020	Qval	1.52	la
nd100 0.77 thr 215							
[DQN-Mod] Ep 928 R	201.99	avgR100	188.23	eps 0.020	Qval	1.93	la
nd100 0.76 thr 237							
[DQN-Mod] Ep 929 R	239.90	avgR100	188.42	eps 0.020	Qval	1.35	la
nd100 0.76 thr 169							
[DQN-Mod] Ep 930 R	218.51	avgR100	187.97	eps 0.020	Qval	0.42	la
nd100 0.76 thr 276							

[DQN-Mod] Ep 931 R	254.71 avgR100	190.51 eps 0.020	Qval	1.92 la
nd100 0.77 thr 252				
[DQN-Mod] Ep 932 R	246.78 avgR100	191.13 eps 0.020	Qval	2.78 la
nd100 0.77 thr 222				
[DQN-Mod] Ep 933 R	17.85 avgR100	190.35 eps 0.020	Qval	3.66 la
nd100 0.76 thr 142				
[DQN-Mod] Ep 934 R	271.51 avgR100	190.84 eps 0.020	Qval	2.89 la
nd100 0.76 thr 176				
[DQN-Mod] Ep 935 R	276.77 avgR100	191.01 eps 0.020	Qval	2.89 la
nd100 0.76 thr 151				
[DQN-Mod] Ep 936 R	-21.26 avgR100	188.40 eps 0.020	Qval	3.92 la
nd100 0.75 thr 99				
[DQN-Mod] Ep 937 R	223.77 avgR100	187.98 eps 0.020	Qval	4.21 la
nd100 0.75 thr 204				
[DQN-Mod] Ep 938 R	281.73 avgR100	187.78 eps 0.020	Qval	3.32 la
nd100 0.75 thr 106				
[DQN-Mod] Ep 939 R	248.26 avgR100	190.06 eps 0.020	Qval	2.86 la
nd100 0.76 thr 246				
[DQN-Mod] Ep 940 R	201.81 avgR100	189.92 eps 0.020	Qval	3.60 la
nd100 0.76 thr 266				
[DQN-Mod] Ep 941 R	240.48 avgR100	192.48 eps 0.020	Qval	4.28 la
nd100 0.77 thr 206				
[DQN-Mod] Ep 942 R	231.80 avgR100	192.03 eps 0.020	Qval	4.58 la
nd100 0.77 thr 288				
[DQN-Mod] Ep 943 R	266.18 avgR100	192.48 eps 0.020	Qval	5.28 la
nd100 0.77 thr 108				
[DQN-Mod] Ep 944 R	214.33 avgR100	192.18 eps 0.020	Qval	4.64 la
nd100 0.77 thr 346				
[DQN-Mod] Ep 945 R	248.87 avgR100	191.91 eps 0.020	Qval	3.03 la
nd100 0.77 thr 216				
[DQN-Mod] Ep 946 R	-12.79 avgR100	189.77 eps 0.020	Qval	4.06 la
nd100 0.77 thr 712				
[DQN-Mod] Ep 947 R	219.55 avgR100	190.37 eps 0.020	Qval	5.54 la
nd100 0.77 thr 201				
[DQN-Mod] Ep 948 R	166.89 avgR100	192.31 eps 0.020	Qval	4.10 la
nd100 0.78 thr 346				
[DQN-Mod] Ep 949 R	229.07 avgR100	191.66 eps 0.020	Qval	4.53 la
nd100 0.78 thr 224				
[DQN-Mod] Ep 950 R	270.36 avgR100	192.04 eps 0.020	Qval	5.37 la
nd100 0.78 thr 218				
[DQN-Mod] Ep 951 R	275.12 avgR100	192.49 eps 0.020	Qval	5.37 la
nd100 0.78 thr 192				
[DQN-Mod] Ep 952 R	253.70 avgR100	192.43 eps 0.020	Qval	5.29 la
nd100 0.78 thr 207				
[DQN-Mod] Ep 953 R	146.91 avgR100	191.58 eps 0.020	Qval	3.40 la
nd100 0.78 thr 321				
[DQN-Mod] Ep 954 R	286.19 avgR100	191.79 eps 0.020	Qval	5.22 la
nd100 0.78 thr 161				
[DQN-Mod] Ep 955 R	201.59 avgR100	191.68 eps 0.020	Qval	5.49 la
nd100 0.78 thr 271				
[DQN-Mod] Ep 956 R	251.54 avgR100	191.66 eps 0.020	Qval	4.25 la
nd100 0.78 thr 230				
[DQN-Mod] Ep 957 R	-15.00 avgR100	191.46 eps 0.020	Qval	3.67 la
nd100 0.78 thr 133				
[DQN-Mod] Ep 958 R	228.92 avgR100	191.63 eps 0.020	Qval	4.01 la
nd100 0.78 thr 175				
[DQN-Mod] Ep 959 R	213.26 avgR100	191.25 eps 0.020	Qval	4.67 la
nd100 0.78 thr 241				
[DQN-Mod] Ep 960 R	231.26 avgR100	191.75 eps 0.020	Qval	5.21 la
nd100 0.78 thr 175				

[DQN-Mod] Ep nd100 0.78	961 R thr 195	223.00	avgR100	191.34	eps 0.020	Qval	4.95	1a
[DQN-Mod] Ep nd100 0.78	962 R thr 293	201.02	avgR100	191.11	eps 0.020	Qval	5.77	1a
[DQN-Mod] Ep nd100 0.78	963 R thr 299	-156.32	avgR100	189.50	eps 0.020	Qval	5.61	1a
[DQN-Mod] Ep nd100 0.79	964 R thr 246	232.36	avgR100	189.47	eps 0.020	Qval	5.98	1a
[DQN-Mod] Ep nd100 0.78	965 R thr 491	52.95	avgR100	187.73	eps 0.020	Qval	6.22	1a
[DQN-Mod] Ep nd100 0.78	966 R thr 247	261.11	avgR100	187.62	eps 0.020	Qval	5.89	1a
[DQN-Mod] Ep nd100 0.77	967 R thr 130	-6.74	avgR100	184.93	eps 0.020	Qval	5.42	1a
[DQN-Mod] Ep nd100 0.78	968 R thr 176	233.14	avgR100	185.33	eps 0.020	Qval	5.97	1a
[DQN-Mod] Ep nd100 0.78	969 R thr 143	288.29	avgR100	185.36	eps 0.020	Qval	7.01	1a
[DQN-Mod] Ep nd100 0.78	970 R thr 179	231.63	avgR100	185.02	eps 0.020	Qval	7.91	1a
[DQN-Mod] Ep nd100 0.79	971 R thr 116	283.89	avgR100	186.58	eps 0.020	Qval	8.26	1a
[DQN-Mod] Ep nd100 0.79	972 R thr 242	240.93	avgR100	186.55	eps 0.020	Qval	8.33	1a
[DQN-Mod] Ep nd100 0.80	973 R thr 166	270.15	avgR100	189.63	eps 0.020	Qval	8.27	1a
[DQN-Mod] Ep nd100 0.81	974 R thr 159	244.20	avgR100	192.44	eps 0.020	Qval	8.17	1a
[DQN-Mod] Ep nd100 0.81	975 R thr 152	244.62	avgR100	192.33	eps 0.020	Qval	8.02	1a
[DQN-Mod] Ep nd100 0.81	976 R thr 216	237.70	avgR100	192.72	eps 0.020	Qval	8.65	1a
[DQN-Mod] Ep nd100 0.81	977 R thr 293	195.12	avgR100	191.71	eps 0.020	Qval	8.89	1a
[DQN-Mod] Ep nd100 0.81	978 R thr 247	250.16	avgR100	191.61	eps 0.020	Qval	10.04	1a
[DQN-Mod] Ep nd100 0.81	979 R thr 306	185.20	avgR100	191.16	eps 0.020	Qval	10.30	1a
[DQN-Mod] Ep nd100 0.81	980 R thr 222	261.06	avgR100	191.32	eps 0.020	Qval	9.79	1a
[DQN-Mod] Ep nd100 0.80	981 R thr 156	236.26	avgR100	190.60	eps 0.020	Qval	10.15	1a
[DQN-Mod] Ep nd100 0.80	982 R thr 124	289.87	avgR100	190.58	eps 0.020	Qval	9.89	1a
[DQN-Mod] Ep nd100 0.81	983 R thr 380	153.80	avgR100	189.67	eps 0.020	Qval	10.35	1a
[DQN-Mod] Ep nd100 0.80	984 R thr 664	-138.12	avgR100	185.85	eps 0.020	Qval	10.03	1a
[DQN-Mod] Ep nd100 0.80	985 R thr 247	207.72	avgR100	185.39	eps 0.020	Qval	11.33	1a
[DQN-Mod] Ep nd100 0.81	986 R thr 317	208.24	avgR100	187.62	eps 0.020	Qval	11.70	1a
[DQN-Mod] Ep nd100 0.82	987 R thr 332	173.50	avgR100	187.68	eps 0.020	Qval	11.88	1a
[DQN-Mod] Ep nd100 0.82	988 R thr 239	243.84	avgR100	187.58	eps 0.020	Qval	11.51	1a
[DQN-Mod] Ep nd100 0.81	989 R thr 211	163.93	avgR100	186.98	eps 0.020	Qval	12.25	1a
[DQN-Mod] Ep nd100 0.82	990 R thr 254	222.70	avgR100	190.47	eps 0.020	Qval	12.90	1a

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[DQN-Mod] Ep 991 | R -20.93 | avgR100 188.39 | eps 0.020 | Qval 11.31 | la
nd100 0.82 | thr 178
[DQN-Mod] Ep 992 | R -15.61 | avgR100 188.94 | eps 0.020 | Qval 11.35 | la
nd100 0.82 | thr 111
[DQN-Mod] Ep 993 | R 251.55 | avgR100 188.78 | eps 0.020 | Qval 13.04 | la
nd100 0.82 | thr 233
[DQN-Mod] Ep 994 | R 257.76 | avgR100 190.13 | eps 0.020 | Qval 12.35 | la
nd100 0.82 | thr 231
[DQN-Mod] Ep 995 | R 180.25 | avgR100 191.10 | eps 0.020 | Qval 11.15 | la
nd100 0.82 | thr 313
[DQN-Mod] Ep 996 | R 73.93 | avgR100 191.89 | eps 0.020 | Qval 12.65 | la
nd100 0.83 | thr 468
[DQN-Mod] Ep 997 | R 221.66 | avgR100 192.33 | eps 0.020 | Qval 12.05 | la
nd100 0.83 | thr 312
[DQN-Mod] Ep 998 | R 230.28 | avgR100 192.28 | eps 0.020 | Qval 14.04 | la
nd100 0.83 | thr 243
[DQN-Mod] Ep 999 | R 60.65 | avgR100 190.80 | eps 0.020 | Qval 13.01 | la
nd100 0.83 | thr 104
[DQN-Mod] Ep 1000 | R 202.00 | avgR100 191.05 | eps 0.020 | Qval 12.25 | la
nd100 0.83 | thr 256
[DQN-Mod] finished 1000 episodes in 3.6 min (final avgR100 = 191.05)

```

```

In [30]: # ---- Configuration 4 of 4: DDQN on the MODIFIED environment ----
RESULTS["DDQN-Mod"] = train_agent(modified=True, double=True, tag="DDQN-Mod")

print("All four configurations trained.")

```

[DDQN-Mod] and100 0.00	Ep 1 thr 41	R	-102.54		avgR100	-102.54		eps 0.995		Qval 0.05		1
[DDQN-Mod] and100 0.00	Ep 2 thr 50	R	-98.27		avgR100	-100.40		eps 0.990		Qval 0.05		1
[DDQN-Mod] and100 0.00	Ep 3 thr 87	R	-307.70		avgR100	-169.50		eps 0.985		Qval 0.05		1
[DDQN-Mod] and100 0.00	Ep 4 thr 75	R	-166.45		avgR100	-168.74		eps 0.980		Qval 0.05		1
[DDQN-Mod] and100 0.00	Ep 5 thr 54	R	-122.17		avgR100	-159.43		eps 0.975		Qval 0.05		1
[DDQN-Mod] and100 0.00	Ep 6 thr 69	R	-281.18		avgR100	-179.72		eps 0.970		Qval 0.05		1
[DDQN-Mod] and100 0.00	Ep 7 thr 58	R	-391.00		avgR100	-209.90		eps 0.966		Qval 0.05		1
[DDQN-Mod] and100 0.12	Ep 8 thr 64	R	-64.07		avgR100	-191.67		eps 0.961		Qval 0.05		1
[DDQN-Mod] and100 0.11	Ep 9 thr 67	R	-240.66		avgR100	-197.12		eps 0.956		Qval 0.05		1
[DDQN-Mod] and100 0.10	Ep 10 thr 80	R	-122.26		avgR100	-189.63		eps 0.951		Qval 0.05		1
[DDQN-Mod] and100 0.09	Ep 11 thr 78	R	-137.70		avgR100	-184.91		eps 0.946		Qval 0.05		1
[DDQN-Mod] and100 0.08	Ep 12 thr 71	R	-336.48		avgR100	-197.54		eps 0.942		Qval 0.14		1
[DDQN-Mod] and100 0.08	Ep 13 thr 93	R	-243.07		avgR100	-201.04		eps 0.937		Qval -0.18		1
[DDQN-Mod] and100 0.07	Ep 14 thr 121	R	-195.35		avgR100	-200.64		eps 0.932		Qval -0.48		1
[DDQN-Mod] and100 0.07	Ep 15 thr 56	R	-172.56		avgR100	-198.76		eps 0.928		Qval -0.60		1
[DDQN-Mod] and100 0.06	Ep 16 thr 80	R	-491.52		avgR100	-217.06		eps 0.923		Qval -0.44		1
[DDQN-Mod] and100 0.06	Ep 17 thr 73	R	-324.77		avgR100	-223.40		eps 0.918		Qval -0.59		1
[DDQN-Mod] and100 0.06	Ep 18 thr 106	R	-275.27		avgR100	-226.28		eps 0.914		Qval -0.73		1
[DDQN-Mod] and100 0.05	Ep 19 thr 87	R	-348.87		avgR100	-232.73		eps 0.909		Qval -0.66		1
[DDQN-Mod] and100 0.10	Ep 20 thr 85	R	-42.96		avgR100	-223.24		eps 0.905		Qval -0.26		1
[DDQN-Mod] and100 0.10	Ep 21 thr 68	R	-136.45		avgR100	-219.11		eps 0.900		Qval -0.49		1
[DDQN-Mod] and100 0.09	Ep 22 thr 89	R	-115.12		avgR100	-214.38		eps 0.896		Qval -0.42		1
[DDQN-Mod] and100 0.09	Ep 23 thr 76	R	-305.09		avgR100	-218.33		eps 0.891		Qval -0.36		1
[DDQN-Mod] and100 0.08	Ep 24 thr 100	R	-184.64		avgR100	-216.92		eps 0.887		Qval -0.46		1
[DDQN-Mod] and100 0.08	Ep 25 thr 62	R	-94.47		avgR100	-212.02		eps 0.882		Qval -0.34		1
[DDQN-Mod] and100 0.08	Ep 26 thr 78	R	-199.04		avgR100	-211.53		eps 0.878		Qval -0.17		1
[DDQN-Mod] and100 0.07	Ep 27 thr 52	R	-284.66		avgR100	-214.23		eps 0.873		Qval 0.04		1
[DDQN-Mod] and100 0.07	Ep 28 thr 66	R	-181.22		avgR100	-213.05		eps 0.869		Qval -0.25		1
[DDQN-Mod] and100 0.07	Ep 29 thr 70	R	-14.17		avgR100	-206.20		eps 0.865		Qval -0.16		1
[DDQN-Mod] and100 0.10	Ep 30 thr 47	R	-20.58		avgR100	-200.01		eps 0.860		Qval -0.01		1

[DDQN-Mod] and100 0.10	Ep 31 thr 69	R	-93.22		avgR100	-196.56		eps 0.856		Qval -0.07		1
[DDQN-Mod] and100 0.09	Ep 32 thr 89	R	-224.29		avgR100	-197.43		eps 0.852		Qval 0.19		1
[DDQN-Mod] and100 0.09	Ep 33 thr 60	R	-250.08		avgR100	-199.03		eps 0.848		Qval -0.15		1
[DDQN-Mod] and100 0.09	Ep 34 thr 100	R	-111.70		avgR100	-196.46		eps 0.843		Qval -0.01		1
[DDQN-Mod] and100 0.09	Ep 35 thr 56	R	-98.64		avgR100	-193.66		eps 0.839		Qval 0.11		1
[DDQN-Mod] and100 0.08	Ep 36 thr 88	R	-161.80		avgR100	-192.78		eps 0.835		Qval -0.25		1
[DDQN-Mod] and100 0.08	Ep 37 thr 61	R	-301.25		avgR100	-195.71		eps 0.831		Qval 0.09		1
[DDQN-Mod] and100 0.08	Ep 38 thr 103	R	-183.17		avgR100	-195.38		eps 0.827		Qval 0.33		1
[DDQN-Mod] and100 0.08	Ep 39 thr 72	R	-90.21		avgR100	-192.68		eps 0.822		Qval 0.04		1
[DDQN-Mod] and100 0.07	Ep 40 thr 63	R	35.66		avgR100	-186.97		eps 0.818		Qval -0.07		1
[DDQN-Mod] and100 0.07	Ep 41 thr 66	R	-162.81		avgR100	-186.39		eps 0.814		Qval -0.00		1
[DDQN-Mod] and100 0.07	Ep 42 thr 58	R	-103.65		avgR100	-184.42		eps 0.810		Qval -0.10		1
[DDQN-Mod] and100 0.07	Ep 43 thr 52	R	-292.50		avgR100	-186.93		eps 0.806		Qval 0.08		1
[DDQN-Mod] and100 0.07	Ep 44 thr 59	R	-123.69		avgR100	-185.49		eps 0.802		Qval 0.09		1
[DDQN-Mod] and100 0.07	Ep 45 thr 94	R	-244.25		avgR100	-186.80		eps 0.798		Qval 0.19		1
[DDQN-Mod] and100 0.07	Ep 46 thr 119	R	-130.69		avgR100	-185.58		eps 0.794		Qval -0.07		1
[DDQN-Mod] and100 0.09	Ep 47 thr 72	R	-54.28		avgR100	-182.78		eps 0.790		Qval 0.49		1
[DDQN-Mod] and100 0.08	Ep 48 thr 73	R	-105.78		avgR100	-181.18		eps 0.786		Qval -0.08		1
[DDQN-Mod] and100 0.08	Ep 49 thr 55	R	-130.06		avgR100	-180.14		eps 0.782		Qval 0.20		1
[DDQN-Mod] and100 0.08	Ep 50 thr 80	R	-132.16		avgR100	-179.18		eps 0.778		Qval 0.20		1
[DDQN-Mod] and100 0.08	Ep 51 thr 54	R	-102.26		avgR100	-177.67		eps 0.774		Qval -0.21		1
[DDQN-Mod] and100 0.08	Ep 52 thr 85	R	-116.45		avgR100	-176.49		eps 0.771		Qval -0.03		1
[DDQN-Mod] and100 0.08	Ep 53 thr 83	R	-218.30		avgR100	-177.28		eps 0.767		Qval 0.01		1
[DDQN-Mod] and100 0.07	Ep 54 thr 63	R	-138.01		avgR100	-176.55		eps 0.763		Qval -0.24		1
[DDQN-Mod] and100 0.07	Ep 55 thr 111	R	-190.75		avgR100	-176.81		eps 0.759		Qval -0.21		1
[DDQN-Mod] and100 0.07	Ep 56 thr 76	R	-119.33		avgR100	-175.78		eps 0.755		Qval -0.02		1
[DDQN-Mod] and100 0.07	Ep 57 thr 95	R	-186.90		avgR100	-175.98		eps 0.751		Qval -0.00		1
[DDQN-Mod] and100 0.07	Ep 58 thr 132	R	-158.05		avgR100	-175.67		eps 0.748		Qval -0.09		1
[DDQN-Mod] and100 0.07	Ep 59 thr 74	R	-103.41		avgR100	-174.45		eps 0.744		Qval 0.10		1
[DDQN-Mod] and100 0.08	Ep 60 thr 117	R	-12.28		avgR100	-171.74		eps 0.740		Qval -0.18		1

[DDQN-Mod] and100 0.08	Ep 61 thr 60	R	-167.60		avgR100	-171.68		eps 0.737		Qval 0.11		1
[DDQN-Mod] and100 0.08	Ep 62 thr 78	R	-77.02		avgR100	-170.15		eps 0.733		Qval -0.06		1
[DDQN-Mod] and100 0.10	Ep 63 thr 53	R	22.71		avgR100	-167.09		eps 0.729		Qval -0.07		1
[DDQN-Mod] and100 0.09	Ep 64 thr 67	R	-107.44		avgR100	-166.16		eps 0.726		Qval -0.03		1
[DDQN-Mod] and100 0.09	Ep 65 thr 109	R	-77.49		avgR100	-164.79		eps 0.722		Qval -0.04		1
[DDQN-Mod] and100 0.09	Ep 66 thr 85	R	-109.24		avgR100	-163.95		eps 0.718		Qval -0.10		1
[DDQN-Mod] and100 0.09	Ep 67 thr 57	R	3.37		avgR100	-161.45		eps 0.715		Qval -0.08		1
[DDQN-Mod] and100 0.09	Ep 68 thr 113	R	-89.51		avgR100	-160.39		eps 0.711		Qval 0.11		1
[DDQN-Mod] and100 0.09	Ep 69 thr 61	R	-178.77		avgR100	-160.66		eps 0.708		Qval -0.24		1
[DDQN-Mod] and100 0.09	Ep 70 thr 77	R	-115.00		avgR100	-160.01		eps 0.704		Qval 0.09		1
[DDQN-Mod] and100 0.08	Ep 71 thr 90	R	-101.79		avgR100	-159.19		eps 0.701		Qval -0.05		1
[DDQN-Mod] and100 0.08	Ep 72 thr 94	R	-50.47		avgR100	-157.68		eps 0.697		Qval -0.01		1
[DDQN-Mod] and100 0.08	Ep 73 thr 115	R	-147.64		avgR100	-157.54		eps 0.694		Qval -0.02		1
[DDQN-Mod] and100 0.08	Ep 74 thr 102	R	-90.37		avgR100	-156.63		eps 0.690		Qval -0.13		1
[DDQN-Mod] and100 0.08	Ep 75 thr 110	R	-81.14		avgR100	-155.63		eps 0.687		Qval -0.22		1
[DDQN-Mod] and100 0.08	Ep 76 thr 96	R	-161.00		avgR100	-155.70		eps 0.683		Qval 0.25		1
[DDQN-Mod] and100 0.08	Ep 77 thr 87	R	-134.27		avgR100	-155.42		eps 0.680		Qval -0.33		1
[DDQN-Mod] and100 0.08	Ep 78 thr 56	R	-135.66		avgR100	-155.17		eps 0.676		Qval -0.13		1
[DDQN-Mod] and100 0.08	Ep 79 thr 70	R	-51.60		avgR100	-153.85		eps 0.673		Qval 0.08		1
[DDQN-Mod] and100 0.07	Ep 80 thr 103	R	-139.33		avgR100	-153.67		eps 0.670		Qval 0.51		1
[DDQN-Mod] and100 0.07	Ep 81 thr 97	R	-145.73		avgR100	-153.57		eps 0.666		Qval 0.05		1
[DDQN-Mod] and100 0.07	Ep 82 thr 117	R	-92.42		avgR100	-152.83		eps 0.663		Qval 0.23		1
[DDQN-Mod] and100 0.07	Ep 83 thr 91	R	-111.68		avgR100	-152.33		eps 0.660		Qval 0.43		1
[DDQN-Mod] and100 0.07	Ep 84 thr 109	R	-96.94		avgR100	-151.67		eps 0.656		Qval -0.37		1
[DDQN-Mod] and100 0.07	Ep 85 thr 117	R	-92.60		avgR100	-150.98		eps 0.653		Qval 0.30		1
[DDQN-Mod] and100 0.07	Ep 86 thr 149	R	3.65		avgR100	-149.18		eps 0.650		Qval 0.63		1
[DDQN-Mod] and100 0.07	Ep 87 thr 59	R	-132.05		avgR100	-148.98		eps 0.647		Qval 0.12		1
[DDQN-Mod] and100 0.07	Ep 88 thr 88	R	-130.52		avgR100	-148.77		eps 0.643		Qval 0.29		1
[DDQN-Mod] and100 0.07	Ep 89 thr 100	R	-93.43		avgR100	-148.15		eps 0.640		Qval 0.34		1
[DDQN-Mod] and100 0.07	Ep 90 thr 92	R	-80.85		avgR100	-147.40		eps 0.637		Qval 0.41		1

[DDQN-Mod] Ep 91 R -126.32 avgR100 -147.17 eps 0.634 Qval 0.36 1
and100 0.07 thr 61
[DDQN-Mod] Ep 92 R -164.52 avgR100 -147.36 eps 0.631 Qval 0.34 1
and100 0.07 thr 89
[DDQN-Mod] Ep 93 R -114.17 avgR100 -147.00 eps 0.627 Qval 0.38 1
and100 0.06 thr 110
[DDQN-Mod] Ep 94 R -168.89 avgR100 -147.24 eps 0.624 Qval 0.29 1
and100 0.06 thr 113
[DDQN-Mod] Ep 95 R -116.45 avgR100 -146.91 eps 0.621 Qval 0.67 1
and100 0.06 thr 122
[DDQN-Mod] Ep 96 R -102.96 avgR100 -146.46 eps 0.618 Qval 0.63 1
and100 0.06 thr 114
[DDQN-Mod] Ep 97 R -179.12 avgR100 -146.79 eps 0.615 Qval 0.26 1
and100 0.06 thr 127
[DDQN-Mod] Ep 98 R -122.55 avgR100 -146.54 eps 0.612 Qval 1.10 1
and100 0.06 thr 100
[DDQN-Mod] Ep 99 R -110.80 avgR100 -146.18 eps 0.609 Qval 0.59 1
and100 0.06 thr 127
[DDQN-Mod] Ep 100 R -119.09 avgR100 -145.91 eps 0.606 Qval 0.46 1
and100 0.06 thr 87
[DDQN-Mod] Ep 101 R -87.23 avgR100 -145.76 eps 0.603 Qval 0.82 1
and100 0.06 thr 80
[DDQN-Mod] Ep 102 R -100.68 avgR100 -145.78 eps 0.600 Qval 1.19 1
and100 0.06 thr 77
[DDQN-Mod] Ep 103 R -54.67 avgR100 -143.25 eps 0.597 Qval 0.97 1
and100 0.06 thr 69
[DDQN-Mod] Ep 104 R -122.99 avgR100 -142.82 eps 0.594 Qval 1.15 1
and100 0.06 thr 112
[DDQN-Mod] Ep 105 R -123.57 avgR100 -142.83 eps 0.591 Qval 0.69 1
and100 0.06 thr 95
[DDQN-Mod] Ep 106 R -165.37 avgR100 -141.67 eps 0.588 Qval 1.08 1
and100 0.06 thr 111
[DDQN-Mod] Ep 107 R -175.55 avgR100 -139.52 eps 0.585 Qval 0.77 1
and100 0.06 thr 101
[DDQN-Mod] Ep 108 R -71.39 avgR100 -139.59 eps 0.582 Qval 1.25 1
and100 0.05 thr 131
[DDQN-Mod] Ep 109 R -31.38 avgR100 -137.50 eps 0.579 Qval 1.61 1
and100 0.05 thr 127
[DDQN-Mod] Ep 110 R -72.94 avgR100 -137.01 eps 0.576 Qval 1.55 1
and100 0.05 thr 72
[DDQN-Mod] Ep 111 R -118.48 avgR100 -136.81 eps 0.573 Qval 1.96 1
and100 0.05 thr 84
[DDQN-Mod] Ep 112 R -107.84 avgR100 -134.53 eps 0.570 Qval 1.71 1
and100 0.05 thr 147
[DDQN-Mod] Ep 113 R -100.39 avgR100 -133.10 eps 0.568 Qval 1.91 1
and100 0.05 thr 75
[DDQN-Mod] Ep 114 R -56.63 avgR100 -131.71 eps 0.565 Qval 2.18 1
and100 0.05 thr 167
[DDQN-Mod] Ep 115 R -69.08 avgR100 -130.68 eps 0.562 Qval 2.02 1
and100 0.05 thr 76
[DDQN-Mod] Ep 116 R -167.97 avgR100 -127.44 eps 0.559 Qval 1.93 1
and100 0.05 thr 107
[DDQN-Mod] Ep 117 R -115.28 avgR100 -125.35 eps 0.556 Qval 1.77 1
and100 0.05 thr 150
[DDQN-Mod] Ep 118 R -48.58 avgR100 -123.08 eps 0.554 Qval 1.81 1
and100 0.05 thr 72
[DDQN-Mod] Ep 119 R -36.03 avgR100 -119.95 eps 0.551 Qval 1.98 1
and100 0.05 thr 96
[DDQN-Mod] Ep 120 R -160.24 avgR100 -121.13 eps 0.548 Qval 2.49 1
and100 0.04 thr 128

[DDQN-Mod] Ep 121 R -99.31 avgR100 -120.76 eps 0.545 Qval 1.93 1
and100 0.04 thr 62
[DDQN-Mod] Ep 122 R -107.79 avgR100 -120.68 eps 0.543 Qval 2.74 1
and100 0.04 thr 96
[DDQN-Mod] Ep 123 R -126.84 avgR100 -118.90 eps 0.540 Qval 2.20 1
and100 0.04 thr 79
[DDQN-Mod] Ep 124 R -92.69 avgR100 -117.98 eps 0.537 Qval 2.65 1
and100 0.04 thr 53
[DDQN-Mod] Ep 125 R -133.46 avgR100 -118.37 eps 0.534 Qval 2.62 1
and100 0.04 thr 92
[DDQN-Mod] Ep 126 R -193.89 avgR100 -118.32 eps 0.532 Qval 2.23 1
and100 0.04 thr 115
[DDQN-Mod] Ep 127 R -94.08 avgR100 -116.41 eps 0.529 Qval 2.90 1
and100 0.05 thr 147
[DDQN-Mod] Ep 128 R -43.32 avgR100 -115.03 eps 0.526 Qval 3.07 1
and100 0.05 thr 162
[DDQN-Mod] Ep 129 R -110.31 avgR100 -116.00 eps 0.524 Qval 2.83 1
and100 0.05 thr 98
[DDQN-Mod] Ep 130 R -85.46 avgR100 -116.64 eps 0.521 Qval 3.25 1
and100 0.04 thr 123
[DDQN-Mod] Ep 131 R -76.78 avgR100 -116.48 eps 0.519 Qval 2.84 1
and100 0.04 thr 103
[DDQN-Mod] Ep 132 R 25.25 avgR100 -113.98 eps 0.516 Qval 3.15 1
and100 0.05 thr 74
[DDQN-Mod] Ep 133 R -75.48 avgR100 -112.24 eps 0.513 Qval 3.22 1
and100 0.05 thr 57
[DDQN-Mod] Ep 134 R 29.23 avgR100 -110.83 eps 0.511 Qval 3.21 1
and100 0.06 thr 172
[DDQN-Mod] Ep 135 R -12.94 avgR100 -109.97 eps 0.508 Qval 3.41 1
and100 0.06 thr 127
[DDQN-Mod] Ep 136 R -74.16 avgR100 -109.10 eps 0.506 Qval 2.97 1
and100 0.06 thr 70
[DDQN-Mod] Ep 137 R -118.75 avgR100 -107.27 eps 0.503 Qval 3.34 1
and100 0.06 thr 145
[DDQN-Mod] Ep 138 R -84.37 avgR100 -106.28 eps 0.501 Qval 4.18 1
and100 0.06 thr 168
[DDQN-Mod] Ep 139 R -119.76 avgR100 -106.58 eps 0.498 Qval 4.24 1
and100 0.06 thr 95
[DDQN-Mod] Ep 140 R -45.30 avgR100 -107.39 eps 0.496 Qval 4.04 1
and100 0.06 thr 74
[DDQN-Mod] Ep 141 R -83.39 avgR100 -106.59 eps 0.493 Qval 4.22 1
and100 0.06 thr 93
[DDQN-Mod] Ep 142 R -101.04 avgR100 -106.57 eps 0.491 Qval 5.16 1
and100 0.06 thr 757
[DDQN-Mod] Ep 143 R -111.42 avgR100 -104.76 eps 0.488 Qval 4.97 1
and100 0.06 thr 99
[DDQN-Mod] Ep 144 R -96.95 avgR100 -104.49 eps 0.486 Qval 5.48 1
and100 0.06 thr 199
[DDQN-Mod] Ep 145 R -67.67 avgR100 -102.72 eps 0.483 Qval 5.45 1
and100 0.06 thr 89
[DDQN-Mod] Ep 146 R -84.92 avgR100 -102.27 eps 0.481 Qval 5.88 1
and100 0.06 thr 174
[DDQN-Mod] Ep 147 R -89.69 avgR100 -102.62 eps 0.479 Qval 6.15 1
and100 0.05 thr 121
[DDQN-Mod] Ep 148 R -4.64 avgR100 -101.61 eps 0.476 Qval 5.94 1
and100 0.05 thr 158
[DDQN-Mod] Ep 149 R -28.52 avgR100 -100.59 eps 0.474 Qval 6.10 1
and100 0.05 thr 76
[DDQN-Mod] Ep 150 R -85.60 avgR100 -100.13 eps 0.471 Qval 6.28 1
and100 0.05 thr 151

[DDQN-Mod] Ep 151 R and100 0.06 thr 122	68.82 avgR100	-98.42 eps 0.469 Qval	6.42 1
[DDQN-Mod] Ep 152 R and100 0.06 thr 86	-91.60 avgR100	-98.17 eps 0.467 Qval	6.53 1
[DDQN-Mod] Ep 153 R and100 0.06 thr 146	-68.46 avgR100	-96.67 eps 0.464 Qval	6.65 1
[DDQN-Mod] Ep 154 R and100 0.06 thr 86	-93.86 avgR100	-96.23 eps 0.462 Qval	6.66 1
[DDQN-Mod] Ep 155 R and100 0.06 thr 160	-10.49 avgR100	-94.43 eps 0.460 Qval	6.56 1
[DDQN-Mod] Ep 156 R and100 0.06 thr 210	-86.88 avgR100	-94.10 eps 0.458 Qval	7.38 1
[DDQN-Mod] Ep 157 R and100 0.06 thr 166	-81.72 avgR100	-93.05 eps 0.455 Qval	7.69 1
[DDQN-Mod] Ep 158 R and100 0.07 thr 94	43.30 avgR100	-91.04 eps 0.453 Qval	8.01 1
[DDQN-Mod] Ep 159 R and100 0.07 thr 87	-55.79 avgR100	-90.56 eps 0.451 Qval	7.56 1
[DDQN-Mod] Ep 160 R and100 0.06 thr 109	-85.51 avgR100	-91.29 eps 0.448 Qval	8.30 1
[DDQN-Mod] Ep 161 R and100 0.06 thr 60	-105.47 avgR100	-90.67 eps 0.446 Qval	7.94 1
[DDQN-Mod] Ep 162 R and100 0.06 thr 75	-21.93 avgR100	-90.12 eps 0.444 Qval	8.13 1
[DDQN-Mod] Ep 163 R and100 0.05 thr 105	-142.84 avgR100	-91.78 eps 0.442 Qval	8.30 1
[DDQN-Mod] Ep 164 R and100 0.05 thr 114	-54.08 avgR100	-91.24 eps 0.440 Qval	8.28 1
[DDQN-Mod] Ep 165 R and100 0.06 thr 274	-58.99 avgR100	-91.06 eps 0.437 Qval	8.81 1
[DDQN-Mod] Ep 166 R and100 0.06 thr 108	-30.81 avgR100	-90.27 eps 0.435 Qval	8.56 1
[DDQN-Mod] Ep 167 R and100 0.06 thr 746	-129.18 avgR100	-91.60 eps 0.433 Qval	9.39 1
[DDQN-Mod] Ep 168 R and100 0.07 thr 141	2.27 avgR100	-90.68 eps 0.431 Qval	10.02 1
[DDQN-Mod] Ep 169 R and100 0.07 thr 158	-23.95 avgR100	-89.13 eps 0.429 Qval	10.14 1
[DDQN-Mod] Ep 170 R and100 0.07 thr 166	-63.07 avgR100	-88.61 eps 0.427 Qval	10.34 1
[DDQN-Mod] Ep 171 R and100 0.07 thr 132	-126.25 avgR100	-88.86 eps 0.424 Qval	10.14 1
[DDQN-Mod] Ep 172 R and100 0.07 thr 147	-31.80 avgR100	-88.67 eps 0.422 Qval	10.29 1
[DDQN-Mod] Ep 173 R and100 0.07 thr 104	-70.83 avgR100	-87.90 eps 0.420 Qval	10.86 1
[DDQN-Mod] Ep 174 R and100 0.07 thr 114	-67.16 avgR100	-87.67 eps 0.418 Qval	10.05 1
[DDQN-Mod] Ep 175 R and100 0.07 thr 107	-48.63 avgR100	-87.35 eps 0.416 Qval	10.88 1
[DDQN-Mod] Ep 176 R and100 0.07 thr 78	-75.60 avgR100	-86.49 eps 0.414 Qval	11.03 1
[DDQN-Mod] Ep 177 R and100 0.07 thr 104	-117.35 avgR100	-86.32 eps 0.412 Qval	10.55 1
[DDQN-Mod] Ep 178 R and100 0.07 thr 143	-36.90 avgR100	-85.33 eps 0.410 Qval	11.20 1
[DDQN-Mod] Ep 179 R and100 0.07 thr 116	-13.23 avgR100	-84.95 eps 0.408 Qval	11.57 1
[DDQN-Mod] Ep 180 R and100 0.07 thr 110	-59.18 avgR100	-84.15 eps 0.406 Qval	11.48 1

[DDQN-Mod] Ep 181 R and100 0.07 thr 142	-17.67 avgR100	-82.87 eps 0.404 Qval 11.41 1
[DDQN-Mod] Ep 182 R and100 0.07 thr 690	-130.97 avgR100	-83.25 eps 0.402 Qval 12.02 1
[DDQN-Mod] Ep 183 R and100 0.08 thr 139	-12.97 avgR100	-82.27 eps 0.400 Qval 12.03 1
[DDQN-Mod] Ep 184 R and100 0.08 thr 257	-78.88 avgR100	-82.09 eps 0.398 Qval 12.43 1
[DDQN-Mod] Ep 185 R and100 0.08 thr 77	-39.56 avgR100	-81.56 eps 0.396 Qval 12.44 1
[DDQN-Mod] Ep 186 R and100 0.08 thr 210	-48.22 avgR100	-82.07 eps 0.394 Qval 12.60 1
[DDQN-Mod] Ep 187 R and100 0.08 thr 127	-10.29 avgR100	-80.86 eps 0.392 Qval 12.92 1
[DDQN-Mod] Ep 188 R and100 0.08 thr 112	-79.38 avgR100	-80.35 eps 0.390 Qval 13.37 1
[DDQN-Mod] Ep 189 R and100 0.08 thr 127	-61.25 avgR100	-80.02 eps 0.388 Qval 13.23 1
[DDQN-Mod] Ep 190 R and100 0.08 thr 290	-81.17 avgR100	-80.03 eps 0.386 Qval 13.59 1
[DDQN-Mod] Ep 191 R and100 0.08 thr 84	-25.65 avgR100	-79.02 eps 0.384 Qval 13.50 1
[DDQN-Mod] Ep 192 R and100 0.08 thr 101	-29.45 avgR100	-77.67 eps 0.382 Qval 14.33 1
[DDQN-Mod] Ep 193 R and100 0.08 thr 106	-47.83 avgR100	-77.01 eps 0.380 Qval 13.19 1
[DDQN-Mod] Ep 194 R and100 0.08 thr 119	-96.53 avgR100	-76.28 eps 0.378 Qval 14.17 1
[DDQN-Mod] Ep 195 R and100 0.08 thr 685	-165.61 avgR100	-76.77 eps 0.376 Qval 13.59 1
[DDQN-Mod] Ep 196 R and100 0.08 thr 110	-53.83 avgR100	-76.28 eps 0.374 Qval 13.72 1
[DDQN-Mod] Ep 197 R and100 0.08 thr 67	-60.09 avgR100	-75.09 eps 0.373 Qval 13.79 1
[DDQN-Mod] Ep 198 R and100 0.08 thr 79	-52.13 avgR100	-74.39 eps 0.371 Qval 14.29 1
[DDQN-Mod] Ep 199 R and100 0.08 thr 67	-51.65 avgR100	-73.80 eps 0.369 Qval 14.07 1
[DDQN-Mod] Ep 200 R and100 0.08 thr 102	-9.04 avgR100	-72.70 eps 0.367 Qval 14.15 1
[DDQN-Mod] Ep 201 R and100 0.08 thr 168	-72.08 avgR100	-72.55 eps 0.365 Qval 14.29 1
[DDQN-Mod] Ep 202 R and100 0.08 thr 529	-59.00 avgR100	-72.13 eps 0.363 Qval 13.52 1
[DDQN-Mod] Ep 203 R and100 0.08 thr 76	-28.73 avgR100	-71.87 eps 0.361 Qval 14.64 1
[DDQN-Mod] Ep 204 R and100 0.08 thr 203	-40.77 avgR100	-71.05 eps 0.360 Qval 14.67 1
[DDQN-Mod] Ep 205 R and100 0.08 thr 156	-77.19 avgR100	-70.58 eps 0.358 Qval 14.95 1
[DDQN-Mod] Ep 206 R and100 0.08 thr 111	-41.72 avgR100	-69.35 eps 0.356 Qval 14.77 1
[DDQN-Mod] Ep 207 R and100 0.08 thr 679	-125.82 avgR100	-68.85 eps 0.354 Qval 15.25 1
[DDQN-Mod] Ep 208 R and100 0.08 thr 107	-52.24 avgR100	-68.66 eps 0.353 Qval 15.18 1
[DDQN-Mod] Ep 209 R and100 0.08 thr 137	-35.90 avgR100	-68.70 eps 0.351 Qval 15.26 1
[DDQN-Mod] Ep 210 R and100 0.08 thr 372	-198.08 avgR100	-69.95 eps 0.349 Qval 15.61 1

[DDQN-Mod] Ep 211 R and100 0.08 thr 94	-62.94 avgR100	-69.40 eps 0.347 Qval 14.83 1
[DDQN-Mod] Ep 212 R and100 0.08 thr 101	-64.47 avgR100	-68.97 eps 0.346 Qval 15.16 1
[DDQN-Mod] Ep 213 R and100 0.08 thr 85	-72.67 avgR100	-68.69 eps 0.344 Qval 14.96 1
[DDQN-Mod] Ep 214 R and100 0.08 thr 700	-133.87 avgR100	-69.46 eps 0.342 Qval 16.00 1
[DDQN-Mod] Ep 215 R and100 0.08 thr 102	-45.17 avgR100	-69.22 eps 0.340 Qval 15.89 1
[DDQN-Mod] Ep 216 R and100 0.08 thr 85	-80.14 avgR100	-68.34 eps 0.339 Qval 15.69 1
[DDQN-Mod] Ep 217 R and100 0.08 thr 101	-54.79 avgR100	-67.74 eps 0.337 Qval 15.71 1
[DDQN-Mod] Ep 218 R and100 0.08 thr 70	-94.72 avgR100	-68.20 eps 0.335 Qval 15.07 1
[DDQN-Mod] Ep 219 R and100 0.09 thr 105	11.95 avgR100	-67.72 eps 0.334 Qval 15.06 1
[DDQN-Mod] Ep 220 R and100 0.09 thr 128	-76.44 avgR100	-66.88 eps 0.332 Qval 15.78 1
[DDQN-Mod] Ep 221 R and100 0.10 thr 278	209.75 avgR100	-63.79 eps 0.330 Qval 15.59 1
[DDQN-Mod] Ep 222 R and100 0.10 thr 92	0.65 avgR100	-62.71 eps 0.329 Qval 15.55 1
[DDQN-Mod] Ep 223 R and100 0.10 thr 93	-52.61 avgR100	-61.96 eps 0.327 Qval 15.80 1
[DDQN-Mod] Ep 224 R and100 0.10 thr 131	-29.93 avgR100	-61.34 eps 0.325 Qval 15.31 1
[DDQN-Mod] Ep 225 R and100 0.10 thr 125	-98.94 avgR100	-60.99 eps 0.324 Qval 15.61 1
[DDQN-Mod] Ep 226 R and100 0.10 thr 92	-83.57 avgR100	-59.89 eps 0.322 Qval 15.60 1
[DDQN-Mod] Ep 227 R and100 0.09 thr 605	-231.85 avgR100	-61.27 eps 0.321 Qval 15.74 1
[DDQN-Mod] Ep 228 R and100 0.09 thr 91	-32.92 avgR100	-61.16 eps 0.319 Qval 14.80 1
[DDQN-Mod] Ep 229 R and100 0.09 thr 119	-75.43 avgR100	-60.81 eps 0.317 Qval 15.47 1
[DDQN-Mod] Ep 230 R and100 0.09 thr 133	-37.27 avgR100	-60.33 eps 0.316 Qval 15.35 1
[DDQN-Mod] Ep 231 R and100 0.09 thr 74	-48.55 avgR100	-60.05 eps 0.314 Qval 14.71 1
[DDQN-Mod] Ep 232 R and100 0.08 thr 115	-42.96 avgR100	-60.73 eps 0.313 Qval 15.19 1
[DDQN-Mod] Ep 233 R and100 0.08 thr 153	-39.14 avgR100	-60.37 eps 0.311 Qval 14.90 1
[DDQN-Mod] Ep 234 R and100 0.07 thr 91	-34.23 avgR100	-61.00 eps 0.309 Qval 15.25 1
[DDQN-Mod] Ep 235 R and100 0.07 thr 426	-371.36 avgR100	-64.59 eps 0.308 Qval 14.46 1
[DDQN-Mod] Ep 236 R and100 0.07 thr 603	-159.88 avgR100	-65.44 eps 0.306 Qval 14.29 1
[DDQN-Mod] Ep 237 R and100 0.07 thr 103	-64.18 avgR100	-64.90 eps 0.305 Qval 14.69 1
[DDQN-Mod] Ep 238 R and100 0.07 thr 133	-0.91 avgR100	-64.06 eps 0.303 Qval 13.87 1
[DDQN-Mod] Ep 239 R and100 0.07 thr 98	-76.36 avgR100	-63.63 eps 0.302 Qval 14.30 1
[DDQN-Mod] Ep 240 R and100 0.07 thr 77	-43.79 avgR100	-63.61 eps 0.300 Qval 14.53 1

[DDQN-Mod]	Ep	241		R	-117.30		avgR100	-63.95		eps 0.299		Qval	14.37		1
and100 0.07		thr 618													
[DDQN-Mod]	Ep	242		R	98.69		avgR100	-61.96		eps 0.297		Qval	13.64		1
and100 0.07		thr 278													
[DDQN-Mod]	Ep	243		R	-85.75		avgR100	-61.70		eps 0.296		Qval	14.37		1
and100 0.07		thr 82													
[DDQN-Mod]	Ep	244		R	-97.77		avgR100	-61.71		eps 0.294		Qval	13.32		1
and100 0.07		thr 64													
[DDQN-Mod]	Ep	245		R	-118.89		avgR100	-62.22		eps 0.293		Qval	14.42		1
and100 0.07		thr 104													
[DDQN-Mod]	Ep	246		R	-20.67		avgR100	-61.58		eps 0.291		Qval	13.19		1
and100 0.07		thr 100													
[DDQN-Mod]	Ep	247		R	211.71		avgR100	-58.56		eps 0.290		Qval	13.27		1
and100 0.08		thr 293													
[DDQN-Mod]	Ep	248		R	-201.04		avgR100	-60.53		eps 0.288		Qval	13.37		1
and100 0.08		thr 627													
[DDQN-Mod]	Ep	249		R	-22.53		avgR100	-60.47		eps 0.287		Qval	13.68		1
and100 0.08		thr 91													
[DDQN-Mod]	Ep	250		R	-17.13		avgR100	-59.78		eps 0.286		Qval	13.12		1
and100 0.08		thr 108													
[DDQN-Mod]	Ep	251		R	-29.19		avgR100	-60.76		eps 0.284		Qval	13.56		1
and100 0.07		thr 121													
[DDQN-Mod]	Ep	252		R	-199.11		avgR100	-61.84		eps 0.283		Qval	12.57		1
and100 0.07		thr 406													
[DDQN-Mod]	Ep	253		R	-18.52		avgR100	-61.34		eps 0.281		Qval	12.96		1
and100 0.07		thr 120													
[DDQN-Mod]	Ep	254		R	-66.87		avgR100	-61.07		eps 0.280		Qval	12.48		1
and100 0.07		thr 75													
[DDQN-Mod]	Ep	255		R	-107.69		avgR100	-62.04		eps 0.279		Qval	13.16		1
and100 0.07		thr 109													
[DDQN-Mod]	Ep	256		R	-21.25		avgR100	-61.38		eps 0.277		Qval	11.97		1
and100 0.07		thr 128													
[DDQN-Mod]	Ep	257		R	-30.21		avgR100	-60.87		eps 0.276		Qval	11.64		1
and100 0.07		thr 100													
[DDQN-Mod]	Ep	258		R	-32.40		avgR100	-61.63		eps 0.274		Qval	12.53		1
and100 0.06		thr 90													
[DDQN-Mod]	Ep	259		R	-91.51		avgR100	-61.98		eps 0.273		Qval	12.80		1
and100 0.06		thr 87													
[DDQN-Mod]	Ep	260		R	-46.13		avgR100	-61.59		eps 0.272		Qval	11.83		1
and100 0.06		thr 108													
[DDQN-Mod]	Ep	261		R	-41.41		avgR100	-60.95		eps 0.270		Qval	11.50		1
and100 0.06		thr 82													
[DDQN-Mod]	Ep	262		R	-163.50		avgR100	-62.36		eps 0.269		Qval	11.43		1
and100 0.06		thr 656													
[DDQN-Mod]	Ep	263		R	-45.24		avgR100	-61.39		eps 0.268		Qval	10.79		1
and100 0.06		thr 89													
[DDQN-Mod]	Ep	264		R	-197.11		avgR100	-62.82		eps 0.266		Qval	10.16		1
and100 0.06		thr 771													
[DDQN-Mod]	Ep	265		R	-42.13		avgR100	-62.65		eps 0.265		Qval	9.58		1
and100 0.05		thr 112													
[DDQN-Mod]	Ep	266		R	-51.78		avgR100	-62.86		eps 0.264		Qval	9.97		1
and100 0.05		thr 645													
[DDQN-Mod]	Ep	267		R	-168.41		avgR100	-63.25		eps 0.262		Qval	9.33		1
and100 0.05		thr 694													
[DDQN-Mod]	Ep	268		R	-139.47		avgR100	-64.67		eps 0.261		Qval	9.10		1
and100 0.04		thr 206													
[DDQN-Mod]	Ep	269		R	-214.91		avgR100	-66.58		eps 0.260		Qval	7.68		1
and100 0.04		thr 731													
[DDQN-Mod]	Ep	270		R	26.13		avgR100	-65.69		eps 0.258		Qval	7.80		1
and100 0.05		thr 98													

[DDQN-Mod]	Ep	271		R	-73.91		avgR100	-65.16		eps 0.257		Qval	8.03		1
and100 0.05		thr 85													
[DDQN-Mod]	Ep	272		R	-110.23		avgR100	-65.95		eps 0.256		Qval	7.76		1
and100 0.05		thr 248													
[DDQN-Mod]	Ep	273		R	-209.50		avgR100	-67.34		eps 0.255		Qval	7.77		1
and100 0.05		thr 84													
[DDQN-Mod]	Ep	274		R	-276.14		avgR100	-69.42		eps 0.253		Qval	5.86		1
and100 0.05		thr 780													
[DDQN-Mod]	Ep	275		R	-23.49		avgR100	-69.17		eps 0.252		Qval	5.79		1
and100 0.05		thr 110													
[DDQN-Mod]	Ep	276		R	-57.11		avgR100	-68.99		eps 0.251		Qval	5.51		1
and100 0.05		thr 62													
[DDQN-Mod]	Ep	277		R	183.62		avgR100	-65.98		eps 0.249		Qval	5.42		1
and100 0.06		thr 373													
[DDQN-Mod]	Ep	278		R	-79.45		avgR100	-66.40		eps 0.248		Qval	5.23		1
and100 0.06		thr 86													
[DDQN-Mod]	Ep	279		R	-42.86		avgR100	-66.70		eps 0.247		Qval	4.79		1
and100 0.06		thr 95													
[DDQN-Mod]	Ep	280		R	-279.43		avgR100	-68.90		eps 0.246		Qval	3.96		1
and100 0.06		thr 613													
[DDQN-Mod]	Ep	281		R	-56.78		avgR100	-69.29		eps 0.245		Qval	3.91		1
and100 0.06		thr 531													
[DDQN-Mod]	Ep	282		R	-44.99		avgR100	-68.43		eps 0.243		Qval	3.79		1
and100 0.06		thr 132													
[DDQN-Mod]	Ep	283		R	-21.11		avgR100	-68.52		eps 0.242		Qval	3.94		1
and100 0.05		thr 76													
[DDQN-Mod]	Ep	284		R	-102.95		avgR100	-68.76		eps 0.241		Qval	2.90		1
and100 0.05		thr 118													
[DDQN-Mod]	Ep	285		R	8.29		avgR100	-68.28		eps 0.240		Qval	1.85		1
and100 0.05		thr 501													
[DDQN-Mod]	Ep	286		R	-24.02		avgR100	-68.04		eps 0.238		Qval	0.69		1
and100 0.05		thr 95													
[DDQN-Mod]	Ep	287		R	-48.26		avgR100	-68.42		eps 0.237		Qval	2.34		1
and100 0.05		thr 109													
[DDQN-Mod]	Ep	288		R	-121.99		avgR100	-68.84		eps 0.236		Qval	2.05		1
and100 0.05		thr 138													
[DDQN-Mod]	Ep	289		R	-129.80		avgR100	-69.53		eps 0.235		Qval	-0.83		1
and100 0.05		thr 643													
[DDQN-Mod]	Ep	290		R	-53.89		avgR100	-69.25		eps 0.234		Qval	0.18		1
and100 0.05		thr 97													
[DDQN-Mod]	Ep	291		R	-109.91		avgR100	-70.10		eps 0.233		Qval	-0.15		1
and100 0.05		thr 224													
[DDQN-Mod]	Ep	292		R	-219.01		avgR100	-71.99		eps 0.231		Qval	-0.49		1
and100 0.05		thr 722													
[DDQN-Mod]	Ep	293		R	-188.32		avgR100	-73.40		eps 0.230		Qval	-1.57		1
and100 0.05		thr 592													
[DDQN-Mod]	Ep	294		R	-59.62		avgR100	-73.03		eps 0.229		Qval	-1.48		1
and100 0.05		thr 74													
[DDQN-Mod]	Ep	295		R	-250.18		avgR100	-73.87		eps 0.228		Qval	-3.34		1
and100 0.05		thr 783													
[DDQN-Mod]	Ep	296		R	-92.10		avgR100	-74.26		eps 0.227		Qval	-3.25		1
and100 0.05		thr 68													
[DDQN-Mod]	Ep	297		R	-78.14		avgR100	-74.44		eps 0.226		Qval	-3.14		1
and100 0.05		thr 110													
[DDQN-Mod]	Ep	298		R	-8.52		avgR100	-74.00		eps 0.225		Qval	-3.32		1
and100 0.05		thr 85													
[DDQN-Mod]	Ep	299		R	-58.55		avgR100	-74.07		eps 0.223		Qval	-3.39		1
and100 0.05		thr 74													
[DDQN-Mod]	Ep	300		R	-50.91		avgR100	-74.49		eps 0.222		Qval	-3.36		1
and100 0.05		thr 112													

[DDQN-Mod]	Ep	301		R	-255.59		avgR100	-76.32		eps 0.221		Qval	-5.29		1
and100 0.05		thr 777													
[DDQN-Mod]	Ep	302		R	-153.47		avgR100	-77.27		eps 0.220		Qval	-5.80		1
and100 0.05		thr 765													
[DDQN-Mod]	Ep	303		R	187.32		avgR100	-75.11		eps 0.219		Qval	-6.43		1
and100 0.06		thr 316													
[DDQN-Mod]	Ep	304		R	-107.67		avgR100	-75.78		eps 0.218		Qval	-6.51		1
and100 0.06		thr 362													
[DDQN-Mod]	Ep	305		R	-132.75		avgR100	-76.33		eps 0.217		Qval	-6.85		1
and100 0.06		thr 387													
[DDQN-Mod]	Ep	306		R	-192.28		avgR100	-77.84		eps 0.216		Qval	-8.06		1
and100 0.06		thr 798													
[DDQN-Mod]	Ep	307		R	-74.88		avgR100	-77.33		eps 0.215		Qval	-8.31		1
and100 0.06		thr 101													
[DDQN-Mod]	Ep	308		R	-172.20		avgR100	-78.53		eps 0.214		Qval	-8.65		1
and100 0.06		thr 715													
[DDQN-Mod]	Ep	309		R	-44.24		avgR100	-78.61		eps 0.212		Qval	-9.06		1
and100 0.06		thr 126													
[DDQN-Mod]	Ep	310		R	-22.60		avgR100	-76.86		eps 0.211		Qval	-10.37		1
and100 0.07		thr 712													
[DDQN-Mod]	Ep	311		R	-79.76		avgR100	-77.03		eps 0.210		Qval	-10.77		1
and100 0.07		thr 73													
[DDQN-Mod]	Ep	312		R	-43.43		avgR100	-76.81		eps 0.209		Qval	-9.73		1
and100 0.07		thr 107													
[DDQN-Mod]	Ep	313		R	-7.94		avgR100	-76.17		eps 0.208		Qval	-10.42		1
and100 0.07		thr 128													
[DDQN-Mod]	Ep	314		R	103.09		avgR100	-73.80		eps 0.207		Qval	-10.47		1
and100 0.08		thr 533													
[DDQN-Mod]	Ep	315		R	-1.83		avgR100	-73.36		eps 0.206		Qval	-11.61		1
and100 0.08		thr 102													
[DDQN-Mod]	Ep	316		R	-181.03		avgR100	-74.37		eps 0.205		Qval	-11.68		1
and100 0.08		thr 751													
[DDQN-Mod]	Ep	317		R	43.55		avgR100	-73.39		eps 0.204		Qval	-11.69		1
and100 0.09		thr 596													
[DDQN-Mod]	Ep	318		R	-35.55		avgR100	-72.80		eps 0.203		Qval	-12.21		1
and100 0.09		thr 99													
[DDQN-Mod]	Ep	319		R	-239.21		avgR100	-75.31		eps 0.202		Qval	-13.90		1
and100 0.08		thr 834													
[DDQN-Mod]	Ep	320		R	-261.41		avgR100	-77.16		eps 0.201		Qval	-14.82		1
and100 0.08		thr 829													
[DDQN-Mod]	Ep	321		R	-263.17		avgR100	-81.89		eps 0.200		Qval	-15.03		1
and100 0.07		thr 831													
[DDQN-Mod]	Ep	322		R	-10.34		avgR100	-82.00		eps 0.199		Qval	-16.23		1
and100 0.07		thr 118													
[DDQN-Mod]	Ep	323		R	-71.76		avgR100	-82.19		eps 0.198		Qval	-16.07		1
and100 0.07		thr 80													
[DDQN-Mod]	Ep	324		R	-218.20		avgR100	-84.07		eps 0.197		Qval	-17.16		1
and100 0.07		thr 816													
[DDQN-Mod]	Ep	325		R	-87.76		avgR100	-83.96		eps 0.196		Qval	-17.06		1
and100 0.07		thr 105													
[DDQN-Mod]	Ep	326		R	-64.99		avgR100	-83.78		eps 0.195		Qval	-16.98		1
and100 0.07		thr 117													
[DDQN-Mod]	Ep	327		R	-219.93		avgR100	-83.66		eps 0.194		Qval	-18.11		1
and100 0.07		thr 840													
[DDQN-Mod]	Ep	328		R	35.98		avgR100	-82.97		eps 0.193		Qval	-19.06		1
and100 0.07		thr 382													
[DDQN-Mod]	Ep	329		R	-223.35		avgR100	-84.45		eps 0.192		Qval	-19.88		1
and100 0.07		thr 737													
[DDQN-Mod]	Ep	330		R	-71.68		avgR100	-84.79		eps 0.191		Qval	-20.79		1
and100 0.07		thr 101													

[DDQN-Mod] Ep and100 0.07	331 R thr 98	-100.14	avgR100	-85.31	eps 0.190	Qval -20.14	1
[DDQN-Mod] Ep and100 0.07	332 R thr 205	-29.71	avgR100	-85.17	eps 0.189	Qval -19.87	1
[DDQN-Mod] Ep and100 0.07	333 R thr 734	-213.82	avgR100	-86.92	eps 0.188	Qval -21.38	1
[DDQN-Mod] Ep and100 0.07	334 R thr 129	-28.40	avgR100	-86.86	eps 0.187	Qval -21.49	1
[DDQN-Mod] Ep and100 0.08	335 R thr 309	171.26	avgR100	-81.44	eps 0.187	Qval -22.19	1
[DDQN-Mod] Ep and100 0.08	336 R thr 112	-74.55	avgR100	-80.58	eps 0.186	Qval -21.94	1
[DDQN-Mod] Ep and100 0.09	337 R thr 579	64.23	avgR100	-79.30	eps 0.185	Qval -23.17	1
[DDQN-Mod] Ep and100 0.10	338 R thr 331	140.78	avgR100	-77.88	eps 0.184	Qval -23.14	1
[DDQN-Mod] Ep and100 0.10	339 R thr 763	-210.96	avgR100	-79.23	eps 0.183	Qval -23.98	1
[DDQN-Mod] Ep and100 0.10	340 R thr 796	-281.50	avgR100	-81.61	eps 0.182	Qval -24.65	1
[DDQN-Mod] Ep and100 0.10	341 R thr 127	-49.57	avgR100	-80.93	eps 0.181	Qval -24.11	1
[DDQN-Mod] Ep and100 0.11	342 R thr 215	271.65	avgR100	-79.20	eps 0.180	Qval -25.35	1
[DDQN-Mod] Ep and100 0.11	343 R thr 87	-41.03	avgR100	-78.75	eps 0.179	Qval -25.56	1
[DDQN-Mod] Ep and100 0.11	344 R thr 883	-330.15	avgR100	-81.07	eps 0.178	Qval -27.12	1
[DDQN-Mod] Ep and100 0.11	345 R thr 97	-94.55	avgR100	-80.83	eps 0.177	Qval -26.83	1
[DDQN-Mod] Ep and100 0.11	346 R thr 821	-271.16	avgR100	-83.34	eps 0.177	Qval -28.24	1
[DDQN-Mod] Ep and100 0.11	347 R thr 496	140.42	avgR100	-84.05	eps 0.176	Qval -28.09	1
[DDQN-Mod] Ep and100 0.11	348 R thr 791	-259.21	avgR100	-84.63	eps 0.175	Qval -31.48	1
[DDQN-Mod] Ep and100 0.11	349 R thr 820	-289.29	avgR100	-87.30	eps 0.174	Qval -30.96	1
[DDQN-Mod] Ep and100 0.11	350 R thr 145	-40.82	avgR100	-87.54	eps 0.173	Qval -31.68	1
[DDQN-Mod] Ep and100 0.11	351 R thr 702	-184.42	avgR100	-89.09	eps 0.172	Qval -32.70	1
[DDQN-Mod] Ep and100 0.12	352 R thr 641	7.10	avgR100	-87.03	eps 0.171	Qval -33.21	1
[DDQN-Mod] Ep and100 0.12	353 R thr 107	-52.51	avgR100	-87.37	eps 0.170	Qval -32.83	1
[DDQN-Mod] Ep and100 0.12	354 R thr 772	-294.13	avgR100	-89.64	eps 0.170	Qval -34.67	1
[DDQN-Mod] Ep and100 0.12	355 R thr 484	97.62	avgR100	-87.58	eps 0.169	Qval -35.61	1
[DDQN-Mod] Ep and100 0.12	356 R thr 826	-237.44	avgR100	-89.75	eps 0.168	Qval -37.46	1
[DDQN-Mod] Ep and100 0.12	357 R thr 824	-318.66	avgR100	-92.63	eps 0.167	Qval -38.59	1
[DDQN-Mod] Ep and100 0.12	358 R thr 164	-20.11	avgR100	-92.51	eps 0.166	Qval -38.64	1
[DDQN-Mod] Ep and100 0.13	359 R thr 572	71.10	avgR100	-90.88	eps 0.165	Qval -37.74	1
[DDQN-Mod] Ep and100 0.14	360 R thr 647	54.52	avgR100	-89.88	eps 0.165	Qval -39.91	1

[DDQN-Mod] Ep and100 0.14	361 R thr 810	-301.75	avgR100	-92.48	eps 0.164	Qval -40.71	1
[DDQN-Mod] Ep and100 0.15	362 R thr 212	295.16	avgR100	-87.89	eps 0.163	Qval -41.35	1
[DDQN-Mod] Ep and100 0.16	363 R thr 569	32.79	avgR100	-87.11	eps 0.162	Qval -42.40	1
[DDQN-Mod] Ep and100 0.16	364 R thr 793	-285.11	avgR100	-87.99	eps 0.161	Qval -42.81	1
[DDQN-Mod] Ep and100 0.16	365 R thr 732	-243.77	avgR100	-90.01	eps 0.160	Qval -43.96	1
[DDQN-Mod] Ep and100 0.16	366 R thr 819	-350.74	avgR100	-93.00	eps 0.160	Qval -44.47	1
[DDQN-Mod] Ep and100 0.16	367 R thr 592	-378.43	avgR100	-95.10	eps 0.159	Qval -44.01	1
[DDQN-Mod] Ep and100 0.16	368 R thr 819	-317.13	avgR100	-96.88	eps 0.158	Qval -45.27	1
[DDQN-Mod] Ep and100 0.16	369 R thr 853	-338.87	avgR100	-98.11	eps 0.157	Qval -47.04	1
[DDQN-Mod] Ep and100 0.15	370 R thr 630	-122.62	avgR100	-99.60	eps 0.157	Qval -45.40	1
[DDQN-Mod] Ep and100 0.16	371 R thr 91	85.44	avgR100	-98.01	eps 0.156	Qval -47.39	1
[DDQN-Mod] Ep and100 0.17	372 R thr 367	198.31	avgR100	-94.92	eps 0.155	Qval -47.21	1
[DDQN-Mod] Ep and100 0.17	373 R thr 98	-88.80	avgR100	-93.72	eps 0.154	Qval -46.82	1
[DDQN-Mod] Ep and100 0.17	374 R thr 113	-34.66	avgR100	-91.30	eps 0.153	Qval -45.94	1
[DDQN-Mod] Ep and100 0.17	375 R thr 664	-169.85	avgR100	-92.77	eps 0.153	Qval -48.95	1
[DDQN-Mod] Ep and100 0.17	376 R thr 279	-194.17	avgR100	-94.14	eps 0.152	Qval -49.13	1
[DDQN-Mod] Ep and100 0.16	377 R thr 805	-313.38	avgR100	-99.11	eps 0.151	Qval -50.33	1
[DDQN-Mod] Ep and100 0.16	378 R thr 127	-16.31	avgR100	-98.47	eps 0.150	Qval -50.93	1
[DDQN-Mod] Ep and100 0.16	379 R thr 91	-50.23	avgR100	-98.55	eps 0.150	Qval -49.44	1
[DDQN-Mod] Ep and100 0.16	380 R thr 686	-402.73	avgR100	-99.78	eps 0.149	Qval -49.62	1
[DDQN-Mod] Ep and100 0.16	381 R thr 107	-7.79	avgR100	-99.29	eps 0.148	Qval -49.75	1
[DDQN-Mod] Ep and100 0.16	382 R thr 774	-230.15	avgR100	-101.14	eps 0.147	Qval -51.08	1
[DDQN-Mod] Ep and100 0.17	383 R thr 532	65.02	avgR100	-100.28	eps 0.147	Qval -52.73	1
[DDQN-Mod] Ep and100 0.17	384 R thr 765	-182.21	avgR100	-101.07	eps 0.146	Qval -52.75	1
[DDQN-Mod] Ep and100 0.17	385 R thr 723	-211.84	avgR100	-103.28	eps 0.145	Qval -53.30	1
[DDQN-Mod] Ep and100 0.17	386 R thr 831	-264.81	avgR100	-105.68	eps 0.144	Qval -54.10	1
[DDQN-Mod] Ep and100 0.17	387 R thr 776	-487.08	avgR100	-110.07	eps 0.144	Qval -54.25	1
[DDQN-Mod] Ep and100 0.17	388 R thr 803	-280.64	avgR100	-111.66	eps 0.143	Qval -55.08	1
[DDQN-Mod] Ep and100 0.17	389 R thr 424	-308.27	avgR100	-113.44	eps 0.142	Qval -56.77	1
[DDQN-Mod] Ep and100 0.17	390 R thr 840	-329.70	avgR100	-116.20	eps 0.142	Qval -56.84	1

[DDQN-Mod] and100	Ep 0.17	R thr 807	-281.55		avgR100	-117.92		eps 0.141		Qval	-56.86		1
[DDQN-Mod] and100	Ep 0.17	R thr 535	-322.07		avgR100	-118.95		eps 0.140		Qval	-57.88		1
[DDQN-Mod] and100	Ep 0.18	R thr 431	167.59		avgR100	-115.39		eps 0.139		Qval	-60.20		1
[DDQN-Mod] and100	Ep 0.18	R thr 126	-63.42		avgR100	-115.43		eps 0.139		Qval	-59.63		1
[DDQN-Mod] and100	Ep 0.18	R thr 772	-225.58		avgR100	-115.18		eps 0.138		Qval	-58.88		1
[DDQN-Mod] and100	Ep 0.18	R thr 817	-342.33		avgR100	-117.68		eps 0.137		Qval	-59.15		1
[DDQN-Mod] and100	Ep 0.18	R thr 239	-140.46		avgR100	-118.31		eps 0.137		Qval	-61.24		1
[DDQN-Mod] and100	Ep 0.18	R thr 210	-83.10		avgR100	-119.05		eps 0.136		Qval	-60.04		1
[DDQN-Mod] and100	Ep 0.19	R thr 248	193.60		avgR100	-116.53		eps 0.135		Qval	-61.13		1
[DDQN-Mod] and100	Ep 0.19	R thr 803	-359.06		avgR100	-119.61		eps 0.135		Qval	-60.44		1
[DDQN-Mod] and100	Ep 0.19	R thr 764	-178.94		avgR100	-118.85		eps 0.134		Qval	-62.22		1
[DDQN-Mod] and100	Ep 0.19	R thr 792	-219.81		avgR100	-119.51		eps 0.133		Qval	-62.51		1
[DDQN-Mod] and100	Ep 0.18	R thr 223	181.63		avgR100	-119.57		eps 0.133		Qval	-63.15		1
[DDQN-Mod] and100	Ep 0.19	R thr 384	176.88		avgR100	-116.72		eps 0.132		Qval	-61.79		1
[DDQN-Mod] and100	Ep 0.19	R thr 89	-26.41		avgR100	-115.66		eps 0.131		Qval	-62.15		1
[DDQN-Mod] and100	Ep 0.19	R thr 350	-197.57		avgR100	-115.71		eps 0.131		Qval	-62.80		1
[DDQN-Mod] and100	Ep 0.19	R thr 724	-238.29		avgR100	-117.34		eps 0.130		Qval	-64.29		1
[DDQN-Mod] and100	Ep 0.20	R thr 325	168.21		avgR100	-113.94		eps 0.129		Qval	-63.72		1
[DDQN-Mod] and100	Ep 0.20	R thr 809	-262.67		avgR100	-116.12		eps 0.129		Qval	-66.01		1
[DDQN-Mod] and100	Ep 0.19	R thr 786	-170.44		avgR100	-117.60		eps 0.128		Qval	-64.42		1
[DDQN-Mod] and100	Ep 0.19	R thr 792	-229.85		avgR100	-119.10		eps 0.127		Qval	-65.16		1
[DDQN-Mod] and100	Ep 0.20	R thr 412	105.95		avgR100	-117.61		eps 0.127		Qval	-66.52		1
[DDQN-Mod] and100	Ep 0.20	R thr 158	-68.42		avgR100	-118.21		eps 0.126		Qval	-64.29		1
[DDQN-Mod] and100	Ep 0.19	R thr 805	-356.00		avgR100	-122.81		eps 0.126		Qval	-63.93		1
[DDQN-Mod] and100	Ep 0.19	R thr 804	-211.69		avgR100	-124.90		eps 0.125		Qval	-66.41		1
[DDQN-Mod] and100	Ep 0.20	R thr 524	77.62		avgR100	-122.32		eps 0.124		Qval	-66.96		1
[DDQN-Mod] and100	Ep 0.19	R thr 833	-273.15		avgR100	-125.48		eps 0.124		Qval	-66.14		1
[DDQN-Mod] and100	Ep 0.19	R thr 106	-57.21		avgR100	-125.70		eps 0.123		Qval	-67.78		1
[DDQN-Mod] and100	Ep 0.20	R thr 410	131.41		avgR100	-121.99		eps 0.122		Qval	-66.37		1
[DDQN-Mod] and100	Ep 0.21	R thr 417	132.07		avgR100	-118.06		eps 0.122		Qval	-66.16		1

[DDQN-Mod] and100	Ep 0.22	R thr 587	52.52	avgR100	-114.90	eps 0.121	Qval	-67.38	1
[DDQN-Mod] and100	Ep 0.22	R thr 783	-173.93	avgR100	-116.54	eps 0.121	Qval	-68.86	1
[DDQN-Mod] and100	Ep 0.23	R thr 460	139.10	avgR100	-114.43	eps 0.120	Qval	-67.00	1
[DDQN-Mod] and100	Ep 0.24	R thr 596	42.77	avgR100	-111.82	eps 0.119	Qval	-67.23	1
[DDQN-Mod] and100	Ep 0.25	R thr 322	158.39	avgR100	-109.36	eps 0.119	Qval	-68.90	1
[DDQN-Mod] and100	Ep 0.25	R thr 781	-211.07	avgR100	-110.82	eps 0.118	Qval	-67.96	1
[DDQN-Mod] and100	Ep 0.25	R thr 673	-163.94	avgR100	-110.26	eps 0.118	Qval	-70.69	1
[DDQN-Mod] and100	Ep 0.26	R thr 423	181.41	avgR100	-108.81	eps 0.117	Qval	-68.30	1
[DDQN-Mod] and100	Ep 0.27	R thr 628	40.80	avgR100	-106.16	eps 0.116	Qval	-68.73	1
[DDQN-Mod] and100	Ep 0.28	R thr 203	235.04	avgR100	-103.10	eps 0.116	Qval	-68.50	1
[DDQN-Mod] and100	Ep 0.28	R thr 696	-164.91	avgR100	-103.74	eps 0.115	Qval	-68.57	1
[DDQN-Mod] and100	Ep 0.29	R thr 325	175.05	avgR100	-101.70	eps 0.115	Qval	-69.78	1
[DDQN-Mod] and100	Ep 0.29	R thr 186	-13.41	avgR100	-99.69	eps 0.114	Qval	-67.67	1
[DDQN-Mod] and100	Ep 0.30	R thr 448	135.25	avgR100	-98.06	eps 0.114	Qval	-68.91	1
[DDQN-Mod] and100	Ep 0.29	R thr 730	-254.33	avgR100	-102.31	eps 0.113	Qval	-70.52	1
[DDQN-Mod] and100	Ep 0.30	R thr 593	67.93	avgR100	-100.89	eps 0.112	Qval	-69.94	1
[DDQN-Mod] and100	Ep 0.29	R thr 779	-248.22	avgR100	-104.01	eps 0.112	Qval	-71.59	1
[DDQN-Mod] and100	Ep 0.29	R thr 489	100.19	avgR100	-104.42	eps 0.111	Qval	-71.74	1
[DDQN-Mod] and100	Ep 0.30	R thr 304	171.90	avgR100	-100.59	eps 0.111	Qval	-72.17	1
[DDQN-Mod] and100	Ep 0.31	R thr 347	183.12	avgR100	-95.94	eps 0.110	Qval	-72.61	1
[DDQN-Mod] and100	Ep 0.31	R thr 738	-186.95	avgR100	-97.32	eps 0.110	Qval	-74.85	1
[DDQN-Mod] and100	Ep 0.31	R thr 472	124.08	avgR100	-98.79	eps 0.109	Qval	-73.02	1
[DDQN-Mod] and100	Ep 0.31	R thr 803	-264.83	avgR100	-101.03	eps 0.109	Qval	-74.70	1
[DDQN-Mod] and100	Ep 0.32	R thr 336	176.89	avgR100	-95.96	eps 0.108	Qval	-75.76	1
[DDQN-Mod] and100	Ep 0.33	R thr 330	174.90	avgR100	-93.27	eps 0.107	Qval	-73.12	1
[DDQN-Mod] and100	Ep 0.34	R thr 589	80.84	avgR100	-89.75	eps 0.107	Qval	-72.90	1
[DDQN-Mod] and100	Ep 0.34	R thr 358	173.35	avgR100	-89.42	eps 0.106	Qval	-74.17	1
[DDQN-Mod] and100	Ep 0.35	R thr 289	225.17	avgR100	-84.57	eps 0.106	Qval	-72.58	1
[DDQN-Mod] and100	Ep 0.36	R thr 456	118.17	avgR100	-80.50	eps 0.105	Qval	-72.27	1
[DDQN-Mod] and100	Ep 0.36	R thr 140	10.72	avgR100	-79.98	eps 0.105	Qval	-72.99	1

[DDQN-Mod]	Ep	451		R	197.54		avgR100	-76.16		eps 0.104		Qval	-72.59		1
and100 0.37		thr 352													
[DDQN-Mod]	Ep	452		R	195.73		avgR100	-74.28		eps 0.104		Qval	-73.26		1
and100 0.37		thr 276													
[DDQN-Mod]	Ep	453		R	179.87		avgR100	-71.95		eps 0.103		Qval	-74.58		1
and100 0.37		thr 236													
[DDQN-Mod]	Ep	454		R	73.29		avgR100	-68.28		eps 0.103		Qval	-73.45		1
and100 0.38		thr 512													
[DDQN-Mod]	Ep	455		R	-198.01		avgR100	-71.24		eps 0.102		Qval	-75.11		1
and100 0.38		thr 765													
[DDQN-Mod]	Ep	456		R	90.82		avgR100	-67.95		eps 0.102		Qval	-75.65		1
and100 0.39		thr 542													
[DDQN-Mod]	Ep	457		R	-214.01		avgR100	-66.91		eps 0.101		Qval	-73.83		1
and100 0.39		thr 705													
[DDQN-Mod]	Ep	458		R	270.88		avgR100	-64.00		eps 0.101		Qval	-74.52		1
and100 0.40		thr 233													
[DDQN-Mod]	Ep	459		R	134.57		avgR100	-63.36		eps 0.100		Qval	-75.80		1
and100 0.40		thr 371													
[DDQN-Mod]	Ep	460		R	158.14		avgR100	-62.33		eps 0.100		Qval	-76.63		1
and100 0.40		thr 401													
[DDQN-Mod]	Ep	461		R	-222.23		avgR100	-61.53		eps 0.099		Qval	-76.04		1
and100 0.40		thr 758													
[DDQN-Mod]	Ep	462		R	275.46		avgR100	-61.73		eps 0.099		Qval	-76.64		1
and100 0.40		thr 208													
[DDQN-Mod]	Ep	463		R	273.62		avgR100	-59.32		eps 0.098		Qval	-74.78		1
and100 0.40		thr 220													
[DDQN-Mod]	Ep	464		R	-174.88		avgR100	-58.22		eps 0.098		Qval	-74.53		1
and100 0.40		thr 726													
[DDQN-Mod]	Ep	465		R	-228.81		avgR100	-58.07		eps 0.097		Qval	-74.52		1
and100 0.40		thr 792													
[DDQN-Mod]	Ep	466		R	89.46		avgR100	-53.66		eps 0.097		Qval	-74.97		1
and100 0.41		thr 568													
[DDQN-Mod]	Ep	467		R	-264.95		avgR100	-52.53		eps 0.096		Qval	-75.06		1
and100 0.41		thr 799													
[DDQN-Mod]	Ep	468		R	-196.37		avgR100	-51.32		eps 0.096		Qval	-75.38		1
and100 0.41		thr 736													
[DDQN-Mod]	Ep	469		R	196.04		avgR100	-45.97		eps 0.095		Qval	-73.37		1
and100 0.42		thr 350													
[DDQN-Mod]	Ep	470		R	198.03		avgR100	-42.77		eps 0.095		Qval	-73.35		1
and100 0.43		thr 331													
[DDQN-Mod]	Ep	471		R	-244.51		avgR100	-46.07		eps 0.094		Qval	-73.08		1
and100 0.42		thr 773													
[DDQN-Mod]	Ep	472		R	-209.09		avgR100	-50.14		eps 0.094		Qval	-73.52		1
and100 0.41		thr 699													
[DDQN-Mod]	Ep	473		R	20.79		avgR100	-49.04		eps 0.093		Qval	-72.70		1
and100 0.42		thr 616													
[DDQN-Mod]	Ep	474		R	-239.06		avgR100	-51.09		eps 0.093		Qval	-73.84		1
and100 0.42		thr 687													
[DDQN-Mod]	Ep	475		R	81.65		avgR100	-48.57		eps 0.092		Qval	-72.95		1
and100 0.43		thr 547													
[DDQN-Mod]	Ep	476		R	139.74		avgR100	-45.23		eps 0.092		Qval	-74.15		1
and100 0.44		thr 458													
[DDQN-Mod]	Ep	477		R	-17.57		avgR100	-42.28		eps 0.092		Qval	-73.31		1
and100 0.44		thr 105													
[DDQN-Mod]	Ep	478		R	62.28		avgR100	-41.49		eps 0.091		Qval	-72.80		1
and100 0.45		thr 541													
[DDQN-Mod]	Ep	479		R	219.09		avgR100	-38.80		eps 0.091		Qval	-73.73		1
and100 0.46		thr 293													
[DDQN-Mod]	Ep	480		R	158.17		avgR100	-33.19		eps 0.090		Qval	-71.17		1
and100 0.47		thr 378													

[DDQN-Mod] Ep 481 R 106.02 avgR100 -32.05 eps 0.090 Qval -72.52 1
and100 0.48 thr 459
[DDQN-Mod] Ep 482 R 124.93 avgR100 -28.50 eps 0.089 Qval -72.45 1
and100 0.49 thr 419
[DDQN-Mod] Ep 483 R -223.15 avgR100 -31.38 eps 0.089 Qval -71.83 1
and100 0.48 thr 746
[DDQN-Mod] Ep 484 R 225.09 avgR100 -27.31 eps 0.088 Qval -71.16 1
and100 0.49 thr 255
[DDQN-Mod] Ep 485 R -14.01 avgR100 -25.33 eps 0.088 Qval -71.05 1
and100 0.49 thr 209
[DDQN-Mod] Ep 486 R 194.58 avgR100 -20.74 eps 0.088 Qval -70.80 1
and100 0.50 thr 300
[DDQN-Mod] Ep 487 R 124.73 avgR100 -14.62 eps 0.087 Qval -72.67 1
and100 0.51 thr 486
[DDQN-Mod] Ep 488 R -24.47 avgR100 -12.06 eps 0.087 Qval -69.29 1
and100 0.52 thr 650
[DDQN-Mod] Ep 489 R -26.87 avgR100 -9.24 eps 0.086 Qval -67.75 1
and100 0.52 thr 124
[DDQN-Mod] Ep 490 R 56.16 avgR100 -5.38 eps 0.086 Qval -70.07 1
and100 0.53 thr 676
[DDQN-Mod] Ep 491 R 196.93 avgR100 -0.60 eps 0.085 Qval -69.24 1
and100 0.54 thr 353
[DDQN-Mod] Ep 492 R -229.89 avgR100 0.32 eps 0.085 Qval -70.57 1
and100 0.54 thr 770
[DDQN-Mod] Ep 493 R 168.01 avgR100 0.33 eps 0.084 Qval -71.96 1
and100 0.54 thr 367
[DDQN-Mod] Ep 494 R 280.46 avgR100 3.77 eps 0.084 Qval -69.91 1
and100 0.55 thr 225
[DDQN-Mod] Ep 495 R 245.82 avgR100 8.48 eps 0.084 Qval -68.19 1
and100 0.56 thr 242
[DDQN-Mod] Ep 496 R -198.48 avgR100 9.92 eps 0.083 Qval -70.29 1
and100 0.56 thr 712
[DDQN-Mod] Ep 497 R 142.76 avgR100 12.75 eps 0.083 Qval -68.61 1
and100 0.57 thr 450
[DDQN-Mod] Ep 498 R 119.35 avgR100 14.78 eps 0.082 Qval -68.66 1
and100 0.58 thr 507
[DDQN-Mod] Ep 499 R 183.40 avgR100 14.67 eps 0.082 Qval -69.09 1
and100 0.58 thr 308
[DDQN-Mod] Ep 500 R -196.05 avgR100 16.30 eps 0.082 Qval -67.00 1
and100 0.58 thr 795
[DDQN-Mod] Ep 501 R 291.70 avgR100 21.01 eps 0.081 Qval -67.95 1
and100 0.59 thr 155
[DDQN-Mod] Ep 502 R -227.49 avgR100 20.93 eps 0.081 Qval -69.34 1
and100 0.59 thr 804
[DDQN-Mod] Ep 503 R 105.54 avgR100 20.17 eps 0.080 Qval -67.31 1
and100 0.60 thr 487
[DDQN-Mod] Ep 504 R 26.77 avgR100 18.67 eps 0.080 Qval -67.31 1
and100 0.60 thr 580
[DDQN-Mod] Ep 505 R -175.63 avgR100 17.18 eps 0.080 Qval -67.28 1
and100 0.60 thr 692
[DDQN-Mod] Ep 506 R 70.93 avgR100 19.86 eps 0.079 Qval -67.18 1
and100 0.61 thr 520
[DDQN-Mod] Ep 507 R 160.38 avgR100 23.85 eps 0.079 Qval -67.74 1
and100 0.62 thr 364
[DDQN-Mod] Ep 508 R 262.91 avgR100 24.80 eps 0.078 Qval -67.74 1
and100 0.62 thr 176
[DDQN-Mod] Ep 509 R 175.22 avgR100 29.18 eps 0.078 Qval -66.78 1
and100 0.63 thr 410
[DDQN-Mod] Ep 510 R -221.86 avgR100 28.66 eps 0.078 Qval -68.07 1
and100 0.63 thr 767

[DDQN-Mod] Ep 511 R 166.50 avgR100 32.63 eps 0.077 Qval -65.86 1	and100 0.64 thr 405
[DDQN-Mod] Ep 512 R 200.06 avgR100 33.57 eps 0.077 Qval -66.92 1	and100 0.64 thr 313
[DDQN-Mod] Ep 513 R 161.39 avgR100 35.86 eps 0.076 Qval -67.93 1	and100 0.65 thr 414
[DDQN-Mod] Ep 514 R 215.37 avgR100 41.58 eps 0.076 Qval -68.21 1	and100 0.66 thr 326
[DDQN-Mod] Ep 515 R 265.76 avgR100 46.35 eps 0.076 Qval -66.05 1	and100 0.67 thr 217
[DDQN-Mod] Ep 516 R 227.23 avgR100 47.85 eps 0.075 Qval -65.58 1	and100 0.67 thr 261
[DDQN-Mod] Ep 517 R 117.63 avgR100 51.76 eps 0.075 Qval -67.54 1	and100 0.68 thr 467
[DDQN-Mod] Ep 518 R 105.08 avgR100 53.38 eps 0.075 Qval -66.64 1	and100 0.69 thr 504
[DDQN-Mod] Ep 519 R -197.52 avgR100 50.09 eps 0.074 Qval -69.00 1	and100 0.68 thr 796
[DDQN-Mod] Ep 520 R 156.64 avgR100 50.34 eps 0.074 Qval -67.07 1	and100 0.68 thr 322
[DDQN-Mod] Ep 521 R 252.15 avgR100 52.33 eps 0.073 Qval -66.98 1	and100 0.68 thr 216
[DDQN-Mod] Ep 522 R 119.57 avgR100 55.27 eps 0.073 Qval -68.77 1	and100 0.69 thr 497
[DDQN-Mod] Ep 523 R 151.66 avgR100 55.39 eps 0.073 Qval -66.54 1	and100 0.69 thr 373
[DDQN-Mod] Ep 524 R -163.21 avgR100 53.33 eps 0.072 Qval -68.92 1	and100 0.68 thr 651
[DDQN-Mod] Ep 525 R 130.94 avgR100 53.06 eps 0.072 Qval -67.59 1	and100 0.68 thr 458
[DDQN-Mod] Ep 526 R -42.74 avgR100 54.74 eps 0.072 Qval -67.97 1	and100 0.68 thr 206
[DDQN-Mod] Ep 527 R -104.56 avgR100 55.34 eps 0.071 Qval -66.72 1	and100 0.68 thr 123
[DDQN-Mod] Ep 528 R 260.80 avgR100 56.13 eps 0.071 Qval -67.96 1	and100 0.68 thr 247
[DDQN-Mod] Ep 529 R 238.55 avgR100 58.11 eps 0.071 Qval -65.64 1	and100 0.68 thr 301
[DDQN-Mod] Ep 530 R 161.96 avgR100 57.38 eps 0.070 Qval -65.68 1	and100 0.68 thr 385
[DDQN-Mod] Ep 531 R 291.89 avgR100 61.94 eps 0.070 Qval -64.07 1	and100 0.69 thr 204
[DDQN-Mod] Ep 532 R 136.25 avgR100 61.56 eps 0.069 Qval -65.65 1	and100 0.69 thr 354
[DDQN-Mod] Ep 533 R -231.90 avgR100 59.37 eps 0.069 Qval -66.76 1	and100 0.69 thr 801
[DDQN-Mod] Ep 534 R -242.15 avgR100 55.60 eps 0.069 Qval -67.48 1	and100 0.68 thr 338
[DDQN-Mod] Ep 535 R 136.07 avgR100 59.50 eps 0.068 Qval -63.97 1	and100 0.69 thr 409
[DDQN-Mod] Ep 536 R -166.49 avgR100 57.16 eps 0.068 Qval -65.28 1	and100 0.68 thr 780
[DDQN-Mod] Ep 537 R 5.32 avgR100 59.69 eps 0.068 Qval -66.11 1	and100 0.69 thr 659
[DDQN-Mod] Ep 538 R 265.46 avgR100 61.35 eps 0.067 Qval -66.66 1	and100 0.69 thr 180
[DDQN-Mod] Ep 539 R 190.77 avgR100 61.53 eps 0.067 Qval -66.03 1	and100 0.69 thr 336
[DDQN-Mod] Ep 540 R 236.76 avgR100 62.07 eps 0.067 Qval -66.81 1	and100 0.69 thr 253

[DDQN-Mod] Ep 541 R and100 0.69 thr 90	-36.62 avgR100	63.57 eps 0.066 Qval -66.72 1
[DDQN-Mod] Ep 542 R and100 0.69 thr 204	248.01 avgR100	64.81 eps 0.066 Qval -67.12 1
[DDQN-Mod] Ep 543 R and100 0.70 thr 356	183.64 avgR100	69.30 eps 0.066 Qval -68.34 1
[DDQN-Mod] Ep 544 R and100 0.69 thr 85	-11.39 avgR100	67.41 eps 0.065 Qval -66.99 1
[DDQN-Mod] Ep 545 R and100 0.69 thr 197	274.51 avgR100	68.41 eps 0.065 Qval -68.33 1
[DDQN-Mod] Ep 546 R and100 0.69 thr 200	256.33 avgR100	70.17 eps 0.065 Qval -68.39 1
[DDQN-Mod] Ep 547 R and100 0.69 thr 229	222.76 avgR100	70.66 eps 0.064 Qval -66.51 1
[DDQN-Mod] Ep 548 R and100 0.69 thr 308	182.90 avgR100	70.24 eps 0.064 Qval -65.55 1
[DDQN-Mod] Ep 549 R and100 0.69 thr 484	103.57 avgR100	70.09 eps 0.064 Qval -66.63 1
[DDQN-Mod] Ep 550 R and100 0.70 thr 410	134.07 avgR100	71.32 eps 0.063 Qval -65.81 1
[DDQN-Mod] Ep 551 R and100 0.70 thr 184	249.24 avgR100	71.84 eps 0.063 Qval -65.33 1
[DDQN-Mod] Ep 552 R and100 0.70 thr 278	209.56 avgR100	71.98 eps 0.063 Qval -66.95 1
[DDQN-Mod] Ep 553 R and100 0.71 thr 197	251.29 avgR100	72.69 eps 0.063 Qval -65.20 1
[DDQN-Mod] Ep 554 R and100 0.71 thr 425	129.78 avgR100	73.26 eps 0.062 Qval -66.67 1
[DDQN-Mod] Ep 555 R and100 0.72 thr 157	299.60 avgR100	78.24 eps 0.062 Qval -66.68 1
[DDQN-Mod] Ep 556 R and100 0.72 thr 216	229.44 avgR100	79.62 eps 0.062 Qval -67.81 1
[DDQN-Mod] Ep 557 R and100 0.73 thr 398	122.90 avgR100	82.99 eps 0.061 Qval -68.47 1
[DDQN-Mod] Ep 558 R and100 0.73 thr 314	191.44 avgR100	82.20 eps 0.061 Qval -67.57 1
[DDQN-Mod] Ep 559 R and100 0.73 thr 258	201.55 avgR100	82.87 eps 0.061 Qval -68.54 1
[DDQN-Mod] Ep 560 R and100 0.73 thr 306	220.04 avgR100	83.48 eps 0.060 Qval -68.61 1
[DDQN-Mod] Ep 561 R and100 0.74 thr 467	103.79 avgR100	86.75 eps 0.060 Qval -65.79 1
[DDQN-Mod] Ep 562 R and100 0.74 thr 296	216.12 avgR100	86.15 eps 0.060 Qval -67.40 1
[DDQN-Mod] Ep 563 R and100 0.74 thr 209	287.53 avgR100	86.29 eps 0.059 Qval -68.33 1
[DDQN-Mod] Ep 564 R and100 0.75 thr 448	131.15 avgR100	89.35 eps 0.059 Qval -69.72 1
[DDQN-Mod] Ep 565 R and100 0.76 thr 376	160.78 avgR100	93.25 eps 0.059 Qval -67.41 1
[DDQN-Mod] Ep 566 R and100 0.76 thr 275	228.52 avgR100	94.64 eps 0.059 Qval -66.55 1
[DDQN-Mod] Ep 567 R and100 0.76 thr 199	193.19 avgR100	99.22 eps 0.058 Qval -68.13 1
[DDQN-Mod] Ep 568 R and100 0.77 thr 351	165.29 avgR100	102.84 eps 0.058 Qval -64.81 1
[DDQN-Mod] Ep 569 R and100 0.77 thr 208	247.58 avgR100	103.35 eps 0.058 Qval -66.47 1
[DDQN-Mod] Ep 570 R and100 0.77 thr 274	260.22 avgR100	103.97 eps 0.057 Qval -65.99 1

[DDQN-Mod] Ep 571 R and100 0.78 thr 676	33.72 avgR100	106.76 eps 0.057 Qval -68.04 1
[DDQN-Mod] Ep 572 R and100 0.79 thr 349	182.97 avgR100	110.68 eps 0.057 Qval -65.69 1
[DDQN-Mod] Ep 573 R and100 0.79 thr 287	230.89 avgR100	112.78 eps 0.057 Qval -65.73 1
[DDQN-Mod] Ep 574 R and100 0.80 thr 254	227.40 avgR100	117.44 eps 0.056 Qval -65.74 1
[DDQN-Mod] Ep 575 R and100 0.80 thr 410	106.02 avgR100	117.68 eps 0.056 Qval -64.24 1
[DDQN-Mod] Ep 576 R and100 0.80 thr 415	161.45 avgR100	117.90 eps 0.056 Qval -63.96 1
[DDQN-Mod] Ep 577 R and100 0.81 thr 226	249.80 avgR100	120.58 eps 0.055 Qval -65.90 1
[DDQN-Mod] Ep 578 R and100 0.81 thr 168	265.53 avgR100	122.61 eps 0.055 Qval -65.08 1
[DDQN-Mod] Ep 579 R and100 0.81 thr 380	155.38 avgR100	121.97 eps 0.055 Qval -65.72 1
[DDQN-Mod] Ep 580 R and100 0.81 thr 333	184.42 avgR100	122.23 eps 0.055 Qval -63.79 1
[DDQN-Mod] Ep 581 R and100 0.81 thr 391	174.68 avgR100	122.92 eps 0.054 Qval -64.26 1
[DDQN-Mod] Ep 582 R and100 0.81 thr 504	100.29 avgR100	122.67 eps 0.054 Qval -64.99 1
[DDQN-Mod] Ep 583 R and100 0.82 thr 204	250.90 avgR100	127.41 eps 0.054 Qval -65.23 1
[DDQN-Mod] Ep 584 R and100 0.82 thr 544	105.36 avgR100	126.22 eps 0.054 Qval -63.64 1
[DDQN-Mod] Ep 585 R and100 0.83 thr 304	176.30 avgR100	128.12 eps 0.053 Qval -63.07 1
[DDQN-Mod] Ep 586 R and100 0.82 thr 187	-49.55 avgR100	125.68 eps 0.053 Qval -61.92 1
[DDQN-Mod] Ep 587 R and100 0.82 thr 526	102.69 avgR100	125.46 eps 0.053 Qval -62.02 1
[DDQN-Mod] Ep 588 R and100 0.82 thr 278	250.19 avgR100	128.20 eps 0.052 Qval -63.40 1
[DDQN-Mod] Ep 589 R and100 0.83 thr 234	242.65 avgR100	130.90 eps 0.052 Qval -63.65 1
[DDQN-Mod] Ep 590 R and100 0.83 thr 425	165.86 avgR100	132.00 eps 0.052 Qval -62.36 1
[DDQN-Mod] Ep 591 R and100 0.83 thr 604	46.88 avgR100	130.50 eps 0.052 Qval -60.43 1
[DDQN-Mod] Ep 592 R and100 0.84 thr 174	239.92 avgR100	135.19 eps 0.051 Qval -59.70 1
[DDQN-Mod] Ep 593 R and100 0.84 thr 333	185.49 avgR100	135.37 eps 0.051 Qval -61.93 1
[DDQN-Mod] Ep 594 R and100 0.84 thr 224	246.04 avgR100	135.03 eps 0.051 Qval -63.97 1
[DDQN-Mod] Ep 595 R and100 0.84 thr 377	164.33 avgR100	134.21 eps 0.051 Qval -61.77 1
[DDQN-Mod] Ep 596 R and100 0.85 thr 217	209.29 avgR100	138.29 eps 0.050 Qval -63.02 1
[DDQN-Mod] Ep 597 R and100 0.85 thr 200	286.30 avgR100	139.72 eps 0.050 Qval -62.93 1
[DDQN-Mod] Ep 598 R and100 0.85 thr 226	246.55 avgR100	141.00 eps 0.050 Qval -60.47 1
[DDQN-Mod] Ep 599 R and100 0.85 thr 211	279.89 avgR100	141.96 eps 0.050 Qval -62.11 1
[DDQN-Mod] Ep 600 R and100 0.86 thr 361	160.55 avgR100	145.53 eps 0.049 Qval -61.34 1

[DDQN-Mod] Ep 601 R and100 0.85 thr 118	-18.03 avgR100	142.43 eps 0.049 Qval -61.72 1
[DDQN-Mod] Ep 602 R and100 0.86 thr 241	204.98 avgR100	146.75 eps 0.049 Qval -60.10 1
[DDQN-Mod] Ep 603 R and100 0.85 thr 869	-233.74 avgR100	143.36 eps 0.049 Qval -63.58 1
[DDQN-Mod] Ep 604 R and100 0.85 thr 414	121.18 avgR100	144.31 eps 0.048 Qval -60.86 1
[DDQN-Mod] Ep 605 R and100 0.86 thr 536	93.16 avgR100	146.99 eps 0.048 Qval -60.22 1
[DDQN-Mod] Ep 606 R and100 0.86 thr 321	209.45 avgR100	148.38 eps 0.048 Qval -61.18 1
[DDQN-Mod] Ep 607 R and100 0.86 thr 255	253.86 avgR100	149.31 eps 0.048 Qval -62.76 1
[DDQN-Mod] Ep 608 R and100 0.85 thr 90	-4.60 avgR100	146.64 eps 0.047 Qval -61.41 1
[DDQN-Mod] Ep 609 R and100 0.85 thr 232	236.73 avgR100	147.25 eps 0.047 Qval -63.91 1
[DDQN-Mod] Ep 610 R and100 0.86 thr 569	91.95 avgR100	150.39 eps 0.047 Qval -59.71 1
[DDQN-Mod] Ep 611 R and100 0.86 thr 626	46.89 avgR100	149.20 eps 0.047 Qval -61.13 1
[DDQN-Mod] Ep 612 R and100 0.86 thr 385	144.04 avgR100	148.64 eps 0.047 Qval -60.60 1
[DDQN-Mod] Ep 613 R and100 0.86 thr 279	242.13 avgR100	149.44 eps 0.046 Qval -61.20 1
[DDQN-Mod] Ep 614 R and100 0.85 thr 263	170.72 avgR100	149.00 eps 0.046 Qval -60.00 1
[DDQN-Mod] Ep 615 R and100 0.85 thr 173	266.38 avgR100	149.00 eps 0.046 Qval -62.05 1
[DDQN-Mod] Ep 616 R and100 0.85 thr 253	208.73 avgR100	148.82 eps 0.046 Qval -61.21 1
[DDQN-Mod] Ep 617 R and100 0.85 thr 208	241.56 avgR100	150.06 eps 0.045 Qval -62.53 1
[DDQN-Mod] Ep 618 R and100 0.85 thr 258	198.13 avgR100	150.99 eps 0.045 Qval -61.95 1
[DDQN-Mod] Ep 619 R and100 0.86 thr 276	225.45 avgR100	155.22 eps 0.045 Qval -62.45 1
[DDQN-Mod] Ep 620 R and100 0.86 thr 236	209.11 avgR100	155.74 eps 0.045 Qval -61.38 1
[DDQN-Mod] Ep 621 R and100 0.86 thr 402	151.02 avgR100	154.73 eps 0.044 Qval -62.14 1
[DDQN-Mod] Ep 622 R and100 0.85 thr 205	220.19 avgR100	155.74 eps 0.044 Qval -61.12 1
[DDQN-Mod] Ep 623 R and100 0.84 thr 270	151.43 avgR100	155.73 eps 0.044 Qval -61.37 1
[DDQN-Mod] Ep 624 R and100 0.85 thr 346	187.83 avgR100	159.24 eps 0.044 Qval -61.52 1
[DDQN-Mod] Ep 625 R and100 0.85 thr 277	222.21 avgR100	160.16 eps 0.044 Qval -60.59 1
[DDQN-Mod] Ep 626 R and100 0.86 thr 340	198.08 avgR100	162.57 eps 0.043 Qval -59.87 1
[DDQN-Mod] Ep 627 R and100 0.87 thr 298	183.44 avgR100	165.45 eps 0.043 Qval -60.06 1
[DDQN-Mod] Ep 628 R and100 0.87 thr 519	98.02 avgR100	163.82 eps 0.043 Qval -60.22 1
[DDQN-Mod] Ep 629 R and100 0.86 thr 328	126.73 avgR100	162.70 eps 0.043 Qval -61.40 1
[DDQN-Mod] Ep 630 R and100 0.85 thr 903	-79.39 avgR100	160.29 eps 0.043 Qval -63.64 1

[DDQN-Mod] Ep 631 R and100 0.85 thr 361	193.72 avgR100	159.30 eps 0.042 Qval -64.63 1
[DDQN-Mod] Ep 632 R and100 0.84 thr 219	-13.30 avgR100	157.81 eps 0.042 Qval -62.08 1
[DDQN-Mod] Ep 633 R and100 0.85 thr 309	198.61 avgR100	162.11 eps 0.042 Qval -61.76 1
[DDQN-Mod] Ep 634 R and100 0.85 thr 116	-25.11 avgR100	164.28 eps 0.042 Qval -60.89 1
[DDQN-Mod] Ep 635 R and100 0.85 thr 505	49.65 avgR100	163.42 eps 0.041 Qval -60.71 1
[DDQN-Mod] Ep 636 R and100 0.86 thr 459	127.82 avgR100	166.36 eps 0.041 Qval -63.68 1
[DDQN-Mod] Ep 637 R and100 0.86 thr 280	188.64 avgR100	168.20 eps 0.041 Qval -63.40 1
[DDQN-Mod] Ep 638 R and100 0.86 thr 218	253.16 avgR100	168.07 eps 0.041 Qval -63.85 1
[DDQN-Mod] Ep 639 R and100 0.86 thr 399	153.45 avgR100	167.70 eps 0.041 Qval -62.92 1
[DDQN-Mod] Ep 640 R and100 0.86 thr 479	123.69 avgR100	166.57 eps 0.040 Qval -63.55 1
[DDQN-Mod] Ep 641 R and100 0.87 thr 337	169.34 avgR100	168.63 eps 0.040 Qval -63.53 1
[DDQN-Mod] Ep 642 R and100 0.87 thr 405	115.65 avgR100	167.31 eps 0.040 Qval -63.69 1
[DDQN-Mod] Ep 643 R and100 0.87 thr 594	47.04 avgR100	165.94 eps 0.040 Qval -64.43 1
[DDQN-Mod] Ep 644 R and100 0.88 thr 420	170.04 avgR100	167.75 eps 0.040 Qval -63.58 1
[DDQN-Mod] Ep 645 R and100 0.88 thr 207	270.51 avgR100	167.71 eps 0.039 Qval -63.19 1
[DDQN-Mod] Ep 646 R and100 0.88 thr 360	179.05 avgR100	166.94 eps 0.039 Qval -62.33 1
[DDQN-Mod] Ep 647 R and100 0.87 thr 377	88.30 avgR100	165.60 eps 0.039 Qval -63.32 1
[DDQN-Mod] Ep 648 R and100 0.87 thr 419	89.99 avgR100	164.67 eps 0.039 Qval -63.41 1
[DDQN-Mod] Ep 649 R and100 0.87 thr 316	195.24 avgR100	165.58 eps 0.039 Qval -63.08 1
[DDQN-Mod] Ep 650 R and100 0.87 thr 291	231.71 avgR100	166.56 eps 0.038 Qval -62.37 1
[DDQN-Mod] Ep 651 R and100 0.87 thr 585	16.40 avgR100	164.23 eps 0.038 Qval -61.83 1
[DDQN-Mod] Ep 652 R and100 0.87 thr 275	220.80 avgR100	164.34 eps 0.038 Qval -62.20 1
[DDQN-Mod] Ep 653 R and100 0.87 thr 248	260.62 avgR100	164.44 eps 0.038 Qval -61.18 1
[DDQN-Mod] Ep 654 R and100 0.87 thr 388	192.33 avgR100	165.06 eps 0.038 Qval -60.63 1
[DDQN-Mod] Ep 655 R and100 0.87 thr 211	233.15 avgR100	164.40 eps 0.038 Qval -60.11 1
[DDQN-Mod] Ep 656 R and100 0.87 thr 292	192.01 avgR100	164.02 eps 0.037 Qval -58.60 1
[DDQN-Mod] Ep 657 R and100 0.87 thr 219	233.22 avgR100	165.13 eps 0.037 Qval -58.80 1
[DDQN-Mod] Ep 658 R and100 0.87 thr 504	79.62 avgR100	164.01 eps 0.037 Qval -60.24 1
[DDQN-Mod] Ep 659 R and100 0.87 thr 629	45.03 avgR100	162.44 eps 0.037 Qval -60.40 1
[DDQN-Mod] Ep 660 R and100 0.87 thr 457	152.79 avgR100	161.77 eps 0.037 Qval -61.48 1

[DDQN-Mod] Ep 661 R and100 0.87 thr 266	227.90	avgR100	163.01	eps 0.036	Qval -59.67	1
[DDQN-Mod] Ep 662 R and100 0.87 thr 263	199.64	avgR100	162.85	eps 0.036	Qval -59.77	1
[DDQN-Mod] Ep 663 R and100 0.87 thr 343	173.52	avgR100	161.71	eps 0.036	Qval -62.13	1
[DDQN-Mod] Ep 664 R and100 0.87 thr 406	119.09	avgR100	161.59	eps 0.036	Qval -62.34	1
[DDQN-Mod] Ep 665 R and100 0.87 thr 389	158.00	avgR100	161.56	eps 0.036	Qval -59.60	1
[DDQN-Mod] Ep 666 R and100 0.87 thr 246	216.78	avgR100	161.44	eps 0.035	Qval -62.70	1
[DDQN-Mod] Ep 667 R and100 0.88 thr 389	165.16	avgR100	161.16	eps 0.035	Qval -60.13	1
[DDQN-Mod] Ep 668 R and100 0.88 thr 630	23.17	avgR100	159.74	eps 0.035	Qval -61.39	1
[DDQN-Mod] Ep 669 R and100 0.88 thr 335	143.73	avgR100	158.70	eps 0.035	Qval -62.79	1
[DDQN-Mod] Ep 670 R and100 0.88 thr 206	258.61	avgR100	158.69	eps 0.035	Qval -60.36	1
[DDQN-Mod] Ep 671 R and100 0.88 thr 407	158.53	avgR100	159.93	eps 0.035	Qval -60.73	1
[DDQN-Mod] Ep 672 R and100 0.88 thr 275	234.46	avgR100	160.45	eps 0.034	Qval -60.93	1
[DDQN-Mod] Ep 673 R and100 0.88 thr 231	251.53	avgR100	160.66	eps 0.034	Qval -60.14	1
[DDQN-Mod] Ep 674 R and100 0.88 thr 371	178.78	avgR100	160.17	eps 0.034	Qval -58.91	1
[DDQN-Mod] Ep 675 R and100 0.88 thr 602	59.12	avgR100	159.70	eps 0.034	Qval -60.58	1
[DDQN-Mod] Ep 676 R and100 0.87 thr 165	-224.67	avgR100	155.84	eps 0.034	Qval -59.99	1
[DDQN-Mod] Ep 677 R and100 0.87 thr 312	197.49	avgR100	155.32	eps 0.034	Qval -60.29	1
[DDQN-Mod] Ep 678 R and100 0.87 thr 340	168.00	avgR100	154.34	eps 0.033	Qval -58.97	1
[DDQN-Mod] Ep 679 R and100 0.87 thr 164	261.74	avgR100	155.40	eps 0.033	Qval -60.22	1
[DDQN-Mod] Ep 680 R and100 0.87 thr 449	154.57	avgR100	155.11	eps 0.033	Qval -61.27	1
[DDQN-Mod] Ep 681 R and100 0.87 thr 451	128.01	avgR100	154.64	eps 0.033	Qval -60.09	1
[DDQN-Mod] Ep 682 R and100 0.87 thr 436	146.60	avgR100	155.10	eps 0.033	Qval -59.61	1
[DDQN-Mod] Ep 683 R and100 0.87 thr 411	135.83	avgR100	153.95	eps 0.033	Qval -59.39	1
[DDQN-Mod] Ep 684 R and100 0.87 thr 305	221.86	avgR100	155.12	eps 0.032	Qval -57.94	1
[DDQN-Mod] Ep 685 R and100 0.87 thr 355	138.27	avgR100	154.74	eps 0.032	Qval -58.22	1
[DDQN-Mod] Ep 686 R and100 0.88 thr 212	255.96	avgR100	157.79	eps 0.032	Qval -59.26	1
[DDQN-Mod] Ep 687 R and100 0.88 thr 342	171.10	avgR100	158.48	eps 0.032	Qval -57.18	1
[DDQN-Mod] Ep 688 R and100 0.88 thr 449	149.67	avgR100	157.47	eps 0.032	Qval -59.87	1
[DDQN-Mod] Ep 689 R and100 0.88 thr 233	232.80	avgR100	157.37	eps 0.032	Qval -59.44	1
[DDQN-Mod] Ep 690 R and100 0.88 thr 269	172.96	avgR100	157.44	eps 0.031	Qval -58.49	1

[DDQN-Mod] Ep 691 R 150.46 avgR100 158.48 eps 0.031 Qval -60.11 1
and100 0.88 thr 370
[DDQN-Mod] Ep 692 R 217.61 avgR100 158.26 eps 0.031 Qval -58.80 1
and100 0.88 thr 217
[DDQN-Mod] Ep 693 R 172.30 avgR100 158.12 eps 0.031 Qval -57.25 1
and100 0.88 thr 321
[DDQN-Mod] Ep 694 R 283.01 avgR100 158.49 eps 0.031 Qval -57.66 1
and100 0.88 thr 128
[DDQN-Mod] Ep 695 R 215.52 avgR100 159.01 eps 0.031 Qval -56.31 1
and100 0.88 thr 280
[DDQN-Mod] Ep 696 R 177.69 avgR100 158.69 eps 0.031 Qval -55.35 1
and100 0.88 thr 356
[DDQN-Mod] Ep 697 R 201.38 avgR100 157.84 eps 0.030 Qval -55.32 1
and100 0.88 thr 318
[DDQN-Mod] Ep 698 R 195.10 avgR100 157.33 eps 0.030 Qval -57.10 1
and100 0.88 thr 311
[DDQN-Mod] Ep 699 R 112.01 avgR100 155.65 eps 0.030 Qval -54.82 1
and100 0.88 thr 484
[DDQN-Mod] Ep 700 R 202.84 avgR100 156.07 eps 0.030 Qval -54.44 1
and100 0.88 thr 285
[DDQN-Mod] Ep 701 R 163.56 avgR100 157.89 eps 0.030 Qval -53.88 1
and100 0.89 thr 336
[DDQN-Mod] Ep 702 R 280.41 avgR100 158.64 eps 0.030 Qval -52.82 1
and100 0.89 thr 187
[DDQN-Mod] Ep 703 R 276.23 avgR100 163.74 eps 0.029 Qval -52.40 1
and100 0.90 thr 160
[DDQN-Mod] Ep 704 R 157.35 avgR100 164.10 eps 0.029 Qval -51.50 1
and100 0.90 thr 356
[DDQN-Mod] Ep 705 R -222.64 avgR100 160.94 eps 0.029 Qval -52.10 1
and100 0.89 thr 929
[DDQN-Mod] Ep 706 R 210.86 avgR100 160.96 eps 0.029 Qval -52.31 1
and100 0.89 thr 325
[DDQN-Mod] Ep 707 R 171.56 avgR100 160.13 eps 0.029 Qval -53.41 1
and100 0.88 thr 302
[DDQN-Mod] Ep 708 R 178.89 avgR100 161.97 eps 0.029 Qval -51.76 1
and100 0.89 thr 308
[DDQN-Mod] Ep 709 R 188.48 avgR100 161.49 eps 0.029 Qval -53.18 1
and100 0.89 thr 294
[DDQN-Mod] Ep 710 R 233.75 avgR100 162.90 eps 0.028 Qval -53.75 1
and100 0.89 thr 212
[DDQN-Mod] Ep 711 R 225.65 avgR100 164.69 eps 0.028 Qval -53.54 1
and100 0.89 thr 264
[DDQN-Mod] Ep 712 R 117.53 avgR100 164.43 eps 0.028 Qval -53.66 1
and100 0.89 thr 386
[DDQN-Mod] Ep 713 R 250.45 avgR100 164.51 eps 0.028 Qval -52.68 1
and100 0.89 thr 257
[DDQN-Mod] Ep 714 R -24.82 avgR100 162.55 eps 0.028 Qval -52.76 1
and100 0.89 thr 137
[DDQN-Mod] Ep 715 R 149.22 avgR100 161.38 eps 0.028 Qval -51.67 1
and100 0.89 thr 389
[DDQN-Mod] Ep 716 R 121.74 avgR100 160.51 eps 0.028 Qval -51.41 1
and100 0.89 thr 502
[DDQN-Mod] Ep 717 R 239.11 avgR100 160.49 eps 0.027 Qval -51.28 1
and100 0.89 thr 278
[DDQN-Mod] Ep 718 R 188.07 avgR100 160.39 eps 0.027 Qval -52.27 1
and100 0.89 thr 324
[DDQN-Mod] Ep 719 R 229.99 avgR100 160.43 eps 0.027 Qval -52.20 1
and100 0.89 thr 233
[DDQN-Mod] Ep 720 R -10.59 avgR100 158.24 eps 0.027 Qval -50.82 1
and100 0.88 thr 105

[DDQN-Mod]	Ep	721		R	236.73		avgR100	159.09		eps 0.027		Qval	-51.04		1
and100 0.88		thr 278													
[DDQN-Mod]	Ep	722		R	-28.29		avgR100	156.61		eps 0.027		Qval	-51.39		1
and100 0.88		thr 104													
[DDQN-Mod]	Ep	723		R	128.06		avgR100	156.38		eps 0.027		Qval	-51.57		1
and100 0.88		thr 377													
[DDQN-Mod]	Ep	724		R	216.08		avgR100	156.66		eps 0.027		Qval	-51.17		1
and100 0.88		thr 300													
[DDQN-Mod]	Ep	725		R	174.20		avgR100	156.18		eps 0.026		Qval	-52.17		1
and100 0.88		thr 378													
[DDQN-Mod]	Ep	726		R	135.01		avgR100	155.55		eps 0.026		Qval	-52.32		1
and100 0.88		thr 387													
[DDQN-Mod]	Ep	727		R	148.16		avgR100	155.19		eps 0.026		Qval	-52.95		1
and100 0.88		thr 328													
[DDQN-Mod]	Ep	728		R	156.97		avgR100	155.78		eps 0.026		Qval	-54.57		1
and100 0.88		thr 361													
[DDQN-Mod]	Ep	729		R	222.79		avgR100	156.74		eps 0.026		Qval	-52.84		1
and100 0.89		thr 287													
[DDQN-Mod]	Ep	730		R	196.12		avgR100	159.50		eps 0.026		Qval	-52.49		1
and100 0.90		thr 268													
[DDQN-Mod]	Ep	731		R	250.80		avgR100	160.07		eps 0.026		Qval	-51.71		1
and100 0.90		thr 264													
[DDQN-Mod]	Ep	732		R	58.62		avgR100	160.79		eps 0.025		Qval	-52.94		1
and100 0.91		thr 508													
[DDQN-Mod]	Ep	733		R	202.67		avgR100	160.83		eps 0.025		Qval	-53.63		1
and100 0.91		thr 317													
[DDQN-Mod]	Ep	734		R	255.65		avgR100	163.64		eps 0.025		Qval	-54.05		1
and100 0.92		thr 275													
[DDQN-Mod]	Ep	735		R	184.36		avgR100	164.98		eps 0.025		Qval	-54.29		1
and100 0.92		thr 308													
[DDQN-Mod]	Ep	736		R	149.88		avgR100	165.21		eps 0.025		Qval	-52.48		1
and100 0.92		thr 390													
[DDQN-Mod]	Ep	737		R	-14.18		avgR100	163.18		eps 0.025		Qval	-52.54		1
and100 0.91		thr 86													
[DDQN-Mod]	Ep	738		R	201.12		avgR100	162.66		eps 0.025		Qval	-53.42		1
and100 0.91		thr 365													
[DDQN-Mod]	Ep	739		R	192.46		avgR100	163.05		eps 0.025		Qval	-52.47		1
and100 0.91		thr 340													
[DDQN-Mod]	Ep	740		R	175.15		avgR100	163.56		eps 0.024		Qval	-53.09		1
and100 0.91		thr 282													
[DDQN-Mod]	Ep	741		R	137.43		avgR100	163.24		eps 0.024		Qval	-54.09		1
and100 0.91		thr 427													
[DDQN-Mod]	Ep	742		R	222.27		avgR100	164.31		eps 0.024		Qval	-53.27		1
and100 0.91		thr 211													
[DDQN-Mod]	Ep	743		R	172.69		avgR100	165.56		eps 0.024		Qval	-50.74		1
and100 0.91		thr 280													
[DDQN-Mod]	Ep	744		R	223.90		avgR100	166.10		eps 0.024		Qval	-53.93		1
and100 0.91		thr 273													
[DDQN-Mod]	Ep	745		R	126.39		avgR100	164.66		eps 0.024		Qval	-53.61		1
and100 0.90		thr 390													
[DDQN-Mod]	Ep	746		R	189.17		avgR100	164.76		eps 0.024		Qval	-54.97		1
and100 0.90		thr 286													
[DDQN-Mod]	Ep	747		R	214.06		avgR100	166.02		eps 0.024		Qval	-54.06		1
and100 0.91		thr 314													
[DDQN-Mod]	Ep	748		R	238.91		avgR100	167.51		eps 0.024		Qval	-54.00		1
and100 0.91		thr 225													
[DDQN-Mod]	Ep	749		R	176.06		avgR100	167.32		eps 0.023		Qval	-53.26		1
and100 0.91		thr 373													
[DDQN-Mod]	Ep	750		R	138.49		avgR100	166.39		eps 0.023		Qval	-52.16		1
and100 0.91		thr 380													

[DDQN-Mod] Ep 751 R 30.16 avgR100 166.52 eps 0.023 Qval -53.43 1
and100 0.90 thr 112
[DDQN-Mod] Ep 752 R 192.19 avgR100 166.24 eps 0.023 Qval -54.75 1
and100 0.90 thr 299
[DDQN-Mod] Ep 753 R 192.26 avgR100 165.55 eps 0.023 Qval -54.16 1
and100 0.90 thr 354
[DDQN-Mod] Ep 754 R 248.76 avgR100 166.12 eps 0.023 Qval -53.04 1
and100 0.90 thr 242
[DDQN-Mod] Ep 755 R -84.53 avgR100 162.94 eps 0.023 Qval -53.45 1
and100 0.89 thr 529
[DDQN-Mod] Ep 756 R 294.18 avgR100 163.96 eps 0.023 Qval -52.15 1
and100 0.89 thr 176
[DDQN-Mod] Ep 757 R 192.30 avgR100 163.55 eps 0.022 Qval -53.14 1
and100 0.89 thr 263
[DDQN-Mod] Ep 758 R 211.34 avgR100 164.87 eps 0.022 Qval -54.44 1
and100 0.89 thr 258
[DDQN-Mod] Ep 759 R 178.21 avgR100 166.20 eps 0.022 Qval -52.95 1
and100 0.89 thr 285
[DDQN-Mod] Ep 760 R 156.25 avgR100 166.24 eps 0.022 Qval -49.97 1
and100 0.89 thr 321
[DDQN-Mod] Ep 761 R 175.55 avgR100 165.71 eps 0.022 Qval -51.69 1
and100 0.89 thr 335
[DDQN-Mod] Ep 762 R 186.15 avgR100 165.58 eps 0.022 Qval -51.88 1
and100 0.89 thr 293
[DDQN-Mod] Ep 763 R 236.10 avgR100 166.21 eps 0.022 Qval -51.74 1
and100 0.89 thr 290
[DDQN-Mod] Ep 764 R 207.15 avgR100 167.09 eps 0.022 Qval -53.03 1
and100 0.89 thr 259
[DDQN-Mod] Ep 765 R 215.49 avgR100 167.66 eps 0.022 Qval -52.90 1
and100 0.89 thr 234
[DDQN-Mod] Ep 766 R 184.57 avgR100 167.34 eps 0.022 Qval -52.62 1
and100 0.89 thr 274
[DDQN-Mod] Ep 767 R 191.27 avgR100 167.60 eps 0.021 Qval -53.11 1
and100 0.89 thr 339
[DDQN-Mod] Ep 768 R 176.81 avgR100 169.14 eps 0.021 Qval -54.97 1
and100 0.89 thr 284
[DDQN-Mod] Ep 769 R 217.73 avgR100 169.88 eps 0.021 Qval -53.82 1
and100 0.89 thr 232
[DDQN-Mod] Ep 770 R -87.82 avgR100 166.41 eps 0.021 Qval -55.31 1
and100 0.88 thr 118
[DDQN-Mod] Ep 771 R 165.70 avgR100 166.48 eps 0.021 Qval -53.84 1
and100 0.88 thr 296
[DDQN-Mod] Ep 772 R 163.74 avgR100 165.78 eps 0.021 Qval -54.43 1
and100 0.88 thr 424
[DDQN-Mod] Ep 773 R 181.26 avgR100 165.07 eps 0.021 Qval -53.93 1
and100 0.88 thr 342
[DDQN-Mod] Ep 774 R 202.55 avgR100 165.31 eps 0.021 Qval -53.58 1
and100 0.88 thr 283
[DDQN-Mod] Ep 775 R 193.86 avgR100 166.66 eps 0.021 Qval -54.29 1
and100 0.88 thr 287
[DDQN-Mod] Ep 776 R 235.08 avgR100 171.26 eps 0.020 Qval -53.61 1
and100 0.88 thr 168
[DDQN-Mod] Ep 777 R 150.22 avgR100 170.78 eps 0.020 Qval -54.66 1
and100 0.88 thr 356
[DDQN-Mod] Ep 778 R 254.03 avgR100 171.64 eps 0.020 Qval -53.48 1
and100 0.88 thr 176
[DDQN-Mod] Ep 779 R 200.09 avgR100 171.03 eps 0.020 Qval -53.19 1
and100 0.88 thr 292
[DDQN-Mod] Ep 780 R 234.90 avgR100 171.83 eps 0.020 Qval -55.66 1
and100 0.88 thr 269

[DDQN-Mod] Ep 781 R 208.46 avgR100 172.64 eps 0.020 Qval -54.56 1	and100 0.88 thr 349
[DDQN-Mod] Ep 782 R 219.77 avgR100 173.37 eps 0.020 Qval -54.29 1	and100 0.87 thr 146
[DDQN-Mod] Ep 783 R 234.21 avgR100 174.35 eps 0.020 Qval -53.31 1	and100 0.87 thr 230
[DDQN-Mod] Ep 784 R 210.18 avgR100 174.23 eps 0.020 Qval -56.24 1	and100 0.87 thr 303
[DDQN-Mod] Ep 785 R 220.57 avgR100 175.06 eps 0.020 Qval -52.42 1	and100 0.87 thr 256
[DDQN-Mod] Ep 786 R -25.96 avgR100 172.24 eps 0.020 Qval -55.95 1	and100 0.86 thr 115
[DDQN-Mod] Ep 787 R 296.72 avgR100 173.49 eps 0.020 Qval -57.08 1	and100 0.86 thr 198
[DDQN-Mod] Ep 788 R 224.17 avgR100 174.24 eps 0.020 Qval -53.88 1	and100 0.86 thr 276
[DDQN-Mod] Ep 789 R 195.07 avgR100 173.86 eps 0.020 Qval -55.75 1	and100 0.86 thr 310
[DDQN-Mod] Ep 790 R 247.45 avgR100 174.61 eps 0.020 Qval -54.98 1	and100 0.86 thr 283
[DDQN-Mod] Ep 791 R 238.50 avgR100 175.49 eps 0.020 Qval -54.44 1	and100 0.86 thr 275
[DDQN-Mod] Ep 792 R 168.95 avgR100 175.00 eps 0.020 Qval -55.75 1	and100 0.85 thr 253
[DDQN-Mod] Ep 793 R 194.71 avgR100 175.22 eps 0.020 Qval -54.25 1	and100 0.85 thr 276
[DDQN-Mod] Ep 794 R 204.44 avgR100 174.44 eps 0.020 Qval -54.24 1	and100 0.85 thr 300
[DDQN-Mod] Ep 795 R 226.24 avgR100 174.55 eps 0.020 Qval -53.98 1	and100 0.85 thr 330
[DDQN-Mod] Ep 796 R 143.24 avgR100 174.20 eps 0.020 Qval -53.33 1	and100 0.85 thr 328
[DDQN-Mod] Ep 797 R 271.05 avgR100 174.90 eps 0.020 Qval -54.07 1	and100 0.85 thr 231
[DDQN-Mod] Ep 798 R 167.34 avgR100 174.62 eps 0.020 Qval -52.25 1	and100 0.85 thr 328
[DDQN-Mod] Ep 799 R 131.07 avgR100 174.81 eps 0.020 Qval -51.71 1	and100 0.85 thr 352
[DDQN-Mod] Ep 800 R 271.74 avgR100 175.50 eps 0.020 Qval -54.92 1	and100 0.85 thr 172
[DDQN-Mod] Ep 801 R 172.08 avgR100 175.59 eps 0.020 Qval -53.26 1	and100 0.85 thr 335
[DDQN-Mod] Ep 802 R 246.52 avgR100 175.25 eps 0.020 Qval -54.52 1	and100 0.85 thr 267
[DDQN-Mod] Ep 803 R 177.50 avgR100 174.26 eps 0.020 Qval -52.68 1	and100 0.85 thr 326
[DDQN-Mod] Ep 804 R 243.75 avgR100 175.12 eps 0.020 Qval -54.11 1	and100 0.85 thr 186
[DDQN-Mod] Ep 805 R -61.13 avgR100 176.74 eps 0.020 Qval -53.68 1	and100 0.85 thr 609
[DDQN-Mod] Ep 806 R 194.34 avgR100 176.57 eps 0.020 Qval -52.96 1	and100 0.85 thr 320
[DDQN-Mod] Ep 807 R 188.22 avgR100 176.74 eps 0.020 Qval -51.92 1	and100 0.86 thr 330
[DDQN-Mod] Ep 808 R 225.59 avgR100 177.21 eps 0.020 Qval -55.24 1	and100 0.86 thr 295
[DDQN-Mod] Ep 809 R 233.87 avgR100 177.66 eps 0.020 Qval -53.06 1	and100 0.86 thr 243
[DDQN-Mod] Ep 810 R 178.16 avgR100 177.10 eps 0.020 Qval -52.40 1	and100 0.86 thr 262

[DDQN-Mod]	Ep	811		R	-26.28		avgR100	174.59		eps 0.020		Qval	-54.18		1
and100 0.85		thr 143													
[DDQN-Mod]	Ep	812		R	-70.47		avgR100	172.71		eps 0.020		Qval	-52.74		1
and100 0.84		thr 684													
[DDQN-Mod]	Ep	813		R	35.44		avgR100	170.56		eps 0.020		Qval	-52.59		1
and100 0.83		thr 593													
[DDQN-Mod]	Ep	814		R	221.64		avgR100	173.02		eps 0.020		Qval	-53.54		1
and100 0.84		thr 246													
[DDQN-Mod]	Ep	815		R	196.62		avgR100	173.49		eps 0.020		Qval	-52.40		1
and100 0.84		thr 270													
[DDQN-Mod]	Ep	816		R	217.21		avgR100	174.45		eps 0.020		Qval	-51.58		1
and100 0.84		thr 235													
[DDQN-Mod]	Ep	817		R	193.18		avgR100	173.99		eps 0.020		Qval	-54.17		1
and100 0.84		thr 244													
[DDQN-Mod]	Ep	818		R	165.57		avgR100	173.76		eps 0.020		Qval	-53.69		1
and100 0.83		thr 249													
[DDQN-Mod]	Ep	819		R	221.50		avgR100	173.68		eps 0.020		Qval	-53.36		1
and100 0.83		thr 237													
[DDQN-Mod]	Ep	820		R	201.47		avgR100	175.80		eps 0.020		Qval	-51.58		1
and100 0.84		thr 278													
[DDQN-Mod]	Ep	821		R	112.12		avgR100	174.55		eps 0.020		Qval	-50.73		1
and100 0.84		thr 417													
[DDQN-Mod]	Ep	822		R	228.65		avgR100	177.12		eps 0.020		Qval	-50.98		1
and100 0.85		thr 186													
[DDQN-Mod]	Ep	823		R	248.17		avgR100	178.32		eps 0.020		Qval	-50.49		1
and100 0.86		thr 284													
[DDQN-Mod]	Ep	824		R	228.21		avgR100	178.45		eps 0.020		Qval	-50.14		1
and100 0.86		thr 258													
[DDQN-Mod]	Ep	825		R	232.02		avgR100	179.02		eps 0.020		Qval	-50.40		1
and100 0.86		thr 259													
[DDQN-Mod]	Ep	826		R	201.29		avgR100	179.69		eps 0.020		Qval	-50.01		1
and100 0.86		thr 260													
[DDQN-Mod]	Ep	827		R	246.71		avgR100	180.67		eps 0.020		Qval	-49.81		1
and100 0.86		thr 205													
[DDQN-Mod]	Ep	828		R	227.41		avgR100	181.38		eps 0.020		Qval	-49.74		1
and100 0.86		thr 226													
[DDQN-Mod]	Ep	829		R	200.21		avgR100	181.15		eps 0.020		Qval	-50.44		1
and100 0.86		thr 253													
[DDQN-Mod]	Ep	830		R	203.21		avgR100	181.22		eps 0.020		Qval	-50.41		1
and100 0.85		thr 233													
[DDQN-Mod]	Ep	831		R	-207.75		avgR100	176.64		eps 0.020		Qval	-50.62		1
and100 0.84		thr 103													
[DDQN-Mod]	Ep	832		R	241.62		avgR100	178.47		eps 0.020		Qval	-50.02		1
and100 0.84		thr 257													
[DDQN-Mod]	Ep	833		R	230.76		avgR100	178.75		eps 0.020		Qval	-50.12		1
and100 0.84		thr 250													
[DDQN-Mod]	Ep	834		R	169.36		avgR100	177.88		eps 0.020		Qval	-50.50		1
and100 0.84		thr 293													
[DDQN-Mod]	Ep	835		R	246.58		avgR100	178.51		eps 0.020		Qval	-50.82		1
and100 0.84		thr 234													
[DDQN-Mod]	Ep	836		R	162.39		avgR100	178.63		eps 0.020		Qval	-51.41		1
and100 0.83		thr 253													
[DDQN-Mod]	Ep	837		R	263.41		avgR100	181.41		eps 0.020		Qval	-51.28		1
and100 0.84		thr 246													
[DDQN-Mod]	Ep	838		R	239.15		avgR100	181.79		eps 0.020		Qval	-51.22		1
and100 0.84		thr 236													
[DDQN-Mod]	Ep	839		R	45.69		avgR100	180.32		eps 0.020		Qval	-50.96		1
and100 0.84		thr 168													
[DDQN-Mod]	Ep	840		R	226.19		avgR100	180.83		eps 0.020		Qval	-51.27		1
and100 0.84		thr 251													

[DDQN-Mod] Ep 841 R 215.25 avgR100 181.61 eps 0.020 Qval -50.52 1
and100 0.84 thr 205
[DDQN-Mod] Ep 842 R 269.03 avgR100 182.08 eps 0.020 Qval -49.52 1
and100 0.84 thr 205
[DDQN-Mod] Ep 843 R 205.16 avgR100 182.40 eps 0.020 Qval -50.74 1
and100 0.84 thr 238
[DDQN-Mod] Ep 844 R 213.51 avgR100 182.30 eps 0.020 Qval -50.27 1
and100 0.84 thr 256
[DDQN-Mod] Ep 845 R 215.91 avgR100 183.19 eps 0.020 Qval -51.05 1
and100 0.85 thr 262
[DDQN-Mod] Ep 846 R 238.22 avgR100 183.68 eps 0.020 Qval -49.48 1
and100 0.85 thr 230
[DDQN-Mod] Ep 847 R 206.11 avgR100 183.60 eps 0.020 Qval -50.54 1
and100 0.85 thr 252
[DDQN-Mod] Ep 848 R 192.68 avgR100 183.14 eps 0.020 Qval -50.27 1
and100 0.85 thr 264
[DDQN-Mod] Ep 849 R 181.49 avgR100 183.19 eps 0.020 Qval -48.91 1
and100 0.85 thr 349
[DDQN-Mod] Ep 850 R 155.74 avgR100 183.37 eps 0.020 Qval -48.96 1
and100 0.84 thr 242
[DDQN-Mod] Ep 851 R 188.07 avgR100 184.95 eps 0.020 Qval -51.08 1
and100 0.85 thr 250
[DDQN-Mod] Ep 852 R 37.59 avgR100 183.40 eps 0.020 Qval -51.78 1
and100 0.84 thr 495
[DDQN-Mod] Ep 853 R 232.45 avgR100 183.80 eps 0.020 Qval -49.95 1
and100 0.84 thr 198
[DDQN-Mod] Ep 854 R 168.89 avgR100 183.00 eps 0.020 Qval -50.31 1
and100 0.83 thr 294
[DDQN-Mod] Ep 855 R 217.82 avgR100 186.03 eps 0.020 Qval -50.25 1
and100 0.84 thr 270
[DDQN-Mod] Ep 856 R 189.51 avgR100 184.98 eps 0.020 Qval -48.93 1
and100 0.84 thr 269
[DDQN-Mod] Ep 857 R 254.48 avgR100 185.60 eps 0.020 Qval -48.52 1
and100 0.84 thr 212
[DDQN-Mod] Ep 858 R 212.74 avgR100 185.62 eps 0.020 Qval -48.86 1
and100 0.83 thr 189
[DDQN-Mod] Ep 859 R -100.44 avgR100 182.83 eps 0.020 Qval -51.05 1
and100 0.82 thr 772
[DDQN-Mod] Ep 860 R 200.04 avgR100 183.27 eps 0.020 Qval -51.30 1
and100 0.81 thr 167
[DDQN-Mod] Ep 861 R 253.27 avgR100 184.04 eps 0.020 Qval -50.93 1
and100 0.81 thr 237
[DDQN-Mod] Ep 862 R 216.87 avgR100 184.35 eps 0.020 Qval -51.16 1
and100 0.81 thr 309
[DDQN-Mod] Ep 863 R 186.83 avgR100 183.86 eps 0.020 Qval -50.66 1
and100 0.81 thr 253
[DDQN-Mod] Ep 864 R 258.95 avgR100 184.38 eps 0.020 Qval -50.38 1
and100 0.81 thr 194
[DDQN-Mod] Ep 865 R 187.63 avgR100 184.10 eps 0.020 Qval -48.91 1
and100 0.80 thr 256
[DDQN-Mod] Ep 866 R 201.88 avgR100 184.27 eps 0.020 Qval -49.92 1
and100 0.80 thr 245
[DDQN-Mod] Ep 867 R 227.38 avgR100 184.63 eps 0.020 Qval -49.15 1
and100 0.80 thr 213
[DDQN-Mod] Ep 868 R 231.79 avgR100 185.18 eps 0.020 Qval -50.23 1
and100 0.79 thr 173
[DDQN-Mod] Ep 869 R 231.07 avgR100 185.32 eps 0.020 Qval -49.51 1
and100 0.79 thr 288
[DDQN-Mod] Ep 870 R 301.06 avgR100 189.20 eps 0.020 Qval -50.18 1
and100 0.80 thr 168

[DDQN-Mod] Ep 871 R 222.17 avgR100 189.77 eps 0.020 Qval -51.72 1
and100 0.80 thr 230
[DDQN-Mod] Ep 872 R 213.27 avgR100 190.26 eps 0.020 Qval -49.93 1
and100 0.80 thr 212
[DDQN-Mod] Ep 873 R -144.33 avgR100 187.01 eps 0.020 Qval -51.07 1
and100 0.79 thr 745
[DDQN-Mod] Ep 874 R 254.47 avgR100 187.53 eps 0.020 Qval -51.38 1
and100 0.79 thr 189
[DDQN-Mod] Ep 875 R -27.60 avgR100 185.31 eps 0.020 Qval -51.90 1
and100 0.78 thr 127
[DDQN-Mod] Ep 876 R 261.38 avgR100 185.58 eps 0.020 Qval -52.13 1
and100 0.79 thr 245
[DDQN-Mod] Ep 877 R 228.70 avgR100 186.36 eps 0.020 Qval -52.58 1
and100 0.79 thr 203
[DDQN-Mod] Ep 878 R 249.63 avgR100 186.32 eps 0.020 Qval -51.08 1
and100 0.79 thr 203
[DDQN-Mod] Ep 879 R 256.48 avgR100 186.88 eps 0.020 Qval -52.16 1
and100 0.79 thr 201
[DDQN-Mod] Ep 880 R 159.39 avgR100 186.13 eps 0.020 Qval -52.59 1
and100 0.79 thr 279
[DDQN-Mod] Ep 881 R 16.28 avgR100 184.20 eps 0.020 Qval -51.26 1
and100 0.78 thr 706
[DDQN-Mod] Ep 882 R 266.99 avgR100 184.68 eps 0.020 Qval -50.97 1
and100 0.79 thr 217
[DDQN-Mod] Ep 883 R 227.24 avgR100 184.61 eps 0.020 Qval -51.61 1
and100 0.79 thr 240
[DDQN-Mod] Ep 884 R 185.08 avgR100 184.36 eps 0.020 Qval -51.89 1
and100 0.79 thr 265
[DDQN-Mod] Ep 885 R 222.87 avgR100 184.38 eps 0.020 Qval -52.94 1
and100 0.79 thr 235
[DDQN-Mod] Ep 886 R 203.15 avgR100 186.67 eps 0.020 Qval -52.95 1
and100 0.80 thr 226
[DDQN-Mod] Ep 887 R 250.25 avgR100 186.20 eps 0.020 Qval -52.12 1
and100 0.80 thr 171
[DDQN-Mod] Ep 888 R 233.98 avgR100 186.30 eps 0.020 Qval -52.20 1
and100 0.80 thr 237
[DDQN-Mod] Ep 889 R 239.38 avgR100 186.75 eps 0.020 Qval -52.20 1
and100 0.80 thr 186
[DDQN-Mod] Ep 890 R 215.10 avgR100 186.42 eps 0.020 Qval -52.09 1
and100 0.80 thr 227
[DDQN-Mod] Ep 891 R 276.10 avgR100 186.80 eps 0.020 Qval -52.09 1
and100 0.80 thr 162
[DDQN-Mod] Ep 892 R 26.20 avgR100 185.37 eps 0.020 Qval -52.78 1
and100 0.80 thr 625
[DDQN-Mod] Ep 893 R 196.45 avgR100 185.39 eps 0.020 Qval -51.99 1
and100 0.80 thr 306
[DDQN-Mod] Ep 894 R 220.11 avgR100 185.55 eps 0.020 Qval -52.98 1
and100 0.79 thr 169
[DDQN-Mod] Ep 895 R 245.74 avgR100 185.74 eps 0.020 Qval -51.97 1
and100 0.78 thr 159
[DDQN-Mod] Ep 896 R 234.18 avgR100 186.65 eps 0.020 Qval -53.38 1
and100 0.78 thr 189
[DDQN-Mod] Ep 897 R 212.84 avgR100 186.07 eps 0.020 Qval -53.18 1
and100 0.78 thr 252
[DDQN-Mod] Ep 898 R 282.84 avgR100 187.22 eps 0.020 Qval -52.65 1
and100 0.78 thr 202
[DDQN-Mod] Ep 899 R 228.50 avgR100 188.20 eps 0.020 Qval -51.13 1
and100 0.77 thr 153
[DDQN-Mod] Ep 900 R 186.56 avgR100 187.35 eps 0.020 Qval -52.84 1
and100 0.76 thr 211

[DDQN-Mod] Ep and100 0.76	901 R thr 285	172.16	avgR100	187.35	eps 0.020	Qval -52.38	1
[DDQN-Mod] Ep and100 0.75	902 R thr 86	-40.03	avgR100	184.48	eps 0.020	Qval -53.13	1
[DDQN-Mod] Ep and100 0.75	903 R thr 281	198.41	avgR100	184.69	eps 0.020	Qval -51.84	1
[DDQN-Mod] Ep and100 0.75	904 R thr 316	178.02	avgR100	184.03	eps 0.020	Qval -52.35	1
[DDQN-Mod] Ep and100 0.75	905 R thr 603	-89.98	avgR100	183.74	eps 0.020	Qval -52.47	1
[DDQN-Mod] Ep and100 0.75	906 R thr 216	247.39	avgR100	184.27	eps 0.020	Qval -51.63	1
[DDQN-Mod] Ep and100 0.74	907 R thr 590	22.46	avgR100	182.62	eps 0.020	Qval -52.07	1
[DDQN-Mod] Ep and100 0.73	908 R thr 198	165.81	avgR100	182.02	eps 0.020	Qval -50.96	1
[DDQN-Mod] Ep and100 0.72	909 R thr 546	-38.85	avgR100	179.29	eps 0.020	Qval -52.48	1
[DDQN-Mod] Ep and100 0.72	910 R thr 324	196.82	avgR100	179.48	eps 0.020	Qval -51.88	1
[DDQN-Mod] Ep and100 0.73	911 R thr 270	209.21	avgR100	181.83	eps 0.020	Qval -52.25	1
[DDQN-Mod] Ep and100 0.74	912 R thr 199	236.72	avgR100	184.91	eps 0.020	Qval -52.24	1
[DDQN-Mod] Ep and100 0.74	913 R thr 745	-104.26	avgR100	183.51	eps 0.020	Qval -52.53	1
[DDQN-Mod] Ep and100 0.74	914 R thr 265	198.45	avgR100	183.28	eps 0.020	Qval -53.21	1
[DDQN-Mod] Ep and100 0.74	915 R thr 201	223.06	avgR100	183.54	eps 0.020	Qval -52.26	1
[DDQN-Mod] Ep and100 0.74	916 R thr 193	248.00	avgR100	183.85	eps 0.020	Qval -54.45	1
[DDQN-Mod] Ep and100 0.73	917 R thr 229	165.00	avgR100	183.57	eps 0.020	Qval -52.73	1
[DDQN-Mod] Ep and100 0.74	918 R thr 278	193.60	avgR100	183.85	eps 0.020	Qval -54.51	1
[DDQN-Mod] Ep and100 0.73	919 R thr 247	202.31	avgR100	183.66	eps 0.020	Qval -52.92	1
[DDQN-Mod] Ep and100 0.73	920 R thr 208	220.04	avgR100	183.84	eps 0.020	Qval -52.87	1
[DDQN-Mod] Ep and100 0.73	921 R thr 254	230.67	avgR100	185.03	eps 0.020	Qval -54.20	1
[DDQN-Mod] Ep and100 0.73	922 R thr 230	209.50	avgR100	184.84	eps 0.020	Qval -53.46	1
[DDQN-Mod] Ep and100 0.73	923 R thr 228	224.07	avgR100	184.59	eps 0.020	Qval -52.86	1
[DDQN-Mod] Ep and100 0.73	924 R thr 218	219.14	avgR100	184.50	eps 0.020	Qval -54.04	1
[DDQN-Mod] Ep and100 0.72	925 R thr 326	168.98	avgR100	183.87	eps 0.020	Qval -53.97	1
[DDQN-Mod] Ep and100 0.72	926 R thr 229	232.26	avgR100	184.18	eps 0.020	Qval -52.99	1
[DDQN-Mod] Ep and100 0.71	927 R thr 255	161.52	avgR100	183.33	eps 0.020	Qval -51.99	1
[DDQN-Mod] Ep and100 0.70	928 R thr 461	23.30	avgR100	181.29	eps 0.020	Qval -51.80	1
[DDQN-Mod] Ep and100 0.70	929 R thr 219	233.07	avgR100	181.62	eps 0.020	Qval -53.33	1
[DDQN-Mod] Ep and100 0.71	930 R thr 206	236.82	avgR100	181.95	eps 0.020	Qval -52.29	1

[DDQN-Mod] Ep and100 0.71	931 R thr 776	-112.29	avgR100	182.91	eps 0.020	Qval -54.20	1
[DDQN-Mod] Ep and100 0.71	932 R thr 184	293.28	avgR100	183.43	eps 0.020	Qval -53.24	1
[DDQN-Mod] Ep and100 0.71	933 R thr 244	217.81	avgR100	183.30	eps 0.020	Qval -53.09	1
[DDQN-Mod] Ep and100 0.71	934 R thr 251	213.58	avgR100	183.74	eps 0.020	Qval -52.21	1
[DDQN-Mod] Ep and100 0.71	935 R thr 227	234.82	avgR100	183.62	eps 0.020	Qval -51.85	1
[DDQN-Mod] Ep and100 0.72	936 R thr 148	260.51	avgR100	184.60	eps 0.020	Qval -52.86	1
[DDQN-Mod] Ep and100 0.72	937 R thr 473	144.69	avgR100	183.41	eps 0.020	Qval -52.93	1
[DDQN-Mod] Ep and100 0.71	938 R thr 266	-85.44	avgR100	180.17	eps 0.020	Qval -52.97	1
[DDQN-Mod] Ep and100 0.71	939 R thr 168	273.58	avgR100	182.45	eps 0.020	Qval -52.98	1
[DDQN-Mod] Ep and100 0.71	940 R thr 239	256.24	avgR100	182.75	eps 0.020	Qval -51.50	1
[DDQN-Mod] Ep and100 0.70	941 R thr 247	171.53	avgR100	182.31	eps 0.020	Qval -50.92	1
[DDQN-Mod] Ep and100 0.70	942 R thr 301	231.96	avgR100	181.94	eps 0.020	Qval -51.37	1
[DDQN-Mod] Ep and100 0.70	943 R thr 199	254.55	avgR100	182.43	eps 0.020	Qval -52.48	1
[DDQN-Mod] Ep and100 0.70	944 R thr 229	223.45	avgR100	182.53	eps 0.020	Qval -52.89	1
[DDQN-Mod] Ep and100 0.70	945 R thr 235	208.25	avgR100	182.46	eps 0.020	Qval -52.40	1
[DDQN-Mod] Ep and100 0.70	946 R thr 215	216.74	avgR100	182.24	eps 0.020	Qval -52.73	1
[DDQN-Mod] Ep and100 0.69	947 R thr 263	160.27	avgR100	181.78	eps 0.020	Qval -52.35	1
[DDQN-Mod] Ep and100 0.69	948 R thr 179	268.58	avgR100	182.54	eps 0.020	Qval -52.62	1
[DDQN-Mod] Ep and100 0.69	949 R thr 163	236.12	avgR100	183.09	eps 0.020	Qval -52.66	1
[DDQN-Mod] Ep and100 0.70	950 R thr 162	259.50	avgR100	184.13	eps 0.020	Qval -52.85	1
[DDQN-Mod] Ep and100 0.70	951 R thr 203	280.13	avgR100	185.05	eps 0.020	Qval -52.72	1
[DDQN-Mod] Ep and100 0.71	952 R thr 350	174.42	avgR100	186.42	eps 0.020	Qval -53.68	1
[DDQN-Mod] Ep and100 0.71	953 R thr 166	302.99	avgR100	187.12	eps 0.020	Qval -52.72	1
[DDQN-Mod] Ep and100 0.72	954 R thr 210	287.61	avgR100	188.31	eps 0.020	Qval -52.72	1
[DDQN-Mod] Ep and100 0.72	955 R thr 211	254.33	avgR100	188.67	eps 0.020	Qval -53.55	1
[DDQN-Mod] Ep and100 0.72	956 R thr 224	256.77	avgR100	189.35	eps 0.020	Qval -53.80	1
[DDQN-Mod] Ep and100 0.72	957 R thr 283	170.20	avgR100	188.50	eps 0.020	Qval -52.52	1
[DDQN-Mod] Ep and100 0.73	958 R thr 237	205.63	avgR100	188.43	eps 0.020	Qval -54.06	1
[DDQN-Mod] Ep and100 0.74	959 R thr 221	210.48	avgR100	191.54	eps 0.020	Qval -54.22	1
[DDQN-Mod] Ep and100 0.75	960 R thr 206	218.06	avgR100	191.72	eps 0.020	Qval -55.21	1

[DDQN-Mod] Ep and100 0.75	961 R thr 274	183.33	avgR100	191.02	eps 0.020	Qval -53.38	1
[DDQN-Mod] Ep and100 0.75	962 R thr 204	260.48	avgR100	191.46	eps 0.020	Qval -56.12	1
[DDQN-Mod] Ep and100 0.75	963 R thr 236	244.33	avgR100	192.03	eps 0.020	Qval -54.92	1
[DDQN-Mod] Ep and100 0.75	964 R thr 205	261.37	avgR100	192.06	eps 0.020	Qval -55.26	1
[DDQN-Mod] Ep and100 0.76	965 R thr 199	244.02	avgR100	192.62	eps 0.020	Qval -55.35	1
[DDQN-Mod] Ep and100 0.76	966 R thr 265	258.75	avgR100	193.19	eps 0.020	Qval -56.71	1
[DDQN-Mod] Ep and100 0.76	967 R thr 217	247.61	avgR100	193.39	eps 0.020	Qval -55.26	1
[DDQN-Mod] Ep and100 0.77	968 R thr 218	252.78	avgR100	193.60	eps 0.020	Qval -55.28	1
[DDQN-Mod] Ep and100 0.77	969 R thr 171	291.99	avgR100	194.21	eps 0.020	Qval -57.04	1
[DDQN-Mod] Ep and100 0.77	970 R thr 247	264.47	avgR100	193.85	eps 0.020	Qval -56.10	1
[DDQN-Mod] Ep and100 0.76	971 R thr 283	152.53	avgR100	193.15	eps 0.020	Qval -53.31	1
[DDQN-Mod] Ep and100 0.76	972 R thr 202	213.26	avgR100	193.15	eps 0.020	Qval -54.00	1
[DDQN-Mod] Ep and100 0.76	973 R thr 168	-39.28	avgR100	194.20	eps 0.020	Qval -53.69	1
[DDQN-Mod] Ep and100 0.76	974 R thr 188	226.58	avgR100	193.92	eps 0.020	Qval -54.30	1
[DDQN-Mod] Ep and100 0.76	975 R thr 654	-67.90	avgR100	193.52	eps 0.020	Qval -54.83	1
[DDQN-Mod] Ep and100 0.76	976 R thr 272	225.81	avgR100	193.16	eps 0.020	Qval -54.73	1
[DDQN-Mod] Ep and100 0.75	977 R thr 194	216.07	avgR100	193.04	eps 0.020	Qval -54.50	1
[DDQN-Mod] Ep and100 0.74	978 R thr 195	150.10	avgR100	192.04	eps 0.020	Qval -54.19	1
[DDQN-Mod] Ep and100 0.74	979 R thr 220	204.17	avgR100	191.52	eps 0.020	Qval -55.58	1
[DDQN-Mod] Ep and100 0.74	980 R thr 255	255.45	avgR100	192.48	eps 0.020	Qval -55.56	1
[DDQN-Mod] Ep and100 0.74	981 R thr 141	-401.59	avgR100	188.30	eps 0.020	Qval -55.94	1
[DDQN-Mod] Ep and100 0.74	982 R thr 177	276.85	avgR100	188.40	eps 0.020	Qval -57.30	1
[DDQN-Mod] Ep and100 0.74	983 R thr 240	222.68	avgR100	188.35	eps 0.020	Qval -57.47	1
[DDQN-Mod] Ep and100 0.74	984 R thr 253	208.86	avgR100	188.59	eps 0.020	Qval -55.43	1
[DDQN-Mod] Ep and100 0.74	985 R thr 179	239.86	avgR100	188.76	eps 0.020	Qval -55.36	1
[DDQN-Mod] Ep and100 0.74	986 R thr 179	244.17	avgR100	189.17	eps 0.020	Qval -57.22	1
[DDQN-Mod] Ep and100 0.74	987 R thr 402	151.43	avgR100	188.18	eps 0.020	Qval -56.89	1
[DDQN-Mod] Ep and100 0.74	988 R thr 329	212.20	avgR100	187.96	eps 0.020	Qval -53.52	1
[DDQN-Mod] Ep and100 0.73	989 R thr 185	219.76	avgR100	187.77	eps 0.020	Qval -54.73	1
[DDQN-Mod] Ep and100 0.73	990 R thr 224	199.05	avgR100	187.61	eps 0.020	Qval -54.97	1

```
[DDQN-Mod] Ep 991 | R 268.02 | avgR100 187.53 | eps 0.020 | Qval -54.73 | 1
and100 0.73 | thr 177
[DDQN-Mod] Ep 992 | R 263.99 | avgR100 189.91 | eps 0.020 | Qval -55.30 | 1
and100 0.74 | thr 150
[DDQN-Mod] Ep 993 | R -33.18 | avgR100 187.61 | eps 0.020 | Qval -55.02 | 1
and100 0.73 | thr 151
[DDQN-Mod] Ep 994 | R 224.21 | avgR100 187.65 | eps 0.020 | Qval -54.93 | 1
and100 0.73 | thr 124
[DDQN-Mod] Ep 995 | R 211.76 | avgR100 187.31 | eps 0.020 | Qval -54.73 | 1
and100 0.74 | thr 238
[DDQN-Mod] Ep 996 | R 255.47 | avgR100 187.52 | eps 0.020 | Qval -55.46 | 1
and100 0.74 | thr 218
[DDQN-Mod] Ep 997 | R 215.38 | avgR100 187.55 | eps 0.020 | Qval -55.43 | 1
and100 0.74 | thr 229
[DDQN-Mod] Ep 998 | R 214.95 | avgR100 186.87 | eps 0.020 | Qval -55.13 | 1
and100 0.74 | thr 228
[DDQN-Mod] Ep 999 | R 147.90 | avgR100 186.06 | eps 0.020 | Qval -55.22 | 1
and100 0.74 | thr 405
[DDQN-Mod] Ep 1000 | R 236.61 | avgR100 186.56 | eps 0.020 | Qval -54.72 | 1
and100 0.75 | thr 221
[DDQN-Mod] finished 1000 episodes in 3.9 min (final avgR100 = 186.56)
```

All four configurations trained.

```
In [31]: # Persist metrics to disk so plots/analysis can be reproduced without retraining
import pickle

_dump = {k: {kk: vv for kk, vv in v.items() if kk != "agent"} for k, v in RESULT
with open("training_results.pkl", "wb") as f:
    pickle.dump(_dump, f)
print("Saved metrics to training_results.pkl")
```

Saved metrics to training_results.pkl

5. Performance Evaluation

We compare the four trained agents on four metrics:

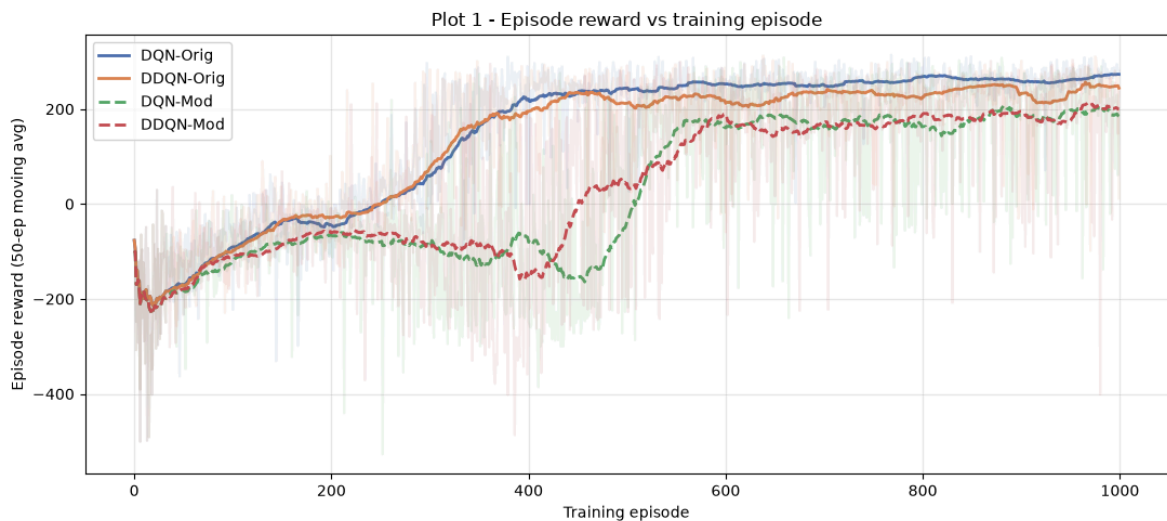
1. **Episode reward** vs training episode.
2. **Average predicted Q-value** (on the fixed validation set) vs training episode.
3. **Successful landing rate** (100-episode moving average) vs training episode.
4. **Average thruster activations per episode** vs training episode.

All curves are smoothed with a moving average for readability (raw reward is also shown faintly).

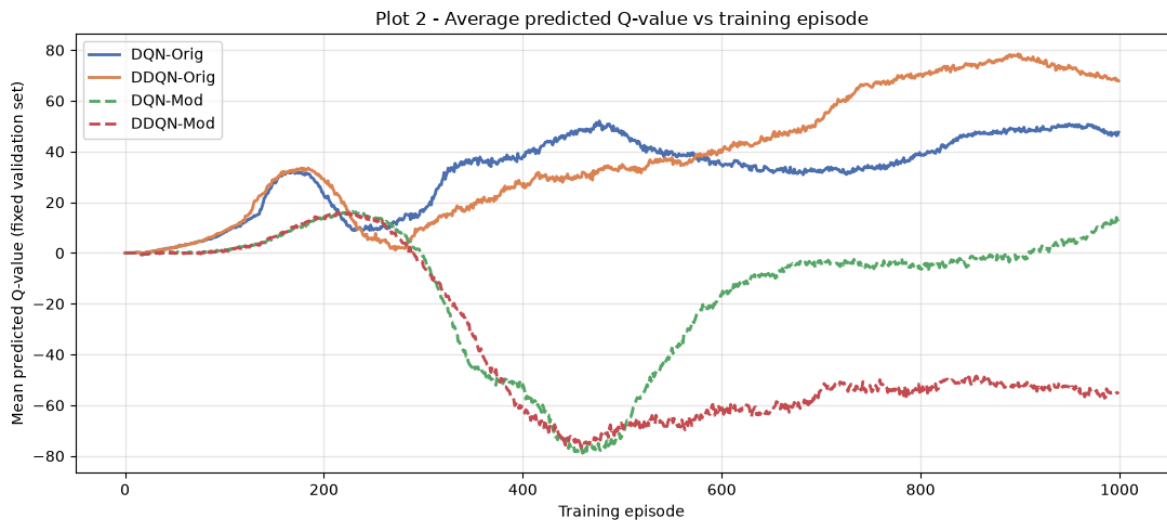
```
In [32]: def moving_average(x, window: int = 100):
    """Return the trailing moving average of ``x`` with the given window."""
    x = np.asarray(x, dtype=np.float64)
    if len(x) < 1:
        return x
    out = np.empty(len(x))
    csum = np.cumsum(np.insert(x, 0, 0.0))
    for i in range(len(x)):
        lo = max(0, i + 1 - window)
        out[i] = (csum[i + 1] - csum[lo]) / (i + 1 - lo)
    return out
```

```
# Consistent colour/style per configuration across all figures.
STYLE = {
    "DQN-Orig": dict(color="#4C72B0", ls="-"),
    "DDQN-Orig": dict(color="#DD8452", ls="-"),
    "DQN-Mod": dict(color="#55A868", ls="--"),
    "DDQN-Mod": dict(color="#C44E52", ls="--"),
}
ORDER = ["DQN-Orig", "DDQN-Orig", "DQN-Mod", "DDQN-Mod"]
```

```
In [33]: # ---- Plot 1: Episode reward vs episode -----
plt.figure(figsize=(11, 5))
for tag in ORDER:
    r = RESULTS[tag]["rewards"]
    plt.plot(moving_average(r, 50), label=tag, **STYLE[tag], lw=2)
    plt.plot(r, color=STYLE[tag]["color"], alpha=0.10) # faint raw signal
plt.xlabel("Training episode")
plt.ylabel("Episode reward (50-ep moving avg)")
plt.title("Plot 1 - Episode reward vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
In [34]: # ---- Plot 2: Average predicted Q-value vs episode -----
plt.figure(figsize=(11, 5))
for tag in ORDER:
    plt.plot(RESULTS[tag]["avg_q"], label=tag, **STYLE[tag], lw=2)
plt.xlabel("Training episode")
plt.ylabel("Mean predicted Q-value (fixed validation set)")
plt.title("Plot 2 - Average predicted Q-value vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
In [35]: # ---- Plot 3: Successful landing rate (100-ep moving avg) -----
plt.figure(figsize=(11, 5))
for tag in ORDER:
    plt.plot(moving_average(RESULTS[tag]["landings"], 100) * 100.0,
             label=tag, **STYLE[tag], lw=2)
plt.xlabel("Training episode")
plt.ylabel("Successful landing rate (%) - 100-ep moving avg")
plt.title("Plot 3 - Successful landing rate vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

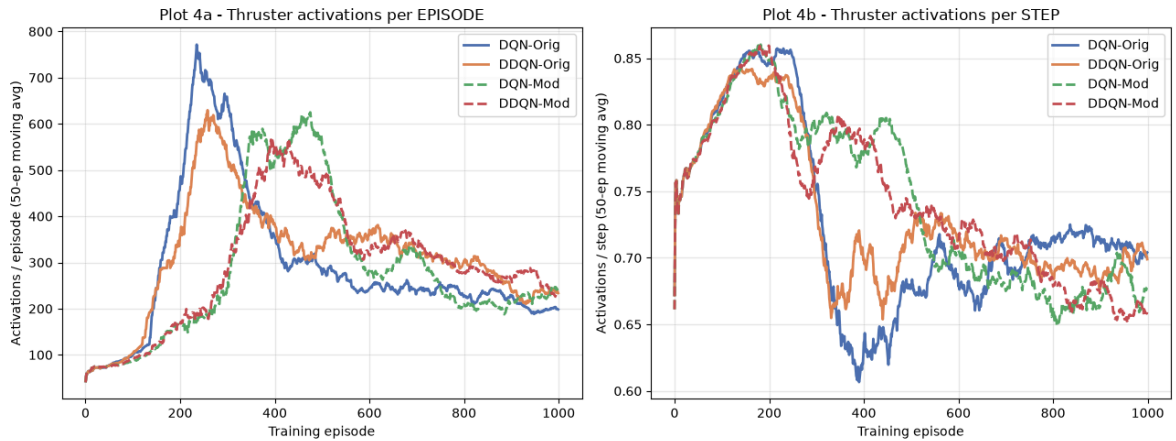


```
In [36]: # ---- Plot 4: thruster activations per episode (required) + per step (insight)
# Left panel : activations per EPISODE (the metric requested by the assignment)
# Right panel : activations per STEP, which normalises out episode length and re
#               the true "firing intensity" of the learned policy.
fig, ax = plt.subplots(1, 2, figsize=(13, 5))
for tag in ORDER:
    thr = np.asarray(RESULTS[tag]["thrusters"], dtype=float)
    steps = np.asarray(RESULTS[tag]["steps"], dtype=float)
    ax[0].plot(moving_average(thr, 50), label=tag, **STYLE[tag], lw=2)
    ax[1].plot(moving_average(thr / np.maximum(steps, 1), 50), label=tag, **STYL
ax[0].set_title("Plot 4a - Thruster activations per EPISODE")
ax[0].set_xlabel("Training episode")
ax[0].set_ylabel("Activations / episode (50-ep moving avg)")
ax[1].set_title("Plot 4b - Thruster activations per STEP")
```

```

ax[1].set_xlabel("Training episode")
ax[1].set_ylabel("Activations / step (50-ep moving avg)")
for a in ax:
    a.legend()
    a.grid(alpha=0.3)
plt.tight_layout()
plt.show()

```



```

In [37]: # ---- Numerical summary table (last-100-episode averages) -----
hdr = f"{'Config':<12}{'AvgReward':>11}{'Land%':>8}{'AvgQ':>9}{'Thr/Ep':>9}{'Steps':>10}"
print(hdr)
print("-" * len(hdr))
SUMMARY = {}
for tag in ORDER:
    r = RESULTS[tag]
    avg_reward = float(np.mean(r["rewards"][-100:]))
    land_rate = float(np.mean(r["landings"][-100:]) * 100)
    avg_q = float(np.mean(r["avg_q"][-100:]))
    thr = float(np.mean(r["thrusters"][-100:]))
    steps = float(np.mean(r["steps"][-100:]))
    thr_per_step = thr / max(1.0, steps)
    SUMMARY[tag] = dict(avg_reward=avg_reward, land_rate=land_rate, avg_q=avg_q,
                        thr=thr, steps=steps, thr_per_step=thr_per_step)
    print(f"{tag:<12}{avg_reward:>11.2f}{land_rate:>8.1f}{avg_q:>9.2f}"
          f"{thr:>9.1f}{steps:>8.1f}{thr_per_step:>10.3f}")
print("-" * len(hdr))

# Key comparative quantities used directly in the discussion below.
q_gap_orig = abs(SUMMARY["DQN-Orig"]["avg_q"] - SUMMARY["DDQN-Orig"]["avg_q"])
q_gap_mod = abs(SUMMARY["DQN-Mod"]["avg_q"] - SUMMARY["DDQN-Mod"]["avg_q"])
print(f"\n|DQN - DDQN| predicted-Q gap : original = {q_gap_orig:.2f}, "
      f"modified = {q_gap_mod:.2f}")
print(f"Gap is {'LARGER' if q_gap_mod > q_gap_orig else 'smaller'} under stochastic "
      f"action failure.")

```

Config	AvgReward	Land%	AvgQ	Thr/Ep	Steps	Thr/Step
DQN-Orig	267.41	89.0	49.00	193.8	275.3	0.704
DDQN-Orig	240.28	85.0	72.60	235.2	339.3	0.693
DQN-Mod	191.05	83.0	5.18	230.7	343.5	0.672
DDQN-Mod	186.56	75.0	-53.78	257.0	394.8	0.651

|DQN - DDQN| predicted-Q gap : original = 23.60, modified = 58.96
Gap is LARGER under stochastic action failure.

Interpretation of the plots

(Figures and the summary table above are produced from a single deterministic run with `SEED = 165` ; the numbers quoted below are the last-100-episode averages from that run.)

- **Plot 1 (Episode reward).** All four agents learn stable, positive policies. The **original**-environment agents reach higher returns (DQN ~267, DDQN ~240) than the **modified** ones (DQN ~191, DDQN ~187): the -0.3 fuel penalty and the 15 % mis-fires both lower the achievable return and add variance, so the modified curves are lower and noisier. DQN and DDQN track each other closely, with no consistent winner.
- **Plot 2 (Predicted Q-value).** The key observation is the **size** of the gap between the two algorithms. On the original environment the DQN and DDQN value estimates are relatively close ($|\text{gap}| \sim 24$), whereas on the modified environment they diverge strongly ($|\text{gap}| \sim 59$): DDQN drives its predicted values far **lower** (even negative) than DQN. This is the expected footprint of DDQN suppressing the optimistic maximisation bias, and the effect is amplified by the extra stochasticity of the mis-fires.
- **Plot 3 (Landing rate).** Success rises steadily for all four agents. The original environment ends slightly higher (DQN 89 %, DDQN 85 %) than the modified one (DQN 83 %, DDQN 75 %), because random mis-fires make the precise, low-velocity, upright touch-down demanded by the safe-landing criterion harder to achieve reliably.
- **Plot 4 (Thruster usage - two views).** *Per episode* (4a), the **modified** agents fire **more** often (~231-257) than the original ones (~194-235). This looks paradoxical for a fuel-penalised agent, but it is an episode-length artefact: modified-environment episodes last **longer** (~344-395 vs ~275-339 steps) because the agent needs more time - and must re-issue commands to make up for the ~15 % that silently fail. *Per step* (4b) the picture reverses: the modified agents fire at a **lower** rate (~0.65-0.67) than the original agents (~0.69-0.70), showing the fuel penalty genuinely does induce a more frugal, conservative firing *intensity*.

6. Discussion

(Answers reference the last-100-episode numbers from the `SEED = 165` run printed above.)

1. Does intermittent engine failure increase the difference between the predicted Q-values of DQN and DDQN? Yes. The magnitude of the gap between the two agents' average predicted Q-values grows from **$|\text{gap}| \sim 24$ on the original environment to $|\text{gap}| \sim 59$ on the modified environment** (Plot 2 / summary table). DQN uses a single `max` for both selecting and evaluating the next action, which over-estimates values;

DDQN decouples selection (online net) from evaluation (target net) and therefore produces markedly lower, more conservative estimates. Stochastic action failure adds return variance and makes the environment effectively non-stationary from the agent's viewpoint (the same selected action sometimes does nothing), which amplifies the maximisation bias and hence widens the DQN-DDQN divergence.

2. Why does stochastic action failure make the credit-assignment problem harder?

Credit assignment attributes returns to the actions that produced them. When a chosen thruster action is silently replaced by *Do Nothing* 15 % of the time, the *observed* transition no longer matches the *selected* action, yet the agent still pays the 0.3 fuel penalty. Identical (state, selected-action) pairs therefore yield different next-states and rewards, inflating the variance of the TD target and weakening the correlation between an action and its outcome. Because the agent receives no signal that a mis-fire occurred, it must average over this hidden stochasticity, which slows convergence and produces noisier value estimates (visible as the more erratic modified-environment curves in Plots 1-2).

3. Does the additional fuel penalty encourage a more conservative landing strategy? Yes, once episode length is accounted for.

Plot 4b shows the modified-environment agents fire at a **lower per-step rate** (~ 0.65 - 0.67) than the original-environment agents (~ 0.69 - 0.70), which is direct evidence that the -0.3 per-press cost makes the policy fire more sparingly *per decision*. The raw per-episode count (Plot 4a) is actually higher on the modified environment, but only because those episodes run longer (~ 344 - 395 vs ~ 275 - 339 steps): the agent needs more time and must re-issue commands to compensate for the ~ 15 % that mis-fire. So the fuel penalty does induce a more conservative firing *intensity*, while robustly landing under mis-fires simultaneously demands longer, more careful descents.

4. Which algorithm performs better under stochastic engine failures? Is this consistent with the theoretical advantage of DDQN over DQN?

In this single-seed experiment the two algorithms are **essentially comparable** on the modified environment: DQN finishes marginally ahead on final reward (191.05 vs 186.56) and landing rate (83 % vs 75 %), while DDQN attains a marginally higher *peak* 100-episode reward (~ 194 vs ~ 193). Crucially, DDQN does deliver its theoretically-predicted effect - **much lower, less-inflated Q-values** under failure (AvgQ ~ -54 vs $\sim +5$) - confirming reduced over-estimation. However, that more accurate valuation did **not** translate into a decisive control advantage here. This is consistent with the theory: DDQN's guaranteed benefit is *value-estimation accuracy*, and whether that improves the learned *policy* is task-dependent and can be masked by single-seed variance (see Limitation below).

5. One limitation and one improvement.

Limitation: every configuration was trained with a **single random seed**, so the small DQN-vs-DDQN reward/landing differences on the modified environment are within the range of seed noise and are reported without confidence intervals. *Improvement:* repeat each configuration over several seeds (e.g. 5-10) and report the **mean +/- standard deviation** (or a shaded confidence band) for every curve, so the comparison becomes statistically robust. Complementary

improvements include a prioritised experience-replay buffer and a short hyper-parameter sweep (learning rate, target-update rate, network width).

Conclusion

We built a correct `gym.Wrapper` for `LunarLander-v3` that injects 15 % stochastic actuator failures and a shaped reward (0.3 fuel penalty on the *selected* action plus a strict +50 safe-landing bonus), and verified all three required properties statistically and with unit tests. Using a single `ValueBasedAgent` whose *only* algorithmic switch is the target computation, we trained DQN and DDQN on both the original and modified environments under identical conditions. The evaluation shows the expected behaviour: the DQN-DDQN gap in predicted Q-values **widens under stochastic action failure** ($|\text{gap}| \sim 24 \rightarrow 59$), with DDQN producing markedly lower, less-inflated values - the signature of reduced over-estimation. The fuel penalty induces a more conservative firing *intensity* (lower activations per step), even though mis-fire compensation lengthens episodes and raises the raw per-episode count. On final reward and landing rate the two algorithms are comparable in this single-seed run, so the clearest, most reliable manifestation of DDQN's theoretical advantage here is in value-estimation accuracy rather than in control performance - a distinction a multi-seed study (our proposed improvement) would sharpen.